UNIVERSITY OF CALIFORNIA

Los Angeles

The Indo-European Primary *t*-Stems

A dissertation submitted in partial satisfaction of the

requirements for the degree Doctor of Philosophy

in Indo-European Studies

by

Aurelijus Vijūnas

2006

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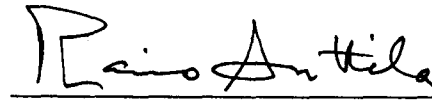
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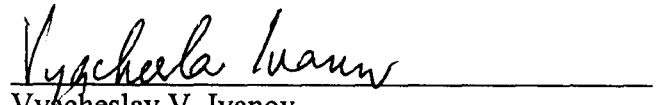
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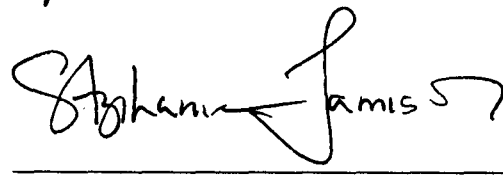
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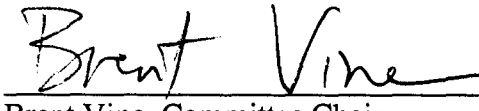
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Any mistakes in this work are either mine or Murphy's.

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ABSTRACT OF THE DISSERTATION

The Indo-European Primary *t*-Stems

by

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The present dissertation is dedicated to the study of Indo-European nominal primary *t*-stems. By "primary *t*-stems" I mean such non-compound formations which from the Indo-European point of view can be divided into an unextended root and a suffix of the structure *-t-* or *-Vt-* (cf. **doh*$_3$*-t-* 'giving' ~ √*deh*$_3$- 'give'; **leuk-ot-* 'shining' ~ √*leuk-* 'shine'), and such compound formations the final member of which consists of a bare root and a suffix *-t-* (cf. **ped-h*$_1$*i-t-* 'going by foot' ~ √*ped-* 'foot' + √*h*$_1$*i-* 'go' + *-t-*; **peri-tḱi-t-* 'dwelling around; surrounding' ~ **peri* 'around' + √*tḱei-* 'dwell' + *-t-*). There exists no secure evidence for primary *t*-stem compounds built with suffixes of the *-Vt-* structure.

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In this dissertation, I study the reflexes of the Indo-European primary *t*-stems in the various Indo-European dialects, reconstructing their earlier structure and semantics. In addition to the reconstructional part of the dissertation, I also observe the development of the Indo-European primary *t*-stems, describing patterns of their development in the individual dialects and in the Indo-European protolanguage itself.

Primary *t*-stems are normally built to verbal roots (cf. the examples shown above), and at least originally denoted verbal abstracts (action nouns). In the course of development, *t*-stems underwent various morphological and semantic changes. Probably as early as late Indo-European, many simplex *t*-stems were reinterpreted as root nouns, whereas *t*-stem compounds were reinterpreted as root compounds (see chapters III – IV). At the dialectal stage, the reflexes of these *t*-stems followed the trends of the development of root nouns and root compounds.

Whereas a certain degree of allomorphy can be reconstructed for the Proto-Indo-European *t*-stems, only its traces can be seen in the dialects (cf. Latin *nox* 'night' ~ Hittite *nekuz* 'of night' ← PIE **nok^{ŭ}-t-/*nek^{ŭ}-t-*; Avestan *nap-āt-əm* 'nephew' [acc. sg.] vs. *nap-t-ō* [gen. sg.] ← PIE **nep-ot-/nep-t-*). In most instances, the original allomorphy was eliminated.

As far as the semantic development is concerned, *t*-stems generally lost their original abstract meaning and developed a more concrete meaning, e.g. La. *dōt-* 'dowry' (< **doh$_3$-t-* 'giving'), Greek δαιτ- 'meal' (< **deh$_2$i-t-* 'distributing') etc.

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I. Introduction

The present dissertation is dedicated to the study of the Indo-European nominal primary simplex and compound *t*-stems. By the term "primary *t*-stem" I primarily mean nominal derivatives built from a root and the suffix *-t-* or *-Vt-*, e.g., IE **nok^{u}-t-* 'night' (< **nog^{u}-* 'be dark' + *-t-*) or IE **leuk-ot-* 'light' (< **leuk-* 'shine' + *-ŏt-*). However, in certain instances it is hard to tell whether one is dealing with a true derivative *t*-stem or a root noun enlarged by a root enlargement *-t-*. This is especially true of the Indo-Iranian substantives of the *stút-* 'praise' type, which theoretically can be analyzed either as original *t*-derivatives or as enlarged root nouns. Also in Greek, where the formant *-t-* appearing in the oblique cases of multiple nouns is often secondary by nature, cf. ὕδα-τ-ος 'water' (gen. sg.; ← IE **ud-n-és*), it is necessary to determine first whether one is dealing with a secondarily added morphologically empty *t*, or an original suffix.

The term "*t*-stem" will be generically used in the present dissertation to refer both to nouns like **nók^{u}-t-* or **leuk-ot-* in which the formant *-(V)t-* can be plausibly interpreted as a suffix, and to compound and simplex nouns of the Indo-Iranian *stút-* and *deva-stút-* type, although, as will be shown in chapter IV below, they can be interpreted in more than one way. However, in such cases where it may be important to make a clearer distinction, more specific terms like "derivative *t*-stem" or "enlarged root noun" will be used.

Technically, also certain root nouns the roots of which end in the consonant *-t-* are *de facto t*-stems, cf. IE **u̯et-* 'year'. However, in the present dissertation, I will only be dealing with substantives in which the formant *-(V)t-* can be separated from the root,

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whether it is a derivational suffix or a root enlargement. In this study, I am not going to discuss secondary and other non-primary *t*-stems either, since the study of all *t*-stems (primary and non-primary) would involve a vast amount of morphologically, derivationally and semantically very diverse data, such as the possessive *ment*-stems, present active participles in *-(o)nt-*, nouns of the Sanskrit *harí-t-* 'tawny mare' or the Indo-European **mel-i-t-* 'honey' (neuter gender) type, decasuative secondary *t*-stems (e.g. Hi. *išpan-t-* 'night' < loc. **k^{u}sp-en-* + *-t-*),[1] numerals (**dek̑m̥-t* '10'), etc., that it would be much more appropriate to split it into two or several dissertations. However, if for any reason it becomes necessary, non-primary *t*-stems will be brought into the discussion.

I.1.

Although the Indo-European nominal *t*-stems have interested scholars for a long time, they still have not received a thorough historical treatment. The somewhat limited scholarship about *t*-stems is scattered around in short articles, book chapters spanning from the nineteenth century until the present time,[2] and is not always easy to come by. Part of the reason for this is that the same thing may be said of *t*-stems themselves, especially non-compound primary formations, the discussion of which will constitute the major part of the present study. *T*-stems of this type are not well represented in most Indo-European languages as a synchronic morphological group. In many languages,

[1] See N. Oettinger (1982: 239), E. Rieken (1999: 146-7), A. J. Nussbaum (2004a).
[2] See G. Curtius (1855), M. Bréal (1875), Rieken (1999), B. S. Irslinger (2002), Nussbaum (2004a and 2004d) etc.

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historical *t*-stems slowly merged with root nouns and/or certain vocalic stems, e.g., *i*-stems, cf. Lithuanian *naktìs* 'night' (< IE **nók^{u}-t-*), which from the point of view of synchronic morphology is no longer distinguishable from *avìs* 'sheep' (< **h$_2$óu̯-i-*), and both are declined like *i*-stems. In Latin, the historical derivative *t*-stem *dōs* 'dowry' (stem **dōt-* < **doh$_3$-t-*) is declined much like the original root noun *vōx* (< **u̯ōk^{u}-*), etc.[3] Even in Greek, where *t*-stems enjoyed a certain productivity, some archaic *t*-stems became obscured by later grammatical reformations, cf. Gk. Περσεφόνη (divine name; older versions are Φερρόφαττα, Περρόφαττα etc.), which is a reformed *t*-stem compound with the final member **-g^{u}hn̥-t-* (~ IE **g^{u}hen-* 'beat, strike').[4] In the Indo-Iranian branch, one faces a different problem: a number of nouns exhibit a formant *-t-*, e.g., Vedic *stút-* 'praise', Avestan *stūt-* 'id.' (~ √*stu-* 'praise'), Ved. *mít-* 'post' (~ √*mi-* 'fix') etc., but since a *t*-enlargement rule was operating in Indo-Iranian, the interpretation of the formant *-t-* in most of these *t*-stems is impossible.

Perhaps never a numerous class, simplex *t*-stems have practically disappeared in most languages, having left only poor traces. Compound *t*-stems are more common, but they mostly occur in Indo-Iranian and Greek. In other languages, they are either rare or not attested at all. As mentioned by Rieken in her brief introduction of Indo-European *t*-stems in her *Untersuchungen zur nominalen Stammbildung des Hethitischen* (1999: 83), *t*-stems are so rare, that many of them occur only in a single language.[5] In an

[3] For a more detailed discussion of the Indo-European noun **nók^{u}-t-* and the history of La. *dōs* see sections III.2 – III.3 and III.5.2 below.
[4] Cf. R. Anttila (2000: 164ff.). The connection with the root **g^{u}hen-* 'swell' (cf. GEW II.517f. and P. Chantraine 1968: 889) is semantically problematic.
[5] Rieken (1999: 83).

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obvious way, this makes their study a difficult task. However, as shown by her subsequent more detailed overview of *t*-stems, at least a part of them obey some general rules and can be fitted into certain well-known Indo-European morphological frames which have already been more or less well established for other nominal formations.

I.2.

In her study, Rieken reconstructed four classes of Indo-European primary *t*-stems: acrostatic, e.g. IE **nók$^{\mu}$-t-/*nék$^{\mu}$-t-* 'night'; hysterokinetic, e.g. Gk. ἕρπ-ητ- 'shingles' (< **serp-ēt-*); proterokinetic, e.g. Gothic *mitade* 'measure' (gen. pl.; < [Proto-Germanic] **metað-*) or La. *teget-* 'mat, cover' (< **teg-ōt-*); and amphikinetic (holokinetic) *t*-stems, e.g. IE **nep-ōt-* 'grandson; nephew; offspring'.[6]

A very similar analysis of the Indo-European *t*-stems can be found in B. Irslinger's *Abstrakta mit Dentalsuffixen im Altirischen* (Irslinger 2002), where she provides a brief overview of Indo-European *t*-stems, for the most part following Rieken's reconstructions and the classification of (non-compound) *t*-stems into acrostatic, protero-, hystero- and amphikinetic.[7] However, unlike Rieken who was primarily concerned with the classification of the various *t*-sems into the Proto-Indo-European ablaut classes on the one hand and primary vs. secondary on the other hand, in her book Irslinger classified the various *t*-stems according to the Indo-European dialects that they were preserved in. Thus, in Irslinger's *Abstrakta* one can find a discussion of Anatolian,

[6] See Rieken (1999: 84-93).
[7] See Irslinger (2002: 45-6).

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Greek, Italic, Germanic, etc. *t*-stems. A detailed discussion is dedicated to the Celtic and especially Old and Middle Irish *t*-stems.

Although Rieken and Irslinger made the first major attempt to discuss various Indo-European *t*-stems in a systematic way and classify them, their analyses and Proto-Indo-European reconstructions are not entirely satisfying. Rieken's reconstruction of the Proto-Indo-European proterokinetic ablaut pattern as R(é)-S(ō)-E(z)/R(z)-S(é)-E(z), unlike the more common reconstruction as R(é)-S(z)-E(z)/R(z)-S(é)-E(z) has already been criticized by P. Widmer.[8] Also the reconstruction of the hysterokinetic pattern as R(e)-S(ē)-E(z), cf. Gk. ἕρπ-ητ- 'shingles' (< **serp-ēt-*) above, is abnormal in that the root exhibits the *e*-grade instead of the generally reconstructed zero grade, which is directly attested in such hysterokinetic formations as Gk. ἀστήρ (< **h₂s-tḗr* 'star' [← 'burner']),[9] or Gk. ὑμήν 'membrane', Hittite *šumanza(n)* 'thread' (< **si̯uH-mḗn*).

A few words need to be said about Rieken's and Irslinger's treatment of the Indo-Iranian *stút-* type nouns. Rieken proposed that these nouns originally may have been inflected in the same way as the acrostatic **nók^{u}-t-* 'night' (weak stem **nék^{u}-t-*, cf. Hi. *nekuz*), but in the course of time adopted the mobile ablaut of root nouns.[10] Thereby, their original ablaut **stóu-t-/*stéu-t-* would have been remade as **stóu-t-/*stu-t-́*, and eventually the weak stem would have been generalized. Although such a theory is in certain ways attractive,[11] it demands certain explanations as well, e.g. it needs to be

[8] See Widmer (2004: 55-6).
[9] Cf. H. Eichner (1978: 161, fn. 76), A. Bomhard (1986: 191-2).
[10] Rieken (1999: 85).
[11] See chapters III – IV below.

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explained how these acrostatic derivative *t*-stem nouns acquired this new quasi-amphikinetic ablaut **stóu-t-/*stu-t-́* with the accent appearing in the root and the ending, but skipping the suffix.[12] When the proposed development for the reconstructed acrostatic **stóu-t-* et alia is compared with the development of other structurally comparable formations, e.g. acrostatic *u*-stems, one can see that the latter tend to be reformed as proterokinetic formations, cf. Skt. *jā́n-u* 'knee' (nom. sg.; < **ĝón-u*) vs. *jñóḥ* 'id.' (gen. sg.; < **ĝn-éu-s* ← ***ĝén-u-s*). Therefore, one should expect that in the course of development, the acrostatic **stóu-t-* should have developed into a protero **stu(u̯)-et-* which, however, is not attested. Rieken's comparison of the *stú-t-* type *t*-stems with root nouns requires special comments too, since root nouns and the derivative *t*-stems built with a suffix are morphologically quite different.

I.3.

Many of the *t*-stems discussed in Rieken (1999) were more recently interpreted in a rather different way by Nussbaum, who proposed a simpler system of primary *t*-stems consisting only of acrostatic and amphikinetic formations, and analyzed many of Rieken's hystero- and proterokinetic *t*-stems as secondary formations built to thematic stems.[13] Thus, comparably to Rieken, Nussbaum interpreted nouns like IE **nók$^{u̯}$-t-/ *nék$^{u̯}$-t-* 'night',[14] **sós-t-* 'sleep (?)'[15] and several others as reflexes of original

[12] Irslinger calls the reconstructed pattern **stóu-t-/*stu-t-ós* "amphidynamic" (i.e., „amphikinetic", see Irslinger 2002: 45). However, the amphikinetic inflectional pattern is generally reconstructed in a different way, namely as R(é)-S(ŏ)-E(z)/R(z)-S(z)-E(é), cf. J. Schindler (1975b: 262-3), Widmer (2004: 54), etc.

[13] See Nussbaum (2004a).

[14] Cf. *e*-grade in Hi. *nekuz* 'of night' (< **nék$^{u̯}$-t-s*, old gen. sg.; Schindler 1967).

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acrostatic *t*-stems, whereas for the Indo-European noun **nep-ot-* 'offspring, grandson (vel sim.)'[16] he reconstructed the amphikinetic type of inflection.[17] However, instead of interpreting such forms as La. *teg-et-* 'mat', *anti-sti-t-* 'priest', Av. *fra-car-āt-* 'moving forth', Gk. ἕρπ-ητ- 'shingles' etc. as **teg-* 'cover' + **-et-*, resp. **pro-kʷe/ol[H]-* + **-ēt-*, resp. **h₂enti-sth₂-* + **-t-*, resp. **serp-* + **-ēt-* etc., Nussbaum explained such formations as secondary *t*-stems built to thematic stems, cf. *teg-e-* + *-t-* (> La. *teget-*), **pro-kʷe/ol[H]-e-* + **-ot-* (> Av. *fra-carāt-*), (possibly) **h₂enti-sth₂-e-* + **-t-* (> La. *anti-stit-*), **serp-e-* + *-ot-* (ἕρπ-ητ-), etc.[18]

Nussbaum's interpretation appears to be more attractive than the older theory, as, first of all, it does not require the reconstruction of primary formations that from the point of view of Proto-Indo-European are irregular, cf. Rieken's "hysterokinetic" **CeC-ēt-* with two *e*-grades or the "proterokinetic" **CeC-ōt-* with the *o*-grade in the suffix. Whereas some of Rieken's proterokinetic *t*-stems are better explained as reflexes of ancient amphikinetic R(é)-ot- formations, cf. Go. *mitaþs* or Old English *mægeð* 'maiden',[19] many others can be plausibly explained as secondary *t*-stems built to (often) verbal thematic stems, cf. the examples shown earlier in this section.

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[15] No *e*-grade **sés-t-* attested (cf. Nussbaum 2004a).
[16] See a more detailed discusssion in section V.10.1 below.
[17] Nussbaum (2004a).
[18] Nussbaum, ibid.
[19] See a more detailed discussion of the history of these nouns in sections V.10.2 and V.10.4 below.

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I.4.

Nussbaum's analysis leaves only a small number of non-compound primary *t*-stems in the Indo-European languages. Among them, only a few are attested in more than one language (e.g., **nók^{u}-t-* 'night' and **nép-ot-* 'offspring, grandson' and several others), whereas several have root etymologies that are not entirely clear (e.g. Ved. *nṛ́t-* 'dance', Gk. θητ- 'day-labourer'), and still others, primarily the Indo-Iranian nouns of the *stút-* type, may be either historical *t*-stems or enlarged historical root nouns.

I essentially agree with Nussbaum's adjustments to Rieken's original classification of the Indo-European *t*-stems, and in this dissertation, I will operate with two major types of primary simplex Indo-European *t*-stems: acrostatic and amphikinetic. Following the Proto-Indo-European ablaut system as described by Schindler (1972b and 1975c) and further elaborated by Nussbaum (2004a and 2004d), the following features should be ascribed to the Proto-Indo-European acrostatic *t*-stems: 1) *ó/é* alternation in the root, 2) the non-alternating suffix *-t-*, and 3) are built to verbal roots (type **nók^{u}-t-/*nék^{u}-t-*).

To the acrostatic class one should probably also attribute various derivative *t*-stems of the Italic **grā-t-* 'praising' type which are built in the same way, except that they exhibit the zero grade in the root, cf. La. *grātēs* 'thank' (< IE **g^{u}ṛH-t-* ~ **√g^{u}erH-* 'praise'), Middle Hittite *karitt-* 'flood' (< **ĝri-t-* ~ **√ĝrei-* 'flow'), Gk. λῖτ- 'linen cloth' (< [probably] **liH-t-* ~ **√leiH-* 'wrap') etc. There are reasons to think that

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originally the *t*-stems of the **grā-t-* type had the *ó/é* alternation as well, and were identical with the *t*-stems of the **nókʷ-t-* type.[20]

In the existing pool of data, no evidence for acrostatic *ḗ/é* type *t*-stems has been found so far. In this respect, primary acrostatic *t*-stems appear to be different from other primary acrostatic stems, such as *i-*, *u-* or *r/n*-stems, all of which exhibit both the *ó/é* and the *ḗ/é* acrostatic types, cf. Gk. δῆρις 'battle' (< **dḗr-i-*), OIr. *fiu* 'worth(y)' (< **Hu̯és-u-*), Gk. ἦπαρ, Ved. *yákr̥-t* 'liver' (< **[H]i̯ḗk-r̥/n-*) etc.[21] Acrostatic *t*-stems also behave differently from acrostatic *s*-stems which form *ḗ/é* acrostats only, cf. Gk. γῆρας 'old age', γέρας 'gift of honour' (< **ĝḗrh₂-es-/*ĝérh₂-es-*), Old Irish *síd* 'divine dwelling' (< **sḗd-es-*), etc.[22]

The Indo-European amphikinetic *t*-stems had the characteristic alternation R(é)-S(ŏ)/R(z)-S(z)-E(é), cf. Younger Avestan *napå̊* 'grandson' (< [ultimately] **nep-ōt-*) vs. *naptō* 'id.' (gen. sg.; < **nep-t-é/ós*).[23] Beside a reflex of an original primary amphikinetic *t*-stem, no thematic verbal stem should be attested in the same dialect, because otherwise the given amphikinetic *t*-stem would become ambiguous: it could be interpreted either as an original primary amphikinetic *t*-stem, or as a secondary de-thematic formation (Nussbaum's *tege-* + *-t-* type, see above). According to these restrictions, Go. *liuhaþ* 'light' should be interpreted as a reflex of a true amphikinetic *t*-stem **leuk-ot-* since no thematic stem **leuk-e/o-* is attested in Germanic.

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[20] See chapter III.
[21] See Widmer (2004: 50-1).
[22] See Widmer (loc. cit.) and Schindler (1975b: 267).
[23] In the paradigm of Avestan *napāt-*, the root allomorph *nap-* with the full grade has been generalized.

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In different cases, e.g. when a thematic stem is attested beside an amphi-looking nominal *t*-stem, this *t*-stem can still be interpreted as an original amphikinetic formation provided it is isolated or built in a non-productive way, cf. Go. *mitaþs* 'measure' (< **med-ot-*; thematic **med-e/o-* is attested, but *mitaþs* belongs to a non-productive class).

In the course of development, the original shape of the Proto-Indo-European *t*-stems changed. Whereas a sub-class of acrostatic *t*-stems developed forms with the zero grade and practically eliminated the *e*-grade, various changes affected the original inflection of amphikinetic *t*-stems as well. The original ablaut alternations were simplified, and occasionally these *t*-stems became thematized.[24]

In addition to this, nouns that originally were not *t*-stems acquired the element *t* secondarily, becoming neo-*t*-stems which could have a totally different ablaut and structure from the original acrostatic and amphikinetic *t*-stems. Among such new formations one could mention Ved. *yákṛt* 'liver' and *śákṛt* 'excrement' (both old *r/n*-stems), Hi. *išpant-* (< **k*^{u}*sp-en-* + *-t-*), Gk. δουρ-ατ- 'tree' (← **dór-u*) etc. Such neo-*t*-stems will not be studied in this dissertation, since they do not reflect original Indo-European primary *t*-formations.[25]

There is a possibility that also a number of Indo-Iranian simplex and compound *t*-stems belong to these neo-*t*-stems. However, some of them probably should be reconstructed as original derivative amphikinetic or acrostatic *t*-stems.[26]

[24] See chapter V.

[25] For the study of Greek neo-*t*-stems of the δουρατ- type see Anghelina (2004).

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I.4.1.

Several words need to be said about the Greek monosyllabic *t*-stems. From the structural point of view, they are indistinguishable from historical root nouns. The nouns of both types can have the lengthened, the normal or the zero grade in their roots, and the same root Auslaut (*-V*, *-R* or *-C*), cf. Gk. φώρ 'thief' (root noun; ~ IE **bher-* 'carry'), θώς 'jackal' (root noun)[27] vs. πλώς 'mullet (type of fish)' (*t*-stem; ~ **pleh*$_3$- 'swim'),[28] λίς/λῖς 'lion' (root noun)[29] vs. λῖ-τ-α 'linen cloth' (acc. sg./pl.; *t*-stem; ~ probably IE **leiH-* 'wrap'),[30] etc. Apparently, the only difference between them is that some of them acquire a -τ- in their oblique stem, whereas others do not.

In this respect, the Greek monosyllabic *t*-stems are different from the Indo-Iranian *t*-stems, since in Indo-Iranian, *t* is obligatory to nominal stems of a particular shape (viz., when they end in *i*, *u*, *ṛ*). Whereas the generalized *t*-addition rule in Indo-Iranian makes a precise historical analysis of a given *t*-stem often impossible, the absence of such a rule in Greek makes the historical analysis of monosyllabic Greek *t*-stems possible, provided the root etymology of a given *t*-stem is known. Unlike neuter historical *men-*, *s*-stems and nouns of several other types, historical root nouns in Greek generally do not acquire a secondary -τ- in the oblique cases, cf. the examples shown above, as well such historical root nouns as μῦς 'mouse' (gen. μυ-ός) and βοῦς 'bull, cow' (gen. βο-ός). Therefore, in those cases, where a synchronic root noun with a clear root

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[26] See chapters IV and V.
[27] The root etymology is not entirely clear, see Schindler (1972a: 85).
[28] See a more detailed discussion in section III.6.4.1 below.
[29] For the discussion of the root etymology see Schindler (1972a: 90) See also É. Masson (1967: 85-7) and P. Chantraine (1968: 643).

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etymology has a -τ- in the oblique stem already in the earliest texts, this -τ- should probably be interpreted as a historical suffix.

I.5.

Compound *t*-stems can be primary and non-primary as well. Having eliminated compounds of the Avestan *fra-carāt-* and Latin *anti-stit-* type as secondary (see section I.2 above), the remaining primary compound *t*-stems can be divided into two groups: in the first group, the roots of the final members reflect the historical zero grade (type *X-R*[z]-). This type is well attested in several branches of the Indo-European family, cf. La. *com-i-t-* 'companion' (< **kom-h_1i-t-* 'going together'), Gk. ὠμο-βρω-τ- 'eating raw flesh' (< **-$g^{u}_{\text{r}}h_3$-t-*), Ved. *pari-kṣí-t-* 'dwelling all around, surrounding' (< **peri-kṗi-t-* ~ Gk. περι-κτί-τ-αι 'neighbours'), Av. *zauuanō-srū-t-* 'listening to the prayers' (< **-k̂lu-t-* ~ Ved. *havana-śrú-t-* 'id.') etc.

In the other group, the final member has the full grade of the root. This type is much rarer, and not all examples are entirely clear. The most secure example of this type is the Latin *t*-stem *sacerdōt-* 'priest' (< **sakro-doh$_3$-t-*),[31] whereas the rest can be interpreted in more than one way, cf. La. *locu-plēt-* 'wealthy' (< **-pleh$_1$-t-* or **-pl[h_1]-e-ot-*, or ← **-pl̥h$_1$-t-*),[32] Gk. ἀ-γνω-τ- 'ignorant' (< **-ĝnoh$_3$-t-* or **-ĝn̥h$_3$-t-*), χερ-νη-

[30] See section III.6.3 below.
[31] See K. Strunk (1995: 226), Nussbaum (1999: 396-8).
[32] For the reconstruction of **-pl(h_1)-e-ot-* see Nussbaum (2004a). Strunk (1995) reconstructed the final member of *locuplēt-* as **-pl̥h$_1$-t-*.

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τ- 'poor' (< *-[*s*]*neh*$_1$-*t*- or *-[*s*]*n̥h*$_1$-*t*-), Avestan *tara-dāt-* 'prevailing' (< *-*dheh*$_1$-*t*- or *-*dhh*$_1$-*e-ot*-)[33] etc.

An exception would appear to be the Hittite noun *kapirt-* 'mouse', which possibly reflects an older **kom-bhēr-t-* with the lengthened grade of the root.[34] However, the form *kapart-* 'id.' which appears beside *kapirt-* can be reconstructed as **kom-bhr̥-t-*, i.e. with the root of the final member in the more common zero grade. This anomalous compound will be treated in section III.4.4 below.

Just as was the case with simplex *t*-stems, the Indo-Iranian *t*-stem compounds that have the final members built to roots ending in the short vowels *-r̥*, *-u*, *-i* ("group I") theoretically can be interpreted in two ways: these final members can either be reconstructed as original derivative *t*-stems or as verbal roots secondarily enlarged by the consonant *t* due to the Indo-Iranian *t*-enlargement rule. For most of these compounds, further analysis is impossible due to the absence of parallels in other Indo-European dialects. However, some of the *t*-stems occurring as the final members of Indo-Iranian compounds are also attested as *t*-stems in other languages, cf. Ved. *pari-kṣí-t-* and Gk. περι-κτί-τ-αι mentioned earlier, also Ved. °*-í-t-* (cf. *div-í-t-* 'going to the heaven') beside La. °*-i-t-* (cf. *com-i-t-*), Ved. °*-bhŕ-t-* (cf. *vajra-bhŕ-t-* 'carrying *vajra*'), Av. °*-bərə-t-* (cf. *ā-bərə-t-* 'water carrier') beside Hi. *kapart-* (most probably < *-*bhr̥-t*-) etc. and may be old.

The ablaut of the final members of compound *t*-stems shows similarities with the ablaut of simplex acrostatic *t*-stems. In both classes, roots of the *CeC* structure exhibit

[33] Nussbaum (2004a) reconstructs a secondary de-thematic *t*-stem.

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the full grade of the root, whereas the roots of the *CeR(C)* structure normally appear in the zero grade. In either class, the element *-t-* does not show any ablaut alternations.

I.6.

One additional important issue that needs to be addressed in this dissertation is the interpretation of the formants *-t-* and *-ot-*. In the course of years, the (pre)history of these formants has been interpreted in several rather different ways.

A. Schleicher (1871), W. Streitberg (1894), A. Meillet (1937) and F. O. Lindeman (1986) claimed that the non-alternating *-t-* derived from sequences **-Vt-* or **-tV-*.[35] In the course of development, the vowel would have been lost by various phonological laws, leaving the bare consonant *-t-*.

According to a different interpretation found in works of M. Bréal (1875), Rieken (1999) et alii, the non-alternating *-t-* has never had a vowel attached to it, i.e. has always been a plain *t*.

According to two still different theories independently developed by J. Kuryłowicz (1964) and Oettinger (2001), the element *-t-* is epenthetic, and was added to certain morphemes for (mor)phonological reasons.

As far as the Indo-European formant *-ot-* is concerned, an alternative historical interpretation has been proposed by J. E. Rasmussen, who claimed that at least in some

[34] See a more detailed discussion in section III.4.4 below.
[35] For a more detailed discussion of these views, see chapter II.

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cases, the Indo-European *-ot-* reflects a pre-Indo-European **-*oc*- with an affricate *[t͡s].[36]

In chapter II, I would like to provide a survey of the major interpretations of the formant *-t-*. After the discussion of the various theories about this formant, I will move on to the study of different types of simplex and compound *t*-stems (chapters III – V).



[36] See Rasmussen (1994) and sections II.9 – II.9.2 below.

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II. The History of the Formants *-t-* and *-ot-*

II.1. Bréal's Interpretation of the Suffix *-t-*

The opinion that the formant *-t-* seen in a large number of Indo-European nominal formations is a non-ablauting suffix existed as early as the 19th century. In his article "Origine du suffixe participial *ant*", Bréal interpreted this formant *-t-* as an original Indo-European present participle suffix, whereas the much more familiar shape of this participle, viz. *-ant-*, according to Bréal, was the result of certain phonetic changes and reinterpretations of morphemic boundaries. According to Bréal, the participial suffix *-t-* is still to be seen in its original shape in multiple compounds, such as Skt. *viśva-jí-t-* 'conquering all', *pari-srú-t-* 'fermenting', Gk. ὠμο-βρω-τ- 'eating flesh', etc. In the course of development, the original suffix *-t-* acquired an *-n-* by means of epenthesis, comparably to the nasal *ṃ* seen in certain Sanskrit neuter formations, cf. *mánas-* 'mind' → *mánāṃsi* (nom./acc. pl.) or *dhánus-* 'bow' → *dhánūṃsi* (nom./acc. pl.).[37] The vowel *-a-* in the suffix *-ant-* was explained by Bréal as the thematic vowel, which would have been attracted to the participial suffix *-nt-* by means of reinterpretation of morpheme boundaries.[38]

Apparently, Bréal must have interpreted the formant *-t-* in the above mentioned compounds as a participial suffix because the meaning of these compounds is comparable to the meaning of active participles. Although this observation raises certain potentially interesting implications concerning the productivity of *t*-stems in certain languages, the similarity of the functions of active participles on the one hand

[37] Bréal (1875: 190).
[38] Bréal (1875: 190-1).

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and *t*-stem compounds on the other hand is only superficial. Adjectival compounds often have participial meaning, as their second (or final) member is often built to a verbal root, cf. Skt. *bhūri-dā́van-* 'giving much' (√*dā-* 'give'), *divi-kṣayá-* 'dwelling in the sky' (√*kṣi-* 'dwell'), *bhadra-vādín-* 'uttering an auspicious cry' (√*vad-* 'speak'), *sutaṃ-bharā́-* 'receiving pressed Soma' (√*bhar-* 'carry'), etc. As can be seen from such examples, in order to have participial meaning, a compound does not need to have the suffix *-t-*.

In deriving the participial suffix *-ant-* from an original *-t-*, Bréal also has to solve the problem of the nasal, which he interprets as epenthetic.[39] If the nasal in the participial suffix *-ant-* is epenthetic, it has to be explained why compounds like *viśva-jí-t-* and ὠμο-βρω-τ- did not acquire it. In addition to this, epenthesis usually has a fairly obvious phonetic reason,[40] but no good reason for the epenthetic insertion of a nasal before the suffix *-t-* can be indicated.

II.2. Schleicher's Weakening Theory

In a different historical interpretation of the suffix *-t-*, A. Schleicher proposed that the original shape of this suffix was *-ti-* (1871: 435), comparing the "alternation" of *-ti-* : *-t-* with that of the suffixes *-tāt-* : *-tāti-*. According to him, the suffix *-ti-* would have lost its *-i-*, thereby becoming a plain *-t-*. However, such an interpretation cannot explain why there still exist so many *ti-* and *tāti*-stems in Indo-European. If the "original" *-ti-* and *-tāti-* had been "weakened" to resp. *-t-* and *-tāt-*, there should exist much fewer *ti-*

[39] Bréal, ibid.
[40] See several examples in section II.8 below.

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and *tāti*-stems; however, if one looks at Vedic alone, one can see many such formations, and their primary stress can fall on practically any syllable, cf. *sóma-pī-ti-* 'Soma sacrifice', *hásta-cyu-ti-* '(quick) motion of the hand', *toká-sā-ti-* 'acquisition of offspring', *devá-tāti-* 'divine service', *paro-gavyū-tí-* 'further than a *gavyūti*', *su-nī-tí-* 'guiding well', etc. As can be seen from these examples, no matter on which syllable the stress falls, the suffix does not undergo any weakening.

II.3. Streitberg's Weakening Theory

W. Streitberg (1894) had an essentialy similar idea to that of Schleicher, the only difference being that instead of *-ti-*, he suggested that the original shape of the suffix was *-to-*. Such a reconstruction with an *o*-grade may well stem from the assumption that in the protolanguage there were in general more full grades than in the attested daughter languages, and that any morphological element (a root, a suffix, etc.), attested in the Indo-European languages in the zero grade, should be reconstructed in the full grade version. Accordingly, Streitberg reconstructed *t*-stem compounds with a suffix *-to-*, cf. Indo-European **medhu̯o-ker-to-* (> Ved. *madhu-kṛ-t-* 'making honey; bee').[41] Streitberg's theory was criticized by H. Hirt, who pointed out that there was no reason for the suffix *-to-* to become weakened to *-t-* in compounds. Just as was the case with compound *ti*-stems and with *tāti*-stems, also here, there exist a large number of compound *ta*-formations with all sorts of stress patterns, cf. Skt. *á-jā-ta-* 'unborn, not yet born', *náva-jā-ta-* 'fresh, new', *devá-jā-ta-* 'race of gods', *puru-jā-tá-* 'variously

[41] Streitberg (1894: 38).

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manifested', *sú-yuk-ta-* 'well-joined', etc., showing both a stressed *-tá-* (< **-tó-*) and a non-stressed *-ta-* (< **-to-*), neither of which is reduced to *-t-*.

Probably the main reasons behind the thought that the formant *-t-* originated in compounds by means of weakening of certain suffixes is that 1) these scholars assumed that weakening *could* happen in this environment, and 2) such weakening could *not* be fabricated for simplex nouns of the IE **nók^{u}-t-*, La. *dō-t-* type. However, well attested *to-* and *ti*-stem compounds already show that the weakening theory is impossible.

II.4. Meillet – Lindeman's Weakening Theory

According to yet another weakening theory, the formant *-t-* derives from an older suffix *-et-*. This interpretation can be found in A. Meillet (1937: 268), and was recently revived by F. O. Lindeman (1986, especially p. 148). Meillet and Lindeman proposed that the suffix *-t-* was a weakened version of the suffix *-et-*. As for the conditions under which the weakening had to take place, Meillet did not provide a clear account for it. However, Lindeman suggested that the suffix **-et-* regularly developed into *-t-*. As an example, Lindeman adduced the development of the Hittite noun *šašt-* 'sleep', which he reconstructed as IE **sós-et-*.[42] But the unstressed vowel *e* did not become lost in Hittite, cf. such examples as Hi. *nepiš-* 'sky' (< IE **nébh-es-*), *pēran* 'in front' (< PA **pérem*)[43] etc. The reconstructed **sós-et-* would have developed into *×šāšit-* vel sim.

[42] "...la chute du vocalisme *posttonique* dans (nom.) **sós + et + s > *sós-t-s*, etc., est parfaitement régulière ("l'ablaut Kretschmérien")..." (Lindeman 1986: 148, fn. 8).
[43] For the latter, see H. C. Melchert (1994: 105).

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On the contrary, if *šašt-* is reconstructed as an originally acrostatic formation **sós-t-*,[44] the Hittite outcome *šašt-* is regular.

II.5. *-T-* as the Suffix of Verbal Abstracts

According to a different interpretation, the non-alternating suffix *-t-* originally was used to form verbal abstracts (primarily action nouns), built from verbal roots.[45] Among such original verbal nouns it is possible to mention the Indo-European noun **nók^{u}-t-* /nóg^{u}-t-/ 'night' (~ IE **neg^{u}-* 'be dark'),[46] La. *dō-t-* 'dowry' (~ IE **deh$_3$-* 'give'), Hi. *šaš-t-* 'rest' (~ IE **ses-* 'sleep'), Gk. δαι-τ- 'meal' (~ IE **deh$_2$i-* 'distribute') etc. Although from the synchronic perspective, these nouns are not action but rather concrete nouns, they are most probably best explained as having originally been action nouns (verbal abstracts). On the one hand, primary nominal derivatives from verbal roots are generally either action nouns or agent nouns. Since none of the *t*-stems listed above can be interpreted as an original agent noun (i.e., not *'darkener' for **nók^{u}-t-*, not *'giver' for **dóh$_3$-t-*, and so forth), one has to accept the other option.[47] On the other hand, action nouns very easily acquire concretized meaning, cf. 'sleeping' → 'sleep, rest', 'giving' → 'gift, what is given' etc.

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[44] Rieken (1999: 131).
[45] Thus Rieken (1999: 84-5), etc.
[46] See also Schindler (1967: 297).
[47] In fact, T. Burrow argued that a number of animate gender action nouns should rather be interpreted either as original neuter action nouns or animate agent nouns, since he believed that in early Proto-Indo-European, agent nouns belonged to the animate gender, and the action nouns were neuter (Burrow 1955: 132ff.). However, although agent nouns indeed tend to belong to the animate gender, a large number of Indo-European action nouns seem to have been animate already at the earliest reconstructable stage of the protolanguage.

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The interpretation of the formant *-t-* in such nouns as a suffix is in many ways attractive, as in such a case they could be interpreted as abstract verbal nouns consisting of a verbal root + a derivational nominalizing suffix *-t-*. It appears that one can also reconstruct a well-established "acrostatic" ablaut pattern for these *t*-stems, viz. the ablaut type in which the vowels *o* and *e* alternate in the root (cf. La. *noct-em* vs. Hi. *nekuz* 'night'), and the suffix (*-t-*) constantly appears in the zero grade.[48]

According to Rieken (1999: 85), also the element *-t-* in the Indo-Iranian root nouns of the *stút-* type should be interpreted as an original suffix rather than a root enlargement. As can be seen from the data, the nouns of the **nók^{u}-t-* type on the one hand and the Indo-Iranian root nouns of the *stú-t-* type on the other hand share a number of structural and semantic similarities. The nouns of both types are built to pure verbal roots and both can be interpreted as original action nouns, cf. La. *dō-t-* 'dowry' (← 'gift' ← '[act of] giving') vs. Skt. *stú-t-* 'praise' (← ' [act of] praising'). Therefore, it is tempting to derive them from a single (pre)historic ancestor.

However, due to the productive *t*-enlargement rule, theoretically any of the Indo-Iranian *t*-stem root nouns built to roots ending in the short vowels *-r̥*, *-u* or *-i* can be enlarged original root nouns. The reconstruction of these Indo-Iranian nouns as original derivative *t*-stems would be possible only if it could be demonstrated that the element *t* in a given *t*-stem predates the *t*-enlargement rule, or if there exist parallel *t*-stems in other Indo-European dialects.

[48] See Schindler (1975b: 262), Rieken (1999: 84), S. Neri (2003: 17ff.).

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II.6. Fick – Oettinger's Ablative Theory

One additional theory about the origin of the formant *-t-* is discussed in Oettinger (1982). This theory was in fact not created by Oettinger himself – originally this idea was suggested by A. Fick as early as the 19th century.[49] However, despite later criticism of Fick's theory by J. Schmidt (1889), Oettinger has tried to revive it in this article.

The core of Fick's original idea was that the formant *-t-* seen in some Greek neuter oblique stems (cf. δουρατ- 'tree', ἡπατ- 'liver') originated in the ablative case which sometimes appears with the ending *-tos* beside the more usual *-os*. According to this theory, as the original Indo-European ablative and genitive merged in the course of the history of Greek, the consonant *-t-* in the old ending *-tos* was misinterpreted as a part of the root, and consequently spread throughout the entire paradigm.

In a critical discussion of the original Fick's idea, Schmidt (1889: 190) argued that if *n*-stems and heteroclites had originally been enlarged by *-t-*, one would expect more such secondary *t*-stems, and thus, for example, besides ἡπα-τ- 'liver' in Greek there would have also existed such formations as *ἄκμα-τ- 'anvil' (instead of the attested ἄκμον-)[50] and *ποίμα-τ- 'herdsman' (instead of the attested ποιμέν-),[51] etc. As for the Vedic *yákṛ-t* and *śákṛ-t*, Schmidt's objection was that the formant *-t-* occurs only in the nominative/accusative, whereas no traces of *-t-* can be seen in the ablative of these

[49] Fick (1880: 183-4).
[50] As if deriving from hypothetical *$*h_2e\hat{k}$-mn̥-t-*.
[51] As if deriving from hypothetical *$*poh_2i$-mn̥-t-*.

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nouns (the supposed source of this *-t-*).[52] If this *t*-enlargement had existed in other cases as well (cf. gen./abl. **ḱek-n̥-t-os* or dat. **ḱek-n̥-t-ei*,[53] in such forms, just like in Greek, one would expect the change **n̥ > a* along with the preservation of *t*, rather than the loss of *t* and the preservation of a consonantal *n*.[54]

Later Oettinger tried to apply the same theory in order to explain the dental enlargement *t* in various other formations, such as Ved. *śīrṣatás* 'in front' (beside *śīrṣṇás* 'head' [abl. sg.]), Ved. *yákṛ-t* and Arm. *lear-d* 'liver' (beside Av. *yākarə* 'id.' with no *t*), La. *mūscer-d-a* 'mouse droppings' (~ Hi. *šakkar*, Av. *sair-iia-* 'dung' with no *t*), Hi. *kanin-t-* 'thirst' (beside *kanin-ant-* 'thirsty') etc.[55]

Oettinger proposed that the reanalysis of the ablative suffix *-tos* had taken place in Proto-Indo-European itself, and that it had been especially prominent among the Indo-European *n*-stems and heteroclites. According to Oettinger, after the reanalysis, the inflection of the Indo-European *n*-stems and heteroclites would have changed in the following way: instead of the original single irregular paradigm with only the ablative exhibiting a formant *-t-*, two parallel regularized paradigms would have developed. In one paradigm, the dental *t* would have been generalized, whereas in the other this *t* would have been eliminated.[56]

Oettinger's theory raises various problems. The evidence for the common occurrence of *-tos* ablative in Proto-Indo-European is virtually non-existent. Further-

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[52] Schmidt, loc. cit.
[53] The historically correct dative and genitive/ablative of the original noun **ḱók-r̥* 'excrement' would have been **ḱék-n-(e)i* resp. **ḱék-n̥-s* (see further in EWAia II.602).
[54] Schmidt, loc. cit.
[55] See Oettinger (1982: 243-5).
[56] Oettinger (1982: 243-4).

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more, the paradigm doublets reconstructed by Oettinger are nowhere attested in the Indo-European dialects. In order to explain their absence in the dialects, Oettinger proposed that these doublets were reduced again, eliminating one or the other paradigm. Thus, Indo-Iranian languages would have eliminated the paradigms with *t*, whereas Greek would have eliminated the paradigms without *t*.[57] However, this leaves unexplained the attested Sanskrit forms with *t*, such as *yákṛ-t* 'liver', *śákṛ-t* 'excrement' and the old *-tos* ablatives like *śīrṣatás* 'at the head, in front' (cf. Ved. *śīrṣán-* 'head'). If the forms with *t* had to be eliminated, these forms should not exist.

At least for *yákṛ-t* and *śákṛ-t*, Oettinger proposed that *t* was preserved in such forms because it followed *ṛ* (whereby they would have been similar to nouns of the *stú-t-* 'praise', *pṛ́-t-* 'battle' type).[58] However, since nouns of the *stú-t-* type take *t* also in Avestan (cf. OAv. *stūt-* 'praise', YAv. *pərət-* 'battle' etc.), one would expect the preservation of *t* also in Av. *yākarə* etc., but this did not happen.

The source of the *t*-enlargement in Indo-Iranian is probably different, and it will be discussed in more detail in chapter IV of this dissertation.

Similar problems arise with the Greek neuter *t*-stems of the ἡπατ- 'liver' type. Since the productivity of *-tos* ablative in Indo-European cannot be demonstrated in the first place, Oettinger's theory about the source of the Greek -τ- enlargement becomes seriously undermined. Alternative sources for this -τ- enlargement have been proposed

[57] Oettinger (1982: 244).
[58] Oettinger, ibid.

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as well. According to H. Rix (1992: 144), the consonant -τ- in the oblique stem of Greek neuters developed by analogy from neuter *nt*-stems.

As far as the Latin, the Anatolian and the Armenian reflexes of the Indo-European *n*-stems and heteroclites are concerned, Oettinger did not provide a detailed account for their history, and it is impossible to tell if these dialects would have generalized or eliminated the paradigms with *t*. If the former had been the case, these dialects would possess far too few forms with *t*. On the contrary, if the paradigms with *t* had been eliminated, one would have to account for the preserved forms exhibiting *t*. Even in Armenian, where, on the basis of *leard* 'liver' (gen. *lerdi*) and *neard* 'sinew' (gen. *nerdi*), one could think that the paradigms with *t* would have been generalized, there are also attested reflexes of old heteroclites that have no traces of *t*, cf. Arm. *jeṙn* 'arm' (< **ĝhes-r/n-*), *garun* 'spring' (< **u̯es-r/n-*), etc.

To summarize the preceding discussion, one could say that Oettinger's theory about the Indo-European *-tos* ablative as the source of the dental *t*-enlargement faces numerous problems. Whereas there is no evidence for the paradigm doublets with and without the *t*-enlargement that Oettinger reconstructed, the origin of the formant *t* in the various dialectal forms (Ved. *yákṛ-t*, Arm. *leard*, Hi. *kanint-* etc.) probably has to be explained in a different way.

II.7. Kuryłowicz's Epenthetic Theory

There also exist two theories according to which the element *-t-* is interpreted as epenthetic. In his *Inflectional Categories of Indo-European*, J. Kuryłowicz proposed

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that at a certain stage of Proto-Indo-European, there existed a rule according to which "between a root ending in a *short* vowel and a following *vocalic* morpheme a union consonant *t* (whatever its origin) was already inserted in I.E. [...] both in flexion and in primary derivation".[59] This rule, according to Kuryłowicz, would account for the appearance of a *-t-* in Skt. °*-ji-t-* 'winning', °*-śru-t-* 'hearing', °*-kṛ-t-* 'doing' as opposed to no enlargement in °*-sad-* 'sitting, dwelling' and °*-ruc-* 'shining'. The insertion of *t* would also have prevented hiatus.[60]

According to this rule, quite a few word-forms should have developed such a *-t-* but never did, e.g. La. *septimus* 'seventh' or *decimus* 'tenth', which reflect theoretical **septm̥-o-* resp. **dek̂m̥-o-*. According to the rule proposed by Kuryłowicz, the original **septm̥-o-* and **dek̂m̥-o-* should have become **septm̥to-* resp. **dek̂m̥to-* and consequently yield La. ˣ*septintus* resp. ˣ*decintus*. However, the development was different. The old **septm̥(m)o-* and **dek̂m̥(m)o-* developed into **septomo-* resp. **dekomo-*, and these forms regularly developed into OLa. *septumus* (> Classical Latin *septimus*) and *decumus* (> *decimus*).[61]

One could say exactly the same things about certain Germanic participial forms, cf. OHG *gi-boran* 'carried; born' (< pre-Proto-Germanic **bhṛ-ono-*), *gi-noman* 'taken' (< IE **nm̥-ono-*). If a *t* had been inserted, the expected Old High German forms would be ˣ*gibordan* and ˣ*ginomdan*. Instead, the Indo-European forms shown above developed into **bhṛr-ono-* resp. **nm̥m-ono-*, whence Proto-Germanic **þurana-* resp. **numana-*, from which the Old High German forms developed regularly.

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[59] Kuryłowicz (1964: 235).
[60] Kuryłowicz, ibid.

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Such a "hiatus-breaking" *t* should have also been inserted in a number of inflected forms of certain nominal formations, cf. IE **mr̥-éi* 'sea' (nom. **mór-i*, originally an *o/e*-acrostat).[62] The expected form would be **mr̥-t-éi*. However, the actual course of development was **mr̥-éi* > **mr̥r-éi* (→ La. *mare*).[63]

One could perhaps argue that in any of the words discussed in the previous paragraphs, a hiatus-breaking *t* was indeed inserted in appropriate inflected forms, but due to pressure from other members of the respective paradigms this *t* was levelled out. However, Kuryłowicz's own hypothesis that redundant features tend to be generalized[64] implies that this *-t-* should not have been levelled out, and, on the contrary, should have been generalized throughout the paradigm.

II.8. Oettinger's Epenthetic Theory

A different account for the formant *-t-* has been recently suggested by Oettinger. In his article which appeared in *Anatolisch und Indogermanisch*, Oettinger switched from the ablative theory (described in Oettinger 1982) to a rather different explanation. In this article, Oettinger compared the Indo-European *t*-formations of various kinds with rather curious Old, Middle and Modern German data. As Oettinger pointed out, a large

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[61] On the development of **mV* in Latin, see B. Vine (1993: 246ff.).
[62] Nussbaum (2004d).
[63] Nussbaum, ibid.
[64] Since, according to Kuryłowicz's *t*-insertion rule, no *t* should be inserted before consonants, this would have resulted in irregular paradigms, some forms exhibiting *t*, and others not. Kuryłowicz's hypothesis about the tendency of languages to generalize redundant features would explain the presence of *t* in those word-forms where it is unexpected. Thus, the dental stop in Ved. *°-śrud-bhis* (< **°-śru-t-bhis*) would have been extended by analogy from those forms where *t* would have been inserted to prevent hiatus, e.g. gen. sg. *°-śru-t-as* (< †*°-śru-as*, cf. Kuryłowicz 1964: 235-6).

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number of German words within the history of the German language were extended by the consonant *-t-*, cf. OHG *nie man* > MHG *niemant* 'nobody' (MoG. *niemand*), OHG *ackes* 'ax' > MHG *ackest* (cf. MoG. *Axt*), MHG *habech* 'hawk' (< Proto-Germanic **haƀukaz*) > MoG. *Habicht*, etc. Also loanwords could be extended by such a *-t-*, cf. MHG *zimment* 'cinnamon' < OHG *cinamin* ← La. *cinnamonum*.[65] Oettinger has noticed that the consonant *-t-* tends to occur after similar sounds in German and in other Indo-European languages. According to his analysis, in German this *-t-* occurs after *s*, *r*, *χ*, *f* and *n*. In other Indo-European languages, as well as Proto-Indo-European itself, *-t-* appears after *n* and *n̥* (Gk. ἡπα-τ- 'liver', Hi. *kanin-t-i* 'thirst' [dat./loc. sg.], *išpan-t-* 'night'), *r* and *r̥* (Ved. *yákr̥-t*, Armenian *lear-d* 'liver', Hi. *ka-pir-t-* 'mouse'), *s* (cf. Hi. *kaš-t-* 'hunger'), as well as the laryngeals (IE ***n̥-dm̥h₂-t-* 'untamed', ***h₂enti-sth₂-t-* 'standing in front') which Oettinger equated to the German velar fricative *χ*.[66] Further Oettinger proposed that in Indo-European, unlike German, the consonant *-t-* could also be epenthetically added to stems ending with the vowels *i* and *u*, cf. IE *mél-i-t-* 'honey', La. *com-i-t-* 'companion', Ved. *stú-t-* 'song of praise', Hi. *ku-tt-* 'wall', and possibly even to those ending with stops, cf. IE ***nók^{u}-t-* 'night'.[67]

Oettinger's theory is different from all the theories mentioned previously in that he interpreted all types of *t*-formants as coming from a single source. According to this theory, the *t*-enlargement is purely phonological and is inherited from Proto-Indo-European. However, Oettinger did not clarify the reason for the epenthesis. Since this

[65] Oettinger (2001: 308).
[66] Oettinger (2001: 308-9).
[67] Oettinger (2001: 309-10).

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theory is phonological, there should be something peculiar about the phonetic quality of the sounds that the consonant *-t-* is added to or inserted between (as in Ved. *pratar-i-t-van-* 'early guest' and G. *öffen-t-lich* 'public', cf. Oettinger 2001: 308), as is often the case when epenthesis happens, cf. Modern Icelandic *slá* 'hit' [ŝ^{d}l̥au:] or *snúa* 'turn' [ŝ^{d}n̥u:a], where an alveolar stop is inserted between a postalveolar /ŝ/ and an alveolar /l̥/ or /n̥/, also or *smár* 'small' [ŝ^{b}m̥au:r̥], in which between a post-alveolar /ŝ/ and a bilabial /m̥/ a bilabial stop is inserted.

In fact, there occur less clear cases of epenthesis, cf. Lith. *tū́kstantis* '1000', which has to be interpreted as *tū-k-stantis*, on the basis of the comparison with OCS тъıсѧшта, Old Icelandic *þús-(h)und* 'id.' and the Indo-European root **teuh$_2$-* 'swell, be strong', the zero grade version of which underlies Lith. *tū-*, OCS тъı- and OIc. *þú-*. Another example could be Lith. *áuksas* 'gold', which on the basis of its comparison with La. *aurum* 'id.' (< **ausom*) has to be analyzed as *au-k-sas*. However, here in both cases the velar stop *k* is at least inserted between a *u* and an *s* (it may also be interesting to point out that with the exception of *k*, all consonants in the two Lithuanian words are alveolar/dentialveolar).

One additional problem is that, as can be seen from Oettinger's examples, the stop *t* can be inserted practically after any sound – among the sounds that "trigger" this epenthesis there are several different vowels (front, rounded, back, unrounded), a nasal, a trill, a lateral liquid, and several types of fricatives (velar, bilabial and dentialveolar). As one can see, most of the sounds should trigger a *t*-epenthesis. However, the consonant *t* appears in relatively few formations.

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Moreover, if a consonant *t* really had been mandatory after all the sounds listed in Oettinger's article (see Oettinger 2001: 308-9), one would expect more words ending with *-t*, and not only nouns, but also verbs, as well as other parts of speech, since one would not expect a *phonetic* change to be restricted to certain *morphological* classes of words. One would also have to assume that this phonologically conditioned *t* could be added to words of any structure and any length, i.e., equally to simplicia and to compounds, to monosyllables and polysyllables. Thus, if we restrict ourselves to Vedic alone, we would have to expect words like **mádhu-t* 'honey' (instead of *mádhu*), **su-gú-t-* 'possessing good oxen/cows' (instead of *su-gú-*), **mand-ú-t-* 'rejoicing' (instead of *mand-ú-*), **hí-t* (emphatic particle; instead of *hí*), **sú-t* 'good (adj.), well (adv.)' (instead of *sú*), **nú-t* 'now' (instead of *nú*), **śubhaṃ-yú-t-* 'rushing to beauty' (instead of *śubhaṃ-yú-*), imperatives in **-hí-t* and **-dhi-t* (instead of *-hi* and *-dhi*), etc. A similar epenthesis would have to be much more common also in the other Indo-European languages, but no such epenthesis is actually attested.

II.9. Rasmussen's Theory about the Early Proto-Indo-European Phoneme */c/

According to a theory which has been put forward by Rasmussen, in a number of instances, the consonant *-t-* appearing in the suffixes *-et-* and *-ot-* derives from an older affricate */c/ (= /ts/) or some other dental sound (e.g., a glottalized /t'/).[68] According to Rasmussen's hypothesis, early Proto-Indo-European possessed an affricate */c/, which disappeared from the phonological system of the protolanguage at an early stage, partly

[68] Rasmussen uses the symbol /c/ (Rasmussen 1994: 325). The symbol /t'/ is used as an alternative by B. A. Olsen (2004: 419).

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becoming /t/, and partly /s/. The development of the affricate /c/ into /t/ or /s/ depended on the position of /c/ in the word. In word-final position, /c/ was assibilated, and in all other positions it became /t/.[69]

Rasmussen assumed that at least some of the nominal formations which are reflected as *t*-stems in the Indo-European dialects in fact had been *c*-stems at the early stages of the protolanguage. He based this assumption on his observation that beside certain *t*-stems, there sometimes also exist *s*-stems possessing a somewhat similar structure and often also a close meaning, cf. IE ***tepn-es-*** 'heat' (> Av. ***tafnah-*** 'id.') vs. ***tepn-ēt-*** (> OIr. ***teine***, gen. sg. ***teined*** 'fire'), ***mḗd-os*** 'measure' (→ Gk. μήδεα 'plans' [pl.]) vs. ***med-ot-*** (> Go. ***mitaþs*** 'measure'), ***ĝhéngh-os*** 'step' (> Ved. ***jáṁhas-*** 'course') vs. ***ĝhengh-eto-*** (> OIr. ***cing***, gen. sg. ***cinged*** 'soldier'), etc.[70]

Thus, according to Rasmussen, the Indo-European /t/ would derive from (at least) two different sources: from the pre-Indo-European ***t*** on the one hand, and the deaffricated pre-Indo-European (or early Proto-Indo-European) affricate ***c***.

However, in spite of later attempts by Olsen to reinforce this theory,[71] it contains a number of problems. Starting with phonetics and phonology, the change from *[t͡s] to [t] in some environments and to [s] in others is quite unlikely. Although both changes, viz. *[t͡s] > [t] and *[t͡s] > [s], are attested in the languages of the world, I have not been able to find a single language/dialect in which both changes in question would have taken place. One should also bear in mind that of the two changes mentioned

[69] Cf. Rasmussen's schema: "**c* > *s* /_#, sonst *t*" (Rasmussen 1994: 329).
[70] See a full list of *c*-stems in Rasmussen (1994: 325-7).
[71] Olsen (2004).

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above, the change *[t͡s] > [t] is extremely rare. The changes *ts > t and *ts > s are possibly attested in several Greek dialects (Attic, Boeotian, etc.), cf. Attic κρείττων 'stronger' vs. Ionic κρέσσων 'id.' (< *kretsi̯ō-? ← IE *kret-i̯ōs); however, the development of the Greek clusters in question may have been more complex. The details of the development of the -ττ- and -σσ- clusters in the Greek dialects are in fact not entirely well understood, and are interpreted differently by different scholars.[72] Unlike Rasmussen's pre-PIE */c/, the affricate underlying the dialectal Greek clusters -ττ- and -σσ- may have been long (geminated), namely, [t͡sts] or [t͡ʃtʃ], which would have entailed a longer closure duration during the articulation of the geminate than that of the non-geminated affricate reconstructed by Rasmussen. Accordingly, the Proto-Greek affricates may have developed in an entirely different way from that of Rasmussen's non-geminated /c/.

II.9.1.

In this context, for typological comparison it might be interesting to consider the development of affricates in the Eskimo languages. According to the present knowledge of the historical phonology of the Eskimo languages, Proto-Eskimo

[72] Hirt assumed an intermediate stage *ts (1902: 163-5). A similar interpretation can be found in M. Lejeune (1972: 106-7), but he assumed that the clusters *ki̯ and *ti̯ developed into *ts at different times. A. Bartoněk assumed the merger of *ki̯ and *ti̯ into a palatal *č which subsequently became *c (ts), and then further developed into ττ or σσ depending on the dialect (1961: 61). Rix assumed the following changes (simplified here): *ti̯ > *tˢi̯ > *sᵗi̯ > ττ, σ or σσ depending on the dialect; *ki̯ > *tˢi̯ > *sᵗi̯ > *ssᵗ > ττ or σσ depending on the dialect (1992: 92). Similarly to Bartoněk, A. L. Sihler reconstructed Proto-Greek palatal affricates as well. However, Sihler reconstructed them as geminates: *ki̯ > *čč > ττ or σσ; *ti̯ > *tˢ > *tˢi̯ > *čč > ττ or σσ (1995: 192).

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possessed both a short, i.e., a non-geminated affricate **c* and a corresponding geminated **cc*.[73] The reflexes of these affricates were different in the various Eskimo dialects. The short **c* has generally yielded *s* in West Greenlandic, cf. WG *sakku* 'weapon, tool' as opposed to the somewhat more conservative in this respect East Greenlandic *cakkeq* 'woman's knife', or Alutiiq Alaskan Yupik *caskuk* 'hunting equipment' < Proto-Eskimo **caðku-*;[74] cf. also WG *pasi-* 'suspect' vs. AAY *paci-* 'blame' < PE **paci-*,[75] etc.

However, the long **cc* is reflected as *ts* in Greenlandic, cf. WG *atsa(k)* 'paternal uncle' < PE **accaɣ-*;[76] WG (Upernavik) *itsaq* 'skin placed on the roof of the house' < Proto-Inuit **itcaq*.[77] In the West Canadian Copper Inuit dialect, the long affricates have suggestively yielded *tt*, WCCI *attak* 'paternal aunt' (cf. WG *atsa[k]* above) resp. *ittat* 'skin tent' (cf. Upernavik Greenlandic *itsaq*).

The result of the deaffrication of the short affricate [t͡s] – if one does not take into consideration the Greek dialects – is most often a sibilant of some sort, usually an alveolar /s/, as in Old High German, Old French and Old Portuguese, cf. OHG *bīzzan* 'bite' [-sː-] < *[biːt͡san] < PGmc. **b̄ītan* 'bite', OF *cent* [sẽt], OPort. *cento* < *[t͡sento] < La. *centum* 'hundred'. In Old Castilian as well as several other Old Spanish dialects,

[73] The following discussion has benefited from my communication with M. Fortescue via email.
[74] M. Fortescue, S. Jacobson & L. Kaplan (1994: 63).
[75] Fortescue/Jacobson/Kaplan (1994: 245).
[76] Fortescue/Jacobson/Kaplan (1994: 2).
[77] Fortescue/Jacobson/Kaplan (1994: 146). The symbols **cc* and **tc* used respectively for the Proto-Eskimo and the Proto-Inuit affricates are provisional, and the dental stop **t* is reconstructed for the Proto-Inuit affricate mostly in order to show that the Proto-Inuit affricate was [–palatalized] (as opposed to the palatalized reflexes of the PE **cc* in Yupik dialects; Fortescue, p. c.).

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the reflex of the ancient affricate [t͡s] is the interdental fricative [θ] which had probably developed from [t͡s] via [s].[78] In Lycian, the Anatolian affricate /z/ is occassionally assibilated to /s/ but never becomes /t/, cf. Lycian *ladas* 'wives' (acc. pl.; < **-ants*), *tideimis* 'sons' (acc. pl.; < **-ints*).[79]

Also in non-Indo-European languages, the result of the deaffrication of [t͡s] tends to be [s]. As was shown above, in West Greenlandic the reflex of the Proto-Eskimo affricate **c* is a sibilant *s*. In North Greenlandic, the outcome of PE **c* is *h* which probably developed via the intermediate stage */s/.[80]

Also the Classical Akkadian alveolar /s/ is thought to have developed from an old affricate /t͡s/, which presumably still existed in Old Akkadian.[81] Assibilation of ancient dental and palatal affricates is also ascribed to some Altaic and Austronesian languages.[82]

The Cornish parallel of *t* becoming *s* word-finally and remaining as *t* elsewhere, adduced by Rasmussen as a "striking parallel" to the alleged Proto-Indo-European phenomenon in question,[83] appears to be not so parallel at a second glance, since whereas in Proto-Indo-European, a single phoneme (*/c/) would have yielded two

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[78] For the development of the affricate /c/ in the Romance languages, see A. Galmés de Fuentes (1962: 63-9, 102-16, 127-57).

[79] Melchert (1994: 323).

[80] For the dialectal reflexes of the Proto-Eskimo affricate **c* see Fortescue/Jacobson/Kaplan (1994: xvi).

[81] R. Stempel (1999: 52-3).

[82] V. Kotvič (1962: 32), O. Dempwolff (1938, passim). Proto-Austronesian itself did not possess affricates, but the sibilant *s*, in many dialects reflecting Proto-Austronesian palatalized **t'*, probably developed via the intermediate stage of an affricate, cf. PAu. **t'abah* 'irrigated rice-field' > **čabah* (?) or **cabah* (?) > Indonesian *saba'*, Javanese, Malay *sawah* 'id.'; PAu. **t'a(ŋ)gu'* 'sago palm' > **č/ca(ŋ)gu'* > Javanese *sagu'*, Indonesian, Tagalog *sago'*.

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distinct sounds (/t/ resp. /s/) depending on the environment, the Cornish situation is entirely different. First, no affricate *c is assumed for the prehistory of Cornish. The Cornish sound change discussed by Rasmussen involves only *t* and *s*. Unlike the alleged Proto-Indo-European change from *c to *s /_#, but to *t otherwise, in Cornish, it is the sound *t* that undergoes assibilation to *s* in word-final position.

II.9.2.

As far as the morphological side of Rasmussen's *c*-stem theory is concerned, the parallel co-existence of various *s*- and *t*-stems, contra Rasmussen's claims, is in most instances not real. To begin with, some of the *t*-stems reconstructed by Rasmussen probably never existed, cf. IE **tep-nōt-s* 'heat', which became a *t*-stem only at a very late, Proto-Celtic stage.[84] In most other cases, the suffixes *-es-/-et-* or *-os-/-ot-*, claimed by Rasmussen to derive from early PIE **-ec-* resp. **-oc-*, can be alternatively explained as independent formations.

The ancient Indo-Iranian words for 'heat' originally were proterokinetic *s*-stems, cf. IIr. **táp-as-* (> Ved. *tápas-*) and **táp-n-as-* (> Av. *tafnah-*). The Proto-Celtic **tepnēt-*, which is also an obviously secondary formation, has been shown by Widmer to have developed from an older *s*-stem.[85]

The suffix *-ōt-* in the protoform of Go. *weitwoþs* 'witness' (noun) has been explained by J. Rau as a nominalizing suffix, independent from the Indo-European

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[83] "Eine ganz schlagende Parallele bietet der kornische Übergang von *t* zu *s* im Auslaut bei Erhaltung von /t/ in anderen Stellungen" (Rasmussen 1994: 329).
[84] See Widmer (1997: 124-6).
[85] Widmer, loc. cit. For Rasmussen's interpretation, see Rasmussen (1994: 326).

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participial suffix -*u̯ōs*-.[86] The formation of Go. *weitwoþs* on the one hand and the active perfect participles on the other hand were entirely different: whereas the Gothic word-form is a secondary amphikinetic derivative from an original *u*-stem (noun or adjective, see a more detailed discussion in section V.9.2 below), perfect participles are built directly to verbal roots.

Also the nouns **léuk-os* 'light', reflected in Av. *raocah-*, etc. and **léuk-ot-* 'id.' (> Go. *liuhaþ*) can be easily explained as independent formations built to the same verbal root. The vowel *o* in the *s*-stem suffix *-os-* is relatively young, and archaic proterokinetic *s*-stems do not show it, cf. Gk. κρέας 'flesh', Ved. *krávis-* 'id.', both reflecting **kréu̯h$_2$-s*, not **kréu̯h$_2$-os*. On the contrary, the vowel *o* in the amphikinetic *t*-stem suffix *-ot-* is old.[87] It also has to be noted that the inflection of the nouns **léuk-os* and **léuk-ōt-* was entirely different.

II.10. Summary

Setting aside for now the Indo-Iranian *t*-stems of the *stút-* type, the most attractive interpretation for the element *-t-* is the one according to which *t* is considered to be a derivational suffix. The reconstruction of primary *t*-stems built with this suffix as original acrostatic formations agrees with the the constant occurrence of the suffix *-t-* in the zero grade. Also the fluctuation between the *o-* and the *e*-grade in the roots of such

[86] Rau (1998: 157-60).

[87] The origin of the vowel *-o-* in the suffix of amphikinetic formations is disputed. It has been suggested by Nussbaum that the amphikinetic inflectional pattern has evolved out of the hysterokinetic pattern at an early stage of Indo-European. The *o*-grade in the amphikinetic suffix could be attributed to the rounding of the unstressed **-è-* (***è* > **o*, as in neuter *s*-stems, cf. ***CéC-ès* > *CéC-os*).

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t-stems is conveniently explained if they are reconstructed as original acrostatic formations, for the earliest stage of which the *ó/é* alternation is reconstructed.

As far as the semantics of the acrostatic *t*-stems is concerned, these formations can be interpreted as original action nouns (verbal abstracts) built from verbal roots and the nominalizing suffix *-t-*.

It has been demonstrated in sections II.9.1 – II.9.2 that the reconstruction of the suffix *-ot-* in several Indo-European formations as a pre-Indo-European or early Proto-Indo-European ***-oc-* is not sufficiently supported by the evidence. In the present dissertation, I will assume that the earliest reconstructable form of this suffix is **-ŏt-*.

The interpretation of the determinative *t* in the Indo-Iranian *t*-stem root nouns of the *stút-* type as an original suffix is attractive, but Rieken's reconstruction of these nouns as old acrostats is not motivated well enough, and needs further investigation.

In the following chapters, I would like to discuss the formation of the primary Indo-European acrostatic and amphikinetic *t*-stems, and their development in the individual dialects. The Indo-Iranian *t*-stems of the *stút-* type will be treated in a separate chapter due to their morphological ambiguity.

II.11. Forms not Included in the Dissertation

In the following sections, I would like to survey forms that will not be included in the present study, either because their etymology is obscure, they are morphologically ambiguous, or are unlikely to be primary *t*-stems in spite of their shape. Nevertheless, some of these forms contain features which may be relevant for the understanding of

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the structure of the Proto-Indo-European *t*-stems, as well as the development of *t*-stems in the Indo-European dialects.

Instead of being an exhaustive study of all ambiguous or irrelevant forms, the following discussion will be focused on selected examples that are meant to illustrate the problems arising in their historical analysis.

II.11.1

The first class of forms contains nouns which have uncertain etymologies and/or exhibit grammatical features not otherwise attested among Indo-European primary *t*-stems:

1. Gk. σταῖς/σταίς 'dough' (neuter):[88] Whereas the reflexes of the Indo-European primary *t*-stems usually are feminine (in Hittite: animate), cf. La. *noc-t-* 'night' (fem.), Gk. δαι-τ- 'meal' (fem.), Hi. *kaš-t-* 'hunger' (animate) etc., σταῖς/σταίς is neuter. It is not likely that σταῖς/σταίς could have become neuter within the (pre)history of Greek, as other reflexes of *t*-stems preserved their original gender well.

Besides the gender problems, also the root etymology of this noun is not entirely clear. The root of σταῖς/σταίς has been reconstructed in several different ways.

[88] In Herodotus, the nominative singular form σταῖς occurs once (II.36.3), and is written with a circumflex. In Aristotle's *Meteorologica*, nom. sg. occurs once, spelled σταίς (386^{b}.14). In his *Problemata*, the nom. sg. form is in all four instances spelled σταῖς (927^{b}.23, 33, 39; 928^{a}.26). The suffix -τ- is seen in the oblique forms, cf. gen. sg. σταιτός (Hippocrates Medicus, Περὶ ἄρθρων, 38.18; Arist., *Problemata*, 928^{a}.17, 22, 28), dat. sg. σταιτί (Hipp. Med., op. cit., 38.16).

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According to one different interpretation, Gk. σταῖς/σταίς is related to several other dialectal Indo-European terms for 'dough' or 'leaven', cf. OHG *deismo*, OE *þǣsma* 'leaven', OCS тѣсто, OIr. *táis* 'dough', *tóis-renn* 'mass', Welsh *toes*, Brit. *toaz* 'dough'.[89] However, scholars disagree as to how the root etymology and the morphology of these formations should be explained. In IEW.1054, the underlying root is reconstructed as **təis-*, and the initial cluster of the Greek stem σταιτ- is explained as an analogy from Gk. στέᾱρ 'stiff fat, tallow; dough'.[90] However, it is not entirely clear what the root **təis-* represents. The element *-s-* has been interpreted as independent from the root.[91] The remaining **təi-* has been interpreted in several ways: Irslinger attempted to associate the radical **ə* with **h*$_2$,[92] but she admitted that such an interpretation would only be possible if **CHi̯-* > *CVi̯-* in Celtic. Moreover, semantically the root **teh*$_2$*i̯-* 'melt' is difficult to associate with 'dough', since melting is not characteristic of dough.

Rasmussen reconstructed the underlying root as **(s)taih*$_3$- 'be compact', and reconstructed σταῖς/σταίς in an entirely different way, analyzing it as a comparative neuter **staih*$_3$*-is* 'more compact', which would refer to the increasing compactness of the dough when kneaded.[93]

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[89] IEW.1054, Chantraine (1968: 1043), Vasmer III.100, Rasmussen (1999a: 302-5), Irslinger (2002: 425), etc.
[90] See also VGkS I.56.
[91] See a recent discussion in Irslinger (2002: 425).
[92] Irslinger writes "*h*$_1$", but Rasmussen (1989: 63) whom Irslinger refers to reconstructed *h*$_2$, cf. **teh*$_2$*i̯-* (Rasmussen, loc. cit.). This root is reconstructed as **teh*$_2$- in LIV.616.
[93] Rasmussen (1999a: 303-5).

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From this discussion it is possible to see that although the noun σταῖς/σταίς is in many ways unclear, from some of its features it is possible to see that this noun is not a primary *t*-stem of the acrostatic or the amphikinetic type. The various etymologies of this noun discussed earlier indicate that the element -τ- in the oblique stem of σταῖς/σταίς is likely to have developed secondarily.[94]

2. Gk. γάλα 'milk' (neut., stem γαλακ-τ-; Homer +), Gk. γλάγος 'id.' (neut.; Hom.; *Iliad*, II.471, XVI.643), κλάγος·γάλα. Κρῆτες (Hesych.), La. *lac* 'id.' (neut.; gen. *lactis*; Ennius +): These nouns face the same gender problem as the the previously discussed Gk. σταῖς/σταίς, as both Gk. γάλα, κ/γλάγος and La. *lac* are neuter as well.

In different works, different etymological and morphological analyses of this dialectal 'milk' word have been proposed. In IEW.400-1, a root **glag-* or **glak-* 'milk' is reconstructed, and Gk. γάλα and La. *lac* are considered to have developed from it independently. The shape of the Greek γάλα is attributed to the irregular development of this word in "Kindersprache".[95]

O. Szemerényi interpreted the Indo-European 'milk' word as a root noun **mĕlĝ-/ mlĝ-* built to the verbal root **melĝ-* 'milk' with a *t*-enlargement.[96] According to Szemerényi's analysis, the root noun **mĕlĝ-/mlĝ-ós* would have developed into Gk. *μελ/*μηλ (nom.) vs. γλαγ- (weak st.), and the weak stem would have been generalized. The consonant -τ- in the oblique forms would have been secondarily

[94] See also Rasmussen (1999a: 304).
[95] IEW, loc. cit.

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introduced by analogy from other nouns, such as μέλι 'honey' (gen. sg. μέλι-τ-ος). All the other Greek forms would have developed from the same original root noun, whereas La. *lac* would have to be interpreted as a Greek borrowing.[97] Despite its surface simplicity, this analysis is not very attractive, because the Indo-European verbal root in question is generally reconstructed as **h₂melĝ-* on the basis of Gk. ἀμέλγω 'milk'.[98] One would expect the laryngeal to have vocalized in the nominal derivative as well.

A still different analysis is argued for by B. Demiraj (1997), where the original root is reconstructed as **ĝl̥h₂-* (1997: 153-4, with further references to earlier scholarship). The root for the word 'milk' is reconstructed in the same way by X. Tremblay (2005), who reconstructs IE **ĝl̥h₂-g̑-t-* as the ancestor of Gk. γάλα and other Greek forms together with La. *lac* (Tremblay 2005: 132-3). The original root noun **ĝl̥h₂-* 'milk' would be reflected by Albanian *dhallë* 'milk'[99] and by the terms for 'milk' in several Nūristāni languages, cf. Aškūn *zǭ*, Waigalī *zōr* etc.[100]

As one can see from this discussion, the historical analysis of the Greek and the Latin words for 'milk' is not entirely clear. According to the reconstruction presented in IEW, **glag/k-t-* would be a primary *t*-stem. However, unlike other archaic Indo-European primary *t*-stems, it is neuter. Furthermore, since no primary verbs with the root **glag/k-* are attested, it is unclear whether the root is nominal or verbal. If the root

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[96] Szemerényi (1958: 174-7).
[97] Szemerényi, ibid.
[98] LIV.279.
[99] Demiraj translates *dhallë* as 'Buttermilch' (1997: 153).
[100] Tremblay, loc. cit.

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is nominal (as implied in IEW.400-1), it would make the reconstruction of an original *t*-stem even more problematic, because all other primary *t*-stems are built to verbal roots.

The alternative interpretation as presented in Demiraj (1997) and Tremblay (2005) is not entirely clear, since it remains to be explained whether *-*ĝ*- is a primary suffix or a root enlargement of some sort.

3. OE *scrūd* 'garment' (neut. *a*-stem; → OIc. *skrúð*):[101] In addition to being neuter, this noun is also etymologically obscure. It most probably is a thematic derivative of some sort, and not an original *t*-stem **skrū-t-*.[102]

4. Gk. Εἵλως 'Helot, Spartan serf' (Herodotus +; stem Εἵλω-τ-): The etymology of this noun is uncertain, and no cognates in any other languages are attested.[103] It is unclear whether the noun Εἵλως was a *t*-stem originally, as synchronic Greek masculine *t*-stems ending in -ως sometimes reflect ancient *s*-stems, cf. ἱδρώς 'sweat' which should be reconstructed as an original *s*-stem on the basis of Homeric ἱδρῶ (acc. sg.; < *-*os-m̥*), La. *sūdor* 'id.' (← *-*ōs*) etc., but is inflected as a *t*-stem ἱδρωτ- already in Hesiod. Also the noun γέλως 'laughter', which still exhibits *s*-stem features in Homeric Greek (cf. dat. sg. γέλῳ [< pre-Gk. *-*ōs-i*?]), in later Greek already is inflected as a *t*-stem γέλωτ-.

[101] See J. de Vries (1977: 504-5).
[102] See Griepentrog (1995: 489-90), Campbell (1959: 252).

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5. Gk. (Ϝ)ἄναξ 'king; master, owner' (masc.; Hom. +; stem ἀναϰ-τ-), Mycenaean *wanakt-* 'master' (cf. *wa-na-ka-te* /uanaktei/, dat. sg.; PY Un 03.1, KN Ga 675, etc.): Several etymologies for this noun have been proposed, but all contain phonological, morphological and semantic problems.

On the basis of the use of this term as an epithet for the Διόσκουροι, J. Puhvel interpreted Gk. ἄναξ as an original compound enlarged by *t*, viz. **u̯a-nak-t-* (< **u̯n̥n-n̥ḱ-*[*t*-]) 'not subject to doom', associating **u̯n̥n-* with OIc. *vanr* 'lacking', La. *vānus* 'empty' etc., and *-n̥ḱ-* with the verbal root **neḱ-* 'disappear' (~ Ved. *náśyati* 'disappears', La. *noceō* 'harm', etc.).[104] The proposed etymology cannot be correct, since the root underlying OIc. *vanr*, La. *vānus* etc. contained both root-initial and root-final laryngeals, and is reconstructed as **h₁u̯eh₂-* 'abandon' (~ Gk. εὖνις 'bereaved of' [< **h₁uh₂-ni-*], Ved. *ūná-* 'deficient, wanting' [< **h₁uh₂-nó-*], La. *vānus* [< **h₁u̯eh₂-no-*]).[105] The zero grade root **h₁uh₂-* could not have developed into Gk. (Ϝ)ἀ-.

I. Hajnal reconstructed Gk. ἄναξ as a *t*-stem compound **u̯n̥-h₂ég̑-t-* 'Person, die Gewinn mit sich führt, erzielt', built to the roots **u̯en(H)-* 'gewinnen, Gewinn' and **h₂ég̑-* 'treiben'.[106] According to Hajnal, the reflex of a genetically related and semantically identical *t*-less compound **u̯n̥-h₂ég̑-* is to be seen in Ved. *vaníj-* 'merchant

[103] GEW I.462, Chantraine (1968: 321). See a more detailed discussion in Fraenkel (1910: 99-103).
[104] Puhvel (1956: 215).
[105] Cf. LIV.254 and Nussbaum (1998: 73f.).
[106] Hajnal (1998: 64ff.).

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(?)'[107] (nom. sg. *vaṇíg* RV, v.45.6d; dat. sg. *vaṇíje* RV, i.112.11a; acc. sg. *vaṇíjaṃ* AV, 3.15.1a).

Hajnal's interpretation of *vaṇíj-* was adopted by S. Scarlata (1999: 16, fn. 21), who also provided a more detailed discussion of the development of the compound **u̯n̥-h₂éĝ-* in Proto-Indo-Iranian. According to Scarlata, **u̯n̥-h₂éĝ-* had an ablauting paradigm, and the root of the final member was in the zero grade in the weak forms, viz. **u̯n̥-h₂ĝ-*. At the common Proto-Indo-Iranian stage, the compound **u̯n̥-h₂éĝ-/*u̯n̥-h₂ĝ-* became reformed as **wan-Haǰ-/*wan-Hǰ-*, and subsequently the weak stem became generalized, yielding Ved. *vaṇíj-*.[108]

The reconstruction of the compounds **u̯n̥-h₂éĝ-t-* and**u̯n̥-h₂éĝ-*, as well as the proposed development in Indo-Iranian face various problems. First, probably due to phonological restrictions, roots of the *CeC*[109] structure occur in the zero grade extremely rarely.[110] This makes it unlikely that the compound **u̯n̥-h₂éĝ-* could have had a weak stem like **u̯n̥-h₂ĝ-*. Second, it is hard to see why the original **u̯n̥-h₂éĝ-/*u̯n̥-h₂ĝ-* should have become reformed in Proto-Indo-Iranian as proposed by Scarlata instead of either of the two stems (presumably the strong stem) being generalized. The expected Sanskrit outcome would then be **vanáj-*.

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[107] Cf. EWAia II.494. Mayrhofer interpreted Ved. *vaṇíj-* as deriving "wohl mit kontextfreiem *ṇ* (< **-n-*) aus **van-íj-* '(Güter) gewinnend'" as well (loc. cit.).

[108] Scarlata (loc. cit.).

[109] Here, *C* = any consonant excluding resonants.

[110] This feature was of importance for the development of Indo-European primary *t*-stems as well, and I will return to the development of *CeC* and *CeR(C)* roots in several places in this dissertation, providing relevant examples. Although the zero grade of *CeC* roots does occur, cf. Skt. *hitá-*, Gk. θετός 'placed' (past ptc.; < **dhh₁-tó-*), in a number of instances it may be analogical after forms built to *CeR(C)* roots. On the distribution of full and zero grades in various *ti-* and *to*-formations see Vine (2004).

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The source of *-ij-* in the noun *vaṇij-* is not clear to me, but it may be that *vaṇij-* is either built like, or is analogical after the substantives *ṛtvij-* 'priest', *uṣij-* and *adhrij-* (divine epithets), *bhurij-* 'hand (?)',[111] all of which are not entirely clear either.[112]

A more serious problem is the structure and the semantics of the compound **u̯ṇ-h₂éĝ-(t)-*. Hajnal reconstructed the meaning of the first member as 'Gewinn' (loc. cit.), presumably interpreting it as a root noun. However, no evidence for a root noun built to the verbal root **u̯enH-* 'desire' exists.[113] The root **u̯ṇ-* in **u̯ṇ-h₂éĝ-(t)-* cannot be interpreted as a pure verbal root either, because compounds with the structure "Verbal root + Verbal root/Verbal Abstract Noun" do not occur.

Thus, the etymology and the morphological analysis of the Greek noun ἄναξ remain unsolved, making ἄναξ useless for the present dissertation.

II.11.2.

The second class comprises nouns and isolated forms that theoretically could be interpreted as *t*-stems of the *R-t-* structure, but most likely are not:

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[111] See EWAia I.67-8, 234-5, 258 resp. II.266.

[112] The interpretation of *ṛtvij-* as **(h₁)ṛ-tu-* 'proper time' + the zero grade of the root *yaj-* 'verehren' (EWAia I.258) is possible, but is suspicious on the basis of the comparison with other Sanskrit compounds built with the root *yaj-*. In all of these compounds, *yaj-* appears in the full grade, cf. *satya-yáj-* 'offering correctly', *divi-yáj-* 'offering in day-time', *pra-yáj-* 'dessen Voropfer (?) kraftvoll ist', *su-yáj-* 'offering well' (see Scarlata 1999: 397ff.).

[113] In the Indo-European dialects, various suffixal formations occur, cf. La. *venus* 'love' (*s*-stem; gen. sg. *veneris*), Skt. *vánas-* 'desire' (*s*-st.), Go. *wens*, OIc. *ván* 'hope' (*i*-st.; cf. Go. gen. sg. *wenais* [Gal 5:5, I Tim 1:1], dat. sg. *wenai* [Col 1:23, Phil 1:20, Rom 12:12, I Thess 5:8, Tit 1:2], OIc. gen. sg. *vánar*, nom./acc. pl. *vánir*), OIc. *vin* 'meadow, oasis' (*jō*-st.; < PGmc. **wen-jō*), etc. (for more examples, see IEW.1146-7).

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1. Go. **spaurds* 'race' (fem.; dat. sg. *spaurd*,[114] I Cor. 9:24; otherwise inflected as an *i*-st.), OE *spyrd*, OHG *spurt* 'id.': The ancestor of these nouns has been interpreted as a *t(i)*-stem **spur-đ(i)-* (< IE **spṛH-t[i]-* ~ **sperH-* 'mit dem Fuß stoßen'),[115] or as a root noun **spurđ-* (< IE **spṛdh-* ~ **sperdh-* 'weglaufen').[116]

Although formally the Proto-Germanic ancestor of the dialectal forms listed above could reflect either a derivative *t*-stem or a root noun, W. Griepentrog (1995: 374-9) has demonstrated that the interpretation of PGmc. **spurđ-* as an old *t*-stem built to the root **sperH-* is both semantically unattractive ("kicking" → "race"?) and not really necessary. On the contrary, the alternative interpretation of PGmc. **spurđ-* as a root noun with the meaning 'running' is semantically much more plausible. Besides, reflexes of the root noun **spṛdh-* are attested in Indo-Iranian, cf. Ved. *spṛdh-* 'contest', Av. *spərəd-* 'zeal' (?). Therefore, the reconstruction of a root noun for Go. *spaurd-* is more attractive than the reconstruction of a primary *t*-stem as well.

2. Go. *dulþs* 'festival' (fem.; dat. sg. *dulþ* in St. John 7:14, but generally *dulþai*; otherwise inflected as an *i*-st.): Based on the dative singular form *dulþ*, this noun could be reconstructed as a *t*-stem **dh[u̯]l̥-t-* vel sim.[117] However, since otherwise it is always inflected as an *i*-stem, Griepentrog (1995: 487) suggested that is *dulþs* probably to be interpreted as an original *ti*-stem with a characteristic secondary zero grade of a *CeRC*

[114] Misspelled as ⟨spraud⟩ in the manuscript.
[115] LIV.585 reconstructs IE **sp^{h}erH-*.
[116] See the most recent discussion in Griepentrog (1995: 373ff., with further refs.).
[117] The root etymology is not entirely clear (see W. P. Lehmann 1986: 97).

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type root.[118] The single instance of the dative form *dulþ* should probably be explained as a scribal error.[119]

II.11.3.

In this section, I would like to discuss several forms that have become ambiguous due to peculiarities of their phonological or morphological development:

1. Go. *brusts* 'breast, heart (fig.)' (fem.; acc. pl. *brusts*, St. Luke 18:13, Col. 3:12, Phil. 12, 20), OHG *brust*, OFr. *brust/burst* 'id.': The ancestor of these nouns has been interpreted as a consonantal stem **þrust-* 'breast'.[120] However, its further analysis is not entirely clear. The Proto-Germanic noun **þrust-* could be interpreted as an old *t*-stem **bhrus-t-* built to the verbal root **bhreus-* 'swell'.[121] This would agree well with the patterns of derivation of *t*-stems, since *t*-stems are normally built to verbal roots. The original meaning of the *t*-stem **bhrus-t-* would have been 'swelling', but a secondary concretization would not be a problem. However, Griepentrog (1995: 470) proposed an alternative interpretation pointing out that the final dental could be reconstructed as an Indo-European */d/, viz. */bhrus-d-/. Even though the voiced stop would have voiced the sibilant /s/ to [z] in Indo-European, the voiced cluster *[zd] would have been devoiced to PGmc. **st* nevertheless, cf. PGmc. **nista-* 'nest' < **ni-sd-o-* *[nizdo-].

[118] See M. Meier-Brügger (2000: 194ff.).
[119] Griepentrog, loc. cit.
[120] See IEW.171, W. Krause (1963: 158).
[121] See Griepentrog (1995: 469-71). For the reconstruction and semantics of the root **bhreus-* (< ***bhreu-s-*) see IEW.170f.

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Although nominal *d*-formations are not numerous, they are securely attested both in Indo-European and Germanic, cf. La. *pecus* 'small animal' (gen. *pecu-d-is*), Gk. κονίς 'nit' (gen. κονί-δ-ος), OIc. *ǫ́lpt* 'swan' (< **albh-i/u-d-* vel sim.), etc. It is also possible that the Proto-Germanic root nouns **χnut-* 'nut' and **χnit-* 'nit' reflect ancient derivative *d*-stems.[122] The suffix *d* apparently could be used to form nouns, although its more precise function remains unclear. As one can see from the examples, in certain instances it has been added to nominal (*peḱu-*) and adjectival (*albh-i/u-*) stems. If **þrust-* is to be interpreted as a *d*-stem, the function of the suffix *-d-* would have been nominalizing as well.

Therefore, the Indo-European ancestor of PGmc. **þrust-* can be reconstructed either as a concretized *t*-stem **bhrus-t-* 'swelling' or as an original *d*-stem **bhrus-d-* 'breast (?)'.

2. Lith. *krūtìs* 'breast': Although the reconstruction of an old primary *t*-stem **kruH-t-*[123] is plausible both from the semantic and the morphological point of view in that it can be interpreted as an old *t*-stem verbal noun with a typical zero grade, it is impossible to determine whether it originally was a *t-* or a *ti*-stem, because *krūtis* can be equally plausibly reconstructed as a *ti*-stem **kruH-ti-*. The genitive plural form *krūtų* is paralleled by gen. pl. *naktų* (*t*-st.), but, as noted in J. H. Larsson (2001: 53-4), after the merger of consonantal and *i*-stems in Baltic, consonantal endings were occasionally added to historical *i*-stems, cf. Old Lith. *išmintų* 'wisdom' (gen. pl.; Daukša, Bretkūnas,

[122] Sic H. Krahe & W. Meid (1967: 175). See also Griepentrog (1995: 471-76).

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etc.;[124] ~ *mintìs* 'thought' [< **mn̥-ti-*]). Also the noun *pažastìs* 'armpit' (gen. sg. *-iẽs*, gen. pl. *-tų̃* or *-čių̃*) is unlikely to be an old consonant stem. It is built in a productive way adding the thematic *žãstas* 'forearm' to the prefix *pa-* 'under'. Such derivatives are very often (*i*)*io*-stems, cf. *tvorà* 'fence' → *patvorỹs* 'place under/at the fence', *ùpė* 'river' → *paupỹs* 'area at the river', etc., and *pažastìs* has most probably developed from the more original *pãžastis* (masc.; *io*-st.).

In addition to what has been said above, the matter is further complicated by Lithuanian nouns of the *sámtis* 'ladle' or *sagtìs* 'buckle' type, which may reflect acrostatic *t*-stems as well, but, unlike *krūtis*, the nouns of this type reflect the *o*-grade of the root.[125]

Thus, to conclude, the noun *krūtis* faces two problems: on the one hand, it could be reconstructed both as a *t*-stem and a *ti*-stem, and on the other hand, if one were to reconstruct this noun as an original *t*-stem, its root shape would be different from other potential reflexes of the ancient Indo-European *t*-stems in Lithuanian.

3. Go. *waihts* 'thing, matter' (generally inflected as an *i*-st., but acc. pl. *waihts* in *Skeireins* II:25): If not for the acc. pl. *waihts*, this noun could be analyzed as an original *i*-stem. The support for such an interpretation could be found in OIc. *vættr/véttr/víttr* (fem.),[126] OE *wiht* 'being' (neut.; < PGmc. **weh-ti-*). Neither of these two nouns

[123] Meid (1982: 198ff.). However, Ch. Stang interpreted Lith. *krūt-* as an unsegmentable root (1966: 223).
[124] See the list of forms in J. Kazlauskas (1968: 251).
[125] See section III.8.5 below.
[126] See A. Noreen (1923: 269-70, and fn. 1 on p. 270).

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should be interpreted as an old consonant stem. Whereas neuter athematic nouns did not exist in Old English,[127] the Old Icelandic pronoun *vettergis* (gen. of *vet[t]ki*, *væt[t]ki* 'nothing') is thought to reflect the frozen *i*-stem genitive singular ending *-er* (< Proto-Scandinavian *$*-ēʀ$ < Proto-Germanic *$*-aiz$).[128]

As in the previous instances, the morphologically ambiguous forms are Gothic. This ambiguity and confusion of inflections may be a sign that the consonantal and the *i*-stem inflections might have slowly started to merge in Gothic. Although the two inflectional types are still kept distinct in Wulfila's translation, certain parts of the consonantal and the *i*-stem paradigms are identical, cf. nom. sg. *anst-s* 'love' (*i*-st.) vs. *baurg-s* 'city' (root noun), acc. sg. *anst* vs. *baurg*, dat. pl. *anst-im* vs. *baurg-im* and gen. pl. *anst-e* vs. *baurg-e*. It may be that the consonantal forms like *waihts*, *dulþ* etc. are hypercorrect, i.e., they belong to historical *i*-stem nouns, but were misintepreted as consonantal. However, this matter calls for a further investigation, which cannot be undertaken here.

[127] A. Campbell (1959: 252).
[128] Noreen, loc. cit.

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III. Primary *t*-Formations with a Non-alternating Suffix *-t-*

III.1. Acrostatic *t*-Stems

According to one common view, Indo-European acrostatic formations can be subdivided into two subclasses: one type of acrostatic formations are characterized by the ablaut pattern R(ó)-S(z)-E(z) in the strong stem and R(é)-S(z)-E(z) in the weak stem, whereas in the other type, the vocalic alternation in the root is **ḗ/é*.[129]

Whereas no primary *t*-stems of the latter type are attested in the Indo-European languages,[130] primary *t*-stems of the *ó/é*-type are rather uncommon. The derivation of acrostatic *t*-stems was also non-productive in Indo-European. Most of such *t*-stems are also attested in not more than one language group, and usually individual *t*-stems of this type are found in a single language. Only one noun, the Indo-European word for 'night' (IE **nók$^{\mu}$-t-s*), has been preserved in most Indo-European languages in one way or another.[131] Otherwise, clear reflexes of the Indo-European acrostatic "Ia" type *t*-stems are somewhat better attested only in Hittite, Greek and Latin.

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[129] See Schindler (1975b: 262), Widmer (2004: 51). In Schindler's nomenclature, the *ó/é*-type is called "Ia", and the *ḗ/é*-type "Ib". Neri (2003) calls the previous type "type I", and the latter "type II". Neri also distinguishes between two subclasses of "type I" (see Neri 2003: 17-25), viz. R(ó)-S(z)-E(z)/R(é)-S(z)-E(z) (type **u̯ód-r̥/*u̯éd-n-*) vs. R(ó)-S(z)-E(z)/R(z)-S(é)-E(z) (type **ĝón-u/*ĝén-u- → *ĝn-éu-*), but the latter class is obviously a reformation of the previous. In some works, no formal distinction between the different subtypes is made, cf. Tichy (2000: 74), Meier-Brügger (2000: 200). Beekes reconstructed the original ablaut as **CéC-R/*CéC-R-s* (Beekes 1995: 176), but see pp. 188-9 below in op. cit. where several instances of the *o*-grade in the strong stem are discussed.

[130] The Vedic *yákr̥-t* 'liver' and Armenian *leard* 'id.' and *near-d* 'sinew' (< **-t*) are all secondary *t*-stems. The element *-t-* was added secondarily to original heteroclitic nouns which should be reconstructed as **(H)i̯ék$^{\mu}$-r̥* 'liver' (Nussbaum 1998: 150) resp. **snéh$_1$-u̯r̥* (cf. G. Þórhallsdóttir 1988: 199) or **sh$_2$-néu̯-r̥* (J. Friedman 2001 and p. c.). In Sanskrit, *t* appears only in the strong forms, whereas the weak stem is *yak-n-*. No traces of *t* are attested in the paradigms of the Avestan *yākarə* 'liver' resp. *snāuuarə* 'sinew'.

[131] For a more detailed discussion of the development of this noun in the various Indo-European dialects see below.

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Recently, the view has been expressed that reflexes of Proto-Indo-European acrostatic *t*-stems are also to be seen in the nouns of the Indo-Iranian *stú-t-* type, viz., Skt. *stú-t-* 'hymn' itself, Avestan *stu-t-/stū-t-* 'id.', and several other nouns.[132] According to this theory, *t*-stems of the *stú-t-* type with their unexpected zero grade in the root originally had the same structure as *t*-stems of the **nók^{u}-t-* type, and had the characteristic *ó/é* ablaut in the root. However, in the course of development, the *stú-t-* type *t*-stems eliminated the full grade in the root altogether, adopting the ablaut of certain root nouns.[133]

This view is in certain ways quite attractive, e.g., it is noteworthy, that both the **nók^{u}-t-* and the *stú-t-* type *t*-stems are primary *t*-formations built to verbal roots and, accordingly, can be interpreted as concretized old action nouns. As was said earlier, according to the present *communis opinio*, the element *-t-* in *t*-stems of the **nók^{u}-t-* type is most plausibly interpreted as a derivational suffix. Accordingly, one would be tempted – together with Rieken – to apply the same analysis for *t*-stems of the *stú-t-* type. However, there are certain problematic issues which Rieken did not discuss in her *Untersuchungen*, and which need to be addressed.

On the one hand, the view has been expressed that the transition from the acrostatic "Ia" inflection, characterized by the constant full grade of the root to a type of inflection in which both the root and the suffix *always* appear in the zero grade, cannot be imagined as a very natural phenomenon.[134] When *t*-stems of the *stú-t-* type are compared to original acrostatic nouns of other suffixal stem classes, it appears that

[132] Rieken (1999: 85), followed by Irslinger (2002: 39-40).
[133] Rieken, ibid.

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other nouns develop in a different way, viz., they either preserve the original full grade in at least a part of the paradigm, cf. Skt. *jắn-u* 'knee' (nom. sg.; < **ĝón-u*) vs. *jñóḥ* 'id.' (gen. sg.; < **ĝn-éu-s* ← ***ĝén-u-s*), or, in the rare cases where the full grade was totally eliminated, cf. La. *mare* 'sea' (← **mṛr-éi*, loc. sg.), it can still be observed in genetically related words in other dialects, cf. Old Irish *muir* 'sea' (< **mór-i*).[135] The story is totally different with the nouns of the *stú-t-* type, where there exists no indication for the existence of any full grade roots anywhere in the nominal system.[136]

On the other hand, since the Indo-Iranian nouns of the *stú-t-* type have no cognates in other Indo-European dialects and the addition of the element *t* to roots of a particular shape was a productive process in Indo-Iranian, the nouns of this type are perfectly ambiguous in that they can be interpreted either as historical root nouns enlarged with a secondary and morphologically empty *-t-*, or as historical derivative *t*-stems (like the **nók^{u}-t-*type). An important issue in the study of the history of the *stú-t-* type nouns is the ultimate source of the *t*-enlargement in Indo-Iranian. Although it may be that this source may will never be discovered, several considerations will be presented in chapter IV.

In the following sections, I would like to study the *t*-stems which can be with more or less plausibility interpreted as reflexes of the ancient Indo-European acrostatic *ó/é t-* stems.

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[134] This issue has been raised by Nussbaum (2004a and p.c.), also J. Friedman (p.c.).
[135] Following Nussbaum (2004d).
[136] The Vedic nouns *srótas-* 'stream' and *rétas-* 'stream', both of which have been explained by Hirt as evidence for full grade **sreu-t-* resp. **rei-t-* will be treated in section IV.7 below.

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III.2. **nók^{u}-t-s* Type (Acrostatic Ia)

The representatives of this type of acrostats are not numerous. The most widely attested in the Indo-European daughter languages are the reflexes of the Indo-European *t*-stem **nók^{u}-t-s* 'night'. The original structure and formation of this noun, as well as its development in the individual dialects will be discussed in more detail in section III.3 below. Otherwise, most of the clear reflexes of Indo-European acrostatic *t*-stems come from Hittite, Greek, and Latin.

In Hittite, besides the noun *nekuz*, the reflex of the frozen genitive singular form of IE **nók^{u}-t-s*, there are also attested the following originally acrostatic nouns: *kašz* 'hunger' (stem *kašt-*),[137] *maišz* 'glittering' (stem *maišt-*), *šašz* 'sleep' (stem *šašt-*) on the one hand, and *kar(a)iz* 'flood' (stem *karaitt-/geret-/karitt-*) and *kuz* 'wall' (stem *kutt-*) on the other hand. The latter two *t*-stems are somewhat different from the former.

In Latin, the number of nouns reflecting original acrostatic formation of this sort is quite small, the only reflexes being *cōs* 'whetstone' (fem.; gen. sg. *cō-t-is*), *dōs* 'dowry' (fem.; gen. sg. *dō-t-is*), *grātes* 'gratitude' (pl.) and *nox* 'night' (fem.; gen. sg. *noc-t-is*).

In Greek, where *t*-stems became extremely prolific and many of them can easily be interpreted as secondary (cf. Gk. δόρυ 'tree', gen. [Attic] δούρα-τ-ος, or κρέας 'flesh', gen. [Attic] κρέα-τ-ος, neither of which was a *t*-stem originally),[138] it is sometimes difficult to tell, which of the nouns that formally look like original acrostats actually reflect ancient Indo-European acrostatic *t*-stem formations. In the studies of historical

[137] A *t*-stem of the same structure is attested in both Tocharian dialects, cf. Toch. A *kaṣt*, Toch. B *kest* 'hunger' (< **Kos-t-*; see Rieken 1999: 84). J. Tischler also adds here the Hieroglyphic Luvian *as-ta-* 'hunger' (Tischler 1990: 90).

[138] The noun δόρυ is an old *u*-stem (IE **dór-u*), and κρέας is an old *s*-stem (IE **kréuh$_2$-s-Ø*).

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Greek morphology, the following nouns have been interpreted as historical *t*-formations with a non-alternating suffix: δαίς 'meal; banquet' (fem.; gen. sg. δαι-τ-ός), νύξ 'night' (fem.; gen. sg. νυκ-τ-ός), πλώς 'mullet, κεστρεύς fish' (masc.; gen. sg. πλω-τ-ός), δώς 'gift' (Hesiod; hapax, fem.), φώς 'man, hero' (masc.; gen. sg. φω-τ-ός), θής 'day labourer' (masc.; gen. sg. θη-τ-ός), *λῖς 'linen cloth' (masc.; only acc. sg. λῖ-τ-α and dat. sg. λῖ-τ-ί are attested) and σταίς (or σταῖς) 'dough' (neut.; gen. sg. σται-τ-ός).[139] Although not all of these nouns are direct reflexes of Proto-Indo-European acrostatic *t*-stems, the history of each of them needs to be discussed individually, in order to demonstrate their differences.

Also the Hittite and the Latin reflexes of acrostatic *t*-stems listed above will receive individual treatment despite the large amount of existing literature about them. This will be done for two purposes: first, in order to bring together the various theories about their history, and second, to evaluate them and to consider several alternative interpretations, as well discuss some unanswered questions.

Due to its widespread attestation in Indo-European, the noun ***nók*$^{\mu}$*-t-s* will be treated in a separate section.

III.3. IE ***nók*$^{\mu}$*-t-s* 'night'

The original Proto-Indo-European structure and the ablaut of type "Ia" acrostatic nouns is now quite well understood, and has been described in a series of works.[140] At

[139] See a detailed study in C. D. Buck (1918). The noun δώς is interpreted as a historical *t*-stem in Rieken (1999: 84). For a brief discussion of σταίς/σταῖς, see section II.11.1 above.
[140] See Schindler (1975b: 262), Meier-Brügger (2000: 199-200), etc.

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the Proto-Indo-European stage, acrostatic *t*-stems of this type had the ictus on the root syllable in all cases except for locative, where it fell on the suffix. In the strong cases, the root had an *o*-grade, whereas in the weak cases it had an *e*-grade. Only in the locative singular the root originally appeared in the zero grade. In the locative, the suffix received the *e*-grade, and was accented.

The original idealized singular paradigm of the *t*-stem **nók^{u}-t-s* 'night' is presented below:[141]

Nom. **nók^{u}-t-s* */nóg^{u}-t-s/ ~ IE verbal root **neg^{u}-* 'be dark'[142]
Acc. **nók^{u}-t-m̥*
Instr. **nék^{u}-t-h$_1$*
Dat. **nék^{u}-t-ei*[143]
Loc. **n̥g^{u}-ét(±i)*
Abl./Gen. **nék^{u}-t-s*[144]

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[141] In a rather different analysis, Szemerényi considered the noun **nók^{u}t-* to have developed from an original **ne-ku-t-* (< **ne* [negation] + **ku-* 'light' + *-t-*; see Szemerényi 1964: 402, and fn. 1), by means of weakening of the unstressed *-u-* to *-u̯-* and subsequent rounding of the vowel **-e-* to **-o-* due to the influence of the labiovelar (see criticism in Eichner 1974: 42). Szemerényi's comparison of **ne-ku-t-* to **ne-pō-t-* 'nephew' is equally ill-advised (see a brief discussion of the problem arising from the interpretation of the noun **nepōt-* as **ne-pō-t-* in Rieken 1999: 92, fn. 431).

[142] Cf. Hi. *ne-ku-uz-zi* (3. sg. pr.) in phrases like *maḫḫan nekuzzi* 'when the evening comes' (KUB 7.5 ii 14 etc.), *kuitman nekuzzi* 'as it becomes evening' (KUB 8.12:7) etc.

[143] Theoretically, one should probably expect an original PIE **nék^{u}-t-i*. However, there is no evidence for such a dative ending in Indo-European. All data seem to point to **-ei*. The dialectal Lithuanian datives ending in *-i*, cf. *piemen-i* 'herdsman', *akmen-i* 'stone', *aus-i* 'ear', *nakt-i* 'night', *šun-i* 'dog', etc. should probably not be interpreted as potentially very archaic forms, preserved since early Proto-Indo-European times (as implied in Kazlauskas 1968: 139-46). They may well have arisen due to certain dialectal "Auslautsgesetze", which, unfortunately, are not thoroughly understood yet. One could imagine that the short vowel *i* in the ending of these datives might have arisen as a result of the weakening or the monophthongization of an original **-ei#* (cf. also the change **ei* > *ī* in southern Žemaitian dialects of Lithuanian). However, the final answer to the question of the Lithuanian *i*-datives is not readily available yet.

[144] Lindeman presents a slightly different reconstruction. According to him, the genitive of **nók^{u}-t-s* was either **nk^{u}-t-é/ós* or **nk^{u}-ét-s*, whereas the locative was either **nk^{u}-t-í* or **nék^{u}-t* (Lindeman 1986: 150). However, no evidence for such forms exists. For the reconstruction of the late (post-Anatolian) Indo-European paradigm, see below in this section.

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This paradigm has nowhere been preserved intact. Probably as early as Proto-Indo-European, the locative *$n̥g^{u}$-*ét*(±*i*) was reshaped to be more like the rest of the paradigm. This would have involved introducing a new *e*-grade into the root, and eliminating one in the suffix, cf. *$n̥g^{u}$-*ét*(±*i*) → *$nék^{u}$-*ti*.[145] There probably were two combinatory reasons for this reformation, first, analogy from the other weak cases, and second, the urge to break up the sequence #$n̥g^{u}$- in word-initial position. The insertion of a secondary *e*-grade into word-initial syllables where the zero grade is expected is quite widely attested in Indo-European, and may have been systematic to some degree.[146]

At a later stage, which would have been either dialectal or at least post-Anatolian, the *o*-grade of the root was generalized throughout the paradigm. Thus, at that stage, the locative would have been *$nók^{u}$-*t*-*i*. Also, the zero grade ablative/genitive ending -*s* most probably was replaced by a full grade -*es* or -*os*. The paradigm that most Indo-European dialects (outside Anatolian) reflect would have looked in the following way:

Nom. *$nók^{u}$-*t*-*s*
Acc. *$nók^{u}$-*t*-$m̥$
Instr. *$nók^{u}$-*t*°
Dat. *$nók^{u}$-*t*-*ei*
Loc. *$nók^{u}$-*t*-*i*
Abl./Gen. *$nók^{u}$-*t*-*e*/*os* or *nok^{u}-*t*-*é*/*ós*[147]

[145] In late Indo-European, locatives probably systematically had the ending -*i*.
[146] See also a brief discussion of this matter in K.-H. Mottausch (2000: 34, 38) and Vine (2004: 366ff.).
[147] The genitive singular ending of late common Indo-European athematic nouns is very difficult to reconstruct due to various complicating factors, such as different and/or phonologically ambiguous outcome in the dialects. Greek has generalized an accented *-*ós*, cf. νυκτός, 'night', δουρός 'tree' (Hom.), γουνός 'knee' (Hom.), whereas Germanic quite often points to *-*es* [±stressed], cf. Olc. *nætr* 'night' (< *$naχt$-*iz* < *-*es*), *merkr* 'boundary' (< *$mark$-*iz* < [virtual] *$morĝ$-*es*). Latin has evidence for both, cf. *pedis* 'foot' (< [virtual] *ped-

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Only the Hittite word-form *nekuz*, which occurs in a frozen phrase *nekuz meḫur* 'night time', shows a preserved *e*-grade in the root, which, as was shown in the previous paradigm, is assumed to have been regular for the weak cases.[148] Thus, the Hittite form also provides important evidence for *e/o* ablaut within the root, which has helped to establish the original ablaut type for the Indo-European noun **nók^{u}-t-* as acrostatic "Ia" type.

The formation of this noun has been explained by Schindler (1967). According to his explanation, the noun **nók^{u}-t-s* originally was an animate verbal noun, built to the root **neg^{u}-* 'be dark'.[149] The early meaning of this verbal noun would have been 'being

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es) and **nominus** 'nominis' (< **-os*, see G. Meiser 1998: 138). The vowel quality of the pre-Indo-Iranian genitive ending is often ambiguous due to the merger of **e* and **o* with **a*. As for the stress, cf. Ved. *paśv-ás* 'animal', *pad-ás* 'foot', Avestan *nəmō* 'house' (< **dm-és*, cf. Schindler 1972b: 32) vs. Ved. *vásv-as* 'good', *krátv-as* 'power', *śún-as* 'dog', etc. Hittite tends to show reflexes of **-os*, cf. *link(a)iḭ-aš* 'oath', *kardiḭ-aš* 'heart', *ginuḭ-aš* 'knee', *kutt-aš* 'wall', but it is unclear whether this ending ever carried the stress. According to Neri (2003: 34), the spelling *tak-na-a-aš* 'earth' (gen. sg.) with plene spelling points to Proto-Anatolian **dagnás* (< **-ós*), but this word is more commonly spelled *ták-na-aš* (see, e.g. *ták-na-aš* [IBoT III 147 3', KUB VII 7 4', KUB XXX 42 IV 3'], *ták-na-aš-ša* [KBo XV 12 11'], *ták-na-aš-ma* [KUB VII 10 7'], etc.) which does not clearly indicate length or provides undisputable information about the stress.

[148] Although earlier it had been suggested by J. Friedrich and later confirmed by C. Watkins that *nekuz* is a nominative form (see Watkins 1965: 355), the genitive interpretation originally proposed by A. Kammenhuber (1961: 271) must be the correct one. The Old Russian word-form **ночь** 'at night' cited by Watkins (1965: 356) can equally be interpreted as a temporal accusative, and this would find support in Lithuanian *naktį* 'at night' (formally, accusative singular). Also Lindeman's claim that *nekuz meḫur* is to be interpreted as an old compound **nek^{u}ti-meHur* (Loc. + Nom.; 1986: 148) is *ad hoc*.

[149] The alternative suggestions by Rix who reconstructed the root as **neg^{u}h-* because of Gk. νύχιος 'nightly' (2000: 213) and by Ch. Peeters who reconstructed **nek-* because of alleged absence of labialization in Gk. νύξ and Proto-Germanic **naχt-* (1974), are based on the misenterpretation of data. For criticism of Rix, see O. Panagl (1971). As for the absence of labialization, one has to notice that in clusters of the type *-C^{u}C-* the labialization of the first consonant tends to disappear, cf. OHG *līht(i)*, Go. *leihts*, OIc. *léttr*, OE *lēoht* 'light' (adj.) < PGmc. **lenχ-ta-* < IE **h$_1$leng^{u}h-*, OIc. *røkkr* 'darkness' < **h$_1$reg^{u}-*, etc.

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dark', which in the course of development acquired a more concrete meaning, 'night' (← 'the dark period of time').

As for the concretization of the original abstract meaning of **nók^{u}-t-* and of various Indo-European abstracta, this is a widely attested phenomenon in Indo-European, and a large number of original verbal abstracta in the course of development lost their abstract semantics in favour of a more concrete meaning, cf. (late) IE **mn̥-tí-s* 'thinking' (~ **men-* 'think') → 'thought', as illustrated by Ved. *matí-*, Lith. *mintìs* 'thought',[150] IE **séh$_1$-mn̥* 'sowing' (~ **seh$_1$-* 'sow') → 'seed', cf. La. *sēmen* 'seed', etc. Such concretization of abstracta is typologically commonplace, and is still productive in many languages, cf. English *building* 'act of building' as well as 'built structure', Lith. *arìmas* 'act of ploughing' and 'ploughed field' (*ar-* 'plough'), etc. Thus, it is easy to imagine how an original abstract noun 'being dark' acquired the meaning 'night'.

To summarize, dialectal evidence points to an original acrostatic noun **nók^{u}-t-/ *nék^{u}-t-*, built from the verbal root **neg^{u}-* 'be dark' and the verbal abstract suffix *-t-*. The original meaning of this verbal noun was 'being dark', which eventually evolved into 'dark period, night'.

III.4. Reflexes of Acrostatic *t*-Stems in Hittite

The most recent and detailed study of the Hittite reflexes of the Indo-European acrostatic *t*-stems is to be found in Rieken's *Untersuchungen* (1999). A brief discussion of the same material can be found in Irslinger (2002: 38-9).

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In her *Untersuchungen*, Rieken reconstructs original acrostatic (type Ia) inflection for a small number of Hittite nouns, viz. *kašt-* 'hunger', *šašt-* 'sleep', *nekuz* 'night', *karaitt-* 'flood', *kutt-* 'wall' and *maišt-* 'glitter, brilliance'.[151] According to Rieken, fairly early within the course of their development, the ancestors of these Hittite *t*-stems split into two groups. The nouns from which later Hittite *nekuz*, *kašt-* and *šašt-* developed continued to be inflected according to the common acrostatic type with the ablaut **ó/é* (type Ia). However, the nouns *karaitt-*, *kutt-* and *maišt-*, according to Rieken, developed in a different way. For *karaitt-* and *kutt-*, Rieken asumed that at an early (Indo-European?) stage of development, they acquired the following ablaut pattern: lengthened *ő*-grade in the nominative, normal *ó*-grade in the accusative and zero grade in the weak forms. For the noun *maišt-*, Rieken offers two choices: according to one, the noun *maišt-* reflects the same inflectional pattern as *karaitt-* and *kutt-* (R[ő/ó] for the strong cases vs. R[ø] for the weak cases). According to the other option, the strong cases would have still had the lengthened resp. normal *o*-grade in the root, whereas in the weak cases, the root would have had *e*-grade.[152]

As for the reconstruction of the *ó/é*-ablaut for the nouns *nekuz*, *kašt-* and *šašt-*, a few observations need to be made. Whereas the nouns *šašt-* and *kašt-* show the *o*-grade of the root throughout the entire paradigm, as seen in a number of from the historical point of view weak forms (cf. the genitive singular form *šašt-aš* or dat./loc. *kašt-i*, etc.), the root of the word-form *nekuz* exhibits a more archaic *e*-grade (< **nék^{u}-t-s*).

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[150] Interestingly, also the English noun *thought* was a verbal abstract originally, and can be reconstructed as Proto-Germanic **þunχ-ti-* 'thinking', a *ti*-derivative of the verbal root **teng-* 'know, think' (cf. La. *tongēre* 'know'; see LIV.629).
[151] See Rieken (1999: 128-39).

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Secondly, whereas *nekuz* also shows a more archaic genitive ending *-s* in zero grade, the genitive form *šašt-aš* shows a secondary **-os*, a very productive genitive marker in the Hittite nominal system.[153] Thirdly, especially the noun *kašt-* is often attested with a long vowel in the root, cf. nom. sg. *ga-a-aš-z[a]* (OH) and *ka-a-aš-za* (OH/NH; beside MH *ga-aš-za*), acc. sg. *ga-a-aš-ta-an* (MH; beside NH *ga-aš-ta-an*), dat./loc. sg. *ka-a-aš-ti* beside *ka-aš-ti* (MH).[154]

As per Melchert (1994: 146-7), *ā* is a regular reflex of **ó*, thus, one could say that forms like *ka-a-aš-za* directly reflect the original **CóC-t-*. However, then one could ask how the forms with the short *ă* should be explained.

In any case, the easiest way to account for the lack of length for the vowel *ă* is via the lack of accent, since the unaccented **o* regularly turned into *ă* in Hittite. However, this would imply that some parts of this originally acrostatic paradigm must have been, or, more precisely, must have *become* oxytone. Later, the short *ă* from the oxytone forms would have been introduced by analogy into the forms with the regular long *ā* (< **ó*).[155]

As far as the forms with *ā* are concerned, the length of the vowel can be explained in two ways. Either one could say that *ā* reflects an older short accented **ó* which, as haas been said earlier, regularly developed to *ā*. However, the situation might have been more complex. For certain reasons it may be worthwhile to consider also a long

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[152] Rieken, loc. cit.
[153] The genitive of *kašt-* has not been preserved, but it would have most probably been **kašt-aš*.
[154] See Rieken (1999: 132). The noun *šašt-* is once attested with *ā*, cf. *ša-aš-dụ-uš* (acc. pl.), but the form occurs in a New Hittite text.
[155] Additional evidence for the reconstruction of oxytone forms in original acrostatic *t*-stems is also discussed in section III.4.1 – III.4.1.1 below.

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*ǒ as the possible source of Hi. *ā* in the attested forms. I shall return to this hypothesis below, but it is possible to draw some preliminary conclusions already: whereas the frozen genitive *nekuz* (which no longer is a part of a living paradigm) has preserved several archaic features, the nouns *kašt-* and *šašt-* (both fully inflected) developed in a different way, reforming both the original ablaut pattern and the original acrostatic accentuation. These reformations can be represented in a following scheme:[156]

Proto-Anatolian			pre-Hittite	Hittite
Nom.	**CóC-t-s*	>	**Cŏ̆C-t-s*	*kăšza* /kăs-t-s/, *šašza* /sas-t-s/[157]
Acc.	**CóC-t-ṃ*	$\overset{?}{>}$	**CóC-t-an*[158]	*šaš-t-an, kăš-t-an*
Gen.	**CéC-t-s*	→	**CoC-t-os*[159]	*šaš-t-aš*
Loc.	**CéC-t-* ?[160]	→	** Cŏ̆C-t-i*	*kăšt-i, šašt-i, maiš-t-i*

Whether this pattern was productive or non-productive may be difficult to judge since the corpus of data is very limited. However, from the attested dative/locative singular form *ma-iš-ti* 'brilliance, glitter' (← orig. **móis-t-/*méis-t-*), one could argue that the noun *maišt-* has been reformed in the same way. In such a case, half of the attested Hittite reflexes of the Indo-European acrostatic *t*-stems would have been

[156] See also section III.4.4 below.
[157] In New Hittite, a thematized *ša-aš-ta-aš* is attested as well (Rieken 1999: 132).
[158] For the question of the development of final *-ṃ in Hittite, see Melchert (1994: 181-2).
[159] For a discussion of the problems related to the reconstruction of the accentuation of late common Indo-European genitives, see section III.3 above.
[160] For the reconstruction of the shape of the idealized early Indo-European locative, see section III.3.1.1 above.

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reformed in the same way, which suggests that this pattern may have been at least somewhat productive.

As for Rieken's reconstruction of the quantitative ablaut **ṓR/óR/R̥* for the nouns *karaitt-*, *kutt-* (and probably *maišt-*),[161] it is based on several phonological and morphological peculiarities of these nouns, namely, the reflection of a lengthened **ṓi* in the diphthong *-ai-* in the Old Hittite nominative form *karaiz* (spelled *ka-ra-i-iz*), the reflection of zero grade in Middle Hittite *kutt-* and *kariz* (spelled *ka-ri-iz*; nom. sg.), and the occurrence of these features among certain Indo-European root nouns, for which quantitative ablaut can be reconstructed quite safely.[162]

Rieken's interpretation that certain *t*-stems (at least *karaitt-* and *kutt-*) developed in a way parallel to root nouns, substituting the ablaut R(ó)-R(z) for the original R(ó)-R(é), is most probably a correct one. However, one needs to clarify how this happened, especially because suffixal acrostatic nominal formations more commonly develop *not* like root nouns, cf. (virtual) **ĝréi-t-s* 'flood' (gen. sg.) → **ĝri-t-os*, but **ĝén-u-s* 'knee' (gen. sg.) → **ĝen-u̯-os* (> Hitt. *genuuaš*) or **ĝn-éu-s* (> Skt. *jñóḥ*), not **ĝn-uu̯-os* or **ĝn̥-u̯-os*, or (late IE, virtual) **nókʷ-t-s* → **nokʷ-t-es* (> PGmc. **naχt-iz*) or **nokʷ-t-os* (> Gk. νυκτός) not **n̥kʷ-t-os*, etc.

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[161] See Rieken (1999: 138-9).

[162] However, also in these root nouns, quantitative ablaut probably does not reflect the original state of affairs. The lengthened grade in the nominative is often due to special phonological development (cf. IE **ḱuu̯ōn* 'dog' < ***ḱuu̯on-s* by "Szemerényi's law") or to analogy (cf. IE **pṓd-s* 'foot', which was probably influenced by words in which the lengthening was phonological). As far as the endings of the weak stem (especially genitive) are concerned, opinions vary, but at least in some cases we may be more or less sure that reconstructable full grade endings are later substitutions for original endings in the zero grade, cf. original IE **dém(h₂)-s* 'home' (gen. sg.) and later **d(m̥)m-és*, etc.

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In the following sections I would like to return to some details of the development of the nouns *karaitt-*, *kutt-* and *maišt-* once again.

III.4.1. Hi. *karaitt-* 'flood'

Hi. *karaitt-* 'flood' is a well attested noun, appearing in Old, Middle and New Hittite texts. The following list shows some of the attested singular forms:[163]

nom. *ka-ra-i-iz*	KUB XXXVI 110 Rs. 19′ (OH)[164]
ge-re-ez-za	KUB XXXIV 10, 10′ (OH/NH)
ka-ri-iz	KUB VIII 27 lk. Rd. a 3 (NH)
dat./loc. *ka-ra-it-ti*	KUB XXXVI 110 Rs. 18′ (OH)

A number of etymologies for *karaitt-* have been proposed, and it has been interpreted both as a derivative *t*-stem and as a root noun (see a survey of various etymologies in HED IV.86).

A few words need to be said about the recent Puhvel's etymology of Hi. *karaitt-*, according to which this noun should be interpreted as root noun **ĝhred-*.[165] Puhvel connected it etymologically with Skt. *hradá-* 'lake, pool', explaining the diphthong *-ai-* as a "hyper-correction" for an underlying *ē*.[166] Such a possibility theoretically cannot be ruled out, since the diphthong *-ai-* indeed occasionally occurs instead of *-ē-* in Hittite,

[163] The forms are adopted from Rieken (1999: 134; see there for a more exhaustive list of singular forms together with plural forms).
[164] As noted by Rieken (1999: 134, fn. 620), the signs ⟨]a-iz-za⟩ in a poorly preserved fragment KUB XV 40, 8′ could represent *ka-r]a-iz-za* (nom. sg.).
[165] HED, loc. cit.
[166] Ibid.

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cf. ⟨a-ip-ta⟩ for *ēpta* 'took' (3. p. pret. sg. act.; KBo V 6 I.11), ⟨ma-iq-qa-uš⟩ for *meggauš* 'large' (acc. pl.; KUB XXVI 1 III.58).[167] Moreover, if the broken form ⟨-r[?]]a-iz-za⟩ is unrelated to *karaitt-*, both forms exhibiting the diphthong *-ai-*occur in the same sentence, which gives scribal error a chance, cf. the following excerpt from KUB XXXVI 110:

18′ ap-pa-li-ia-al-la-aš É[-ir-še-it]	'the house of an ensnarer
19′ ka-ra-it-ti pi-e-ra-an ú-[e-ta-an]	[is] built in the path of a flood:
20′ ka-ra-i-iz la-a-ḫu-[a-a-i]	the flood pours,
21′ na-at pa-ra-a šar-ta-i ša[?]-ku[?]-[ ] a-ru-na [	sweeps it forth and [...] to the sea'[168]

However, even though a scribal error cannot be entirely ruled out, the connection of *karaitt-* with Skt. *hradá-* and the reconstruction of **ĝhred-* is less attractive than the reconstruction of the root **ĝrei-* and the etymological connection with Skt. *jráyas-* 'expanse', *jráyati* 'extends' (3. sg.; < **ĝréi̯-e-*), Av. *zraiiah-* and Old Persian *drayah-* 'sea' (< **ĝréi-es-*) as per B. Čop (1954: 162). First, the history of Skt. *hradá-* itself is not entirely clear, and it may reflect either **ĝhrod-ó-* or **ĝhlod-ó-*.[169] Secondly, if the original meaning of the *t*-stem **ĝroi-t-* is reconstructed as verbal abstract 'stretching, extending', the semantic change to 'flood' in Hittite becomes quite similar to the change that has taken place in the development of Common Iranian **zraia̯h-* 'sea' (←

[167] Friedrich (1960a: 26).
[168] HED IV.86.
[169] See EWAia I.822.

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*ĝrei̯-es- 'expanding').[170] In addition to this, it becomes unnecessary to assume that text is corrupted by a scribal error.

III.4.1.1.

Rieken is most probably right in assuming a lengthened *ó*-grade for the root in the nominative form, which is at least once spelled *ka-ra-i-iz*. To assume a long diphthong **ói* seems to be the only way to account for the Hittite diphthong *ai*, as the short diphthong **ŏi* probably would have monophthongized to *e* in this position.[171] However, the form *ge-re-e-ez-za* with a monophthong *e* ([e:]?) would reflect a short diphthong **ŏi*.[172] The existence of both a lengthened grade and a normal grade in a single paradigm has caused some scholars to believe that originally this noun must have had a lengthened *ō*-grade in the nominative singular, and a normal *o*-grade in the accusative, i.e. had features usually reconstructed for root nouns.[173] Finally, the occurrence of spellings with a vowel *i* in the root, cf. *ka-ri-iz* (Middle Hittite; nom. sg.) suggests that this noun also had forms with the root in the zero grade, cf. **ĝri-t-*.[174] The zero grade would have originally occurred in the weak forms, but eventually would have spread to the strong forms as well (cf. the MH/NH nominative singular *kariz*). On the basis of the

[170] Cf. LIV.170.
[171] Cf. S. E. Kimball (1999: 215). Also J. Catsanicos assumed that **oi* > *e* /_*t* in Hittite, cf. *mitā(i)-* 'fasten' < **h₂moit-* (1986: 156 and further). Melchert (1994) considered a theoretical possibility that *oi/ai* is preserved before *t*, just as it is preserved before other coronals (1994: 148), but in Melchert (1998), he adopted the earlier interpretation by Catsanicos.
[172] This nominative form may have beeen influenced by the accusative, cf. Rieken (1999: 135), also Melchert (p. c.).
[173] See Rieken (1999: 135; with further refs.).
[174] Cf. Rieken (1999: 134-5), Nussbaum (2004a), apparently already Schindler (1972b: 35) as well.

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attested forms of *karaitt-*, the original paradigm of this noun has been reconstructed in the following way: nom. sg. **ĝrṓi-t-s*, acc. sg. **ĝrói-t-ṃ*, gen. sg. **ĝri-t-és*.[175]

Now, as was said earlier, such an ablaut pattern seems to be virtually unknown among suffixed nouns. The lengthened grade of the root in the nominative case is attested among root nouns, cf. Gk. φώρ 'thief' (< **bhōr-*), Ved. *vā́c-*, Av. *vāxš* and La. *vōx* 'voice' (< **u̯ṓkᵘ-*), Ved. *gaú-*, La. *bōs*, Gk. βοῦς (Doric βῶς) 'ox' (< **gᵘṓu̯-*), Ved. *pā́t*, Gk. πούς (Doric πῶς [Hesych.]), Proto-Germanic **fōt-* 'foot' (< **pṓd-*), etc. The only suffixal nouns that habitually exhibit roots with the lengthened grade in the strong forms, are those of the **h₁ég*ᵘ*h-i-* type, and they have not an *ō*-grade, but an *ē*-grade.[176] The ablaut pattern of the nouns that belong to the **h₁ég*ᵘ*h-i-* type is totally different from the root nouns of the **bhōr-* type discussed above.

As far as the weak stem is concerned, the pattern R(ø)-S(ø)-E(e/o) resp. R(ø)-E(e/o) is attested among suffixal hystero- and amphikinetic formations resp. root nouns, but generally not among suffixal acrostatic formations.

In order to explain the lengthened grade in the root of the noun *karaitt-*, one most probably has to assume an analogy from root nouns of the **bhōr* type, as per Rieken (1999: 84-5) who had adopted Schindler's theory of the ablaut in the root nouns with roots of the *-oR(C)-* type. According to Schindler, acrostatic root nouns which have

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[175] Rieken, ibid. Schindler reconstructs the genitive singular ending as **-ós* (1972b: 35). Differently Rasmussen, assuming a *t/s*-stem (or a "*c*-stem") **ĝróih₁-os/ot-* (strong stem?) vs. **ĝrih₁-és/ét-* (weak stem?; 1994: 327). Not to mention the awkward ablaut pattern and the *ad hoc* reconstruction of h_1, there is no necessity to assume an original single PIE *c*-stem of this sort instead of two different nouns, built with different suffixes and exhibiting different ablaut patterns, viz. a proterokinetic *s*-stem **ĝréi̯-es-* (> Indo-Iranian **ǰrai̯-as-*) and an acrostatic *t*-stem **ĝrói-t-/*ĝréi-t-*.

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roots of the *-oR(C)-* shape tend to develop root zero grade in the weak stem, and full grade in the endings.[177] One would have to assume that the *t*-stem **ĝrói-t-* would have developed in the same way.

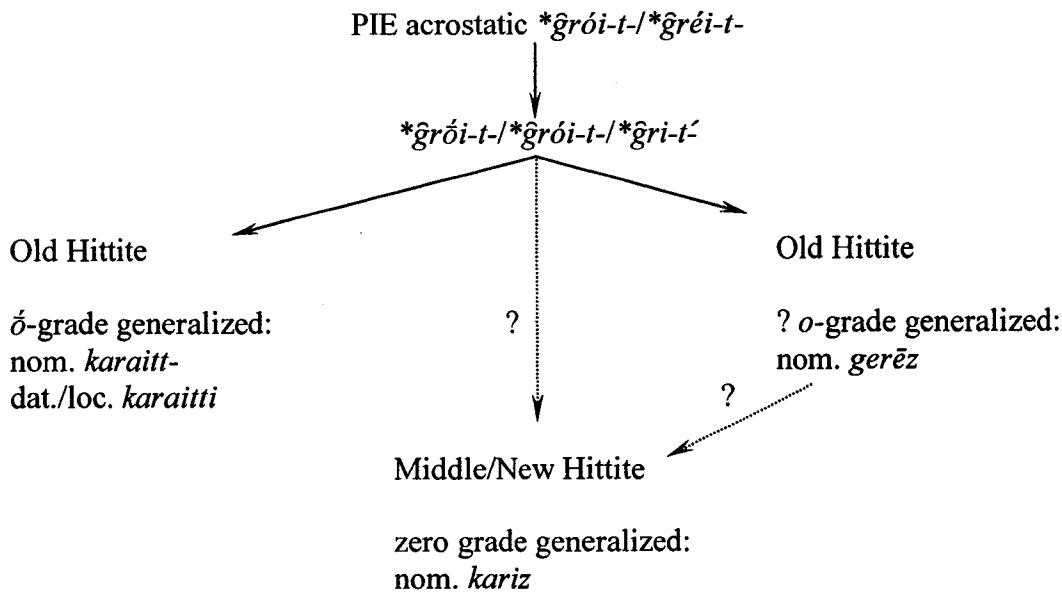
The question that one can raise now is how and why the suffixal noun **ĝrói-t-* adopted the ablaut of *root* nouns. This question becomes especially important when one looks at the development of other suffixal nouns, which has been touched upon in section III.4 above. From the examples adduced there, one could expect that the originally acrostatic (type "Ia") noun **ĝrói-t-* should have developed in the following way: **ĝrói-t-/*ĝréi-t-* → **ĝrói-t-* (strong st.) vs. **ĝrei-t-os* (gen. sg.), or **ĝrói-t-* (strong st.) vs. **ĝrii̯-et-(o)s* (gen. sg.). However, the noun **ĝrói-t-* apparently developed in a different way.

The development of Hi. *karaitt-/karitt-* from the Indo-European acrostatic **ĝrói-t-/ *ĝréi-t-* via the intermediate **ĝrṓi-t-/*ĝrói-t-/*ĝri-t-* (as per Rieken 1999: 134-5) must have involved a number of reformations that have not been very clearly described.[178] According to Rieken, in Old Hittite, both the lengthened grade of the nominative spread to the weak forms (cf. OH dat./loc. *karaitti*) and the normal *o*-grade was introduced by analogy into the nominative (cf. OH/NH *ge-re-e-ez-za*). Eventually, all these forms were replaced by forms with the zero grade in the root (*gri-t-*). Rieken's account could be represented in a following scheme:

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[176] See Nussbaum (1998: 150, fn. 179), Widmer (2004: 50-1). For an alternative view and some references see Neri (2003: 26-7).
[177] Schindler (1972b: 32-6).
[178] See Rieken, loc. cit.

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The picture drawn by Rieken leaves several questions unanswered. First, the ablaut pattern **ĝrŏi-t-/*ĝrói-t-/*ĝri-t-* does not really look "irregular",[179] as such ablaut appears to have been quite common among root nouns. After the ancestor of the noun *karaitt-* became concretized from *'stretching, extending' to 'flood',[180] the suffix *-t-* must have lost its function, and the noun became a virtual root noun.

Rieken's explanation of the origin of the lengthened grade in the Old Hittite dat./loc. *karaitti* is quite problematic, too. If one assumes analogy from the nominative (as per Rieken), this would imply that the lengthened grade must have been generalized throughout the *entire* paradigm. However, in such a case it would be difficult to explain where the stem **ĝri-t-* with the zero grade comes from, unless one assumed *two* paradigms for the noun *karaitt-* in pre-Hittite (or Old Hittitte).

[179] See Rieken (1999: 135).
[180] Already in the oldest texts, the noun *karaitt-* means 'flood' (see HED IV.85-6).

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The most problematic form in this case apparently is the awkward dative/locative *karaitti*. If not for this form, the development of *karaitt-* from an *ó/é* acrostat to *ó/z* type (~ root nouns), as presented by Rieken, would posit no particular problems. However, the unexpected lengthened grade in the dat./loc. *karaitti* either suggests that at some point of its development the paradigm of this noun must have had the highly implausible shape shown below:

Nom. **ĝrṓi-t-s*
Acc. **ĝrói-t-m̥*
Dat. **ĝri-t-éi*
Instr. **ĝri-t-éh*$_1$ (vel sim.)
Loc. **ĝrṓi-t-i* (or **ĝrōi-t-i*?)
Gen/Abl. **ĝri-t-és*[181]

or that the development of this noun in fact must have been different, and perhaps somewhat more complex.

III.4.1.2.

The lengthened grade *ōi* in the dative/locative form *karaitti* cannot be historical, as there is no evidence for the existence of such datives or locatives in Proto-Indo-European. The lengthened grade is only attested in the root of the locatives of acrostatic root nouns, but they have the *ē*-grade, cf. **pḗd* from **pód-/*péd-* 'foot', **sḗm* from **sóm-/*sém-* 'one', etc.

The lengthened grade *ōi* in the Hittite noun *karaitti* is most probably analogical, but the source of this analogy was not the nominative **ĝrṓi-t-s* as was assumed by Rieken.

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I think that the source of analogy were namely the acrostatic root nouns of the **pód-/ péd-* type. As the noun **ĝrói-t-* became associated with root nouns (on this, see below in this section), it probably also developed a locative with a lengthened grade, comparable to that of the root nouns. However, instead of the expected **ĝréit(± -i)* it would have been **ĝróit(± -i)*. The answer to the question why the locative would have had the *o*-grade instead of the expected *e*-grade is to be sought in other Hittite word-forms which reflect ancient acrostatic weak forms, e.g. dat./loc. *maiš-t-i* 'glittering' (< **móis-t-* or **mōis-t-*, instead of ***méis-t-*, see also below in section III.4.3), gen. *šaš-t-aš* 'rest' (< **sos-t-os* instead of ***sés-t-s*), dat./loc. *ša-aš-ti* 'id.' (< **sos-t-* instead of ***sés-t-*), abl. *šạ -aš-ta-za* 'id.' (< **sos-t-* instead of ***sés-t-*), dat./loc. *ka-(a)-aš-ti* 'hunger' (< **Kŏs-t-* instead of ***Kés-t-*), abl. *ka-aš-ta-za* 'id.' (< **Kos-t-* instead of ***Kés-t-*).[182] Apparently, instead of the expected *e*-grade, all of them show a secondarily generalized *o*-grade. Therefore one could argue that the development of **ĝrói-t-* would have been as follows: at a fairly early stage, the *o*-grade would have been generalized throughout the paradigm, and when the new lengthened grade locative was formed by analogy from root nouns, it acquired an *ō*-grade and not an *ē*-grade, because the rest of the paradigm had an *o*-grade as well. This locative with a lengthened grade would have survived into Old Hittite and yielded the attested *karaitti*.

Now the question must be raised where the zero grade **ĝri-t-* comes from. Here I essentially agree with Rieken, that the quasi-root noun **ĝróit-* eventually developed

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[181] As per Rieken (1999: 135).
[182] All forms taken from Rieken (1999).

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weak forms with a zero grade, viz. **ĝrit-*, which in the Middle Hittite period started spreading even to the strong forms, gradually ousting the older forms with a full grade.

The suggested development of the noun *karaitt-* can be represented in a following scheme:

I. IE and PA **ĝrói-t-/*ĝréi-t-*[183]

↓

II. pre-Hi. **grṓit-/*gróit-*?
(nom. **grṓit-s*, loc. **grōit-i*, cf. OH dat./loc. *ka-ra-it-ti*)

↓

III. (late Old Hittite and early Middle Hittite) *gráit-/*grit-*

↓

IV. *grit-* generalized
(cf. MH nom. sg. *ka-ri-iz*)

As Rieken points out, in some forms of this noun either the accent needed to be on the endings, or the root needed to be in the zero grade, in order to prevent the lenition of the suffix *-t-*.[184] Since the zero grade in the root has to be relatively recent, it implies that the weak stem must have been unaccented already in Proto-Anatolian.[185]

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[183] In the idealized reconstructed Proto-Indo-European paradigm, the locative singular would have been **ĝriį-ét* (cf. Av. *drao-ca* 'tree' [loc. sg.; < **dr-éu* belonging to original **dór-u/*dér-u-*] etc., see Friedman [2004]), but it would have been remade into **ĝréi-t-* at an early stage.

[184] Rieken (1999: 135).

[185] Rieken assumes accented endings in the weak stem as well, but, unlike my interpretation, she assumes zero grade in the root of the weak stem already in Indo-European. Given the lengthened grade of dat./loc. *maišti* and *karaitti*, I think that zero grade in the weak stem cannot be of Indo-European antiquity.

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Now one could return to the question why **ĝrói-t-* did not develop like other suffixal acrostatic stems, e.g. like nouns of the **ĝón-u-* type, introducing a new full grade into the suffix. The reason is probably to be sought in the shape of the suffix on the one hand and the shape of the root on the other hand. The suffix in **ĝón-u-* is (semi-)vocalic, in some forms appearing as a vowel *u* and in others as a continuant *u̯* (or vowel + continuant, *-uu̯-*). At least in a part of the paradigm, it formed an independent syllable. On the contrary, after the allomorph ***-et-* had been analogically removed from the locative singular, the generalized purely consonantal suffix *-t-* did not form its own syllable any more, which probably made *t*-stems of this type appear more similar to root nouns than to suffixal nouns with (semi-)vocalic suffixes, or other suffixes with the structure *-CV-* or *-VC-*.

To summarize, one can say that the Hittite noun *karaitt-* very likely reflects an original acrostatic *t*-stem **ĝrói-t-/*ĝréi-t-*. From the semantic point of view, it would have been an action noun originally, meaning 'stretching, extending, expanding'. In the course of development, the original meaning changed to '(over)flowing', 'flooding' and was eventually concretized to 'flood'.

The ablaut of this originally acrostatic *ó/é* type noun developed via several different stages: first, the *o*-grade was generalized throughout the paradigm, and subsequently the noun *karaitt-* adopted the ablaut of root nouns.

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III.4.2. Hi. *kutt-* 'wall'

The noun *kutt-* is attested from the Old Hittite period on, cf. the following list of singular forms:[186]

nom.	*ku-uz-za*	IBoT I 36 I 10, I 16 (MH)
	ku-ú-uz-za	KUB XXIX 4 IV 27 (MH/NH)
acc.	*ku-ut-ta-an*	KBo XXXII 14 Rs. 45 (MH), KUB XXIX 4 IV 24, 26, 38 (MH/NH)
gen.	*ku-ut-ta-aš*	KBo XVII 11(+) IV 30′ (OH), KBo I 44+ Rs. 10 (NH)
dat./loc.	*ku-ṷt-ti*	KBo I 44″ (OH)

As noted in Rieken, the stem **kutt-* 'wall' also existed in Luvian, cf. the Luvoid ^{NA_4}*ku-ut-ta-aš-ša-ri* 'orthostat' (dat./loc. sg.) and Hieroglyphic Luvian (SCALPRUM) *kutasra/i* 'id.'.[187]

The root of Hi. *kutt-* is generally reconstructed as **ĝheu-* 'pour' (~ Gk. χέω 'pour', Ved. *ju-hó-ti* 'id.' etc.), and the noun itself is analyzed as a derivative *t*-stem.[188]

In Rieken's *Untersuchungen*, the noun *kutt-* receives the same treatment as the noun *karaitt-*. Rieken reconstructs the following ablaut pattern for *kutt-*: nom. **ĝhṓu-t-s*, acc. **ĝhóu-t-ṃ*, gen. **ĝhu-t-és* (1999: 137). According to Rieken, *kutt-* has taken a similar course of development to that of *karaitt-*, except that no forms reflecting a full or lengthened grade root (**ĝhou-* or **ĝhōu-*) are attested in Hittite.

Despite the fact that the nominative form **kauz* (vel sim.; cf. *karaiz* < **ĝrṓi-t-s*) is not attested in (Old) Hittite, which makes the reconstruction of the nominative singular **ĝhṓu-t-s* only a speculation, Rieken's assumption that *kutt-* originally was an

[186] Adopted from Rieken (1999: 136; see there for the plural forms), see also HED IV.297.
[187] See also Melchert (1993: 113).

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acrostatic abstract noun, and that the historical development of *kutt-* was parallel to that of *karaitt-* is most probably correct. Since the noun *kutt-* is formed in a non-productive way[189] and the verbal root is not attested in the Hittite verbal system, *kutt-* is likely to be an archaism. Rieken's interpretation also works well semantically, as the noun *kutt-* can be interpreted as an old verbal noun built to the verbal root **ĝheu-* 'pour'. The original meaning of this noun would have been 'pouring (earth, etc.)', which later would have acquired a more concrete meaning, viz. 'wall'.

From the morphological point of view, one can assume exactly the same development as in the case of *karaitt-*, i.e. that the original stem **ĝhou-t-* would have eventually be reinterpreted as *gaut-*, and a weak stem (= root) *gut-* was created beside it. In the case of *kutt-*, too, I assume that the original acrostatic **ĝhóu-t-/*ĝhéu-t-* would have undergone a development comparable to that of *karaitt-* as described in the previous section: first, **ĝhóu-t-/*ĝhéu-t-* would have been remade as **ĝóut-/*ĝout-́*, after that as **góut-/*gut-́*, and eventually the weak stem *gut-* was generalized (cf. Middle Hittite nom. sg. *ku-uz-za*). Assuming that this indeed was the course of development, the absence of forms with a full grade in the root could be explained in such a way that by the time of the earliest attestation of the noun *kutt-* in Middle Hittite, the weak stem with a zero grade had became generalized throughout the paradigm.

Also in the case of *kutt-* it is probably necessary to assume with Rieken that in the weak stem the accent would have fallen on the endings.

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[188] Schindler (1967: 297), Watkins (1978: 160), HED IV.298, Rieken (1999: 136-7), etc.

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III.4.3. Hi. *maišt-* 'glitter, brilliance'

The study of the history of this word is complicated by its poor attestation. The noun *maišt-* is attested twice only, once in dative/locative, cf. *ma-iš-ti* (Middle Hittite, see Rieken 1999: 138), and once probably in the nominative singular case, cf. *mạ-iš-za-* (Old Hittite).[190] The verbal root **meis-* to which Hi. *maiš-t-* has been built is further attested in Hi. *miš-ri-u̯ant-* 'glittering, brilliant', and perhaps Ved. *miṣáti* 'opens the eyes'.[191] The original meaning of the root **meis-* could have been 'glitter' or 'move', and the ancestor of the Hittite *maišt-* would have been a verbal abstract.

For the noun *maišt-* Rieken reconstructs two possible versions of the original ablaut, one with the vowel alternation **ṓi* : **ói* : **i*, and the other with the vowel alternation of the following type: **ṓi* : **ói* : **éi* (1999: 138-9). The former pattern resembles the ablaut she had reconstructed for the nouns *karaitt-* and *kutt-*, while the latter is a sort of compromise between the former and the ablaut usually reconstructed for *nekuz* (i.e., *ó/é*-ablaut).

As for the reconstruction of the original ablaut with the alternation **ṓi* : **ói* : **i*, Rieken must have done it simply because of the similarity between the roots of the noun *maišt-* on the one hand and nouns *karaitt-* and *kutt-* on the other hand, all three having roots with an original diphthong. However, despite the extremely poor

[189] In Hittite, verbal abstracts could also be formed with a synchronic suffix *-att-*, which could be added both to thematic and athematic stems, cf. *tarna-* 'distribute' → *tarnatt-* 'distribution', *karš-* 'cut' → *karšatt-* 'separation', etc. (see Rieken 1999: 106-17).
[190] The interpretation of this form is not entirely clear (see Rieken 1999: 137-8).
[191] In IEW.714 and EWAia II.379-80, the Hittite forms are not mentioned.

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attestation of this noun, it seems that *maišt-* developed not like *karaitt-* and *kutt-*, but rather like *kašt-* and *šašt-*.

As Rieken correctly notices, in the course of the development of the nouns *karaitt-* and *kutt-*, by the Middle Hittite period the root allomorphs with a zero grade, viz. *grit-* resp. *kutt-* were generalized, pushing out the older forms with a full grade. Since the noun *karaitt-* was attested already in Old Hittite, those archaic forms could still be seen. However, by the time the noun *kutt-* appeared in texts in the Middle Hittite period, the full grade of the root had already been levelled out. Now the dative/locative form *maišti* is from the Middle Hittite period as well (see Rieken 1999: 138), and therefore, if one assumed that the noun *maišt-* had undergone the same course of development as *karaitt-* and *kutt-*, one would expect this dat./loc. form to be **mišti*, i.e. show the zero grade of the root (**mis-*). However, the attested *maiš-t-i* points to the *o*-grade **mois-* or, possibly even **mōis-*.

It would hardly be correct to assume that the attested *maišti* instead of the expected **mišti* is an archaic form, somehow preserved until the Middle Hittite period. I think, it might be better to think that *maišt-* developed not like *karaitt-* and *kutt-*, but rather like *kašt-* and *šašt-*. As I wrote in section III.4 above, the latter two nouns reformed the original *ó/é* ablaut by generalizing the *o*-grade throughout the paradigm. If one assumes the same course of development for *maišt-*, both attested forms can be explained in an easy way, namely, as reflecting the *ŏ*-grade.

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III.4.4. Hi. *kapi/art-* 'rodent/mouse' and Lydian *kabrdokid* 'steals' (3. p. sg.)

Several words need to be said about the Hittite noun spelled *kapirt-* and *kapart-* in different texts[192] and most probably meaning 'rodent' or 'mouse', as well as the Lydian verb form *kabrdokid* 'steals' which by some scholars is believed to be related to the Hittite noun mentioned earlier. Several different etymologies have been proposed for Hi. *kapi/art-*,[193] but I find Oettinger's original suggestion that *kapi/art-* 'rodent/mouse' is a primary *t*-stem compound built from the preposition **kom* 'together' and the verbal root **bher-* 'carry' the most attractive semantically, cf. **kom-bher-t-* 'carrying together' → 'mouse (carrying grains to its storage)'.[194] The Lydian verb form *kabrdokid* would then contain the denominal stem *kabrd-* *'carrying together' (→ 'stealing').

The compound which I for now will reconstruct as **kom-bher-t-* must be fairly old, since the verbal root **bher-* 'carry' is otherwise not used in the Anatolian languages. However, no reflexes of such a noun are attested in any other Indo-European language, and it is impossible to determine whether it is an Indo-European dialectism, a calque from some neighbouring non-Indo-European language, or anything else.

The ablaut of the second member of this compound can be interpreted in several ways. Whereas the Hittite forms with the vowel *-a-* can continue an earlier zero grade

[192] In a New Hittite copy of an Old Hittite text, an acc. sg. form *ga-pár-ta-an* is attested (KBo X 37 III 54; see Rieken 1993: 152-3). Otherwise, this noun is attested in Middle and (mostly) New Hittite, including New Hittite copies of Middle Hittite texts. The three times this noun occurs in a Middle Hittite text, it is written *kapart-*, cf. nom. sg. *ka-pár-za* (KBo IX 114 14', MH/MH), acc. sg. ^{PÉŠ}*ga-pár-ta-na-* (KUB XVII 10 + III 58'; MH/MH) and ^{PÉŠ}*ga-pár-ta-an*(-) (KBo XV 10 + III 49' etc.; MH/MH). In New Hittite and the New Hittite copies of Middle Hittite texts, the noun is spelled *kapirt-* (see the examples in Rieken, loc. cit.).

[193] See an overview of various etymological interpetations in Rieken (1999: 153-4).

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*-*bhr̥*- or the full grade *-*bher*- (or even *-*bhor*- [– accent]), the forms with the vowel -*i*- at least at first sight indicate an underlying *ē*-grade. The Lydian -*br*- possiby reflects *-*bhr̥*-.[195]

Based on her comparison with various *t*-stem compounds and simplex nouns, Rieken proposed that originally the compound **kom-bher-t*- had an ablauting paradigm. According to Rieken, the nom. sg. form of this compound would have been **km̥-bhér-t*(-*s*), whereas the oblique forms would have been **km̥-bher-t*- or **km̥-bhr̥-t*-.[196] Rieken interpreted the lengthened grade of *-*bhēr*- to be an archaic feature preserved nowhere else in Indo-European except in Hittite *kapirt*-. According to Rieken, the very common zero grade in the -*eRC*- type roots of second members of compounds (cf. Gk. προ-βλη-τ- 'forthstretching' < *-g^{u}ḷh_1- 'throw', La. *com-i-t*- 'companion' < *-h_1*i*- 'go', Skt. *deva-stú-t*- 'praising the gods' < *-*stu*- 'praise' etc.) was a later innovation of all the Indo-European branches.[197] This, however, is problematic. First, the zero grade in the second members of compounds is very old, as is clearly demonstrated by the laryngeal reflexes in Greek.[198] Also the agreement between Indo-Iranian, Greek and Italic suggests that the pattern probably goes back to the protolanguage. However, the reconstruction of the "archaic" lengthened grade is based on a single Hittite noun which also occurs only in late Hittite (New Hittite and New Hittite copies of Middle Hittite texts). As was said earlier, in original Middle Hittite texts and a New Hittite copy of an

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[194] Oettinger (1995: 44-6).
[195] For the discussion of various issues concerning the development of *r̥ in Lydian, see Melchert (1994: 363).
[196] See Rieken (1999: 154).
[197] Rieken, ibid.
[198] See more examples in section III.6.5 below.

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Old Hittite text this noun is spelled with the vowel *a*, which could reflect the full grade or, which is more likely, the zero grade. The zero grade is probably also indicated by the Lydian *ka-brd-*.

If one reconstructs the protoform of this Anatolian compound as a non-ablauting **kṃ-bhṛ-t-*, it becomes much more like *t*-stem compounds in the other Indo-European languages. However, some explanation needs to be found for the late Hittite forms with the vowel *i*. I do not think it reflects a very archaic feature as suggested by Rieken, both due to lack of parallels and because the form *kapirt-* occurs only in late texts. The forms with the vowel *i* need a special investigation and some different explanation.

III.4.5. The Summary of the Development of Acrostatic *t*-Stems in Hittite

The development of original acrostatic *t*-stems in Hittite seems to have been fairly complex, and not a single Hittite *t*-stem that can be reconstructed as an earlier acrostat, has preserved the original inflection intact. Changes have affected most parts of the paradigms, but this happened to a different extent in different words.

Reflexes of the ancient *e*-grade, which originally occurred in the weak stem are only preserved in the frozen genitive *nekuz*, which ceased to be a part of a living paradigm at a very early stage. In all other Hittite *t*-stems, the *e*-grade was eliminated. Certain details of the development of the Hittite *t*-stems imply that as the *e*-grade was being removed from the paradigms in pre-Hittite, it was everywhere replaced by the *o*-grade at first.[199] Later these nouns underwent a further reformation from static to

[199] One cannot claim, though, that all acrostatic suffixal nouns were remade in such a way in pre-Hittite. Although the generalization of the *o*-grade seems to be the trend among the attested

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mobile inflection, moving the accent to the endings in the weak forms. At this stage, the development of *t*-stems took two different paths: the *t*-stems of the *šašt-* type, i.e., those having *CeC* roots stayed as such, whereas nouns with roots of the *CeRC* type eventually developed roots with the zero grade, cf. *CR̥C*. In Hittite, two nouns – *karaitt-* 'flood' and *kutt-* 'wall' – have undergone this further reformation.

When the proposed trends of development are compared with the trends of development of other ancient acrostatic formations in Hittite, comparable details can be observed, as the *o*-grade has been generalized also among nouns that belong to other morphological classes, cf. Hi. *pa-at-ta-ni* 'basket; sieve' (dat./loc.) < **poth₂-* (orig. **póth₂-r̥/*péth₂-n-* acrostatic *r*/*n*-stem),[200] or *ḫa-a-an-ti* 'in front' (dat./loc.) < **h₂e̯ónt-*, not **h₂e̯ént-* (orig. **h₂ónt-/*h₂ént-* acrostatic root noun) etc. Such development is by no means irregular, as comparable generalizations are also attested elsewhere, cf. Gk. ποῦς 'foot' (nom. sg.; < **pṓd-s*) – ποδός (gen. sg.; ← **péd-s*), or Gk. νυκτός 'night' (gen. sg.; < **nok[ʷ]tós* ← **nékᵘ-t-s*) etc.

However, on the basis of certain nominative forms of root nouns, such as **bhṓr* (< ***bhór-s* via "Szemerényi's law") and especially forms like **pṓd-s* or **klṓp-s* which could not have undergone "Szemerényi's law" and acquired the lengthened grade of the root vowel by analogy, it is fair to speculate whether one could not reconstruct the

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t-stems, other old acrostats behave in different ways, e.g. *tāru* 'tree, wood' (< **dór-u/*dér-u-*) shows the *o*-grade, but *gēnu* 'knee' (< **ĝón-u/*ĝén-u-*) shows the *e*-grade. The noun *u̯ātar* 'water' reflects the *o*-grade in the strong forms (cf. **u̯ód-r̥*), but in the weak forms it shows a reflex of the *e*-grade, cf. *u̯etenaš* (gen.; ← **u̯éd-n̥-s*). Because of the lack of uniformity in the development of acrostatic nouns in Hittite, it is impossible to claim that at first all of them generalized the *o*-grade of the root. However, at least the existing corpus of Hittite *t*-stems suggests that this class of nouns followed namely this course of development.

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lengthened grade **ó* instead of **ó* also for the protoforms of certain Hittite roots noun nominatives, cf. Hi. *ḫa-a-aš* 'ash; soap' < **h₂ṓs(-s)* (← ***h₂ós-s*).[201] Furthermore, the lengthened grade in the nominative of acrostatic formations could have also been extended to the acrostatic *t*-stems of the *kašt-* type, and thus, the ancestor of the Hittite nominative *ga-a-aš-za* could also be reconstructed as **Kṓs-t-s* (← ***Kós-t-s*).[202] In a similar fashion, *ša-aš-za* may have possessed a more archaic nominative **šāšz*, reflecting **sṓs-t-s* (← ***sós-t-s*).

The development of the acrostatic Hittite *t*-stems, which most probably was quite similar to that of ancient acrostatic root nouns, can be represented in a following scheme:

Proto-Anatolian			pre-Hittite
Nom.	**Có(R)C-t-s*	>	**Cṓ(R)C-t-s*
Acc.	**Có(R)C-t-m̥*	>	**Có(R)C-t-an*
Gen.	**Cé(R)C-t-s*	→	**Co(R)C-t-os*[203]
Loc.	**Cé(R)C-t-* ?	→	**Cṓ(R)C-t-i*

Nom. **CṓC-t-s*	**Cṓ(R)C-t-s*	→	**CR̥C-t-s*
Acc. **CóC-t-an*	**Có(R)C-t-an*	→	**CR̥C-t-an*
Gen. **CoC-t-ós*	**Co(R)C-t-os*	→	**CR̥C-t-ós*
Loc. **CṓC-t-i*	**Cṓ(R)C-t-i*	→	**CR̥C-t-i*

[200] See Rieken (1999: 298).
[201] In fact, in this particular noun, one could also assume the simplifaction of the *-ss-* cluster followed by a compensatory lengthening of preceding vowel.
[202] See section III.8.1 for the development of the noun **Kós-t-* in Tocharian.
[203] Accent neglected (see section III.3).

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However, as was said above, this scheme represents a general trend, and it had occassional exceptions. Among root nouns, such an exception would be the noun *ḫant-* 'front side', briefly discussed earlier in this section. Since this noun has the root with the structure *CeRC*, its weak stem should have become *CR̥C-*. However, the attested dative/locative *ḫa-a-an-ti* points to a full, or possibly even a lengthened grade. Among the Hittite *t*-stems, the noun *maišt-* was not reformed in such a way either, even though it has the root structure *CeRC-*. One could to try to argue that the noun *maišt-* was not reformed, unlike *karaitt-* and *kutt-*, because the stem (= root) shape of *maišt-* was different from that of *karaitt-*, the former being *CoRCC-*, and the latter being *CoRC-*. However, this is hardly very persuasive. It may well be that *maišt-* adopted the course of development of nouns like *kašt-* and *šašt-* for some entirely different reason. Possibly, the choice of this course may have been conditioned by a more common ocurrence of nouns of the *šašt-* type in Hittite, or because the stem of *maišt-* ended in *-št-*, just like in *kašt-* and *šašt-*. However, due to very limited resources, it is hard to draw definitive conclusions.

III.5. Reflexes of Acrostatic *t*-Stems in Latin

In Latin there are a small number of inherited monosyllabic nouns with a non-ablauting suffix *-t-*: *cōs* 'whetstone' (fem.; gen. sg. *cō-t-is*), *dōs* 'dowry' (fem.; gen. sg. *dō-t-is*; cf. also *sacer-dō-t-* 'priest'), *nox* 'night' (fem.; gen. sg. *noc-t-is*), *lac* 'milk' (neut.; gen. sg. *lac-t-is*).[204] In addition to these, one should mention the plurale tantum

[204] M. Leumann (1977: 372). On *lact-*, see section II.11.1.

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Latin noun *grātes* 'thanks, gratitude' which also has an Oscan parallel, **brateís** 'favour, Gunst' (gen. sg.).[205] Although all of these *t*-stems (with the exception of *lac*) have already been quite plausibly interpreted as reflexes of ancient Indo-European acrostatic *t*-stems,[206] certain details of their development will be discussed in the following sections.

III.5.1. La. *nox* 'night'

As has been demonstrated by Schindler, the Latin noun *nox* 'night' (Plautus +) reflects the Indo-European acrostatic *t*-stem **nók^{u}-t-/*nék^{u}-t-*. The original *o/e* ablaut alternation has not been preserved in in the Latin paradigm. Just like other Indo-European dialects, Latin shows the generalization of the *o*-grade which originally belonged to the strong forms.

It is not possible to tell precisely when the *o*-grade was generalized in the paradigm of the noun **nók^{u}-t-*. Although the *o*-grade has pushed the *e*-grade out in all dialects (except Anatolian), there is no clear evidence that the generalization of the *o*-grade would have occurred at the pre-dialectal common Indo-European stage. On the contrary, the preservation of the *e*-grade in such originally acrostatic formations as La. *ignis* 'fire' (← **h$_1$ógni-/*h$_1$égni-*), *pēs* 'foot' (← **pód-/*péd-*), etc. (see more examples in the following section) would suggest that acrostatic formations would have retained their characteristic *o/e* alternation in the root at least until the dialectal stage, and either

[205] See Rix (2000) and J. Untermann (2000: 150-1).
[206] See Schindler (1967), M. Peters (1993), Rieken (1999), Janda (2000), Irslinger (2002), Nussbaum (2004a).

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the *o*- or the *e*-grade was generalized in particular words individually in the separate dialects.

III.5.2. La. *dōs* 'dowry'

The Latin noun *dōs* 'dowry' (Plautus +) has been interpreted as a reflex of an ancient derivative *t*(*i*)-stem *dō-t*(*i*)- built to the verbal root *dō-* 'give'.[207] In terms of the laryngeal theory, the Proto-Indo-European verbal root is reconstructed as **deh*$_3$- (cf. Gk. δοτός < **də*$_3$*-tó-*), whereas the question whether the noun *dōs* should be reconstructed as a *ti*-stem or a *t*-stem most probably should be resolved in favour of the *t*-stem **dóh*$_3$*-t-*/**déh*$_3$*-t-* as per Rieken (1999: 84), Rix (2000: 213), Nussbaum (2004a) et alii, since the vowel *i* in the suffix of the alleged *ti*-stem **doh*$_3$*-ti-* would have probably been preserved. As noted in Nussbaum (2004a), *-ti-* is deleted in the nominative singular of *ti*-stems built to roots ending either in a resonant (cf. *mors* 'death' [< **mortis* < IE **mr̥-ti-s*], *mēns* 'mind' [< **mentis* < IE **mn̥-ti-s*]), or a post-tonic long vowel (cf. *quiēs* 'rest' [< **kwii̯ētis* < IE **k*^{u}*i̯éh*$_1$*-ti-*; ~ Av. *šāiti-* 'joy', Old Persian *šiyāti-* 'happiness'] and perhaps *praegnās* 'pregnant' [< ?**-ĝn̥h*$_1$*-ti-*]).[208] On the contrary, after an accented vowel or roots ending in a non-resonant, the suffix *-ti-* is preserved, cf. La. *sítis* 'thirst' (< **dhg*^{u}*hi-ti-*), *cútis* 'skin' (< **[s]ku[H]-ti-*; ~ OE *hȳd* 'id.'), *vestis* 'garment' (< **u̯es-ti-*), *hostis* 'stranger/guest' (if < **ghos-ti-*),[209] etc.

[207] Grundriß 426, Walde/Pokorny (1938: 360), A. Ernout & A. Meillet (1959: 179), Leumann (1977: 275), Schindler (1967: 297), Rieken (1999: 84), Irslinger (2002: 42).
[208] Nussbaum (2004a).
[209] See Vine (2004: 374).

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The original meaning of the reconstructed Indo-European noun *$dóh_3$-t-/*$déh_3$-t- would have been abstract, viz. '(act of) giving', but in the course of time this *t*-stem acquired a more concrete meaning, becoming a resultative (or patient) noun 'gift' (< 'what is given'). The meaning 'gift' is probably preserved in the compound *sacro-dō-t-* 'priest', which originally may have been a bahuvrīhi.[210] The more specific meaning of the Latin noun *dōs*, viz. 'dowry', must have developed from 'gift' secondarily.

The reconstruction of an original acrostatic inflection for this noun is quite plausible, as it works out well at the phonological, derivational, and semantic levels. The structure and the semantics of the reconstructed Indo-European noun *$dóh_3$-t-/ *$déh_3$-t- can be compared to the noun *$nók^{u}$-t-/*$nék^{u}$-t- 'night', as well as the Hittite reflexes of Indo-European acrostatic *t*-stems, discussed in sections III.4 – III.4.3 above. It has to be mentioned, though, that the acrostatic *ó/é* ablaut can be reconstructed for this noun only on structural grounds, i.e., because of its similarity to *$nók^{u}$-t-/*$nék^{u}$-t- (and the like), the *ó/é* ablaut of which is fairly secure. The *e*-grade of the verbal root *$déh_3$- is unattested, since *deh_3- cannot appear in the *e*-grade in the daughter languages due to the colouring effect of the laryngeal h_3. Therefore, already at the common Indo-European stage after the colouring effect of the laryngeals, the noun *doh_3-t- did not possess the characteristic *o/e* alternation, and its weak and the strong stem looked exactly alike, both being *$dóh_3$-t-.

[210] See Strunk (1995: 228ff.), Nussbaum (1999: 396-8).

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III.5.2.1. Exursus on Gk. δώς 'gift' or 'giver'

The noun δώς occurs once only in the nominative singular form in Hesiod's *Works and Days*, line 356, and appears to be a (divine) feminine name, cf. the following passage: Δὼς ἀγαθή, Ἄρπαξ δὲ κακή, θανάτοιο δότειρα 'D. is good (fem.), but H. is bad (fem.), a giver of death'.

The interpretation of this noun is not entirely clear. As far as its semantics is concerned, δώς is commonly interpreted as meaning 'gift, giving'.[211] However, West argued for an agentive interpretation of δώς, translating it as 'giver' and comparing it with the Vedic agent root noun *dā́-* 'id.' (hapax; RV, vi.16.26).[212] As West noted, the noun ἅρπαξ generally has agentive meaning as well, cf. its usage in Aristophanes' *Knights*, 136, Xenophon's *Memorabilia*, III.1.6, etc.

The morphological analysis of the noun δώς is equally problematic. The reconstruction of δώς as **dō-t-*[213] and its comparison with La. *dōt-* is primarily based on the translation of δώς as an action noun 'giving' (or the concretized 'gift'). However, first of all, the translation 'giving' is not secure. Secondly, no oblique forms of the noun δώς are attested, so there is no secure evidence for -τ-. In several works, δώς is reconstructed as either a root noun **dō-* or a *t*-stem **dō-t-*.[214]

As mentioned in Chantraine (loc. cit.), Schindler (loc. cit.) and West (loc. cit.), this divine name is most probably an *ad hoc* formation, an "Augenblickbildung" created by

[211] IEW.225, LSJ.465, R. Lattimore (1959: 61), E. Schwyzer (1959: 499), Schindler (1972a: 82), Rieken (1999: 84), Nussbaum (2004a). Chantraine (1933: 266) translated it as 'dowry'.
[212] M. L. West (1978: 246).
[213] Sic IEW, loc. cit., Schwyzer, loc. cit., Rieken, loc. cit., etc.
[214] GEW I.389, Chantraine (1968: 280).

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Hesiod for this particular passage. To Schindler, this suggested that δώς should probably be interpreted as a root noun. However, a decisive conclusion may never be reached due to the poor attestation of this noun.

III.5.3. La. *cōs* 'whetstone'

Also the Latin noun *cōs* 'whetstone' (gen. sg. *cō-t-is*; Cicero +) has been interpreted as a *t*-stem,[215] either an old acrostatic formation **ḱóH-t-/*ḱéH-t-*,[216] or a decompositional noun from an old bahuvrīhi *°ṇsi-ḱóH-t-* *'(Möglichkeit zur) Schärfung des Schwertes bietend' vel sim.[217]

La. *cōs* is genetically related to multiple words in other Indo-European languages, cf. Avestan *saēni-* 'point' (< Proto-Indo-Iranian **śain-*), OIc. *hein*, OE *hān* 'whetstone' (< Proto-Germanic **hain-*), Skt. *śitá-* 'whetted, sharp' (perf. ptc.), (late) Skt. *śāna-/ śāṇa-* 'whetstone, grindstone', La. *catus* 'shrewd' (← 'sharp'), OIr. *cath* 'wise, sharp-minded', Gk. κῶνος 'pine cone',[218] OIc. *hella* 'flat stone' (< **hallijōn*, cf. Finnish loanword *kallio* 'rock' [see more in de Vries 1977: 221 with further refs.]).[219] All these words ultimately are derivatives of an Indo-European verbal root which is sometimes reconstructed as **ḱeH-*, cf. Nussbaum (2004a), and sometimes more specifically as **ḱeh$_3$(i)-*, cf. LIV.319-20, also Rix (2000: 214) and Irslinger (2002: 42). The reconstruction of a laryngeal is required by a number of reasons; however, the quality

[215] Walde/Pokorny assume that it was a *ti*-stem (1938: 184), but Ernout/Meillet analyze it as a *t*-stem (1959: 145). Irslinger (2002: 42). Recent discussion by Nussbaum (2004a).
[216] Nussbaum (2004a).
[217] Rix (2000: 213, fn. 26). The decompositional analysis was accepted by Irslinger, loc. cit.
[218] Also seen in Hesychian κωνῆτες·θύρσοι.
[219] Ernout/Meillet (1959: 145), A. Walde & J. B. Hofmann (1938: 184-5).

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of the laryngeal is difficult to determine. The scholars who reconstruct *h_3 generally refer to the vowel *ō* of the Latin *cōs*, assuming that it reflects *-eh_3-.[220] This, however, is hard to prove. If the noun *cōs* is reconstructed as an original acrostatic formation with an alternation of the vowels *ó/é* in the root (as per Nussbaum, loc. cit.), no matter what laryngeal one would like to reconstruct, one could say that the long vowel *ō* could have easily arisen in the strong forms, in which any laryngeal would have caused the lengthening of the original short **ŏ* to **ō*.[221] The decompositional theory (as per Rix, loc. cit.) is not helpful in any way either, since there is no evidence for the existence of the bahuvrīhi °*n̥si-k̂óH-t-* (see above).

One might try to determine the quality of the laryngeal in some less direct ways, viz., by studying the trends of development of other Latin reflexes of the ancient Indo-European acrostatic nominal formations. If it were possible to detect certain trends within the development of more transparent acrostatic nouns, this might shed some light on the development of the noun *cōs* as well. In this particular case, it would be important to find out how the ablaut of the ancient acrostatic formations with the *ó/é* alternation developed in Latin. As is often the case (cf. the development of acrostatic nouns in Hittite), one would expect that also in the history of Latin, the original

[220] If I understand Rix correctly (see LIV.320, fn. 1); also Rix (2000: 214, fn. 28). Armenian *sowr* 'sword, dagger', which is sometimes adduced as additional evidence, does not provide clear clues to the quality of the laryngeal (see Olsen 1999: 55 and Nussbaum 2004a).

[221] One could argue that the reconstruction of h_2 would be excluded, since in Sanskrit, h_2 often is reflected as aspiration also in historical zero grade forms, cf. *sthitá-* 'stand' (perf. passive ptc.) < **stə₂-tó-*; *duhitár-* 'daughter' < */d^{h}ug^{h}itar-/ <**dhugə₂ter-*, etc. However, the voiceless laryngeal (h_2) did not cause the aspiration of $\hat{k}$, and therefore the participle *śitá-* can equally reflect any of the following, **$\hat{k}h_1$-*, **$\hat{k}h_2$-*, or **$\hat{k}h_3$-*, since all of them would have yielded *ś* anyway.

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ablauting paradigms would have been reformed in such a way that a particular shape of the root/stem would have been generalized. If the trend were to generalize the *o*-grade (the grade of the strong stem), as was said previously, it would be impossible to determine the quality of the laryngeal in the root **ĥeH-*. However, if it turned out to be that in the history of Latin the *e*-grade tended to be generalized, this would support the reconstruction of h_3, as was proposed in LIV (loc. cit.).

When one studies the corpus, it turns out that both the *e*-grade and the *o*-grade could be generalized in Latin, cf. the *e*-grade in *ignis* 'fire' (← **h₁ég-n-i-/*h₁ég-n-i-*), *pēs* 'foot' (← **péd-/*péd-*), *auris* 'ear' (← **h₂éus-/*h₂éus-*), *cleps* 'thief' (← **klóp/*klép-*) vs. *o*-grade in *nox* 'night' (← **nókᵘ-t-/*nékᵘ-t-*), *ovis* 'sheep' (← **h₂óu-i-/*h₂éu̯-i-*), *bōs* 'cow' (← **gᵘóu-/*gᵘéu-*),[222] *vōx* 'voice' (← **u̯ókᵘ-/*u̯ékᵘ-*), *ocris* 'peak' (← **h₂ók̂-r-i-/*h₂ék̂-r-i-*).

As one can see from these examples, even nouns belonging to the same stem class could generalize different ablaut grades, cf. the root nouns *bōs* and *vōx* (*o*-grade) vs. *cleps* (*e*-grade), or the *i*-stems *ovis* and *ocris* (*o*-gr.) vs. *ignis* (*e*-gr.). Although the reason for the generalization of either the *o*- or the *e*-grade is not quite clear to me, one can notice that the same discrepancy can be observed also in other languages, cf. Hi. *tāru-* 'tree' (*o*-gr.; < **dór-u/*dér-u-*) vs. *gēnu-* 'knee' (*e*-gr.; **ĝón-u/*ĝén-u-*).

In any case, it seems quite plausible to derive La. *cōs* from an earlier acrostatic verbal noun with the stem **ĥóH-t-/*ĥéH-t-* 'sharpening', which would be structurally parallel to the Indo-European 'night' word. It should only be borne in mind that in case

[222] The long vowel *ō* is analogical after the accusative singular form *bōm*, in which it had regularly develop by "Stang's law" (**gᵘóu̯m* > **gᵘómm* > **gᵘōm* > *bōm*).

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the laryngeal in the root *$\hat{k}$*éH*- was h_2 or h_1, one should assume that the noun *cōs* has generalized the *o*-grade of the root. However, if h_3 is to be reconstructed, the development of *cōs* would have been the same as that of *dōs* discussed in the preceding section.

III.5.4. La. *grātēs* 'gratitude' and Oscan **brateís** 'favour'

The two Italic nouns have been in an attractive way interpreted by Rix as reflexes of a single ancient *t*-stem built to the Indo-European verbal root *g^{u}*erH*- 'praise'.[223] According to Rix, the original meaning of this *t*-stem would have been 'praising', but in the course of development, it would have been remade into 'gratitude' in Latin resp. 'favour' in Oscan.

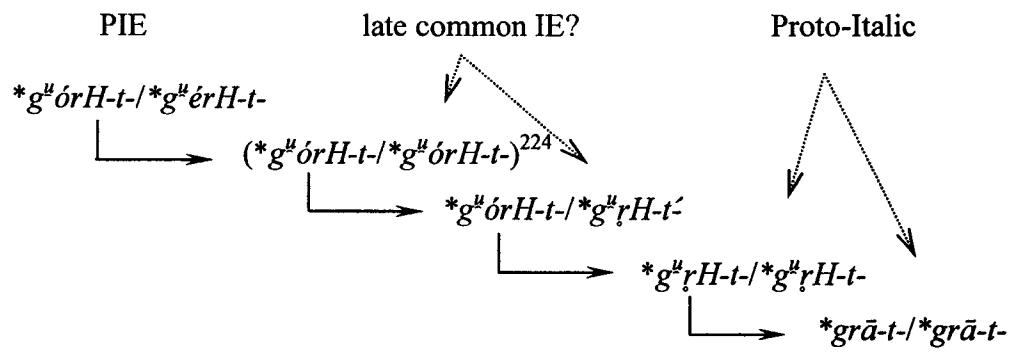
As stated by Rix, both Italic nouns reflect the zero grade version of the root, viz. *g^{u}*r̥H*- (> regularly Proto-Italic **grā*-). In this respect, the stem *grāt*- differs from the rest of the Latin reflexes of the Indo-European acrostatic *t*-stems which, as has been shown in the previous sections, preserve the more archaic full grade of the root, cf. *noc-t*- (< **nok^{u}*-*t*- 'night'), *dō-t*- (< **doh*$_3$-*t*-) and *cō-t*- (< *$\hat{k}$*oH-t*-). I can refer back to the discussion of the Hittite and the Greek reflexes of the Indo-European acrostatic *t*-stems, where a similar situation was observed. In both languages, the ancient *t*-stem acrostats developed in two different ways: whereas nouns with roots of the *Ce*(*C*) structure preserved the original full grade into the dialectal stage, cf. Hi. *kašt*-, *šašt*-, Gk. νύξ, φώς, δώς, etc., nouns with the roots of the *CeR* structure developed a new zero grade

[223] See LIV.210-11. Untermann reconstructs the root as *g^{u}*erh*$_2$- (2000: 150-1). However, it is hard to find any evidence for h_2.

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(*CR̥*), cf. Hi. *kutt-* (~ IE **ĝheu-*), *karitt-* (~ IE **ĝrei-*), Gk. λῑτ- (~ late IE **lei-*). In Italic, apparently, the same thing has happened. Whereas all the *t*-stems with *Ce(C)* roots show the full grade, the stem *grāt-* derives from the zero grade of a root in *which* the vowel is immediately followed by a resonant, cf. **g*^{u}*erH-*. Although, strictly speaking, the root **g*^{u}*erH-* has the structure *CeRH*, the presence of the laryngeal did not prevent the reformation of the ablaut. With the only two exceptions of Gk. δαίς and Hi. *maišt-*, the original *ó/é* ablaut was reformed to *z/z* (via the intermediate *ó/z*) in all roots which contain the sequence *-eR-*, cf. IE **ĝheu-*, **ĝrei-*, **leiH-*, **g*^{u}*erH-*.

Accordingly, the development of the Italic noun **grāt-* can be presented in the following scheme:



It is difficult to tell precisely when the reformations of the ablaut shown above took place, as the Italic data consists of this single example. The comparable formations in

[224] No direct evidence for this step can be found, but one could speculate that before the switch of the original *ó/é* ablaut to *ó/z*, the *o*-grade may have been generalized throughout the paradigm (cf. also the discussion of Hi. *karaitt-* in section III.4.1 above). It has to be noted, that this step would have had to happen *after* "Saussure's law", because otherwise the laryngeal would have been deleted.

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Hittite and Greek are also extremely few and their development implies certain differences. From the Anatolian evidence (based solely on *karaitt-*), one can draw the tentative conclusion that roots with the structure *CeR(H)* preserved the full grade until early Old Hittite. However, Greek λῑτ- which does not possess any reflex of a virtual **loiH-t-* implies that the full grade disappeared from the ancient paradigm **loiH-t-/ *liH-t-* at the pre-Greek stage of affairs.

As far as the Italic **grāt-* is concerned, the stem **$g^{u}_{\text{r}}rH$-t-* with the root in the new zero grade must have existed as early as the Proto-Italic stage, as the phonological change **$_{\text{r}}rH$ > *rā* is common Italic.[225] However, it is impossible to tell whether the original ablaut alternation *ó/é* was reformed as *ó/z* and ultimately as *z/z* at the Proto-Italic stage (but *before* the laryngeal vocalization), or whether it happened in a late pre-Italic Indo-European dialect. In any case, judging from the available evidence, Latin appears to agree more with Greek than with Anatolian.

III.5.5. Excursus to Latin *t*-Stem Compounds

The following Latin *t*-stem compounds have been interpreted as compounds with a non-alternating suffix *-t-*: *ped-i-t-* 'going on foot; infantry man' (< **ped-h_1i-t-*), *com-i-t-* 'companion' (< **kom-h_1i-t-* 'going together'), *sacer-dō-t-* 'priest',[226] *anti-sti-t-* 'high priest', *prae-sti-t-* 'guardian', *super-sti-t-* 'standing over', *locu-plē-t-* 'rich', *mān-suē-t-* 'tame', *inter-pre-t-* 'go-between'.[227] In a recent study, P. Weiss has argued that the

[225] See Meiser (1998: 108).
[226] See section III.5.2 above.
[227] Leumann (1977: 372), Strunk (1995: 224-7), Rix (2000: 214), etc.

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Latin compound *salaputium* 'purification of salt' provides indirect evidence for the existence of a compound **sal-pū-t-* 'salt purifier' which ultimately was Oscan.[228]

The ablaut grade of the second members of these compounds is not always the same: on the one hand, whereas *-dō-t-* in *sacer-dō-t-* reflects the full grade of a laryngeal-final root (cf. **-doh₃-t-*), *-sti-t-* in *anti-sti-t-* etc. reflects the zero grade of another laryngeal-final root (cf. **-sth₂-t-*). On the other hand, whereas the *eR(H)* type roots **h₁ei-* 'go' and **peuH-* 'clean' occur in the zero grade (cf. *ped-i-t-*, *com-i-t-*, **sal-pū-t-*), the root **per-* 'go' underlying *-pre-* in *inter-pre-t-* shows some kind of "Schwebeablaut". Finally, the syllable *-plē-* in *locu-plē-t-* could reflect either **-pleh₁-* (*e*-gr.) or, according to Strunk, **-pl̥h₁-* (zero gr.).[229]

In a recent attempt by Nussbaum to reduce these compounds to a rule, some of them were explained as secondary *t*-stems, built to already existing thematic stems.[230] To this class Nussbaum attributed the compound *inter-pre-t-*, analyzing it as a secondary *t*-stem and reconstructing it as a virtual **enter-por-e-t-*.[231] According to his theory, also *locu-plē-t-* should be interpreted as a secondary "amphi-style" nominal *t*-stem compound, a virtual **°-pl(h₁)-e-ot-*.[232] One should probably also attribute to this class the compounds with the second member °*-sti-t-* (*anti-sti-t-*, *prae-sti-t-* etc.),

[228] For the discussion of La. *salaputium*, see Weiss (1996).
[229] See fn. 227 for references. If one reconstructs the zero grade, the vowel *ē* has to be interpreted as secondary.
[230] See section I.3 above.
[231] Nussbaum (2004a).
[232] Ibid. The laryngeal could have been lost in a second member of a compound.

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reconstructing *-*sth*$_2$-*e*-*t*-.[233] This way, the asymmetry between *-*doh*$_3$-*t*- (full grade; IE √**deh*$_3$-) vs. *-*sth*$_2$-*t*- (zero grade; IE √*steh*$_2$-) could be eliminated.

According to Nussbaum's analysis, the ablaut grades of the second members of primary Latin compounds have the following distribution: *eR*(*C*) type roots appear in the zero grade *R̥*(*C*), cf. *-i-* (< *-*h*$_1$*i*-), whereas the roots of the *CeC* type[234] appear in the full grade, cf. *-dō-* (< *-*doh*$_3$-). Since the original colour of the root vowel in the root *-dō-* is obscured by the colouring effect of *h*$_3$, theoretically one could recontruct either *-*doh*$_3$- or *-*deh*$_3$-. The *e*-grade is very rarely attested among *t*-stems, but evidence for the *e*-grade in the final members of *t*-stem compounds can probably be seen in Greek.[235] If we were to reconstruct the *e*-grade, it would in fact be possible to reinterpret the compound *locu-plē-t-*, reconstructing it as an original *°-*pleh*$_1$-*t*-.

In fact, the reconstruction of *°-*pleh*$_1$-*t*- may be the most plausible of all the options available. According to Nussbaum's analysis, the thematic stem that *-plēt-* could have been built to can be observed in La. *mani-pulus* 'handful' (< **manu-pl*[*h*$_1$]-*o*-; Nussbaum 2004a). The secondary *t*-stem compound *locu-plēt-* would then have been built to the thematic **loko-pl*(*h*$_1$)-*e*- (cf. La. *loc-u-lu-s* 'cash box' < **lok-o-lo*-) by means of the addition of the nominalizing suffix -(*o*)*t*-. The original meaning of the compound *locu-plēt-* (< **loko-pl*[*h*$_1$]-*e-ot*-) would have been 'filling one's coffer'.[236]

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[233] The "weak stem" of *-*sth*$_2$-*e-ot*- (cf. Nussbaum 2004a).
[234] C = any consonant excluding resonants and glides.
[235] See further in section III.6.5.
[236] Nussbaum (2004a). In Ernout/Meillet (1959: 365), *locuplēt-* is interpreted as 'riche en terre', cf. also Cicero's *De re publica* (II.ix.16): *...tum erat res in pecore et locorum possessionibus, ex quo pecuniosi et locupletes vocabantur...* 'wealth at that time consisted of domestic animals (*pecus*) and the ownership of places (*loci*) and from these two kinds of property we get our

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However, based on comparison with La. *manipulus* (< **manu-pl*[h_1]*-o-*) which originally must have meant 'filling the hand', one should probably reconstruct a comparable meaning for **stloko-pl*(h_1)*-o-*, viz. 'filling the *locus*'.[237] The formation of the amphi-style *t*-stem compound **stloko-pl*(h_1)*-e-ot-* with exactly the same meaning then becomes difficult to justify.[238] The alternative solution would to reconstruct an athematic *t*-stem compound with the suffix *-t-* added directly to the root. Because of the *CReC* structure of the root **pleh*$_1$-, the reconstruction of **stloko-pleh*$_1$*-t-* 'filling the *locus*'[239] becomes the most plausible.

The remaining compound *man-suēs* 'tame' is quite obscure. Its second member has been connected with the reflexive pronoun **su̯e*, but the lengthened grade *ē* is perhaps analogical after the verbs *suēscere* 'get used to', *suēre* 'be used to', and may ultimately go back to the instrumental **su̯e-h*$_1$.[240] The original structure of the second member *-suēs* (*-suēt-*) cannot be determined, as *-suēs* can theoretically continue more than one formations, e.g. a *ti*-stem (cf. *quiēs*, *in-quiēs*). If one were to reconstruct a *ti*-stem compound **man-su̯ē-ti-*, it could have been a bahuvrīhi.

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words "wealthy" (*pecuniosus*) and "rich" (*locuples*)' (translation: C. W. Keyes). In such a case, the first member of the compound should be reconstructed as **stloko-* (cf. IEW.1019).

[237] The meaning 'filling the land' sounds awkward. Perhaps *locus* in the compound *locuplēs* is to be understood as 'dwelling, abode' (→ *locuplēs* 'filling one's dwelling, household').

[238] Cf. Nussbaum op. cit. on the lack of reason for the reconstruction of an amphikinetic **di̯eu̯-ot-* 'sky' beside **di̯-eu-* 'id.'.

[239] Semantics as per Nussbaum (op. cit.).

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III.5.6. The Summary of the Development of Acrostatic *t*-Stems in Latin

Although the reflexes of ancient Indo-European acrostatic *t*-stems are very sparse in Latin, their development presents a relatively clear picture. Three of them, all possessing *CeC* roots,[241] in one way or another have generalized the ancient *o*-grade which agrees well with the generalization of the *o*-grade in many other Indo-European dialects. The noun *grāt-* which is built to a root of a different structure (*CeRH*), has in the course of its development reformed its ablaut eliminating the full grade and introducing a new zero grade.

The only problematic question is the chronology of the elimination of the ancient *o/e* alternation in the Latin reflexes of the Indo-European acrostatic *t*-stems. Although, as has been mentioned previously, dialectal reflexes of the Indo-European 'night' word generally show the generalized *o*-grade, the preservation of the original *e*-grade in a number of Latin acrostatic nouns would suggest that at least in nouns with *CeC* roots, the original *o/e* alternation was preserved until the split of common Indo-European into the individual branches. In the case of nouns with *CeR(H)* roots, the ablaut was reformed already in Proto-Italic.

III.6. Reflexes of Acrostatic *t*-Stems in Greek.

The number of *t*-stems with a structure /R-t-/ is larger in Greek than in the preceding language groups. The following Greek nouns have been attributed to this class: δαίς 'meal; banquet' (Homer +; fem.; gen. sg. δαι-τ-ός), νύξ 'night' (fem.; gen.

[240] P. Schrijver (1991: 141), Strunk (1995: 227). Schrijver's opinion seems to be that the instrumental **sue̯h₁* ultimately came to be used as a verbal root.

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sg. νυκ-τ-ός), φώς 'man, hero' (Hom. +; masc.; gen. sg. φω-τ-ός), θής 'day labourer' (Hom. +; masc.; gen. sg. θη-τ-ός), *λῖς 'linen cloth' (Hom. +; masc.; only acc. sg./pl. λῖ-τ-α and dat. sg. λῑ-τ-ί are attested), πλώς 'mullet, κεστρεύς fish'[242] (Epicharmus +; masc.; gen. sg. πλω-τ-ός), δώς 'gift' (hapax; Hesiod, *Op.*356.; nom. sg. fem.),[243] σταίς/σταῖς 'dough' (Herodotus +; neut.; gen. sg. σται-τ-ός), ἄναξ 'king' (Hom. +; gen. sg. ἄνακ-τ-ος), γάλα 'milk' (Hom. +; gen. sg. γάλακ-τ-ος). However, these nouns have diverse background. Some of them, viz. νύξ, δαίς, φώς, πλώς,*λῖς and (by some) δώς have been interpreted as reflexes of ancient Indo-European acrostatic *t*-stems.[244] Whereas for νύξ and δαίς, the reconstruction of an ancient acrostatic inflection is quite secure, and σταίς/σταῖς, ἄναξ and γάλα are likely to be non-primary *t*-stems, the question whether φώς, πλώς, *λῖς, δώς and θής reflect ancient acrostatic *t*-stems or not has not been answered in a satisfactory way yet. Among the questions that arise in the comparison of the nouns of the latter group with those like νύξ, one could ask the following: first, whereas acrostatic *t*-stems originally were action nouns, how should a reconstruction of an original acrostatic *t*-stem be reconciled with the fact that the noun πλώς 'mullet' (ultimately deriving from **pleh*$_3$- 'swim') is better interpreted as an original agent noun (cf. 'mullet' ← 'swimmer')? Quite obscure is the noun θής, the root etymology of which is not well understood (see a more detailed discussion in

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[241] Here, *C* = any consonant including laryngeals, but excluding resonants.
[242] Probably the black mullet, see section III.6.4 below.
[243] Rieken (1999: 84): δώς 'Gabe' < **dóh*$_3$*-t-*/**déh*$_3$*-t-*.
[244] See Schindler (1967), Peters (1993), Rieken (1999), M. Janda (2000), Irslinger (2002), Nussbaum (2004b).

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section III.6.4.2 below). As for the noun *λῑ́ς, several different root etymologies have been proposed, and they will be discussed in section III.6.3 below.

Certain questions arise also regarding the nouns νύξ and δαίς although, as was said above, the reconstruction of original acrostatic inflection for them is fairly secure. As one compares the shape of the roots of the noun νύξ on the one hand and δαίς on the other hand, one notices that whereas the former has generalized the *o*-grade which originally belonged to the strong stem, the latter appears to have generalized the *e*-grade (δαιτ- < IE **deh₂i-t-*).[245] The noun δαίς is also interesting from another point of view, viz., when it is compared with the noun *λῑ́ς which has been interpreted as a reflex of an Indo-European acrostatic *t*-stem as well.[246] As one can see, the root of the noun δαίς reflects an earlier full grade, whereas the root of *λῑ́ς reflects what seems to be a zero grade (**liH-* vel sim.).[247] Although the roots reconstructed for these nouns originally did not have the same structure (cf. **deh₂i-* [> δαίς] with a laryngeal standing between the vowels *e* and *i* vs. **leiH-* [> *λῑ́ς], in which a laryngeal follows a diphthong), after the loss of the laryngeals they would have become structurally similar, and one wonders if these two nouns should not have developed in the same way, namely, both generalizing the *e*-grade of the root (like δαίς), or, vice versa, both generalizing the zero grade (like *λῑ́ς).

In the following sections I am going to discuss the development of these Greek nouns individually, trying to find answers to the questions raised in the previous

[245] Rieken (1999: 84), Nussbaum (2004a).

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paragraphs. However, I am not going to discuss the noun νύξ separately, as several words about this noun have already been said in sections III.1 – III.3 above. Some additional details have also been mentioned in section III.5.1.

Also the history of the Greek hapax δώς 'gift' has been discussed previously in section III.5.2.1.

III.6.1. Gk. φώς 'man, hero'

The noun φώς (Homer +; gen. sg. φω-τ-ός, acc. sg. φῶ-τ-α, dat. sg. φω-τ-ί etc.)[248] occurs commonly, but predominantly in the poetic language, and has the following meanings: 'man (as a hero, warrior)', 'man (as a mortal, opposed to a god)', 'man (as opposed to a woman)'. It has been interpreted by K. Brugmann (and later followed by a number of scholars) as etymologically related to a number of nouns in several languages meaning 'light' as well as some other semantically related concepts, cf. Gk. φάος 'light' (Homer; Attic φῶς), Ved. *bhás-* 'light', Avestan *bā-* 'light', cf. also Cuneiform Luvian *piḫa-* 'luminosity, splendour, might' (occurs in personal names, cf. Melchert 1993: 176), etc. All these nouns have been built to the Indo-European verbal root **bhā-* 'shine' (in the laryngeal terms, **bheh*$_2$-).[249] If one accepts this etymology,

[246] Irslinger (2002: 41).
[247] Sic Irslinger, loc. cit. See also below in section III.6.3.
[248] See also below in this section, where several excerpts from Homer are quoted.
[249] Cf. Grundriß 536, GEW II.1060, and more recently G. E. Dunkel (1997: 32), Janda (2000: 246; see also Schmitt 2002: 75), and Irslinger (2002: 41).

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the Indo-European ancestor of the Greek noun φώς would have originally meant something like 'the shining one' (cf. Janda 2002, based on Peters 1993: 104-7).[250]

Theoretically, one could also derive the noun φώς from another root, viz. **bheh₂-* 'speak'.[251] In such a case, φώς would have originally meant 'the speaking one', i.e. 'human being, man' (as opposed to animals which do not have the gift of speech). One could support such an interpretation with a semantical parallel from Icelandic, cf. the Icelandic compound noun *mál-leysingi* which means 'animal' (lit. 'the speechless one'). However, such an interpretation of the noun φώς would most probably be incorrect. If the noun φώς were to be interpreted as 'the speaking one, speaker' (> 'human being'), one would expect it to be used of both men and women. However, the noun φώς appears to be used of men only, which make its connection with the root **bheh₂-* 'speak' less attractive. On the contrary, the derivation of φώς from the verbal root **bheh₂-* 'shine' is for a number of reasons quite attractive. Although the semantics of the noun φώς 'man, hero' on the one hand and the verbal root **bheh₂-* 'shine' on the other hand at first sight seem quite distant, the semantic transfer from 'shining' to 'hero, valiant warrior' is quite easy (cf. also Peters 1993: 101-6, and see below in the present section).

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[250] Chantraine expressed doubts about the relationship of these two nouns (1968: 1238).

[251] According to the interpretation of Dunkel, originally the two homonymous roots **bheh₂-* were variants of a single root, the meaning of which was 'radiate energy, give off vibrations' (1997: 31). However, in the course of development a semantic split took place, and two homonymous roots developed.

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There are two ways in which the noun φώς could be reconstructed.[252] Since φώς always appears with the element -τ- in the oblique stem already in Homer, it might be tempting to assume that this noun always contained a *-t-*, and therefore probably is an ancient *t*-stem. If this were the case, one could imagine that in Indo-European there existed a *t*-stem acrostatic action noun **bhóh₂-t-/bhéh₂-t-* 'shining, radiance', which would have acquired the concrete meaning 'man, hero' by means of personification, cf. original 'shining, radiance' (action noun) → 'the one who is characterized by shining, radiance; who radiates, a shiner' → 'hero'. A parallel development can be observed in the "agentivization" of the *ti*-stem **teh₂-ti-* 'theft' in Old Church Slavic and Old Irish, in both of which this original action noun has secondarily acquired an agentive meaning 'thief', cf. OIr. *táid*, OCS тать 'thief' (< **tā-ti-*).[253]

Theoretically, one could also consider a different possibility. Since sometimes already in Homeric Greek the element -τ- appearing in the oblique stem of nouns is an innovation, cf. χρώς 'skin; flesh', gen. χρω-τ-ός (beside the more archaic χρο-ός), δόρυ 'tree', gen. δούρ-α-τ-ος (beside the more archaic δουρ-ό-ς < **dor-u̯-ós*), etc., one could think that perhaps also in the noun φώς the suffix -τ- is secondary. If this -τ- were a later Greek addition, the noun φώς could then also be interpreted as an original agent root noun of the well-attested φώρ 'thief' type (← **bher-* 'carry [away]').[254] If φώς really was a root noun, it should be reconstructed as **bhóh₂-s* (or **bhṓh₂-s*). The

[252] Both mentioned in Peters (1993: 104-5), but no discussion of the formation is provided.
[253] Cf. Vasmer III.82. The OCS noun тать is a *io*-stem from the synchronic point of view, not an *i*-stem (as per Vasmer, loc. cit.). However, it is a later thematization of an old *ti*-stem. Respectively, the Old Irish form *táid* has been reformed as an *nt*-stem (OIG.209).
[254] Cf. also Peters (1993: 104).

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meaning of this noun would have been 'shiner, the shining one, the radiant one', which could have developed into a more concrete 'hero' as well.

However, the latter interpretation is not likely to be the right one. Although, as was said above, in certain instances, like χρω-τ-ός, δούρ-α-τ-ος etc. already Homeric Greek shows a new (i.e., post-Indo-European) suffix -τ-, the secondary character of this -τ- is easy to see, because beside the τ-forms shown above, there also exist τ-less forms, like χρο-ός, δουρ-ός, which are more archaic. In the case of φώς, no such archaic τ-less forms exist. Already in Homeric Greek, the quite commonly attested noun φώς always has the stem φω-τ-, which probably indicates that the suffix -τ- is inherited.

One could also imagine a still different scenario for the development of the agentive semantics for the noun φώς. Although, as was said in the preceding paragraph, the noun φώς is most probably not a historical root noun, from the synchronic point of view it is hardly distinguishable from root nouns like φώρ 'thief', κλώψ 'id.', θώψ 'flatterer', σκώψ 'owl' (← 'looker'), etc. The consonant -τ- which appears in its oblique forms must not have caused any problems in the interpretation of the noun φώς as a root noun, since the same consonant was becoming more and more widespread in the oblique forms of various nominal formations, despite their origin.

Therefore, one could imagine that as the ancestor of the noun φώς became absorbed by the root nouns, it also acquired a new agentive meaning, quite common among nouns which belong to this class. It is possible, that for some time it had both the agentive and the action noun meanings, as, in fact, the distinction between the two is

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not always sharp among the root nouns themselves, cf. the noun φλόξ 'fire', which was interpreted by Schindler as an action noun, 'burning',[255] but perhaps could be equally plausibly interpreted as an agent noun, 'burner'. The same could perhaps be said of the root noun *$\mu\acute{o}k^{\mu}$- 'voice', which could be interpreted both as 'speaking' and 'speaker'. Thus, also the noun φώς could have for a certain period of time possessed two meanings, but the agentive meaning eventually pushed out the more original action noun meaning.

Regarding the oldest meaning of the noun φώς in Greek as well as its semantic development, the reconstruction of the meaning 'shining' for the Proto-Indo-European ancestor of this noun implies that the original meaning of the noun φώς was namely 'hero', since the concepts 'shining' and 'radiance' can be easier associated with it than with the other meanings of the noun φώς, viz. 'man' and 'mortal'. The latter two meanings, 'man' and 'mortal' must have secondarily developed out of the original meaning 'hero'.[256] The textual evidence is neutral towards this view, neither unambiguously supporting nor denying it, as the noun φώς appears in various meanings already in Homer. In some instances the noun φώς probably should only be interpreted as 'valiant warrior, hero', cf. the formula ἰσόθεος φώς 'godlike warrior, hero', used of Euryalos (*Iliad*, II.565) and Odysseus (*Odyssey*, I.324),[257] or where the noun φώς is used to denote such heroic figures as Achilles (*Il.*, II.239-40: ...ὃς καὶ νῦν Ἀχιλῆα, ἔο

[255] Schindler (1972b: 36).

[256] Peters arrives at essentially the same conclusion, assuming that the original meaning would have been 'hero, warrior', and the meaning 'man' would have evolved via "semantic weakening" ("Bedeutungsabschwächung", 1993: 101-2).

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μέγ' ἀμείνονα φῶτα, ἠτίμησεν '...that has now done dishonour to Achilles, a man [= hero?] far better than he [Agamemnon] is' [translation: Murray]), Menelaus (*Il.*, III.52-3: οὐκ ἂν δὴ μείνειας ἀρηΐφιλον Μενέλαον; γνοίης χ' οἵου φωτὸς ἔχεις ὐαλερὴν παράκοιτιν 'wilt thou indeed not abide Menelaus, dear to Ares? Thou wouldest learn what manner of warrior he is, whose lovely wife thou hast'), etc. However, in other cases the noun φώς seems to mean simply 'man, person', cf. *Il.*, III.219, where Odysseus is described as ἀϊδρεϊ φωτὶ ἐοικώς 'in resemblance of a man of no understanding', or *Od.*, VI.187, where Odysseus is addressed by Nausikaa in the following way: "...οὔτε κακῷ οὔτ' ἄφρονι φωτὶ ἔοικας" 'thou seemest to be neither an evil man, nor a witless', etc. In a number of instances, either interpretation is possible, cf. *Od.*, I.355: πολλοὶ δὲ καὶ ἄλλοι φῶτες ὄλοντο 'many others (men?/heroes?) likewise perished', or *Il.*, V.214: ...αὐτίκ' ἔπειτ' ἀπ' ἐμεῖο κάρη τάμοι ἀλλότριος φώς '...then may some alien (man?/hero?) forthwith cut my head from me', etc.

To summarize, one could say that the noun φώς is a fairly good candidate to inherited acrostatic *t*-stems. It can be explained as a derivative of the verbal root **bheh*$_2$- 'shine', the original shape of the noun having been **bhóh*$_2$*-t-*/*bhéh*$_2$*-t-* (the *o*-grade generalized, cf. νύξ). The original meaning of this noun would have been 'shining, radiance', but in the course of development it would have either undergone a concretization to 'hero' by means of personification, or the agentive meaning was

[257] This formula is used in Peters' argumentation for the reconstruction of the original meaning (1993: 102, fn. 71).

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substituted for the original action noun semantics as the ancestor of the noun φώς became absorbed by root nouns.

As for the semantic development, it is likely that the original meaning of the noun φώς was 'hero', and not 'man, mortal'. However, this interpretation primarily rests on the derivation of this noun from the root **bheh₂-* 'shine'. The textual evidence does not provide any decisive clues for the reconstruction of the original semantics.

III.6.2. Gk. δαίς 'meal; banquet'

The Greek noun δαίς 'meal; banquet' (Hom. +; stem δαι-τ-) has been interpreted as a *t*-derivative of the verbal root **deh₂i-* 'to distribute', reflected also by the Greek verbs δαίω 'divide', δαίομαι 'distribute', Ved. *dáyate* 'distributes' (< **deh₂i-i̯e-*), further δαίνῡμι 'to give a banquet'.[258] The stem *dai-t-* is also attested in several personal names, both alphabetic Greek and Mycenaean personal names, cf. Δαίτ-ων, Δαίτ-ιχος, Δαίτ-ις, *da-te-wa* /Daitēu̯ās/ (nom. sg.;[259] KN V 147.4), and possibly *e-u-da-i-ta* /E$^{(h)}$u-daitās/ (KN Dl 47.1) and *a-pi-da-ta* /Amp^{h}i-daitās/ (KN KN Vc 175).[260] At least in the alphabetic Greek names and the Mycenaean *da-te-wa*, the historical stem *dait-* functions as an indivisible root to which the endings/suffixes -ις, -ιχος, -ων, *-ēwās* (cf. alphabetic Greek Οἰν-έας) and *-ās* (cf. Gk. Διοκλῆς [< **-kleu̯-ās*]) are added. The perception of *-t-* as a part of the root in all these personal names indicates that *dait-* and δαίς are rather old formations.

[258] Rieken (1999: 84), Nussbaum (2004a), etc.
[259] Probably (cf. F. Aura Jorro 1985: 159).

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According to the interpretation presented in Rieken (1999), Irslinger (2002) and Nussbaum (2004a), δαίς is to be reconstructed as an Indo-European acrostatic *t*-stem verbal abstract **dóh₂i-t-/déh₂i-t-* 'distribution, act of distributing'. Originally (i.e., at the Indo-European stage), the noun **dóh₂i-t-* might have been used in various contexts, but in the course of development of the pre-Greek dialect and Greek itself, the usage of **dóh₂i-t-* (or δαίς)[261] was confined to the distribution of food, wherefrom the very specific meaning 'meal; banquet' has developed.

It has to be noted that the noun δαίς shows unexpected ablaut. With its generalized the *e*-grade of the root (IE **deh₂i-*), the noun δαίς deviates from the general tendency of Greek reflexes of the Indo-European acrostatic nouns to generalize the *o*-grade of the root, cf. φώρ 'thief' (gen. φωρ-ός; ← IE **bhór-/*bhér-*), κλώψ 'id.' (gen. κλωπ-ός; ← IE **klóp-/*klép-*), νύξ 'night' (gen. νυκ-τ-ός; ← IE **nókᵘ-t-/*nékᵘ-t-*), βοῦς (Doric βῶς) 'bull' (gen. βο-ός; ← IE **gᵘóu-/*gᵘéu-*), φλόξ 'flame' (gen. φλογ-ός; ← IE **bhlóg-/*bhlég-*),[262] γόνυ 'knee' (gen. γούνατος;[263] ← IE **ĝón-u/*ĝén-u-*), δόρυ 'tree' (gen. δούρατος;[264] ← IE **dór-u/*dér-u-*), etc.[265] Neither does the noun δαίς with its *eRC*-type root look like the noun *λῑ́ς 'linen cloth', which has a *eRC*-type root as well, but has generalized the zero grade, viz. **liH-* in Indo-European terms.

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[260] Cf. A. Leukart (1994: 101-3), J. L. García-Ramón (2005: 32).
[261] For the phonological details, see below.
[262] The plain velar of the related Sanskrit *s*-stem *bhárgas-* 'radiance, splendour' (as per Stüber 2002: 75; see also IEW.124-5) indicates a preform with a velar **-g-*, i.e. **bhleg-*, and not **bhleg-/bhleĝ-*, as per LIV.86.
[263] In Homer, also γουν-ός.
[264] In Homer, also δουρ-ός.
[265] The noun δώς cannot be used as already in Common Indo-European it did not have the *o/e* alternation any more due to the colouring effect of *h_3.

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The occurrence of the *e*-grade in δαίς is probably best explained by analogy from the ancestor of the verb δαίω 'divide' (< **deh₂i-i̯e/o-*). Since neither the reflexes of the theoretical *o*-grade variant **dóh₂i-t-* nor the zero grade †*dih₂-t-*[266] are attested in Greek, Myceanean or elsewhere, the analogy from the verb and the generalization of the *e*-grade must have taken place at an early stage.

III.6.3. Gk. *λῑ́ς 'linen cloth'

The element -τ- occurs in the Greek word-forms λῑ̃-τ-α (acc. sg.; Hom., *Iliad*, VIII.441; *Odyssey*, I.130; acc. pl. in *Od.*, X.353) and λῑ-τ-ί (dat. sg.; Hom., *Il.*, XVIII. 352; *Il.*, XXIII.254), two oblique forms of an otherwise unattested Greek noun *λῑ́ς 'linen cloth'. The history of this noun is not well understood, and several opinions about its etymology have been put forward.

In LSJ, there is a possibility mentioned that the noun *λῑ́ς is related to λίνον 'flax, anything made of flax'.[267] From the semantic point of view, this sounds quite attractive, but the historical connection of these two nouns contains a large number of complications. The immediate problems are the following: first of all, the etymology of λίνον itself is not entirely clear, although the noun which could be reconstructed as **līnom* or **linom* is attested in multiple Indo-European languages, and therefore would

[266] For structural reasons, I assume that in the course of development beside the full grade *doh₂i-t-* a zero grade stem †*dh₂i-t-* would have developed. However, laryngeal metathesis to †*dih₂-t-* would be expected in such environment.
[267] LSJ.1054a.

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seem to be a part of the ancient Indo-European lexicon.[268] Second, λίνον and *λῑ́ς differ in the length of the root vowel, cf. -ĭ- vs. -ī-. Furthermore, the appearance of *λῑ́ς (stem λῑ-τ-) would seem to suggest that the word 'flax' in fact should be interpreted as **li-n-om/*lī-n-om* or perhaps **li-nó-m/*lī-nó-m*.[269] If this is true, one would have to explain the function of the suffix (?) *-*n*- (or *-*nó*-), as well as the meaning and the structure of the root **li-/*lī-*, which otherwise seems to be unattested.[270]

H. Petersson claimed that *λῑ́ς and λίνον derived from a common ancestor which he reconstructed as an ancient **lī-t*, gen. **li-n-és*.[271] This formation, according to Petersson, was also the source of La. *linteum* 'material', the shape of which supposedly was a result of multiple reformations,[272] as well as OIc. *linnr* (< PGmc. **linþ-*),[273] *lindi* 'belt' (< **linđī-*), etc. Whereas an etymological connection between the roots of La. *linteum* and Gk. λίνον is probable, the reconstruction of **lī-t* (gen. **li-n-és*), as well as its development from Indo-European to the daughter languages as presented by

[268] See a brief discussion in GEW II.126, IEW, p. 691, Masson (1967: 20), Chantraine (1968: 642), Th. V. Gamkrelidze & V. V. Ivanov (1995: 568-9), Mallory/Adams (1997: 206), among others.

[269] In IEW.691, the reconstructed form is *li-no-m*.

[270] Any relationship between λίνον and *λῑ́ς is rejected in GEW II.129. However, see below on a new attempt to connect the two words by Irslinger (2002: 41).

[271] See Petersson (1921: 174).

[272] Petersson assumed the following development: under the influence of the nominative, the genitive **li-n-és* became **li-t-n-és*, then the latter became **lintnés*, and from this form, ultimately the nominative *linteum* developed by means of, apparently, analogy (although Petersson does not state this clearly himself, see loc. cit.).

[273] Petersson translates OIc. *linnr* as 'Gürtel' ('belt, girdle'), but the attested meanings of this word are 'serpent', 'sword' and 'fire' (the latter two only in the poetic language, cf. *linn-vengi* 'serpent pillow', i.e. 'gold').

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Peterssman is *ad hoc*. Also their genetic relationship with the Icelandic nouns *lindi* and *linnr* is extremely doubtful.[274]

Fraenkel argued that the noun *λῖς was connected to such words as λῑτός 'simple' and λεῖος 'smooth', also La. *lēvis* 'even'.[275] From the semantic point of view, there perhaps are no major problems, as linen clothes can certainly be "simple", "smooth" and "even". Because of the long vowel ῑ Fraenkel reconstructed the root as **lēi-/*lī-*, which in laryngealistic terms could indicate **leHi-* or **leiH-*. According to this theory, the noun *λῖς would have generalized the zero grade version of the root, viz. **lī-* (or **liH-*), to which a suffix *-t-* would have been added. Such a theory does not seem to create any major phonological difficulties. However, since the meaning of the proposed root **lēi-/*lī-* is not clear, a further interpretation of the history of the noun *λῖς, as well as its formation from this root becomes impossible.

Recently Irslinger attempted to combine the noun *λῖς together with the noun λίνον again, deriving both from the Indo-European verbal root **leiH-* 'wrap up; hide', otherwise observed in Ved. *láyate* 'clings, nestles; hides' (3. sg.), Hi. *u-lešta* 'hid' (3. sg. pret.), etc.[276] According to her interpretation, from the verbal root **leiH-* there were derived the noun λίνον and the *adjective* λῖς, the latter eventually becoming nominalized as 'linen cloth'.[277]

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[274] For a different account of their history, see de Vries (1977: 358).
[275] For the discussion of *λῖς and related matters see Fraenkel (1910: 88-95). Fraenkel's interpretation was followed by Buck (1918: 75).
[276] LIV.405, Irslinger (2002: 41).
[277] Irslinger, loc. cit.

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To my mind, this interpretation contains a lot of complications. First of all, the stem shape of the adjective λίς is not known, since it only occurs twice (both times in the Iliad), and the attested form is nominative, cf. λὶς πέτρη 'smooth rock' (*Iliad*, XII.64) and πέτρη γὰρ λίς ἐστι 'for the rock is smooth' (*Iliad*, XII.79). Therefore, there is no way to know if the stem of this adjective ends in a -τ- or not. Furthermore, I cannot see how one can derive an adjective 'smooth, even' from a verbal root with the meaning 'wrap up'.

Also the derivation of the noun 'flax' (Gk. λίνον, etc.) from this verbal root sounds nearly impossible to me. If one imagines that from the verbal root **leiH-* a verbal adjective with the suffix *-nó-* could be formed, cf. **liH-nó-* vel sim., the only imaginable meaning that it could acquire would have been 'wrapped up'. This could not have developed into 'flax' (as a plant name) even if one assumed that 'wrapping up' would describe the structure of the stem of the plant. The stem of any grass-like plant is in a way a wrapped up sprout or leaf, and therefore the more appropriate original meaning for the putative **liH-nó-* would have been 'stem'. However, this could have hardly been the original name for the flax, since there is solid evidence that originally flax was grown not for textile purposes, but for the oil of its oleaginous seeds.[278]

The derivation of the noun λίνον from the root **leiH-* also faces phonological problems. First of all, instead of the attested λίνον, the Proto-Indo-European **liH-nó-* should have yielded Greek ˣλῑνόν with a long vowel of the root and the accented ending. Similar problems arise for the Balto-Slavic 'flax' word, where instead of the

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expected Lith. ˣ*lynas*, Latv. ˣ*līni* (pl.), OCS ˣлинъ or Ru. ˣлин, the forms actually attested are Lith. *lìnas* and Latv. *lini*, OCS льнъ and Ru. лён, all reflecting **lĭ-no-*.

Conversely, in those Indo-European languages in which one would expect a more or less regular shortening of the long vowel **ī* in this position by means of the so-called "Dybo's law",[279] the situation is exactly the opposite. Dybo's law more or less regularly operated in Proto-Italic, Proto-Celtic and Proto-Germanic, and has been formulated in the following way:

**IH/*VH > *Ī/*V̄ > *Ĭ/*V̆ /_RV́* (Schrijver 1991: 356-7)[280]

As the result of the operation of this phonological law, the protoform **lī-nó-* (< **liH-nó-*) should have become shortened to **li-nó-* in Proto-Celtic, Proto-Italic and Proto-Germanic. However, all these dialects curiously show the long vowel *ī*, cf. La. *līnum*, OIr. *lín*, OIc. *lín*, Go. *lein* [li:n], OE/OS/OHG *līn*.

Thus, the reconstruction **liH-nó-* apparently does not match any of the attested dialectal forms. For these reasons, I am inclined to think that the derivation of λίνον from the verbal root **leiH-*, as proposed by Irslinger, is quite implausible.

However, Irslinger's derivation of the noun *λῖς from the root **leiH-* is perhaps more fortunate. The root **leiH-* does not seem to form any verbs in Greek, but the meanings 'wrap oneself' and 'linen cloth' are perhaps easy to connect semantically.

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[278] H. Helbæk (1959: 103ff.).
[279] See a discussion of "Dybo's law" in Schrijver (1991: 334-57).
[280] *I* = *i/u*, *V* = *e/a/o*, *R* = resonant.

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Much more complicated is the reconstruction of the original shape of this noun. Just as was the case with the noun φώς (see section II.6.1 above), since the noun *λῑ́ς is already attested in Homer, there are two ways in which one could interpret the stem final consonant -τ-. One could either assume that this -τ- had always been there, whereby one would have to interprete the noun *λῑ́ς as an original *t*-stem, or, alternatively, one could think that the consonant -τ- was inserted in post-Indo-European times.

If one assumed an original *t*-stem, one would probably have to reconstruct a verbal abstract **lóiH-t-/léiH-t-* 'wrapping up', which could later become concretized to 'cloth' and semantically narrowed to '*linen* cloth'. Semantically, such a development is not problematic at all. However, in order to account for the zero grade of the root of the noun (i.e., **liH-t-* in terms of Indo-European), one would have to assume that the original *ó/é* ablaut was reformed as *ó/z*, and eventually the zero grade was generalized throughout the paradigm.

Such reformations of acrostatic *t*-stems are attested in other Indo-European languages, cf. the development of Hi. *kutt-* 'wall' and *karaitt-* 'flood', and therefore the reformation of **loiH-t-/*leiH-t-* as **liH-t-* would appear quite normal. The Greek noun δαίς with its generalized *e*-grade is not problematic at all, as it can be explained away as analogical.

It has to be noted that the strong stem **loiH-t-* was subject to laryngeal loss via "Saussure's law", whereby already at the Proto-Indo-European stage the strong and the

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weak stem would have looked quite different, cf. **lói-t-* vs. **léiH-t-*. How should the attested Greek stem λῑ-τ- have developed out of these ancient stems?

One can describe the development in the following way: first, the laryngeal could have been analogically re-introduced into the strong stem, whereby **lói-t-* → **lóiH-t-*. Then, the sequences **loiH-t-* and **leiH-t-* would have developed as follows: instead of the development into the phonologically expected **loi̯V-t-* and **lei̯V-t-*, the *ó/é* alternation in the stems **lóiH-t-/*léiH-t-* would have been remade as *ó/z*, viz. **lóiH-t-/*liH-t-*, and then the weak stem **liH-t-* would have been generalized in Proto-Greek before **loiH-t-* would have developed into **loi̯V-t-*. The attested λῑτ- would have been the regular phonological outcome of **liH-t-* in Greek.

Theoretically, one could also assume that the noun *λῑ́ς originally was a root action noun, as such a type existed in Indo-European as well.[281] One could imagine that the original shape of the noun would have been **léiH-/liH-*. From such a root noun, it would be very easy to get *λῑ́ς by assuming the generalization of the zero grade of the weak stem. The suffix -τ- could perhaps be explained as a later addition.

To summarize, I would like to say that of all the theories discussed above, the derivation of the noun *λῑ́ς from the root **leiH-* (as also implied by Irslinger 2002) seems the most plausible to me. However, the details of the development of this noun are not entirely clear. Theoretically, it can be reconstructed both as an original *t*-stem verbal abstract, and as an original root verbal abstract.

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III.6.4. Unclear Greek *t*-Stems

III.6.4.1. Gk. πλώς 'mullet (type of fish)'[282]

In Greek there exists a noun πλώς 'mullet' (Epicharmus, Xenocrates), the oblique stem of which ends in -τ-, cf. nom. pl. πλῶ-τ-ες 'mugiles pisces, κεστρεῖς (ThGL.7693). The noun πλώς is generally connected with the verb πλώω, the Ionic version of the verb πλέω 'swim', although the details of derivation are usually left unspecified.[283] These details partially depend on one's historical interpretation of the Ionic verb πλώω.

The verb πλώω has been interpreted in two different ways. At the pre-laryngealistic stage of Indo-European studies, the root of the Ionic πλώω was commonly reconstructed as **plō-*.[284] In more recent times, it was suggested by J. A. Harðarson that instead of **plō-* one should reconstruct **pleh₃-*.[285] The Ionic form πλώω could then be reconstructed as **pleh₃-i̯e/o-*vel sim.[286]

Based on such an interpretation of πλώω, the noun πλώς has recently been interpreted by Irslinger as a reflex of an ancient acrostatic *t*-stem **plóh₃-t-/pléh₃-t-* (2002: 41). The reconstruction of an original acrostatic inflection implies that the original meaning of the noun **plóh₃-t-/pléh₃-t-* would have been 'swimming', since, as has been said earlier, *t*-stems of this type originally were action nouns (verbal

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[281] Schindler (1972b), Nussbaum (2004a).
[282] Possibly, the black (striped) mullet (*Mugil cephalus*).
[283] See Chantraine (1939: 266) and Chantraine (1968: 919), GEW II.565, Schwyzer (1959: 499).
[284] IEW.836, Krahe/Meid (1969: 52), Chantraine (1968: 920).
[285] Harðarson (1993: 183).
[286] Harðarson, ibid., LIV.485.

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abstracts). However, the synchronic meaning of the noun πλώς is 'mullet' (concrete noun ← 'swimmer'?), not 'swimming' (verbal abstract). Therefore, in order to account for the agentive meaning of πλώς, either one has to assume that in the course of its evolution, this noun underwent the same semantic development as the noun φώς, from an action noun becoming an agent noun.

However, the reconstruction of the root **pleh*$_3$- has been criticized by J. H. Jasanoff as a less plausible option to account for the long vowel -ω- in the verb πλώω, especially because all other Indo-European languages point to **pleu-* or **plou-*. The root **pleu-* has to be reconstructed also for the non-Ionic Greek πλέω.[287] As shown by Jasanoff, the Ionic form could also reflect an older lengthened grade of the usual Indo-European 'swimming' root **pleu-* 'swim', cf. **plōu-*. According to such an interpretation, the glide of the diphthong **ōu* would have been lost between vowels, and the outcome πλώω would be phonologically regular, cf. IE **plōu̯-ō* > πλώω.[288] Thereby the root **pleh*$_3$- would become unnecessary. As a consequence, also the reconstruction of the *t*-abstract **plóh*$_3$*-t-* would become impossible.

If one assumed (together with Jasanoff) that Indo-European possessed only one 'swim' root, namely **pleu-*, it would be tempting to interpret the noun πλώς in an entirely different way. The original meaning of the noun πλώς could have namely been 'swimmer', whereas the meaning 'mullet' would have developed out of it secondarily. It is not uncommon that animals and plants are called after a certain characterizing

[287] Jasanoff (2003: 224), and p.c.

[288] Jasanoff, ibid. NB that Jasanoff's плавити 'swim' is to be read as плавати. The verb плавити is a causative of плоути.

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action and possess agentive/iterative sense, cf. IE *$h_3$ér-ōn 'eagle' (< 'the rising one' ~ *$h_3$er- 'rise'), Ru. кукýшка 'cuckoo' (< 'the cuckooing one'), Lith. *plaukìkas* 'northern phalarope' (*Phalaropus lobatus*, lit. 'swimmer', cf. *plauk-* 'swim'), *súopis* 'buzzard' (*Buteo*, lit. 'complainer', called thus after its call, cf. *suop-* 'shout, complain'), Icel. *stein-bítur* 'wolf-fish, catfish' (*Anarhichas lupus*, lit. 'stone-biter', called thus after its feeding habits),[289] etc. One could therefore argue that the original agent noun πλώς either is deverbative, and was formed from the stems of the thematic verbs πλέω or πλώω, or was formed according to some different pattern of derivation of agent nouns. As was shown above, the meaning 'mullet' could be explained as the result of a secondary semantic development, and the hypothetical verbal abstract *'swimming' could be altogether eliminated.

In Chantraine's treatment, the noun πλώς is considered to be a derivative of the Ionian πλώ-ω without further analysis.[290] However, it seems that the agent noun πλώς cannot be built to the thematic stem of the verb πλώω. If one considers other Greek *t*-stem agent nouns built to thematic verbs, one notices that these deverbative nouns take the suffix -ητ-, cf. πλάν-ητ- 'wanderer' (← πλανάω 'roam'), ἕρπ-ητ- 'shingles' (← ἕρπω 'creep'), πέν-ητ- 'daily labourer' (← πένομαι 'toil'), etc. Therefore, one would probably expect that the agent noun 'swimmer', built to the verb πλώω would have become *πλώης vel sim. (stem *πλω-ητ- vel sim.), which would not be the attested form πλώς. In fact, any attempt to derive the noun πλώς from the thematic stem of the

[289] The wolf-fish feed on molluscs, crushing their shells with large and powerful teeth.
[290] Chantraine (1933: 266). Same in Rix (1991: 195).

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non-Ionian πλέω would be equally fruitless, since an agentive derivative from this thematic verb would have probably become *πλέης (or *πλῆς, by contraction).

How, then, should the noun πλώς be explained?

One could also try to derive πλώς directly from the verbal Indo-European root **pleu-*. As was mentioned in section III.6.1, quite a few Greek root nomina agentis were formed according to this pattern, cf. κλώψ 'thief' (< 'stealer' ← **klep-* 'steal'), φώρ 'id.' ('carrier away' ← **bher-* 'carry [away]'),[291] σκώψ 'owl' (← 'looker' ← **spék̂-* 'watch'). As one can observe from the examples adduced, the pattern of derivation of agent nouns from verbal roots is the following: verbal R(é) → agent noun R(ṓ). Therefore, perhaps also the noun πλώς could have been formed in this way from the verbal root **pleu-*.

However, despite its superficial attractiveness, this theory cannot be correct. First, if one assumes the derivation of πλώς according to a pattern no longer productive in Greek, viz. verbal R(é) → agent noun R(ṓ), as shown above, one must asume that the noun πλώς is a rather old, probably pre-Greek formation. In such a case, the expected nominative singular form of the derived agent noun in question would have to be reconstructed as **plṓu-s* (cf. **klṓp-s*), and it would have yielded Gk. ˣπλοῦς. Accordingly, the weak stem would have been **plou-*, and it would be difficult to get the necessary πλω- from it as well. In fact, it would not be possible to get the morph πλώ- even in the accusative singular (cf. **plóu-m*), where one would expect to see the effect of "Stang's law", because the putative accusative singular form **plóu-m* would have

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developed into *πλῶν (instead of the desired πλών), cf. βῶν 'ox, cow' (< *g^{u}*óu-m*) or Ζῆν 'Zeus' (< *d_{i}*-éu-m*).

As a result, also the derivation of the noun πλώς from the verbal root **pleu-* turns out to be impossible.

One could think of one additional way in which the *t*-stem πλώς could have been built. It could have also been a root noun **plŏh₃-s*, built to the root **pleh₃-* 'swim'. If πλώς is an original root noun, the agentive meaning 'swimmer' could be original (cf. Gk. φώρ, etc.). Since πλώς is attested relatively late, it could have received its consonant -τ- in the oblique stem by analogy, cf. also several other masculine Greek nouns ending in -ως, which have a secondary -τ-: γέλως 'laughter' (Att. acc. sg. γέλωτα vs. γέλων or Hom. γέλω[ν]), sic also ἔρως 'love', χρώς 'skin', ἱδρώς 'sweat'.

However, if -τ- in πλώς is inherited, the most plausible interpretation becomes the one discussed at the beginning of this section, according to which πλώς was interpreted as an original action noun, built to the root **pleh₃-* 'swim'.

III.6.4.1.1. An Excursus to the Proto-Germanic **flōđu-* 'flood'

The plausibility of the reconstruction of the root **pleh₃-* is further supported by the *ō*-grade of the Proto-Germanic word for 'flood', cf. PGmc. **flō-đu-*, reflected in Go.

[291] Semantics as per Nussbaum (2004a).

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flōdus, OE *flōd* 'flood' (*u*-st.), and indirectly by OHG *fluot* 'id' (*i*-st. < **u*-st.)[292] and possibly OIc. *flóð* (*o*-st.; neut.).[293]

There is no direct way to derive the Proto-Germanic noun **flōđu-* from the Indo-European root **pleu-*. It would not be feasible to derive **flōđu-* from the Proto-Germanic verb **flōwan* 'swim' (< **plōu-*, as per Jasanoff 2003: 224, or **pleh*$_3$-, as per Harðarson 1993: 183), since primary *tu*-stems were not a productive class in Germanic. It would not have been possible to build a primary *tu*-stem **plōu-tu-*, since *tu*-stems tend to be proterokinetic, and a regular Indo-European *tu*-stem built to the root **pleu-* should have been **pléu-tu-*.

Although the alternative derivation of the noun **flōđu-* from an original **pléh*$_3$*-tu-* raises a certain question as well (see below), when the development of **flōđu-* is presented in such a way, it both becomes less complex and fits well into the pattern of deriving verbal nouns from verbal roots with the help of the suffix *-tu-*. One could ask, though, why the Proto-Germanic form is reconstructed as **flō-đu-* and not **flō-þu-*, as one should expect from a proto-form like **plṓ-tu-* (< **pléh*$_3$*-tu-*). It must be noted, that there really are no strict rules as to which of the two allomorphs, the allomorph **-þu-* with the voiceless fricative *þ* or the allomorph **-đu-* with the voiced *đ* is generalized after roots ending in vowels and resonants, cf. Go. *liþus*, OE *liþ*, OHG *lid* and OS *liđ*

[292] W. Braune (1953: 202, fn. 1).

[293] Krahe assumed that OIc. *flóð* reflected an earlier **flōđu-*, which would have lost *u* after a long syllable (Krahe/Meid 1969: 131). Although the syncope of the short vowel *u* cannot be doubted in this position, the transition of non-neuter *u*-stems with a long root to neuter *o*-stems is by no means a regular development, cf. OIc. *bógr* 'shoulder, upper arm' (masc., *u*-st.; dat. sg. *bógi*) < PGmc. **þōguz*, which has apparently retained its *u*-stem inflection. Although it is very likely that also OIc. *flóð* reflects an ancient *tu*-stem, its transition to the neuter *o*-stem class was governed by some different processes, and not those proposed by Krahe.

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'member' (< *li-þu-), but Go. *sidus*, OE *sidu*, OHG *situ* 'custom' (< *si-đu-), Go. *wulþus* 'glory' (< *wul-þu-), but OIc. *stuldr* 'theft' (< *stul-đu-), etc.[294] In the course of its development, the Proto-Germanic noun *flō-đu- happened to generalize the voiced suffix *-đu-.

III.6.4.2. Gk. θής 'day labourer'

The noun θής (stem θη-τ-) 'day labourer, slave' is attested fairly early, occurring already in Homer (θῆτες, nom. pl.; *Iliad*, IV.644) and Hesiod (θῆτα, acc. sg.; *Opera*, 602). It is also attested in Hesychius' glossary, where it is spelled differently, cf. θᾶτας.[295] θῆτας, τοὺς δούλους. Κύπριοι.

The noun θής does not have a widely accepted etymology. According to Brugmann's early interpretation, this noun is a derivative of the verbal root *dheu- 'run', the meaning of the noun being 'runner' vel sim.[296] Although from the semantic point of view, this interpretation is interesting, it works neither phonologically nor morphologically.[297]

Later it was suggested by A. Assmann that θής is a loanword from Semitic. Assmann derived the noun θής from the Hebrew adjective *taḥti* 'lowest'.[298] It would seem potentially interesting for several reasons. First, as pointed out by Assman, θής is used to denote low-paid workers, who did not enjoy equal rights with people of the

[294] Bammesberger, ibid.
[295] This form is accusative plural. The expected nom. sg. would have been *θᾶς.
[296] Notation mine. See Brugmann (1906: 388-90). This etymology was accepted by Boisacq (1923: 344).
[297] For criticism, see Fraenkel (1910: 87-8, fn. 2.), E. Kretschmer (1929: 80).

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higher ranks (such as πεντακοσιομέδιμνοι, ἱππεῖς and ζευγίται). Secondly, the change of initial Hebr. *t* to Gk. θ also occurs in several other loanwords, cf. Gk. Θάμαρ 'Thamar' (PN) < Hebr. *tmr*, Θάρα 'Thare' (PN) < *trḥ*. Thirdly, as pointed out by Assmann as well, Hebrew *ḥ* disappears in Greek, cf. Gk. Βηθλεέμ 'Bethlehem' < *bθlḥm*.

However, the proper nouns mentioned above are all very late (Biblical). On the contrary, the noun θής is attested as early as in Homer (*Odyssey*, IV.644). The consonant *t* of early Semitic loanwords appears to be reflected in a *different* way in Greek, cf. Gk. χιτών 'chiton, tunic' < Phoenician *ktn*,[299] further λιβανωτός 'frankincense, gum of the plant *Boswellia Carterii*' < Old Arabic *lbnt*.[300] The noun τύμπανον 'kettle-drum', which is considered to be a Semitic loanword by H. Lewy, cf. Hebr. *tōp*, Aramaic *tuppā*, and is attested already in Herodotus (Lewy 1895: 166), shows a *t* as well.[301] Lewy adduces one more example, Gk. Ἰάπετος (PN; *Iliad*, VIII.479), which derives from Hebr. *Jepet*.[302] Thus, it turns out that indeed only in late Semitic borrowings do we see Semitic *t* change into a Greek θ.[303] Therefore, Assmann's idea can hardly be correct.

Chantraine speculated that the noun θής derives from the Indo-European root **dheh*$_1$- 'put, place', which is some languages came to mean 'do', cf. La. *facere* 'do,

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[298] Assmann (1918: 96).
[299] Also attested in Mycenaean, cf. *ki-to*, *ki-to-ne* (Masson 1967: 27-8).
[300] Masson (1967: 53-4).
[301] Masson considers it to be a "probable Semitic borrowing" (Masson 1967: 95).
[302] See Lewy (1895: 243).
[303] Cf. one additional late example adduced by Masson, viz. θῖβις 'basket' < Hebr. *tēbā* 'case, chest' (Exodus; Masson 1967: 76).

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make' (< *dhh_1-*k*-). Thus, one could interpret the noun θής as an agent noun, 'maker, doer', i.e. 'worker'.[304] However, first of all, the root *$dheh_1$- never developed the meaning 'do' in Greek. Secondly, there is no way to derive the Cypriote form θᾶτας from this root.

One more interpretation has been put forward by Rix (1991). According to Rix's interpretation, the noun θής is an agent root noun, built to the root *$dheh_2$- 'tire, exhaust', which can also be observed in the Latin words *famēs* 'hunger' and *fatīgō* 'exhaust'.[305] The original meaning of the noun θής in such a case would have been 'who tires, exhausts himself (by labour)'.[306]

This interpretation is in partial agreement with the traditional historical interpretation of the above-mentioned Latin words, *famēs* and *fatīgō*. In the standard etymological dictionaries, both are considered to derive in one way or another from the Indo-European verbal root *$dhē$-/*$dhə$-.[307] The root *$dheh_2$-, as reconstructed by Rix, could be easily taken as a "modernized" version of the former. The *e*-colour of the root variant with a lengthened grade, viz. *$dhē$- instead of the expected *$dhā$- (< *$dheh_2$-), is not a big problem, since the lengthened grade is not attested in Latin. However, the reconstruction of *h_2 is essential, if one wants to reconstruct this root for the Greek noun θής, and especially the Cypriote *θᾱ́ς. The reconstruction of *h_2 is not

[304] Cf. Chantraine (1933: 266).
[305] Rix (1991: 195).
[306] Rix, ibid.
[307] Walde/Hofman (1938: 463), IEW.239, Ernout/Meillet (1959: 215). See also Dunkel (2000: 94).

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problematic for *famēs* and *fatīgō*, as any laryngeal would have been vocalized as *a* in this position.[308]

Rix's interpretation has some attraction, but it also raises a question (see below). First, let us discuss the semantics of θῆς, assuming that its root should be reconstructed as **dheh₂-*. The difference between the meaning 'dwindle, lessen, hinschwinden' as reconstructed for the root **dhē-/*dhə-* in IEW vs. 'tire, exhaust oneself' as reconstructed by Rix is not really problematic, as the two reconstructed meanings can be combined into 'exhaust, lessen (food/strength)'. The Latin noun *famēs* could then be interpreted as a concretized action noun, 'exhaustion, dwindling of food', whereas the semantic development of the Latin verb *fatīgō* becomes obvious.

As far as Rix's interpretation of the noun θής as an agent noun built to the root 'exhaust (oneself)' is concerned, although the terms for 'worker' are generally built to the verbal root meaning 'work', it may be that the original meaning of θής was *'poor, miserable one' (< *'the one with exhausted resources/strength'). A semantic parallel can be found in Lithuanian, where the noun *pavargėlis* 'poor, miserable one' is built from the verb *pavargti* 'to become tired'. In the case of Gk. θής, the meaning 'worker, day-labourer' could have developed secondarily within Greek.

The reconstruction of the Indo-European protoform of θής is more problematic. The agentive function (as per Rix) was common among root nouns at the early stages of the protolanguage. However, the occurrence of -τ- the oblique forms of θής already in

[308] See Schrijver (1991: 85ff.), Meiser (1998: 107).

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the earliest texts suggests that the ancestor of θής may have been a genuine *t*-stem. Moreover, if θής was indeed built to the otherwise unattested verbal root **dheh₂-*, it would more likely be an archaism.

If an old *t*-stem has to be reconstructed, one would first think of a verbal abstract **dhóh₂-t-/*dhéh₂-t-* 'being tired, exhausted (→ 'being miserable, wretched')'. Such a verbal abstract could become concretized to 'poor one' and eventually to the attested 'labourer'. This semantic development is somewhat reminiscent of the development of the noun φώς described in section III.6.1 above. However, the reconstruction of an old acrostatic *t*-stem raises an ablaut problem, since unlike most other Greek reflexes of the ancient acrostatic formations θής would have generalized the *e*-grade. It is impossible to determine, whether the *e*-grade is analogical (like in δαίς) or not, because no verbs with the root **dheh₂-* are attested in Greek.

III.6.5. An Excursus to Greek *t*-Stem Compounds.

In Greek, there are attested quite a few primary-looking compounds with a non-alternating suffix *-t-*: ἀνδρο-θνη-τ- 'killing men' (< **-dhn̥h₂-t-* ~ IE **dhenh₂-* 'run away'),[309] ἐπι-βλη-τ- 'fitting into', προ-βλη-τ- 'forthstretching' (< **-gᵘl̥h₁-t-* ~ **gᵘelh₁-* 'throw'), ὠμο-βρω-τ- 'eating raw flesh' (<**-gᵘr̥h₃-t-* ~ **gᵘerh₃-* 'eat, devour'), ἀμφι-τρη-τ- 'pierced from end to end' (< **-tr̥h₁-t-* ~ **terh₁-* 'drill, rub'), χερ-νη-τ- 'poor' (< **ĝhes-r-*[*s*]*n̥h₁-t-* or **ĝhes-r-*[*s*]*neh₁-t-* 'spinning with hands' ~

[309] The original meaning of this root changed to 'die' in Greek, see LIV.144-5.

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*[*s*]*neh*$_1$- 'spin'),[310] ἀ-γνω-τ- 'not knowing, ignorant' (< *-*ĝn̥h*$_3$-*t*- or *-*ĝneh*$_3$-*t*- ~ **ĝneh*$_3$- 'know'), πρό-βα-τ-α 'sheep, cattle' (pl.; < ?collective **pro-g*^{u}*m̥-t-h*$_2$ 'going forward'),[311] περι-κτί-τ-αι 'neighbours' (pl.; ← ?coll. **peri-tk̂i-t-h*$_2$; ~ Ved. *pari-kṣí-t-* 'living around, surrounding'), etc. built to the same verbal roots as the ones already shown above.

From this list one can see that the roots of the second members that have the structure *-eR(C)-*[312] routinely appear in the zero grade. This is reminiscent of the Latin situation, where the compounds *com-i-t-* and *ped-i-t-* show the zero grade of the underlying *-eR* type verbal root **h*$_1$*ei-*. Essentially the same situation can be observed in Indo-Iranian, where, just like in Greek, primary *t*-stem compounds with second members built to verbal *-eR(C)-* roots are the dominant type e.g., *deva-stú-t-* 'praising the gods' (~ IE **steu-* 'praise'), *apsu-kṣí-t-* 'dwelling in the waters' (~ IE **tk̂ei-* 'live,

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[310] There also exist different analyses of this Greek adjective, cf. χερνητ- < *χερ-αρν-ητ- (~ ἄρνυμαι 'gain, win'; see Schwyzer 1928: 105 and GEW II.1089), or χερνητ- ← **ĝher-no-* (~ **ĝher-* 'take pleasure, Gefallen finden'; Meier-Brügger 2003). The former interpretation raises morphological problems: since the nasal ν belongs only to the present active stem (cf. subj. ἄρηται, inf. ἀρέσθαι, ptc. ἀρόμενος), one would expect the pure root -αρ- to be used in nominal derivatives, cf. also Gk. ῥήκ-της 'breaker' instead of ˣῥήγν-ης, ˣῥηγνή-της vel sim. (cf. pres. ῥήγνῡμι, but aor. ἔρρηξα etc.). In those cases, where ν does occur in the nominal derivatives of *n-* or *nu*-formations (cf. κινητής 'agitator' [← κῑνέω 'move' < **kih*$_2$*-neu/nu-*; LIV.346], δυναστής 'ruler' [← δύναμαι 'be able' < **du-né/n-h*$_2$-; LIV.123]), from the synchronic point of view it is a part of the root. Despite Frisk's doubts in GEW (loc. cit.), the traditional interpretation of χερνητ- as 'spinning with hands' is semantically not really problematic, and compounds containing a seemingly redundant word 'hand' are attested elsewhere, cf. E. *hand-book, manuscript* (← late La. *manu-scriptus* 'hand-written'), Gk. χειρο-γραφέω 'report in writing', Lith. *rañk-darbis* 'handicraft' (lit. 'hand-work'), Icel. *hand-öxi* 'small axe' (lit. 'hand-axe') etc.

Meier-Brügger's connection of χερνητ- with the root **ĝher-* 'take pleasure' and the comparison with such words as χαίρω 'rejoice', χορός 'dance', and semantically quite different χείρων 'inferior, meaner' raises semantic issues. Thererefore, in the present study I will assume the traditional analysis of χερνητ- as **ĝhes-r-* + **[s]neh*$_1$*-t-* 'spinning with the hands'.

[311] The thematic singular form πρόβατον can be a back-formation.

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dwell'), *satya-srú-t-* 'hearing the truth' (~ IE **k̂leu-* 'hear, listen'). The interpretation of the Greek compounds with the final members built to roots of the *(C)ReH* type (cf. **[s]neh₁-* and **ĝneh₃-* above) is not entirely clear, as they can either show the full or the zero grade. If these roots had the full grade, they can be structurally compared with La. *sacer-dō-t-* 'priest', the final member of which is built to a *CeH-* root **deh₃-* 'give'. However, in such a case, **ĝhes-r-[s]neh₁-t-* would provide us with an obvious *e*-grade in the root of a final compound member. As was said in section III.5.2 above, the original colour of the vowel in the root of *-dō-t-* in the Latin compound *sacer-dō-t-* is obscured by the rounding effect of h_3.

In order to find out whether it is more plausible to reconstruct the stems °-νη-τ- and °-γνω-τ- with the zero or the full grade, it might be important to consider the behaviour of other roots, especially roots with the *(C)ReC* structure. The studies of the root structure of simplex acrostatic *t*-stems, root nouns (see Schindler 1972a, 1972c and Nussbaum 2004c and 2004d), and proterokinetic *ti*-stems (see Vine 2004) have yielded the following results: *CeC* and *(C)ReC* roots tend to appear in the full grade throughout the paradigm, and whereas the acrostatic formations show the *o*-grade characteristic of the strong forms, the mobile formations respectively show the *e*-grade which originally was at home in the strong forms as well (cf. **bhlog-* [> Gk. φλόξ], **prok̂-* [> Ved. *práś-*], **tu̯ek-* [> Ved. *tvác-*], **nokʷ-t-*, **sos-t-*, **meh₁-ti-*, **u̯es-ti-*). On the contrary, roots of the *eR(C)* type tend appear in the zero grade (cf. **pik̂-*, **dr̥k̂-*, **liH-t-*, **gʷr̥H-t-*,

[312] C = any consonant, including laryngeals but excluding resonants.

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mn̥-ti-, *mr̥-ti-*).[313] The *o*-grade of the acrostatic formations built to *CeC* and *(C)ReC* roots would lead one into the assumption that perhaps also the final members of *t*-compounds should exhibit the *o*-grade when they have the *(C)ReC* root structure. This would apparently not be convenient for Gk. χερ-νη-τ- which cannot reflect *-[*s*]*noh*$_1$-*t*-. However, since -νη- has the root structure *(C)ReC*, it is unlikely to reflect the zero grade *-[*s*]*n̥h*$_1$-*t*-. Therefore, if χερνητ- really contains the Indo-European 'spin' root, it is likely to reflect the protoform **ĝhes-r*-[*s*]*neh*$_1$-*t*- with the final member exhibiting the *e*-grade of the root.

Greek *t*-compounds must be fairly old. As one can see from the laryngeal reflexes in the roots of the other *t*-stems, these *t*-stems apparently were built when the Indo-European laryngeals were still around and were phonetically distinct from each other, cf. Gk. °-τρη-τ- (< **-tr̥h*$_1$-*t*-), °-θνη-τ- (< **-t*^{h}*nā-t*- < **-dhn̥h*$_2$-*t*- ~ θάνα-τος 'death'), °-βρω-τ- (<**-g*^{u}*r̥h*$_3$-*t*-). Thus, Greek provides additional, and particularly good evidence for the existence of *t*-stem compounds in Indo-European.

III.6.6. The Summary of the Development of the Acrostatic *t*-Stems in Greek

As one can see from the previous sections, Greek *t*-stems are not homorganic. Although some of the *t*-stems discussed in these sections indeed reflect earlier

[313] Root nouns built to roots of the *-eR-* shape often preserve the full grade, cf. Ved. *tán-* 'extension' (< **ten-*), *ván-* 'forest', Av. *nam-* 'fief', Gk. φορ-ός 'thief' (gen. sg., nominative has *ṓ*, cf. φώρ), etc. However, the full grade in some of these forms may have been restored, in order to avoid the ultra-short and morphologically opaque stems like †*ta-* (< **tn̥-*), †*va-*, †*na-*, whereas the full grade in Gk. φορ- may have either been restored in the same way as in the Indo-Iranian forms shown above in order to avoid †φαρ-, or it may be analogical after other root nouns (cf. φλογ-, κλοπ- etc.).

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acrostatic *t*-stem action nouns, the historical interpretation of several others is not entirely clear, cf. θής and πλώς.

As far as the preservation/reformation of the original ablaut in the monosyllables is concerned, the following conclusions may be drawn. Whereas on the one hand, the nouns νύξ and φώς and have generalized the *o*-grade of the original strong stem agreeing with other Indo-European languages (Hittite, in particular), on the other hand, the noun *λῑ́ς has generalized the zero grade, which agrees with other Indo-European data as well (cf. Hi. *kutt-*, *karitt-*, La. *grāt-*).

One should also notice that in Greek, the historical non-compound acrostatic *t*-stems with roots of the shape **CeH-* or **CeC-* tend to generalize the *o*-grade of the root, originally characteristic only of the strong forms.

Greek also possesses various reflexes of *t*-stem compounds. Whereas some are still athematic, cf. the compounds ending in °-βρω-τ-, °-βλη-τ-, others were thematized, cf. περι-κτί-τ-αι, πρό-βα-τ-α. From the semantic point of view, the Greek *t*-stem compounds are agentives.

When the ablaut of the Greek *t*-stem compounds is compared with that of the simplex *t*-stems, one can observe both similarities and differences. Whereas roots of the *eR(C)* structure generally appear in the zero grade (with few exceptions, cf. δαι-τ- < **deh₂i-t-*), the roots of the *CeC* and *CReC* type seem to behave differently. Whereas simplex nouns with roots of this shape generalize the *o*-grade of the root, the *e*-grade of χερ-νη-τ- suggests that the final members of compounds may have generalized the *e*-grade.

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III.7. Miscellannea

III.7.1. Tocharian B *kest*, Tocharian A *kast* 'hunger'

Both Tocharian dialects share the word for 'hunger', cf. Toch. B *kest* and Toch. A *kaṣt* 'hunger'. These two nouns reflect a *t*-stem **Kŏs-t-*, and have an exact parallel in the Hittite *kašza* 'id.' (**Kós-t-/*Kés-t-*).[314] Like in Anatolian, *t*-stems of this type are not a productive in Tocharian, and the attestation of this *t*-stem in both branches makes it possible to reconstruct an old Indo-European formation, a primary acrostatic *t*-stem.

Unlike Anatolian, no traces of the *e*-grade are to be seen in Tocharian. Just like the other languages, Tocharian has generalized the *o*-grade of the root.

III.7.2. Several Remarks about IE **h₂énh₂-t-* 'duck'

The very problematic Indo-European word for 'duck' in the course of time has been discussed in a number of works.[315] This noun has generally been interpreted either as a root noun **h₂énh₂t-*, **anət-*, **anh₂t-* vel sim. or as a *t*-stem, occasionally reconstructed as **h₂ónh₂-t-*, cf. Rix (1991), but more usually as **h₂énh₂-t-* (gen. sg. **h₂n̥h₂-t-és*).[316] The reconstruction of laryngeals is demanded by several factors, but in such a case, the reconstruction of a root noun **h₂énh₂t-* becomes less attractive due to the awkward consonant cluster **-nh₂t-*. However, when this noun is analysed as a *t*-stem, only a part

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[314] See section III.4 above.

[315] For the most recent discussion and references to earlier studies see Griepentrog (1995: 299) and Katz (2004).

[316] For now, I will be using the stem **h₂énh₂-t-* as default. The morphological structure of the forms **anh₂t-* and **n̥h₂tih₂* reconstructed in Griepentrog (1995: 299 and fn. 21) is unclear and is not discussed in his book. Both **anh₂t-* and **n̥h₂tih₂* would have develop into the attested dialectal forms; however, they look like phonological *ad hoc* reconstructions with no attempt made to explain their morphology.

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of the problems is solved, since, although the consonant cluster shown above is split, neither the meaning of the root *$h_2énh_2$- nor the original meaning of the derivative itself – which may have been other than 'duck' – are known. Besides these problems, the reconstructed root *$h_2énh_2$- has the $C_1e(R)C_1$ shape which is extremely rare in Indo-European.

The noun *$h_2énh_2$-*t*- developed in different ways in the separate Indo-European branches. As has been argued by Rix (1991), and later confirmed by J. T. Katz (2004),[317] the noun *$h_2énh_2$-*t*- underwent the most radical changes in the prehistory of Greek. According to Rix, the original root *$h_2énh_2$- was first replaced by a more familiar root *$sneh_2$- 'swim', and then, via the influence of the ancestor of the Greek verb νήχω 'swim', the original *$h_2énh_2t$- became pre-Greek *$snākh$-$i̯a$, which regularly yielded the attested Greek forms.[318] The semantics of the reformed *$snākh$-$i̯a$ would have been 'swimmer', vel sim., which for obvious reasons can further develop into 'duck'.

In his later treatment of the Indo-European 'duck' word, Katz suggested that the substitution of *$snākh$-$i̯a$ for an earlier *$snāt$-$i̯a$ (< *[*s*]neh_2-*t*-ih_2)[319] was not absolutely necessary, since the *t*-stem *$sneh_2$-*t*- (gen. *$sn̥h_2$-*t*-*és*) which would have replaced the original *$h_2énh_2$-*t*-, could already have an agentive meaning, viz. 'swimmer'.[320]

Katz's suggestion, however, contains difficulties. As I have shown in the previous sections, the original meaning of primary *t*-stems was that of verbal action nouns.

[317] Katz has suggested a simplified version of the original scenario that had been proposed by Rix. Katz's suggestions will be commented upon below.
[318] See a more detailed discussion in Rix (1991).
[319] Katz (2004: 198).

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Wherever such *t*-stems acquire an agentive meaning, this must have happened secondarily, in all likelihood after the suffix *-t-* had lost its original function and the respective *t*-stems were re-interpreted as root nouns, among which both agentive and verbal abstract meanings are well attested.

I do not doubt that the original root of the noun **h₂énh₂-t-* was replaced by the root **sneh₂-* in the prehistory of Greek, but, unlike Katz, I would suggest that instead of the old *t*-stem a new *t*-stem was not built. Nouns of this type were a recessive class everywhere in Indo-European, and they were steadily merging with root nouns. It would be more plausible to think that the original *t*-stem **h₂énh₂-t-* which by the pre-Greek stage most probably already functioned as a *root noun* (**h₂énh₂t-*), had been replaced by another *root noun*. This would solve several problems: first, one would not need to assume unproductive derivation, secondly, a (re)construction of a semantically implausible agentive *t*-stem could be avoided. Therefore, I would rather accept the earlier Rix's suggestion that the "root noun" **h₂énh₂t-* was ultimately replaced by an etymologically more transparent agentive "root noun" **snākh-*. The scenario could then be imagined as follows: IE **h₂énh₂-t-*[321] → **sn̥h₂t-* > **snāt-* → **snākh-* → νῆσσα.

Thus, as appears from the development of the Indo-European noun **h₂énh₂-t-* in Greek, it has been reformed so extensively, that the synchronic Greek νῆσσα has preserved nothing from the original Indo-European word.

What was the development of **h₂énh₂-t-* in the other Indo-European languages?

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[320] Katz, ibid.

[321] See the discussion of the shape of the stem below in this section.

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IE *$h_2énh_2$-t- could have regularly developed into the attested dialectal forms, cf. La. *anat-*, Lith. *ántis*, OPr. *antis*, OCS *ѫтъı*, OIc. *ǫnd*, OHG *anut*, OE *æned*, etc.[322] However, *$h_2énh_2$-t- is not very likely to have been the oldest shape of the Indo-European base-word. The reconstruction *$h_2énh_2$-t-/*$h_2n̥h_2$-t-és, which is commonly assumed, is problematic, as it does not fit well into the commonly accepted ablaut system as described in Schindler (1975b). Although the existence of an ablaut type, in which the accented *é*-grade moves between the root in the strong forms and the ending in the weak forms has been proposed and dubbed "amphikinetic II" and by other names, all the examples which supposedly reflect this type, can be explained using the Schindler's system mentioned above.[323]

Originally, the *e*-grade could appear in the strong root of athematic nouns only in proterokinetic and amphikinetic formations. Whereas no secure examples of proterokinetic *t*-stems are known,[324] there is no evidence either that *$h_2énh_2$-t- should be reconstructed as an amphikinetic *t*-stem, viz. **$h_2énh_2$-ŏt-. The lengthened grade of the suffix *-ōt-* tends to be preserved at least somewhere, cf. Lith. *nepuotis* 'grandson' < **nep-ōt-*, Go. *witoþ* 'law' < **u̯id-ōt-*, etc.[325] However, there is no evidence for **anōt-.

[322] The vocalization of the interconsonantal laryngeal is not needed for Germanic, as both in Old High German and in Old English, prop vowels are often inserted secondarily, cf. OHG *elah* 'elk' (*US*: 'moose') beside the older *elh*, OE *furuh* 'furrow' beside the older *furh*, *wylif* 'she-wolf' (< *wylf*), etc. The presence of a prehistoric **u* in the protoform of OIc. *ǫnd* is not needed either, since the *u*-umlaut is analogical after *ō*-stems like *sǫk* 'guilt', etc.

[323] See a critical discussion in Widmer (2004: 56-60).

[324] For a different interpretation of Rieken's proterokinetic *t*-stems see Nussbaum (2004a).

[325] See more examples and a detailed discussion in chapter V below.

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The accented *é*-grade also occurred in the weak stem of Indo-European acrostatic formations, and theoretically one could also reconstruct an old acrostatic *t*-stem *$*h_2ónh_2$-t-/*$*h_2énh_2$-t-*. How would it have developed in Indo-European?

In order to account for the shape of the attested dialectal forms, a fairly complex development must be posited. First of all, the root-final h_2 should have been lost in the strong forms due to "Saussure's law". If the *aniṭ*-version of the root had been analogically extended to the weak stem, this would have become problematic for Latin, since there would remain no way to account for the vocalism of La. *anat-*.[326] Therefore, one must assume that the laryngeal was not lost in the weak stem. On the contrary, it may have been analogically re-introduced into the strong forms.

The second problem is the justification for the derivation of the various dialectal forms from the weak stem *$*h_2énh_2$-t-* as is generally assumed, since, due to the specific structure of the root, the ablaut of this noun should have been reformed already in Indo-European in such a way that namely the *e*-grade would have been eliminated, cf. also La. *grātes* < *$*g^u_ṛH$-t-* (← orig. *$*g^u$órH-t-/*g^uérH-t-*). This would imply that the attested dialectal forms should either reflect the *o*-grade *$*h_2ónh_2$-t-* or the zero grade *$*h_2n̥h_2$-t-*.

As far as the Germanic, the Baltic and the Slavic reflexes are concerned, all of them can reflect the strong stem *$*h_2ónh_2$-t-*. Moreover, such a protoform would also account for the accentuation of Lith. *ántis*. Germanic, in which the old consonantal inflection of the original word has been preserved faithfully, cf. OIc. nom. pl. *endr* (< PGmc. **anđ-iz*), also provides evidence for the transfer of the accent to the desinence, witnessed by

[326] In fact, also Ved. *ātí-* 'duck', which is generally considered to be related despite the different suffix (see EWAia I.163), would also require the reconstruction of a *seṭ*-root.

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the Vernerized *-*đ*- instead of the expected *-*þ*- (< IE *-*t*). The voiced fricative -*đ*- developed in the weak forms of a paradigm with a mobile accent, as in the weak forms, the stress fell on the final syllable, cf. gen. sg. *h_2onh_2-*t*-*és* > *anþ-és > *ánđ-iz.[327] The preserved *o*-grade of the root instead of the expected zero grade, seen in Anatolian, Indo-Iranian, Greek and Italic *t*-stems with a comparable root structure should probably be considered an archaic feature of Baltic, Slavic and Germanic.

A more serious problem is the development of the La. *anat*-. It cannot reflect IE *h_2onh_2-*t*- despite the claims that *h_2o yielded *a* in Latin.[328] That the regular reflex of IE *h_2o was La. *o* is shown by words like *os* 'bone' (< *h_2ost-), *ocris* 'jagged peak' (< *$h_2o\hat{k}r$-*i*-), etc.[329] The other option is to derive La. *anat*- from the zero grade stem *$h_2\underset{\circ}{n}h_2$-*t*-.

Although the development of *#H$\underset{\circ}{R}$H* sequences in Latin is not very well understood due to lack of clear examples, according to Schrijver it cannot be ruled out that *anat*- actually reflects *$h_2\underset{\circ}{n}h_2$-*t*-.[330] If this is really is the case, then Latin together with Baltic, Slavic and Germanic would provide evidence for an older ablauting paradigm *$h_2$ónh$_2$-

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[327] The Old Icelandic gen. sg. *andar* reflects not Proto-Germanic **anđ-ōz*, but is an analogical reformation of the original **anđ-iz*, as was common practice among root nouns, cf. OIc. *markar* 'forest' (gen. sg.; beside *merkr*), etc. In many nouns of this type, the original genitive ending *-r* has been replaced by the ending *-ar*, cf. *strandar* 'shore' (but nom. pl. *strendr* or [later] *strandar/strandir*), *síldar* 'herring' (nom. pl. *síldr* [→ *síldar/síldir*]), etc. The old consonantal inflection can be seen better in the nom./acc. plural form *endr* (see main text).

[328] For references, see M. Mayrhofer (1986: 135, fn. 156).

[329] For more examples, see Schrijver (1991: 51).

[330] See Schrijver (1991: 311). Possible additional evidence is La. *animus* 'spirit, soul' (< **anamos*), which has been reconstructed by Peters as PIE *$h_2\underset{\circ}{n}h_1$-*mo*- (1980: 2, fn. 1). However, even though *mo*-stems were generally built to roots in the zero or the *o*-grade in Indo-European, there is also evidence for *mo*-stems with the *e*-grade in the root, cf. Gk. θερμός 'warm' < *$g^{\mu}her$-*mo*-. If the root vowel in θερμός is inherited, La. *animus* could also be reconstructed as *h_2enh_1-*mo*-.

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t-/**h₂n̥h₂-t-* with mobile accentuation.[331] As was said earlier, such paradigms with *ó/z* ablaut really existed, and they were an intermediary step in the regular reformation of the original *ó/é* acrostatic formation with *eR(H)*- roots, cf. the following schematic representation: PIE *ó/é* acrostatic (→ *ó/ó*) → *ó/z* → *z/z*.

To summarize, one could say that although the new analysis of the Indo-European noun **h₂énh₂-t-* proposed in this section rests on a number of assumptions and requires certain analogical reformations, to my mind, it provides a more attractive account for the original structure of this *t*-stem than the traditional reconstruction **h₂énh₂-t-* which has no structural parallels and does not fit into the traditional ablaut system of athematic nouns.

In the following section, I would like to discuss the semantics of the reconstructed acrostatic *t*-stem **h₂ónh₂-t-*/**h₂énh₂-t-*.

III.7.2.1.

The accurate semantic interpretation of the noun **h₂ónh₂-t-* is made difficult by the fact that the meaning of the root **h₂énh₂-* is not really known. However, if my proposed analysis of this noun as an acrostatic *t*-stem is correct, the root must be verbal, since *t*-stems of this type are always built to verbal roots. One could also say that since the root **h₂énh₂-* only occurs in this particular word meaning 'duck', it must refer to some characteristic feature of ducks. The meaning 'swim', or 'swim in a particular way' could certainly be considered, cf. also the history of Gk. νῆσσα 'duck'. However, one

[331] In Baltic, Slavic and Germanic, the strong stem would have been generalized.

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could also think of the meaning 'walk like a duck, wobble', since wobbly walking is characteristic of ducks – birds with relatively short legs and long heavy bodies. Certainly, this can be even more true of grebes (*Podiceps*) or loons (*Gavia*), which are even poorer walkers than ducks, but these birds are much rarer than ducks, and, living a more secretive and solitary life, they are not as familiar to the humans as ducks. In addition to this, they could be named after some different features, for example, OIc. *lómr* 'loon', Ru. rarápa 'id.' and Lith. *krãgas* 'grebe' were probably named thus after their call,[332] whereas the Lithuanian *nãras*, Modern German *Meer-taucher* and Modern Greek βουτηχτάρα 'loon' literally mean 'diver', and the loon indeed is an excellent diver.

Therefore, if the original *$h_2$ónh$_2$-t- had meant sth. like 'swimmer' or 'wobbler', the transfer of this word for the duck would not appear abnormal. However, as was concluded in the previous sections, acrostatic *t*-stems originally were verbal abstracts, and not agentives, as would be required by the hypothetical 'swimmer' or 'wobbler'. If *$h_2$ónh$_2$-t- really is the correct interpretation of the Indo-European word for 'duck', how should one explain the semantic development from some original verbal abstract to an agentive noun and ultimately to the concrete noun 'duck'?

This problem can probably be solved in a relatively easy way. As shown in sections III.6.1 and III.6.4 above, the same semantic problem arose for the Greek nouns φώς 'man, hero' and πλώς 'swimmer; mullet'. In those sections, it was concluded that the ancestors of these nouns denoted verbal abstracts, but in the course of development

[332] For Old Icelandic, cf. de Vries (1977: 365).

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acquired a more concrete agentive meaning, cf. 'shiner' resp. 'swimmer', which subsequently became narrowed down to 'hero' resp. 'mullet' (type of fish). I argued in those sections, that those action nouns developed their new agentive meaning after they had been reinterpreted as root nouns.

Essentially, I would propose the same scenario for the noun *$h_2$ónh$_2$-t-, which, according to this analysis, originally would have meant something totally different from 'duck'. Instead, it would have been a verbal abstract meaning 'swimming', 'wobbling', vel sim. depending on the original meaning of the root. As the original *$h_2$ónh$_2$-t-/ *$h_2$énh$_2$-t- started to merge with root nouns, becoming reformed as *$h_2$ónh$_2$t-/*h_2n̥h$_2$t- with a mobile accentuation, it developed a parallel agentive meaning 'swimmer/ wobbler' (vel sim.), and was subsequently attached to the familiar swimmer/wobbler – the duck.

As one can see from the discussion in the preceding sections, the history of the Indo-European 'duck' word contains a number of difficulties, some of which are very serious, e.g., the fact that the meanings of the root *$h_2$enh$_2$- discussed in the preceding paragraphs are mere guesses. As was said earlier, also the structure of the root itself is problematic. However, if the root *$h_2$enh$_2$- really existed and meant 'swim' or sth. similar, my new analysis of the original structure, the original semantics and the development of this word, according to which the Indo-European word was an acrostatic *t*-stem verbal abstract, provides a plausible analysis of this noun. The structural and the semantic development of this word, as presented in my discussion, also have parallels elsewhere in Indo-European.

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III.7.3. Old Irish *clí* 'house-post' and Old English *studu* 'id.'

In his treatment of the Old Irish noun *clí* 'house-post' in *Ériu* 29 (1978), Watkins reconstructed it as an ancient Indo-European *t*-stem, **k̂li-t-s* (gen. **k̂li-t-os*), built to the Indo-European verbal root **k̂lei-*'lean, etc.'. Watkins also pointed out a similar formation occurring in a Sanskrit compound *kṛchre-śrít-* 'providing support in need' (RV, vi.75.9),[333] the final member of which can reflect **k̂li-t-* as well.

In the same paper, Watkins proposed a similar analysis for the Old English noun *studu* 'stud' (< **stu-t-*), Hi. *ku-(ú)-uz-za* 'wall' (< **ĝhu-t-*) and Skt. *mít-* 'post' (< **mei*[h_2]*-t-*),[334] interpreting them all as ancient construction terms, built as *t*-stems.[335]

In 1996, Tremblay proposed a slight modification of Watkins' reconstruction of the original paradigm for the noun **k̂li-t-s* by providing it with *e*-grades, cf. nom. **k̂lei-t-s*, gen. **k̂li-t-es* (i.e., **k̂léi-t-s*/**k̂li-t-és*?).[336]

Although I essentially agree with Watkins' and Tremblay's interpretation of these nouns as *t*-stems, there are several details that need to be mentioned. As far as the semantics is concerned, the meaning 'house-post' or 'prop' is most probably secondary. The study of acrostatic *t*-stems has revealed that those *t*-stems that are reconstructable for Proto-Indo-European originally must have been verbal abstracts. Therefore, the original meaning of **k̂li-t-* would have been 'leaning, supporting', and the more concretized meaning 'prop' would have developed secondarily. The noun **k̂li-t-* is

[333] Similarly K. F. Geldner, cf. 'die Zuflucht in der Not' (lit. 'shelters in need', III.177). Grassmann translates this compound as 'venturing into danger' ('in Gefahr sich begebend', 1872: 346). However, the former interpretation fits better into the context.
[334] Nowadays, the last verbal root is sometimes reconstructed with a root-initial laryngeal, viz. **$h_{2/3}$mei-* (see LIV.426, fn. 1).
[335] Watkins (1978: 158-60).

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possibly quite old, as it is attested in two different branches, and already occurs in the Rigveda. The original meaning 'supporting, support' is probably still seen in the compound *kṛchre-śrít-* which may have originally been a bahuvrīhi.

The Old English *studu* must be an archaism, too, as it is built in a way that was no longer productive.

As for the ablaut, also the "amphikinetic II" paradigm (Tremblay's "update" of the old Watkins' paradigm) needs to be re-considered since, as was said in the preceding section, the "amphikinetic II" type hardly existed in early Proto-Indo-European. Since structurally the ancestors of OIr. *clí* and OE *studu* correspond well to acrostatic *t*-stem verbal abstracts, I would propose that their original paradigms should be reconstructed accordingly, namely as **k̂lói-t-/*k̂léi-t-* 'leaning, supporting' resp. **stó(h₂)u-t-/ *sté(h₂)u-t-* 'supporting, standing' (?).[337]

Since both roots after the loss of the laryngeals structurally became *-eR*, they would have systematically replaced the full grade of the root with the zero grade.

III.7.4. OIr. *né/nía* 'hero, champion'.

The history of the Old Irish noun *né/nía* 'hero, champion' (gen. *néth/níath/níad*) is complicated by the fact that at an early stage it became confused with the Celtic reflex of the amphikinetic Indo-European *t*-stem **népōt* 'offspring' (→ 'sister's son' in Old Irish), and are formally indistinguishable already in the Ogamic inscriptions.[338] The

[336] See Tremblay (1996a: 34, fn. 9).
[337] Originally, the vowel *u* must have been an enlargement or a suffix of some sort (see various examples with this *u*-formant in IEW.1008-9).
[338] For a recent discussion, see D. McManus (1997: 109-10) and Irslinger (2002: 52).

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paradigms of the two originally distinct nouns became contaminated into one, and as a result, some forms of the Old Irish paradigm of *né/nía* reflect the old forms of **népōt* (e.g., gen. sg. *níath* < **nepotos*),[339] whereas others, e.g., the nom. sg. *né* reflect the inflected forms of a noun which has been reconstructed as **nei(H)-t-* or **neiH-ōt-*, built to the verbal root **neiH-* 'lead' (~ Ved. *náyati*, YAv. *naiieiti* 'leads' [3. p. sg.], Hi. *nē[i̯]anzi* 'turn' [3. p. pl.]).[340] Irslinger considered all these reconstructions problematic: first, since the meaning of the putative noun **neiH-t-* was agentive, i.e., 'leader', she expected the suffix to be in the full grade (cf. OIr. *cinget-*).[341] Second, in case the Indo-European ancestor of **neiH-t-* were to be reconstructed as an acrostatic **nóiH-t-/ *néiH-t-*, Irslinger expected the generalization of the *o*-grade, as well as considered the loss of the laryngeal problematic. As an alternative solution, Irslinger proposed that OIr. *né/nía* may also reflect an amphikinetic **neiH-ōt-* which might or might have not been influenced by **nepōt-*.[342]

The issues raised by Irslinger are perhaps not really problematic. First, *t*-stems with a non-alternating suffix *-t-* and agentive semantics are attested, e.g., Gk. φώς 'hero' (< **bhóh₂-t-*) or πλώς 'fish; swimmer' (< **plóh₃-t-*). The semantic development of the putative noun **noiH-t-* 'leading' (→ 'leader' → 'hero') could then be compared with that of **bhoh₂-t-* 'shining' (→ 'shiner, the shining one' → 'hero').

If one reconstructs the ancestor of *né/nía* as an acrostatic formation, the laryngeal loss is not unexpected, since the conditions for the "Saussure's law" are fulfilled, cf.

[339] See sections V.10.1.1 – V.10.1.2 below.
[340] See Irslinger (2002: 52-3).
[341] Irslinger, ibid.
[342] Irslinger, ibid.

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**noiH-t-* > **noi-t-*. The strong stem **noi-t-* could have subsequently influenced the weak stem, whereby **neiH-t-* would have lost its laryngeal as well.

The final issue is the generalization of the *e*-grade instead of the more usual *o*-grade (or even zero, since the root has the *-eR-* structure, cf. also OIr. *clí* < **k̂liH-t-*).[343] Here one can turn to the contamination with the noun **nepōt-* mentioned at the beginning of this section. As will be discussed in section V.10.1.4 below, in the course of its development, the Indo-European noun **nepōt-* preserved the *e*-grade of the root in all the dialects. The *e*-grade of **nepōt-* may have been preserved due to phonotactic reasons.[344] If the nouns **nepōt-* and **noi-t-/*nei-t-* were contaminated at the stage when the latter noun still had qualitative ablaut, it would have not been unnatural to generalize the *e*-grade which would have appeared throughout the paradigm of **nepōt-* and in all the oblique forms of **noi-t-/*nei-t-*. It only remains to decide when the contamination of the two paradigms would have taken place.

Since the two nouns are confused already in the Ogamic texts, the merger must have happened fairly early. However, it probably followed the loss of **p* in Proto-Celtic. As to how exactly the noun **noi-t-/*nei-t-* would have looked at that stage of development, it is impossible to tell, and the comparison with the trends of development in other Indo-European dialects could be misleading, as there seems to be no agreement among them either. Whereas in Italic and Indo-Iranian roots of the *eR-* type generally exhibit the zero grade, Old Hittite *karaitt-* points to the *o*-grade and

[343] See section III.8.3 above.
[344] Cf. section III.3 above.

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Greek θής reflects the *e*-grade (if really < **dheh₂-t-* as per Rix).[345] Finally, the *o*-grade is exhibited by a number of Lithuanian forms which may continue Indo-European acrostatic formations as well.[346] One could say that the original acrostatic *t*-stem **noi-t-* /**nei-t-* had preserved its qualitative ablaut till the Proto-Celtic stage and then the *e*-grade (and not the *o*- or the zero grade) was generalized in the process of contamination of this noun with the *p*-less Proto-Celtic reflex of IE **nepōt-*.

III.7.5. Lithuanian Nouns of the *sámtis* 'ladle' Type

In Lithuanian, there are attested a small class of *tį̆o*-stem nouns with the structure *R*(*o*)-*tį̆o*- and the radical accent. They are built to verbal roots and have instrumental sense, e.g., *sámtis* 'ladle' (verb *sémti* 'scoop'), *dañgtis* 'lid' (~ *deñgti* 'cover'), *dãgtis* 'wick' (~ *dègti* 'burn'), *kamštis* 'cork' (~ *kimšti* 'stuff'), *raĩštis* 'band' (~ *rìšti* 'tie'), etc.[347] As one can see, the ablaut grade of the root generally differs from that of the corresponding verb. No matter whether the verbal root shows the zero or the *e*-grade, the noun will have the *o*-grade. This implies that this pattern of derivation is fairly old, i.e., it goes back to the time when ablaut was still actively used in derivation.

Beside some of these thematic stems, there also exist structurally close athematic feminine *ti*-stems, which also have a different accentual pattern and in rare cases also a

[345] See section III.6.4.2 above.
[346] See the following section.
[347] The Lithuanian noun *giñčas* 'quarrel, dispute' cited in Brugmann (Grundriß 186) is not a *tį̆o*-stem, as it should be ˣ*ginčias* (cf. *kurčias* 'deaf' < Proto-Baltic **kurtį̆a-*). P. Skardžius proposed the analysis **gint-š-a-*, cf. also *ilg̃-š-as* 'tall person' (~ *ilg̃-* 'long'), *jùk-š-as* 'who laughs at everything' (~*juõk-* 'laugh') etc. (1941: 315).

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slightly different meaning, cf. *dangtìs* 'lid' or 'covering' (gen.-*iẽs*)[348] but *dãgtis* resp. *dagtìs* 'wick', *lañktis* resp. *lanktìs* 'reel, swift (in textile)' (~ *leñkti* 'bend').

However, it is not very likely that these *tį̆o*-stem nouns really were built as such in Proto-Indo-European. There are attested a small number of archaic *tį̆o*-stems in the Indo-European dialects, but they are often built to various particles, cf. Go. *niþjis* 'relative', Skt. *ní-tya-* 'one's own' (< **h₁ni-* 'in' + *-tį̆o-*), Skt. *ápa-tya-* 'offspring' (~ *ápa* 'away, off'), etc. Certain adjectives that have a similar shape, cf. Skt. *jā́tya-*, Gk. γνήσιος 'noble' (< virtual **ĝn̥h₁-t-į̆o-*), Go. *alþeis* 'old', are not genuine *tį̆o*-stems because they are *į̆o*-stems built from past passive participles in *-to-*. Only La. *spatium* 'space; race course' could perhaps be interpreted as a *tį̆o*-stems noun built to verbal root in the zero grade (i.e., virtual **sph₂-tį̆o-*).

The Lithuanian nouns of the *samtis* type show some similarity to the Latin noun *spatium* shown above, yet the Lithuanian nouns routinely show roots in the *o*-grade.

How are these Lithuanian *tį̆o*-stems to be analysed?

Skardžius believed that these *tį̆o*-stems have developed from older *ti*-stems.[349] This may to some extent be true, since indeed, whenever a given *tį̆o*-stem has a parallel *ti*-

[348] Both meanings are attested in Bretkūnas, cf. *...Nesa io Rubas ira wienatija Dangtis jo odos*... 'for his clothes is his only (fem.) covering (fem.)' (Exodus 22:27). The gender of the accusative form *Dangti* is ambiguous in the clauses *Jr padare Dangtį ant Tabernakulo isch raudonų Awino adų* 'and they made a covering/lid for the tabernacle from the red skins of ram[s]' and *Jr ant to kita wel Dangti isch Apschra adų* 'and on that one [they made] another covering/lid from the skins of badger[s]' (Exodus 36:19), but the accusative plural form *Danktis* in Exodus (35:11) which refers to the same lids as the ones mentioned in Ex. (36:19) implies *ti*-stem inflection.

[349] See Skardžius (1941: 328).

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stem, the latter is generally obsolete. Also, as is well known, there was some interaction between *tį̆o-* and *ti*-stems in Lithuanian.[350]

However, if the nouns of the *samtis* type are reconstructed as ancient *ti*-stems, i.e., as IE **som(h_2)-ti-* etc., this creates a new problem, since, setting aside the obscure noun **ghosti-* 'guest, stranger' which may or may not be a *ti*-stem,[351] no *ti*-stem formations with acrostatic features are known.

One could in fact go one step further. Since some of these Lithuanian *tį̆o*-stems indeed possess *ti*-stem variants and these variants tend to belong to the more archaic language, one is forced to think that these *ti*-stems are relatively old. However, since *ti*-stems sometimes co-exist with *t*-stems, there even is a possibility that also the *ti*-stem inflection like *dangtis, -ies* is not original and has developed from a more archaic consonantal inflection.

The reconstruction of the nouns of the *samtis* type as original *t*-stems would make certain sense for various reasons. First of all, the *o*-grade of the root is well attested among acrostatic *t*-stems. Second, all these *tį̆o*-stems are built to verbal roots, and can easily be analysed as concretized abstract nouns, cf. **som(h_2)-t-* 'scooping' → Li. *samtis* 'scoop, ladle', **dhog$^{\mu}$h-t-* 'burning' → *dagtis* 'wick', etc. Third, consonantal stems often become *i*-stems in Baltic.

However, reconstructing these nouns as original *t*-stem acrostats, we face a still different problem. With the exception of *dagtis* which has a *CeC* root, the rest have *CeR(H)* or *CeRC* roots, and, as was concluded from the observation of the development

[350] On the history of Lithuanian *į̆o*-stems, see Kazlauskas (1968: 178ff.).

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of *t*-stems with such roots in other language groups (Indo-Iranian, Anatolian, Latin, etc.), roots with resonants generally appear in the zero grade. Therefore, instead of *samtis*, *dangtis*, *raištis* etc. one should expect **simtis*, **dingtis*, **ništis* and so forth. The full grade would be regular only in *dagtis*, since its root is *CeC*.

A still different problem is why these nouns became *tį̆o*-stems (masculine), whereas the noun *naktìs* 'night', a sure reflex of an ancient *t*-stem, and possibly the noun *ántis* 'duck' remained as feminine *ti*-stems.

This latter problem can be explained away in several ways, although none of them is easy to prove and all are speculative. The feminine gender of *naktis* and *antis* could equally be a preserved archaic feature or, in the case of *naktis*, analogical after *dienà* 'day' (fem.), and *ántis* could have retained (or regained) its feminine gender because a lot of bird names are feminine in Lithuanian.

On the contrary, the tool names of the *samtis* type may have been influenced by other tool names which were *į̆o*-stems, e.g., *kir̃vis* 'axe' or *peĩlis* 'knife', or back-formed from compounds, which usually have *į̆o*-stem inflection in Lithuanian no matter what inflection its second member had as a simplex noun, cf. *dantìs* 'tooth' (gen. *-ties*) vs. *reta-dañtis* 'gap-toothed' (gen. *-čio*), *dienà* 'day' vs. *pusiáu-dienis* 'noon, midday', also *sámtis* vs. *kiaurã-samtis* 'strainer' (gen. *-čio*), *raĩštis* 'band' vs. *galvã-raištis* 'headband', etc. There may also exist other ways to explain the transfer of the nouns like *samtis* over to the masculine *tį̆o*-stems, although, as was said before, none of them is totally satisfactory.

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[351] See IEW.453. Several problematic nouns reflecting structures like /R(o)-ti-/ are briefly discussed in Vine (2004: 374-6).

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III.8.5.1.

The more serious and more difficult to deal with problem is the ablaut difference between Lithuanian and the other Indo-European dialects. As was said earlier, where most Indo-European dialects show the zero grade, Lithuanian has the *o*-grade. This raises a dilemma: since the *o*-grade originally *did* occur in acrostatic *t*-stems with *CeR(C)* roots, should the *o*-grade of the *samtis* type nouns be interpreted as an archaic feature preserved by Baltic or do these nouns reflect something different, whatever it may be?

Although at first sight the preservation of the original *o*-grade only in Baltic and nowhere else may appear unlikely, it is interesting to note that also the noun *ántis* 'duck' for which I suggested an original acrostatic inflection shows the *o*-grade as well. Such „irregularities" are also attested elsewhere, notably in Hittite, cf. the Hittite noun *maišt-* 'glittering', which, according to the general pattern of the development of ablaut in acrostatic *t*-stems should have become ***mišt-***.[352] Also the Old Hittite *karaitt-* 'flood' shows a surviving archaic *o*-grade in a noun with a *CeR(C)* root.

Since, as was concluded earlier, the preservation of the *o*-grade in these Hittite forms (especially OH *karaitt-*) implies that the full grade was pushed out by the zero grade in *CeRC* roots at a relatively recent stage, one could say that the Lithuanian nouns of the *samtis* type reflect Indo-European acrostatic *t*-stems with a very archaic *o*-grade which originally occurred in the strong cases as well. The difference between Lithuanian and other the Indo-European dialects should be interpreted as a dialectal feature.

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However, one could also consider a different interpretation of the *o*-grade in the Lithuanian nouns of the *samtis* type. Their *o*-grade could also be interpreted as analogical after old *t*-stems with *CeC* roots, such as *dagtis* 'wick', in which the preservation of the *o*-grade was regular. Although nouns of this type are very rare in Modern Lithuanian, the situation was different at the earlier stages of development, i.e., when the alleged analogy would have taken place.

As has been said earlier, the thematic masculine inflection for the nouns of the *samtis* type most probably is an innovation, and before that they were inflected as *ti*-stems (and ultimately – as *t*-stems). Before the nouns of the *samtis* type became thematized, there existed more comparable athematic nouns with *CeC* roots, e.g., *naktìs* 'night' (< **nók$^{\#}$-t-*), and *sagtìs* 'buckle' (~ *seg-* 'button, fasten, attach'), which could probably be interpreted as an old *t*-stem **sóg-t-* as well.[353] It may well be that the *t*-stems with the *CeC* roots and the regularly preserved *o*-grade influenced the *t*-stems with *CeRC* roots, forcing them to adopt their ablaut.

III.8.6. Gk. ἄρτι 'just, just now' and Several Related Formations

Several words need to be said about the Greek adverb ἄρτι 'just, just now' (Theognis Elegiacus, Aeschylus +) and its relatives in the other Indo-European dialects. Traditionally, ἄρτι has been interpreted as a fossilized locative; however, scholars disagree whether this locative belonged to an old *ti*-stem or a *t*-formation of some sort.

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[352] See section III.4.3 above.

[353] The voiced consonant *g* in *sagtis* should be explained as analogical after the verb, cf. *seg-ù* 'button, fasten' (1. p. pr. ind.), or other nouns that exhibit this root, cf. *sag-à* 'button'.

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Frisk (GEW I.115), Schwyzer (1959: 266) and Chantraine (1968: 118) analyzed ἄρτι as ἄρ-τ-ι, although none of them tried to compare it with other Indo-European *t*-stems. In IEW.57, ἄρτι is interpreted as a locative as well; however, the base noun is assumed to have been a *ti*-stem. Such an interpretation may in a way be supported by the attestation of a *ti*-stem in Sanskrit, cf. Ved. *ṛtí-* 'Art, Weise'. Also the Latin noun *ars* 'art' (stem *art-*) could reflect a *ti*-stem, since the suffix *-ti-* seems to be regularly lost after roots of the *-eR-* type.[354] The Lithuanian adverb *artì* 'near' and Armenian *ard* 'just now' could perhaps reflect a comparable *ti*-stem as well.

The root of the underlying *ti*-stem has been reconstructed in two different ways, either as **h₂er-* or **(h₁)ar-* 'join', also seen in Ved. *ṛṇvánti* 'join' (3. pl.; < **Hṛ-nu̯-*), Gk. ἦρσα 'joined' (aor.) and ἀραρίσκω 'join' (pr.) and perhaps Hi. *āra* 'right' (adv.), (LÚ)*āra* 'friend' etc.[355] If the Hittite material is related, it would be preferable to reconstruct **(h₁)ar-*.

The original meaning of the *ti*-stem **(h₁)ar-ti-* (or **[H]Vr-ti-*) would have been 'joining' vel sim., from which the attested dialectal words would have developed.

The interpretation according to which Gk. ἄρτι, Lith. *artì* etc. were interpreted as reflexes of an old *ti*-stem has been recently adopted by J. Gorbachov (2005), who added an additional Phrygian preposition *art* 'around; near', attested in an Old Phrygian inscription at Vezirhan.[356] However, Gorbachov analyzed these forms as reflexes of an old instrumental **h₂er-ti-h₁*.

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[354] See Nussbaum (2004a) and section III.5.2 above.
[355] LIV.269-70, HED I.116-21.
[356] Gorbachov, loc. cit. with a reference to G. Neumann's earlier work.

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Gorbachov's intepretation raises several problems. First, it is difficult to imagine how this instrumental which would have meant 'by joining' vel sim. would have developed into 'just' or 'near'. Second, in order to account for the short word-final -ῐ, one needs to assume laryngeal loss. However, there are attested quite a few fossilized instrumentals in Greek, and they often exhibit a final long vowel, e.g. ἠρεμί̄ 'gently' (Aristophanes +), ἁμαρτή̄ 'together, at the same time' (Hom. +) etc.[357] Both problems could be avoided if one interpreted ἄρτι as a reflex of an old locative.

The reconstruction of an original *ti*-stem is problematic, too, since one would ideally expect an assibilated *ἄρσι in Greek. Although an obvious *ti*-stem can be observed in Ved. *ṛtí-*, this can be an independent creation. The reconstruction of a *ti*-stem is not very convenient for the Lithuanian *arti*, because Lith. *ar-* can only go back to a full grade, and *ti*-stems built to *-eR-* roots generally show the zero grade of the root (cf. Ved. *ṛtí-*).[358] The full grade in the root of the Lithuanian noun *artis* 'proximity' (gen. *arčio* [*io*-stem] or *arties* [*i*-stem]) is quite useless, as this noun is secondarily built from the adverb *arti* just like *tõlis* 'distance' was built from the corresponding adverb *tolì* 'far'.

A different option is to reconstruct an old *t*-stem *$(h_1)ar$-t- or *$(H)Vr$-t-[359] beside the *ti*-stem mentioned earlier and reflected in Sanskrit. Since *t*-stems originally were action nouns, the original meaning of this *t*-stem would have been 'joining', and the adverbial sense would be secondary.

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[357] See more examples in Schwyzer (1959: 550 et passim).
[358] See a brief discussion of several exceptional-looking Greek forms in Vine (2004: 357-9).
[359] Cf. Schwyzer and others' analysis of ἄρτι as ἄρ-τ-ι described above.

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Reconstructing a *t*-stem, one can solve the problem of the lack of assibilation in Greek, as it could be attributed to analogy from other cases in the paradigm, in which the conditions for assibilation were never fulfilled.

The existence of an old *t*-stem is also indicated by the Greek adverb ἁμαρτή mentioned earlier. This adverb has been interpreted as an old compound of **sm̥-* 'together' and a nominal derivative of the verbal root **ar-* (as in ἀραρίσκω 'join'), and is a frozen instrumental singular in **-ē* (< **-éh₁*).[360] No matter how one would like to reconstruct the underlying verbal root (**h₁ar-*, **ar-* or **h₂er-*), the immediate ancestor of the compound ἁμαρτή should probably be reconstructed as **sm̥m-r̥-t-ē* (< **-éh₁*), as the laryngeal would have probably been lost in the second member **-[H]r̥-t-*.

Now it only remains to clarify the development of the attested dialectal forms ἄρτι and *arti* from the original locative. As was concluded in sections III.4.5, III.5.6, and III.6.6, *t*-stems built to verbal roots of the *-eR(C)-* type normally generalize the zero grade of the root at some point of their development. The locative **(H)r̥-t-i* could be reflected as Gk. ἄρτι, but not as Lith. *arti*, which, as was said earlier, has to go back to a full grade.

One could select between two options. According to one option, one could say that the Lithuanian *arti* reflects the *o*-grade just like the nouns of the *samtis* type discussed in the previous section. In such a case, Lith. *arti* and Gk. ἄρτι would reflect the same

[360] Schwyzer (1959: 550); GEW I.87; Chantraine (1968: 70-1). The adverb ἁμαρτή is spelled in several different ways in Greek. In Homer, it is attested twice, and is once spelled ἁμαρτῇ in *Iliad*, V.656. The other instance is from *Odyssey*, XXII.81 where this adverb is spelled ἁμαρτῆ. The form ἁμαρτή is first attested in Aristarchus Grammaticus, but is thought to be the correct form (GEW, loc. cit.; Chantraine, loc. cit.).

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locative reformed in two different ways in the two dialects. However, since in both languages it is an archaism, one could also consider a single ancestor, which in this case should be reconstructed as loc. sg. *(*H*)*Vr-t-i* (i.e., *h_2*er-t-i* or *[h_1]*ar-t-i*).





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IV. Primary *t*-Stems in Indo-Iranian

IV.1. General Overview

In Sanskrit and Avestan, a small number of monosyllabic root nouns enlarged by a formant *-t-* are attested, cf. Ved. *stú-t-* 'praise' and OAv. *stū-t-* 'id.' (~ √*stu-* 'praise'), Ved. *hrú-t-* 'stumbling-block, pit, trap' (~ √*hvṛ-* 'be crooked'),[361] *mí-t-* 'post' (~ √*mi-* 'support'), *vṛ́-t-* 'hostile array' (~ √*vṛ-* 'cover, surround'), Av. *xšnŭ-t-* 'reward' (~ √*xšnu-* 'aneifern'), *i-t-* 'going, course' (~ √*i-* 'go'), Ved. *rí-t-* 'flowing' or 'stream' (~ √*ri/rī-* 'flow').[362] To these, one should also add the Sanskrit noun *nák-t-* 'night' which has correspondences in other Indo-European languages, as well as *pṛ́-t-* 'battle' and YAv. *pərə-t-* 'id.' which probably are derivatives of the Indo-European verbal root **per-* 'beat'.[363] Also the Sanskrit noun *nṛ́t-* 'dancing, dance' (AV) ends in *t*, but its etymology is not clear.[364] In addition to these, also the Avestan noun *cāt-* 'well' (?) occurring in two passages of Vendīdād[365] has been treated as a *t*-formation of some sort by several scholars,[366] although according to a new interpretation suggested by

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[361] See EWAia II.824-5.

[362] The translation 'flowing' is the most common, and can be found in H. Grassmann's Rigvedic dictionary, as well as K. F. Geldner's German translation of the Rigveda, etc., although R. T. H. Griffith translated *rí-t-* as 'stream'. See a more detailed discussion below in section IV.9 below.

[363] See section IV.6 below.

[364] See EWAia II.21-2, where a connection to the root **h₂ner-* 'be strong, possess vitality' is considered. If this is correct, then *nṛ́t-* could be analyzed historically either as a derivative *t*-stem **h₂nṛ-t-* or as a root noun **h₂nṛ-* + *t*-enlargement.

[365] Cf. Vendīdād 13,37: *...yezi nōiṯ spā ahąm.baoδəmnō maeγe vā cāiti vā vaēmi vā urūiδi vā apō vā nāuuaiiå paiδiiāite...* '...if not, the witless dog may wander into a hole, or a [...], or a crevice, or a river-bed of water navigable...', and V. 15,6: *yezi.ca aēša gaδβa maēγe vā cāiti vā vaēmi vā urūiδi vā apō vā nāuuaiiå paiδiiāite...* '...and if this dog may wander into a hole, or a [...], or a crevice, or a river-bed of water navigable...'.

[366] H. Reichelt (1909) and J. Kellens (1974) interpreted it as an enlarged root noun, Schindler called it a "consonant stem" (1972a: 60).

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Tremblay (2004) it should be interpreted as an acrostatic root noun with the root **ket-* instead.[367] The interpretation of the forms *cāiti* as locatives of a root noun **kót-/*két-* is quite possible. First, the locative of this acrostatic root noun would have been **két(± i)*, and it would have regularly developed into the attested Av. *cāiti*.[368] Second, as has been shown by Tremblay, in other Indo-European dialects there are attested several nouns built to the root **ket-/*kot-*, and they have relatable semantics, cf. Go. *heþjon* 'chamber' (acc. sg., Matt 6:6; nom. **heþjo* < **kēt-*), Gk. κοτύλη 'cup', Tocharian B *kotai* 'hole' (obl. sg.; nom. **koto*), etc.[369] Although only the Tocharian form has quite close meaning to the Av. *cāiti*, the semantics of the Greek and the Gothic forms imply some sort of 'cavity', and perhaps can be related with the Avestan and the Tocharian forms without serious difficulties. However, in such a case one should probably assume that the original meaning of the root **ket-* would have been 'cavity' vel sim., and that the meanings of the attested dialectal words had been modified.

[367] See Tremblay (2004: 585-6). It has to be mentioned here, that the meaning 'hole' proposed by Tremblay for Av. *cat-* is less preferable than the meaning 'well' which has been traditionally assigned to this word, since the word *maeye* which precedes *cāiti* in the passages quoted above is thought to mean 'hole' as well (see Kellens 1974: 80-1, cf. also Pahlavi *my* 'hole, pit').
[368] See Tremblay (loc. cit.), also Friedman (2004).
[369] Tremblay (2004: 586, fn. 61-2).

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IV.2. Several Additional Observations about the Indo-Iranian Simplex and Compound *t*-Stems

Unlike the other Indo-European dialects, the monosyllabic Indo-Iranian *t*-stems routinely exhibit the zero grade of the root, and with the exception of *nák-t-*, all of them are built to roots ending in the short vowels *-i*, *-u*, and *-ṛ*.[370]

Beside the monosyllabic *t*-stems, Sanskrit and Avestan possess a large number of primary *t*-stem compounds, e.g. Ved. *deva-stú-t-* 'praising the gods', *pathi-kṛ́-t-* 'path-making', *divi-kṣí-t-* 'dwelling in the sky', *aśva-jí-t-* 'winning horses', Av. *ātrə-kərə-t-* 'making the fire', *zauuanō-srū-t-* 'listening to the prayer' (Ved. *havana-śrú-t-* 'id.'), *dunmō-fru-t-* 'flying in the clouds' etc.[371]

The roots of the second members of the *t*-stem compounds in most instances are in the zero grade and end in the short vowels *-i*, *-u*, *-ṛ*, and *-a* (< **ṇ/ṃ*) as well. There exist three exceptional compounds in Avestan, in which *t* may have been added to the root *dā-* 'place, bestow' (< **dheh*$_1$- 'place'), cf. Av. *dąma-dātəm* (acc. sg.; Vendīdād 19,15) or *dāmi-dātəm* (V. 19,35), *tara-dātō* 'prevailing (?)'[372] (gen. sg.; 5x; Y. 22, 24; Yt. 2,1 etc.),[373] *para-δātąm* '?' (gen. pl.; V. 20,1).[374] However, all three forms contain various problems and have been interpreted in more than one way.[375]

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[370] The length of the vowel is sometimes obscured by inaccurate spelling in Avestan, but in most cases it can be securely reconstructed.

[371] For more Vedic examples, see A. A. Macdonell (1910: 186-8) and S. Scarlata (1999), for Avestan, see J. Kellens (1974).

[372] Semantics as per Nussbaum (2004a).

[373] There are attested spelling variants *tarəδātō* and *taraδātō*.

[374] There are attested thematic variants *para-δātanąm* and *pərə-δātanąm* in several manuscripts (see Kellens 1974: 264).

[375] See section IV.2 below for a more detailed discussion.

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The number of *t*-stem compounds largely exceeds that of simplex primary *t*-stems both in Sanskrit and in Avestan. In many instances, when a second member of a given *t*-stem compound occurs as simplex it takes the suffix *-ti-*, e.g. compound °*-kṛ́-t-* 'making X' vs. simplex *kṛ-tí-* 'making', °*-jí-t-* 'winning X' vs. *ji-tí-* 'winning, victory', °*-kṣí-t-* 'dwelling in X' vs. *kṣi-tí-* 'dwelling, abode', °*-bhṛ́-t-* 'carrying X' vs. *bhṛ-tí-* 'carrying; support' etc. In several instances, there are attested two simplex nouns built both with the suffix *-t-* and *-ti-*. The meanings of these simplex nouns sometimes are the same, but sometimes different, cf. *stú-t-* and *stu-tí-* 'praise', *nṛ́-t-* (AV) and *nṛ-tí-* 'dance' (RV, 2x), but *mí-t-* 'post' and *mi-tí-* 'erecting',[376] *vṛ́-t-* 'troop, host' (RV, 6x) and *vṛ-ti-* 'surrounding' (late). Only *hrú-t-* and *pṛ́-t-* do not occur as simplex *ti*-stems.

A similar situation can be observed in Avestan, where a number of verbal roots take the suffix *-t-* when they occur as second members of compounds, but if they occur in simplex nouns, they take the suffix *-ti-*, cf. °*-srŭ-t-* 'hearing' vs. *srŭ-ti-* 'performance, reciting' (YAv.; ← *'listening'), °*-bərə-t-* vs. *bərə-ti-* 'carrying' (YAv.), °*-ka/ərə-t-* vs. *kərə-ti-* 'making'.

IV.3. An Excursus to Three Avestan Compounds

A few words need to be said about the three Avestan compounds *dāmi-dātəm/ dąma-dātəm*, *tara-dāt-* and *para-δāt(a)-* mentioned in the previous section. An

[376] Both occur only once in the Rigveda, cf. the following passages: *sahásram míta úpa hí śráyantām* 'one thousand posts shall be put up' (RV, x.18.12b); *śáṃ náḥ svárūṇāṃ mitáyo bhavantu* 'for our good fortune shall be the erectings of the sacrificial posts' (RV, vii.35.7c).

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interpretation of these compounds as *t*-stems has been proposed;[377] however, at least for *dāmi-dātəm*, *dąma-dātəm* and *para-δāt-* also a different interpretation is possible, viz. as *to*-stems, because whereas the former two compounds occur only in the accusative singular form,[378] the latter occurs in two variants: one athematic *para-δātąm*, and one thematic *para-δātanąm* (both forms gen. pl.).[379] The compound *tara-dāt-* which has a clearly athematic genitive singular form *tara-dātō*, has been interpreted in two ways as well: according to Kellens (1972: 259ff.), the final member of this compound could be reconstructed as °*-dhā-t-* (< IE **-dho/eh*$_1$-), but recently Nussbaum proposed a de-thematic amphikinetic derivation, reconstructing the final member as **-dhh*$_1$*-e-ot-* (Nussbaum 2004a).

IV.3.1. *dāmi-dātəm* and *dąma-dātəm*

The forms *dāmi-dātəm* and *dąma-dātəm* occur in the following passages: (V. 19,15) *vaxšəm mē asąsaţ zaraϑuštrō nizbaiiemi ahurō mazdå aşauua dąma.dātəm nizbaiiemi miϑrəm vouru.gaoiiaoitīm*...'my words repeated Zarathustra: I invoke the holy Ahura Mazda [nom. sg.] *dąma.dātəm* [acc. sg.], I invoke Mithra possessing the wide pastures/hunting grounds...'; (V. 19,35) *vaxšəm mē asąsaţ zaraϑuštrō nizbaiiemi ahurō mazdå aşauua dāmi.dātəm nizbaiiemi ząm ahuraδātąm āpəm mazdaδātąm*... 'my words repeated Zarathustra: I invoke the holy Ahura Mazda [nom. sg.] *dāmi.dātəm* [acc. sg.], the earth created by Ahura, the water created by Mazda...'.

[377] See Kellens (1972: 247-51, 259-65).
[378] See the following section.
[379] See section IV.2.3. below.

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Both forms are accusative singular and both can theoretically be interpreted either as *t*- or *to*-stems, as in the accusative singular they would have looked the same. The *to*-stem *dāmi-dā-ta-* is well attested in Avesta and has the sense of a past participle, 'créé par le createur' (cf. Y. 10,10 and 61, etc.).[380] The *t*-stem *dāmi-dā-t-* is not attested anywhere else, but if these accusative forms are to be interpreted as *t*-stems, based on semantic comparison with other *t*-stem compounds, the *t*-stems *dāmi-dā-t-* and *dąma-dā-t-* should have most likely had an agentive meaning 'creating the creation' vel sim.[381]

According to Kellens, the *to*-stem interpretation is preferable to the *t*-stem interpretation at least in V. 19,35, since in this passage, "tous les termes qui succèdent à *dāmi-dātəm* désignent à l'evidence des éléments de la création".[382] However, as Kellens himself admitted, if the preceding nominative forms *ahurō mazdå aṣauua* in both passages are to be interpreted as mistakes for expected accusatives,[383] they could be modified by the following compounds *dāmi-dātəm* and *dąma-dātəm*. In the latter case, it would be preferable to interpret them as agentive *t*-stems. The passages could then be interpreted in the following way: "... I invoke the holy Ahura Mazda, the creator of the creation, I invoke Mithra, the possessor of the wide pastures/hunting grounds..." (V. 19,15) and "...I invoke the holy Ahura Mazda, the creator of the creation, I invoke the earth created by Mazda, the water created by Ahura, ..." (V. 19, 35).

[380] Kellens (1972: 247-51).
[381] Bartholomae (1904: 737).
[382] Kellens (1972: 249-50).
[383] Or, in Kellens' words, the nominative forms *ahurō mazdå aṣauua* could be due to "emploi abusif [...] qui s'est généralisé dans l'ensemble de la tradition" (Kellens 1972: 249).

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If this is correct, the second member of these compounds could be reconstructed as *-*dheh*$_1$-*t*- and structurally compared with *-*doh*$_3$-*t*- reconstructed for the second member of the Latin compound *sacer-dō-t-*.[384] The full grade of the root contrary to the zero grade of the root in the final member of many other *t*-stem compounds would be preserved because of the structure of the root **dheh*$_1$- (-*eC*- root).

IV.3.2. *tara-dāt-*

Although the athematic inflection for the Avestan compound *tara-dāt-* is secured by the genitive singular form *tara-dātō* (Y. 22,24, Yt. 2,1, F. 18, S. 1,2 and 21), its further morphological analysis is not entirely clear. Ch. Bartholomae (1904) and Kellens (1972) analyze it as a *t*-stem compound *tara-dā-t-*. According to Bartholomae, this compound has the agentive meaning 'hinsehend über'.[385] When morphologically and semantically interpreted in this way, this compound becomes comparable to *dāmi-dā-t-* discussed in the preceding section, and one could reconstruct its final member as IE *-*dheh*$_1$-*t*- as well. The semantics of the original compound could be reconstructed as 'setting aside', and it could have later developed into 'prevailing' vel sim.[386]

The reconstruction of *tara-dāt-* as a *t*-stem compound becomes especially attractive when *tara-dāt-* is compared with other Indo-Iranian compounds that employ the roots **dhā-* 'place' or **dā-* 'give', but do not have the *t*-enlargement, cf. Av. *Mazdā-* (~ Ved.

[384] Theoretically, also *-*dhoh*$_1$-*t*- would have yielded -*dā-t-* in Avestan, but the *e*-grade in Gk. χερ-νη-τ- suggests that the final members of the Indo-European *t*-stem may have had the *e*-grade.

[385] Bartholomae (1904: 640). However, according to Kellens, this meaning "ne peut être q'approximé" (1972: 259).

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medhá- 'wisdom' < **maz-dhaH-* < **mn̥s-dheh*$_1$-), Av. *zraz-dā-* 'loyal' and Ved. *śrad-dhá-* 'faith, confidence' (< virtual **k̂red-dheh*$_1$-),[387] Av. *puθrō-dā-* 'giving sons', OAv. *vaŋhu-dā-* 'bestowing good' (YAv. *vaŋhu-δā-*), *gaiiō-dā-* 'giving life' (<**-doh*$_3$-) etc. Whereas the compounds with unenlarged final members are historical root noun compounds, the Avestan compound *tara-dāt-* could be interpreted as a historical derivative *t*-stem compound.

A different morphological interpretation for the compound *tara-dāt-* was recently suggested by Nussbaum, who proposed that it should be interpreted as a de-thematic amphikinetic formation, viz. **-dhh*$_1$*-e-ot-*.[388] Nussbaum based such an interpretation, on the one hand, on the comparison with the Vedic compound *ratna-dhá-* 'bestowing wealth' (< **-dhh*$_1$*-ó-*), and on the other hand, on the comparison with other secondary *ot*-stems built to thematic stems, such has Av. *fra-carāt-* 'going forth' (< **pro-k*$^{\mu}$*el*[*H*]*-e-ot-*; cf. thematic *fra-car-a-* 'id.'), Gk. ἀργητ- 'bright' (< **-e-ot-*; cf. ἀργ-ό-ς 'shiny') etc.[389]

Such an interpretation for *tara-dāt-* is theoretically possible, however, it may be not the most attractive one. Whereas the amphi-style *t*-stems like *fra-carāt-* are built to old thematic stems which go back to the protolanguage, the situation with Av. *tara-dāt-* may be different. On the one hand, it is not entirely clear whether Ved. *ratna-dhá-* is an old Indo-European formation, or is back-formed from *ratna-dhā-* 'id.' (< **-dheh*$_1$-, not ×**-dhh*$_1$*-e-o-*). In Vedic, there are attested more such cases when a root occurring as a

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[386] Here I am adopting Nussbaum's semantic interpretation, although his morphological analysis is different (see Nussbaum 2004a and below in this section).
[387] The initial *z* in Av. *zrazdā-* is probably due to influence from the noun *zərəd-* 'heart'.
[388] Nussbaum (2004a).

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final member of compounds exhibits either a long vowel or a short vowel, cf. °*-sthắ-* 'standing' (← IE **-steh₂-*; Ved. *rathe-ṣṭhắ-* 'standing in a chariot'), vs. °*-sthá-* (*rathe-ṣṭhá-* 'id.'), °*-jắ-* 'born' (< IE **-ĝṇh₁-*; *pūrva-jắ-* 'born before') vs. °*-já-* (*pūrva-já-* 'id.'), °*-bhű-* 'appearing' (< IE **-bhuH-*; *puru-bhű-* 'appearing in many places') vs °*-bhú-* (*puru-bhú-* 'id.') etc.[390] As one can see especially from the pairs °*-bhū-* vs. °*-bhu-* and °*-jā-* vs. °*-ja-*, the roots of with the short vowel cannot be old, since they cannot regularly derive from the underlying Indo-European roots. The roots with the short vowel are most probably analogical, and the source of this analogy could be compounds of the *rathe-ṣṭhắ-* and *rathe-ṣṭhá-* type, which more or less regularly reflect two independent Indo-European derivation patterns. The compound root *rathe-ṣṭhắ-* is an athematic root noun compound with the structure *X-R(e)-*, whereas *rathe-ṣṭhá-* has a thematic second member, and its structure is *X-R-e/o-*. The thematic character of °*-ṣṭhá-* (< **-sth₂-e-*) is compatible with the thematic inflection of the verb *ti-ṣṭha-* 'stand'.

However, the verbal root **dhā-* 'place' forms athematic verbs in both Sanskrit and Avestan, cf. Ved. *da-dhā-/da-dh-*, Av. *da-dā-/da-d-*, and given the presence of the athematic compounds like Av. *Mazdā-*, Ved. *medhắ-* (< **-dheh₁-*) etc. shown above, the interpretation of Ved. *ratna-dhá-* as a dialectal analogical formation becomes perhaps more attractive that the reconstruction of an archaic **-dhh₁-o-*.

On the other hand, the reconstructed de-thematic **-dhh₁-e-ot-* 'setting' would face the same semantic problem as **-pl(h₁)-e-ot-* discussed in section III.5.5 above. Since the underlying thematic stem most probably would have already had an agentive

[389] Ibid. See also chapter I of this dissertation.

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meaning, the creation of an additional agentive compound would have become unnecessary. Therefore the final member of the Avestan compound *tara-dāt-* should probably be reconstructed as an athematic *t*-stem with the full grade of the *CeC* type root, viz. **-dheh*$_1$*-t-*.

IV.3.3. *para-δāt-*

The last compound, *para-δāt-* shows an athematic ending only in one passage, viz. V. 20,1: *kō paoiriiō maşiianqm δamnaŋ^{v}hatqm varəcaŋ^{v}hatqm yaoxštiuuatqm yātumatqm raēuuatqm taxmanqm paraδatqm...* 'qui fut le premier des hommes attentifs, énergiques, malicieux, sages, riches, hardi, *p.*, ...'.[391] Otherwise, it more commonly occurs in the proper name *Haošiianha Paraδāta*, and is thematic (Yt. 5,21; 15,7 etc.).[392]

Although beside the athematic *para-δātqm*, there exist two variae lectiones, thematic *para-δātanqm* (Jp1, Mf2) and *pərə-δātanqm* (L1.2, Br1), the athematic *para-δātqm* is obviously lectio difficilior, and may be the original form, whereas the thematic *para-δātanqm* may have been influenced by the proper name or by the preceding thematic *taxmanqm*. The athematic compound *para-δāt-* would be comparable to *dāmi-dāt-* and *tara-dāt-* discussed in the previous sections, and its original meaning may have been *'prevailing', *'victorious' (← *parǎ-* 'away, aside' + *dā-* 'place') or *'glorifying, pious' (← *parǎ-* 'before' + *dā-* 'place'). In such a case, one could reconstruct an

[390] See Scarlata (1999).
[391] Adopted from Kellens (1972: 264).
[392] See Kellens, loc. cit.

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agentive *t*-stem compound with the final member **-dheh$_1$-t-* like in the preceding sections. However, one should also note Kellens' suggestion that *para-δāt-* might perhaps be a misspelling for *tara-δāt-* 'prevailing'.[393]

IV.3.4. Summary.

From the preceding discussion, one can see that the interpretation of the three Avestan compounds involves various problems, and all of them have or can be interpreted in more than one way. If all three really represent *t*-stems, I find the reconstruction of the final members as **-dheh$_1$-t-* the most attractive, since, on the one hand, their interpretation as de-thematic formations would cause semantic problems. On the other hand, beside the *t*-stems with **-dheh$_1$-t-* as the final member, there are also attested a number of compounds with an unenlarged final member **-dheh$_1$-* both in Iranian and Indic.

The Vedic compound *ratna-dhá-* 'bestowing wealth' which theoretically could be reconstructed as thematic **-dhh$_1$-o-* in fact may be a secondary analogical formation, comparable to *puru-bhú-* and *pūrva-já-* mentioned earlier.

IV.4. Two Theories about the Historical Interpretation of the Indo-Iranian *t*-Stems

The historical interpretations of the Indo-Iranian *t*-stems can be roughly classified into two groups: according to one common theory, these *t*-stems are root nouns enlarged by a morphologically empty formant *-t-* the origin of which was explained

[393] Kellens (1972: 265).

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differently by different scholars.[394] Accordingly, the *t*-stem compounds are interpreted as enlarged root noun compounds comparable to the root noun compounds of the Sanskrit *havir-ád-* 'eating the oblation' type with no enlargement.

According to the other theory, the Indo-Iranian *t*-stems are historical derivative *t*-stems built with a special suffix *-t-*.[395]

The former theory according to which the Indo-Iranian *t*-stems of the *stú-t-* type are considered root nouns is based on the facts that from the synchronic point of view the *stú-t-* type *t*-stems function as root nouns, and that the formant *-t-* does not seem to add any additional semantic nuances to these nouns, cf. Ved. *stú-t-* 'praise' (~ √*stu-* 'praise') vs. *dvíṣ-* 'hate' (√*dviṣ-* 'hate') etc. It is significant, too, that these roots never occur without an enlargement, at least when they are used in nominal formations.[396] Moreover, the verbal roots that take the *t*-enlargement when used as substantives, are enlarged by this *-t-* also in future passive participles (gerundives) and when they occur in as second members of compound gerunds, e.g. Ved. °*-stú-t-ya* (√*stu-*), °*-jí-t-ya* (√*ji-*) as opposed to °*-úc-ya* (√*vac-*), °*-nī́-ya* (√*nī-*), etc.

Since the formant *t* is added to similar roots in Indic and Iranian, there is a good reason to assume that this rule was already active at the common Indo-Iranian stage. Yet the (semi-)regular addition of the formant *-t-* to specific roots must be an Indo-

[394] See sections II.1– II.8 above.

[395] Rieken (1999), Irslinger (2000). Also Brugmann considered the formant *-t-* to be a formant of feminine abstracta, although otherwise his description of this formant is rather vague (see Grundriß 422). Nouns of the *stú-t-* type were not included in Schindler's study of the Indo-European root nouns (1972a).

[396] However, several other monosyllabic roots ending in *-u* do not take the *t*-enlargement in Sanskrit, cf. Ved. *pṛthú-ṣṭu-* 'having a broad tuft of hair' (RV, x.86.8b), *vanar-gú-* 'wandering in the woods' (RV, i.145.5a), *ádhri-gu-* 'irresistible' (RV, 15x).

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Iranian innovation, since such a rule never operated in other Indo-European languages.[397]

The shortcoming of the "enlarged root noun" theory is that it does not explain the source of this enlargement.

The merit of the other theory about the historical interpretation of the Indo-Iranian *t*-stems is that it tries to suggest a possible origin of the formant *-t-*. According to this theory, the *t*-enlargement originally was a derivational suffix similar to the suffix *-t-* seen in the nouns of the **nók^{u}-t-* type discussed in chapter III. Based on this comparison, in her discussion of the Indo-European *t*-stems, Rieken reconstructed the Indo-Iranian *stú-t-* type nouns as original derivative acrostatic *t*-formations. She suggested that at the Proto-Indo-European stage, the nouns of the *stú-t-* type had a typical acrostatic ablaut pattern *R(ó)-t-/*R(é)-t- which eventually was reformed to *R(ó)-t-/*R(z)-t- under the influence from root nouns. Ultimately, the zero grade was generalized throughout the paradigm, whence the attested Ved. *stú-t-*, *mí-t-*, *hrú-t-* etc.[398] The generalized zero grade in the roots of the Indo-Iranian *t*-stem root nouns can be compared with the very common zero grade in unenlarged root nouns built to *eR(C)* roots, cf. Ved. *vṛ́dh-* 'growth', *dvíṣ-* 'hate', *kṣúdh-* 'hunger', *gír-* 'praise' [< IE **g^{u}ṛH-*], *gṛ́bh-* 'seizing', *dṛ́ś-* 'look', *yúdh-* 'fight', Av. *bərəz-* 'high; hill', *gar-* 'welcome' [< **g^{u}ṛH-*] etc.

[397] See already Hirt (1913: 273). The addition of *-t-* in Greek is an independent and different development.
[398] See Rieken (1999: 84-5). This analysis was adopted by Irslinger (2002).

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Rieken's theory proposes an attractive interpretation of the *t*-enlargement, in that according to this theory Indo-Iranian nouns of the *stú-t-* type can be structurally compared with primary derivative *t*-stems of other languages, cf. La. *dō-t-*, *noc-t-*, Gk. δαι-τ- etc.[399] However, the interpretation of the Indo-Iranian simplex *stú-t-* type nouns as original derivative *t*-stems is complicated by several factors. Since a productive *t*-addition rule was operating in Indo-Iranian and, with the exception of *nák-t-*, virtually all the Indo-Iranian *t*-stem root nouns meet the requirements for the operation of this rule,[400] it is difficult to see clearly whether the attested *stú-t-* type nouns are historical derivative *t*-stems or enlarged historical root nouns. No support for either interpretation can be found in other Indo-European dialects due to the absence of cognates.

If some of these *t*-stems should be interpreted as original root nouns, this would also open up more options for the reconstruction of their original ablaut. Whereas historical *t*-stems with a non-alternating suffix *-t-* normally reflect Indo-European acrostatic formations, root nouns with the generalized zero grade of the root can continue either reformed *ó/é* acrostats (**dóm*[h_2]-/**dém*[h_2]- type) or original *é/z* mobile root nouns (**$h_2$nér-*/**h_2n̥r-* type).[401]

[399] Rieken, ibid.
[400] See the list of *t*-stems at the beginnning of section IV.1. With the exception of *nákt-*, all these *t*-stems have roots ending in *-r̥*, *-u* or *-i*.
[401] See Schindler (1972b, 1975c), Nussbaum (2004c).

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IV.5. The Evaluation of the Two Theories about the Historical Interpretation of the Indo-Iranian *t*-Stems

Despite the weaknesses of both theories mentioned in the previous section, important observations about the characteristics of Indo-Iranian *t*-stems are made in both theories, too, viz. that the formant *-t-* has no apparent function from the synchronic point of view, and that the structure and the semantics of the Indo-Iranian *t*-stems resembles that of the derivative *t*-stems of other Indo-European dialects. As was discussed in chapter III, derivative *t*-stems originally were verbal abstracts built with the suffix *-t-*, but in the course of development became concretized, and the suffix *-t-* lost its original function. The Indo-Iranian nouns of the *stú-t-* type can be interpreted in exactly the same way, e.g. *stú-t-* 'praise' (← 'praising' ~ √*stu-* 'praise'), *mí-t-* 'prop' (← 'setting up, erecting' ~ √*mi-* 'fix, set up') etc. It should also be emphasized here that the suffix *-t-* in the derivative *t*-stems of the non-Indo-Iranian languages has no synchronic function either. Therefore, from the synchronic point of view, both the Indo-Iranian *t*-stems of the *stú-t-* type and the *t*-stems of the other languages technically are "enlarged root nouns". The main difference between Indo-Iranian and the other dialects is that whereas in Anatolian, Latin etc. the formant *-t-* can be plausibly explained as an original derivative suffix, in Indo-Iranian the origin of *t* in most cases has been obscured by the *t*-enlargement rule.

Further parallels between the *t*-stems in Indo-Iranian and the other Indo-European dialects can be observed in the realm of *t*-stem compounds. In all the dialects, *t*-stem compounds are built according to the same principles: the final members of these compounds are verbal roots, and they generally reflect a historical zero grade, cf. Ved.

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deva-stú-t- 'praising the gods', *aśva-jí-t-* 'winning horses', Av. *ātrə-kərə-t-* 'making the fire' on the one hand and Gk. ὠμο-βρω-τ- 'eating raw flesh' (< *-$g^{u}_{\text{̥}}rh_3$-*t*-), προ-βλη-τ- 'forthstretching' (< *-$g^{u}_{\text{̥}}lh_1$-*t*-) etc. on the other hand. In rare instances, the root of the second member appears in the full grade, cf. La. *sacer-dō-t-* 'priest' (< *-*doh*$_3$-*t*-) and Av. *dāmi-dā-t-*, *tara-dā-t-* and *para-δā-t-* (< *-*dheh*$_1$-*t*-). The primary *t*-stem compounds also have comparable semantics, and most of them are agentive. In those instances where the meaning is concrete, it can be explained as secondary concretization, e.g. La. *com-i-t-* 'companion' (← *'going together') or Gk. χερ-νη-τ- 'poor' (← **ĝhes-r-*[*s*]*neh*$_1$-*t*- 'spinning with hands') etc.

The structural and the semantic parallels between the compound and simplex *t*-stems in Indo-Iranian and the other Indo-European dialects make the interpretation of the Indo-Iranian determinative *-t-* as an original derivative suffix quite attractive, and primary *t*-stems may well be the ultimate source of this formant. If Indo-Iranian really possessed derivative *t*-stems, one could postulate that after the suffix *-t-* had lost its original function, it became added as a "root enlargement" to all roots of particular structure. However, in such a case it would be ideal to find such nouns among the Indo-Iranian *t*-stems for which the reconstruction of an original derivative *t*-stem would be more plausible than any other reconstruction.

The interpretation of at least some of the Indo-Iranian *t*-stem compounds as possessing derivative *t*-stems as their final members raises a different issue. These *t*-stem compounds are semantically adjectival, and traditionally are classified together

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with determinative root noun compounds of the *havir-ád-* type.[402] In terms of synchronic grammar the *havir-ád-* and the *deva-stú-t-* types agree from the accentual, the semantic and the structural point of view. However, if the second members of the *t*-stem compounds are interpreted as original *t*-stem nouns with a historical suffix *-t-*, structurally they become closer to bahuvrīhi compounds. The interpretation of the Indo-Iranian (and Indo-European) *t*-stem compounds as original bahuvrīhis with reformed accent has been proposed earlier, e.g. see I. Gerschewitsch (1938: 150-61), Rix (2000: 213, fn. 26); however, the details of this reformation need to be clarified. First, it needs to be explained why only the *t*-stem bahuvrīhis underwent this reformation whereas others, such as *urvy-ūti-* 'giving wide aid', *áśvaṃ-iṣṭi-* 'desiring horses' etc. retained their original bahuvrīhi status. Second, it needs to be explained what the reason for this reformation would have been.

In addition to the problems raised above, it is also necessary to investigate the disproportional distribution of the *t*-stem compounds vs. the *t*-stem simplex nouns in Indo-Iranian. As was said earlier, the number of Indo-Iranian *t*-stem compounds largely exceeds that of non-compound *t*-stems, and in many instances *t*-stem compounds pair up with *ti*-stem simplex nouns.[403] It is perhaps significant, that a similar situation can be observed in Greek where, on the one hand, *t*-stem simplex nouns and *t*-stem compounds are built to different roots are do not overlap at all, and on the other hand, Greek *t*-stem compounds pair up with simplex *ti-* or *men*-stems, cf. °-βλη-τ- vs. βλῆμα 'throw, cast' (< **-mn̥*), °-βρω-τ- vs. βρῶμα 'food', °-νη-τ- vs. νῆσις 'spinning' (< **-ti-*), °-

[402] Macdonell (1910: 159); AiGr II.1, 174ff., Scarlata (1999).

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γνω-τ- vs. γνῶσις 'knowledge', etc. In the following sections, I will attempt to see if this distribution of *ti*- and *t*-stem simplex nouns vs. *t*-stem compounds in Indo-Iranian and Greek is in any way significant for the under-standing of the development of *t*-stems overall.

In addition to the issues raised in the previous paragraphs, several words need to be said about the ablaut of the Indo-Iranian *t*-stems as, unlike other Indo-European dialects where derivative *t*-stems occasionally show the full grade of the root, virtually no full grades are to be seen among the Indo-Iranian *t*-stems of the *stú-t*- type.

IV.6. The Historical Interpretation of Vedic *nákt*- 'night'

I would first like to investigate the question whether any of the Indo-Iranian simplex or compound *t*-stems can be plausibly interpreted as original derivative *t*-stems.

The most likely candidate for original derivative *t*-stems is Ved. *nák-t*- 'night', which has good parallels in many other Indo-European dialects, and can be securely reconstructed as an original derivative acrostatic *t*-stem **nók^{u}-t-/*nék^{u}-t-*. In Sanskrit, this noun probably did not have a full paradigm any more, and only three forms occur: the nominative singular form *nág* (< *nák* < **nákts* due to sandhi; RV, vii.71.1a: *ápa svásur uṣáso nág jihīte*), the accusative singular form *náktam* which is used adverbially and means 'at night' (33x in RV),[404] and in the dual, cf. *náktā* (always beside *uṣásā*).[405]

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[403] See section IV.1 above.

[404] The thematic neuter *nákta*- 'night' is not attested in the Rigveda, and the form *náktam* in RV, i.90.7a (*mádhu náktam utóṣáso*) is to be analysed not as nom. sg. (contra Grassmann, p. 703) but as temporal adverb as well (following Geldner, I.115).

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The dual form *náktā* can only be analyzed as athematic and most probably is feminine, like in the other Indo-European languages.

The noun *nák-t-* also exhibits the full grade of the root, which is in agreement with all the other Indo-European dialects. The preservation of the full grade is normal for this noun, because its root has the *-eC-* structure.[406]

The historical interpretation of other Indo-Iranian non-compound *t*-stems is much more problematic since they have no parallels in other Indo-European dialects. This greatly compromises the study of their history, because even in those instances when a given *t*-stem is attested in both Sanskrit and Avestan, e.g. Ved. *stút-* and OAv. *stūt-* 'praise', it can be ambiguous because the *t*-enlargement rule dates back to Proto-Indo-Iranian. In such a case, the only conclusion one can draw is that the noun **stu-t-* may have existed already at the common Indo-Iranian stage. Whether it was a *t*-stem in Proto-Indo-European or is a root noun with a determinative *t*, one can only speculate.

However, several substantives with an element *-t-* imbedded in their roots may provide additional evidence for the existence of historical *t*-stems in Indo-Iranian.

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[405] The dual form *náktā* occurs five times in the dvandva *náktoṣāsā*, ten times in *uṣāsānáktā* and twice the duals of *nákt-* and *uṣás-* occur separated (RV, i.73.7c: *náktā ca cakrúr uṣásā vírūpe* and RV, vii.42.5c: *ā́ náktā barhíḥ sadatām uṣāsā-*).

[406] Ved. *aktú-* 'ointment; light; darkness, the last part of night' probably reflects the zero grade of the root (cf. EWAia I.40). Although the etymology of the root is not entirely clear (**neg^{u}-* 'be dark' or **$h_{2/3}$eng^{u}-* 'anoint'? Cf. EWAia, loc. cit. and p. 54; LIV.267), *aktú-* may be a historical *tu*-stem. The zero grade may be analogical after the weak forms, since *tu*-stems commonly had proterokinetic inflection.

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IV.7. Indo-Iranian **pṛt-* 'battle'

One potentially old derivative *t*-stem may be reflected in the Vedic root noun *pṛ́t-* 'battle'[407] which has an Avestan parallel, cf. YAv. *pərət-* 'id.'. The root *pṛt-* is further attested in a number of words among which one could mention the Younger Avestan verb form *pərətəṇte* 'fight' (3. pl. pr. mid.; < **pṛt-éntoi*), as well as the Vedic feminine noun *pṛ́tanā-* 'battle' and a corresponding *pəšanā-* 'id.' in Avestan, both of which derive from common Proto-Indo-Iranian **pṛ́t-anā-* (< IE **-e-neh₂-* ~ Old Saxon *thekina* 'cover' [< virtual **(s)tog-e-neh₂*] or Lith. *lùpena* 'peel' [< virtual **lup-en-eh₂*] etc.).[408]

From the existing Sanskrit and Avestan evidence, one can reconstruct a Proto-Indo-Iranian root **pṛt-* which would have either been nominal and meant 'battle' or verbal and meant 'to fight'.

The Indo-European etymology of **pṛt-* has been explained in several ways. In an article published in 1940, J. Pokorny interpreted **pṛt-* as an indivisible root, comparing it with Welsh *Prydain* 'Britain' and Gk. Πρεταvική νῆσος 'id.', both of which he considered to be of Illyrian origin. Pokorny reconstructed the underlying Illyrian form as **Prittenni* or **Prittinni*, and connected it with Skt. *pṛ́tanā-* 'battle', reconstructing a common Indo-European protoform **pṛt-eno-*.[409] However, the connection between the toponyms shown above and the nouns **pṛt-* or **pṛt-anā-* seems to be based only on their similar phonological shape, whereas the semantic link between 'Britain' and 'battle' was not explained by Pokorny.

[407] Commonly attested in the Rigveda, but only in loc. pl. *pṛtsú* (30x) and *prtsúṣu* (RV, i.129.4e).

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In IEW, Pokorny connected the Indo-Iranian **pṛt-* with the verbal root **per-* 'hit, beat', based on the comparison with Lith. *peřti/pérti* 'beat, thrash', OCS пьрати 'thrash', Arm. *ehar* 'struck').[410] The original meaning of the noun **pṛt-* could then be reconstructed as 'beating, fighting'.

In an article dedicated to the study of the Vedic noun *pṛ́tanā-*, E. P. Hamp dismissed the derivation of *pṛt-* from the verbal root **per-* 'beat', claiming that the evidence for this root is insufficient,[411] and proposed an etymological connection with **pṛtu-* 'passage' instead (< **pṛ-tu-*; ~ **per-* 'cross, pass'), evidently because there exists some mythological evidence for combats that took place in/at fords and bridges.[412] However, first of all, the existence of the root **per-* 'beat' is well supported by evidence (see above). Second, the mythological combat scenes at fords that Hamp refers to do not prove that the two words are necessarily etymologically related. Fords, bridges, fjords (especially narrow, winding ones) are for obvious reasons strategically good places to have a battle or to set up an ambush, and it is no wonder that battles actually happened in such places (cf. the Stirling Bridge battle in Scotland in 1297).

In addition to this, there are further problems involving morphology, since, whereas the noun **pṛ-tu-* (> Av. *pərətu-* 'ford') is a *tu*-stem (cf. also OIc. *fjǫrðr* 'fjord' < **ferd/þu-*), the Indo-Iranian **pṛt-* is a *t*-formation.

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[408] For more Indic and Iranian derivatives with the root *pṛt-*, see EWAia II.160.
[409] Pokorny (1940: 115). See also AiGr II,2.595 and KEWA II.331.
[410] See IEW.818.
[411] Hamp (1982: 66).
[412] See Hamp (1982: 64).

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Semantically the most attractive etymology for **pr̥t-* is to derive it from the verbal root **per-* 'beat' (despite Hamp 1982). In such a case, the noun **pr̥t-* can be interpreted as an original action noun 'beating, thrashing' (→ 'battle, fight'). The consonant *t* would have to be explained either as an enlargement, or as a derivational suffix.

The occurrence of *t* in the roots of the Proto-Indo-Iranian noun **pŕ̥t-anā-*, the Avestan verb *pərətaṇte* and other words imply that the addition of *t* must have taken place so early that by the time the noun **pŕ̥t-anā-* and the ancestor of *pərətaṇte* etc. were formed, *t* was already perceived as an unsegmentable part of the root **pr̥t-* (< IE **pr̥-t-*). If the consonant *t* had been added to a putative root noun **pr̥-* more recently, i.e. after the enlargement rule had begun to operate, this enlargement would have most probably neither been inserted into **pŕ̥t-anā-* nor into the Avestan *pərətaṇte* etc. The derivation of nouns with the complex suffix **-e-neh₂-* is a well established Indo-European pattern, and this suffix is generally added directly to roots, cf. also Ved. *vadh-ánā-* 'slaughter' (~ √*vadh-* 'kill'), *jar-aṇá-* 'old age' (~ √*jr̥̄-/jur-* 'waste away'), etc. Therefore, instead of **pŕ̥t-anā-* one should probably expect †*par-anā-*. Similarly, if the *t*-enlargement in the root noun **pr̥t-* were relatively recent (i.e. Proto-Indo-Iranian), instead of Av. *pərətaṇte* one should probably expect †*parəṇte* (like YAv. *barəṇte* from √*bar-* 'carry').

The appearance of the "alien" *t* in the Indic and Iranian verbal and nominal forms discussed in the previous paragraphs suggests that this *t* was added to the root *pr̥-* and became reanalyzed as a part of at the pre-Indo-Iranian stage, i.e. at some stage within the Indo-European protolanguage itself. One could perhaps say that the element *t* could

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still be a root enlargement, in this case a Proto-Indo-European enlargement like *-k-* and *-d-* in the roots **pleu-k-* and **pleu-d-* (> Lith. *plauk-iù* 'swim', OIc. *fljúga* 'fly' resp. Lith. *pláudžiu* 'wash' [< **plou-d-i̯ō*], OIc. *fljóta* 'float').[413] However, no evidence for such an enlargement can be found outside of Indo-Iranian.[414] Therefore, for the Indo-European stage of development, the most plausible way is to interpret *t* as a derivative suffix, in the case of the noun *pṛ́t-* added to the verbal root **per-* in its zero grade variant **pṛ-*. The morphological analysis of the Proto-Indo-Iranian (and Indo-European) noun *pṛ́t-* should in such a case be **pṛ́-t-*. From the formal perspective, this *t*-stem agrees with the *t*-stems of other Indo-European dialects which are built to roots of the *-eR(C)-* type and exhibit the zero grade as well, cf. La. *grāt-*, Gk. λῑτ-, etc. These parallels allow for the reconstruction of the idealized Indo-European ancestor of the Indo-Iranian root noun **pṛ́t-* as an derivative acrostatic *t*-stem **pór-t-/*pér-t-* 'beating' (→ 'fight, battle'), which regularly became **pṛ-t-* because of its root structure.

IV.8. Ved. *rétas-* 'stream; semen', *srótas-* 'stream', **vétas-* 'reed'

Reflexes of ancient derivative *t*-stems are probably to be seen in the Sanskrit nouns *rétas-* 'stream' and 'semen' (RV), *srótas-* 'stream' (RV) and **vétas-* 'reed' (attested in the compound *vetas-vant-* 'filled with reeds' [Pāṇini] and ultimately also in the thematized *vetas-á-* 'ratan, reed, stick' [RV], *vetra-* 'cane, reed' [Kauś +], *vaitasá-* 'made of reed; stick; membrum virile' [RV]).

[413] LIV.488.
[414] Lith. *pirtìs* 'sauna' most probably is a *ti*-stem **pṛ-ti-*.

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These *s*-stems are ultimately built to the verbal roots *ri-*, *sru-* 'flow' and IE **u̯eih₁-* or **u̯ih₁eh₁-* 'twist, plait';[415] however, all three have an unexpected shape, because regular neuter *s*-stems built to these roots should have become †*ráy-as-*, †*sráv-as-* resp. †*váy-as-*. From the shape of the attested Sanskrit forms, one can reconstruct their immediate ancestors as **raitas-*, **srautas-* and **waitas-*, in all of which a consonant *t* is inserted between the root and the normal *s*-stem suffix *-as-* (< **-e/os-*).

The history of this *t* has been explained in several different ways. In AiGr II,2.615-6, these nouns are interpreted as *tas*-derivatives, and the suffix *-tas-* is considered to be inherited from the protolanguage, since several nouns reflecting formations of the same kind are attested in other Indo-European languages, cf. Gk. χῆτος 'need, want' (gen. sg. -εος), κλεῖτος 'slope',[416] Lith. *sraũtas* 'torrent' (→ *o*-st.). Hirt (1913: 276-7) analyzed the suffix *-tas-* as *-t-* combined with the *s*-stem suffix *-as-* (< **-e/os-*). J. Manessy-Guitton considered the formant *t* of the noun **vétas-* inherited, but she thought that in the nouns *rétas-* and *srótas-*, *-t-* was analogical after the root noun *rít-* and the enlarged °*-sru-t-* from compounds.[417]

Although due to the examples attested in several branches, the reconstruction of an undivisible Indo-European suffix *-tas-* as was suggested in AiGr (loc. cit.) may appear attractive, the nouns built with such a suffix are both extremely few and most of them are isolated. In addition to this, since one could also split the suffix *-tas-* further into *-t-*

[415] See Manessy-Guitton (1963: 29) and LIV.695.
[416] The noun κλεῖτος may in fact not reflect anything old. It is attested once in the plural form κλείτεα, and may simply be a variant of κλίτος 'id.' (Lycophron +), influenced by the vocalism of the *tu*-stem κλειτύς 'slope, hillside' (Hom. +).
[417] Manessy-Guitton (1963: 29-30).

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and *-as-* (< **-e/os-*) as per Hirt and Manessy-Guitton, it may well be that all these *tas*-formations are *s*-stems with some kind of *t*-infix. In such a case, one would have to find out what this *t*-infix in these *s*-stems represents.

In his study, Hirt only considered the nouns *rét-as-* and *srót-as-*, and interpreted them as secondary *s*-stems built to older *t*-stems **sro-t-* and **re-t-* otherwise attested in Sanskrit with the zero grade of the root, cf. *rí-t-* 'stream' or 'flowing',[418] and °*-srú-t-* 'flowing'.[419]

Hirt's analysis is different from Manessy-Guitton's interpretation shown earlier in this section, but I think that Hirt's reconstruction of ancient derivative *t*-stems is more plausible than Manessy-Guitton's assumption that the consonant *-t-* in *rétas-* and *srótas-* is analogical. First, there are attested Sanskrit *s*-stems with no *-t-* inserted, even though comparable enlarged root nouns are attested as well, cf. Skt. *śráv-as-* 'fame' (not ×*śró-t-as-*) beside the enlarged °*-śru-t-*, or *hvár-as-* 'crookedness' (not ×*hvárt-as-*, ×*hórt-as-* vel sim.) beside the *t*-stem *hrú-t-*. In addition to this, no comparable source of analogy could be indicated for the Greek noun χῆτος or for Lith. *sraũtas* 'torrent' (< **srou-*) which originally probably was an *s*-stem comparable to Skt. *srótas-*.[420] Finally, the old age of the *-t-* in *vétas-* is indicated by the occurrence of this dental in the etymologically related words, e.g. Skt. *vét-ra-* 'reed', OIc. *víðir* 'willow' (masc.; < PGmc. **wīđ/þ-ija-* < **u̯[e]ih₁-t-*), OHG *wīda* 'id.' (fem.; < PGmc. **wīþ-* < **u̯[e]ih₁-t-*) etc. Etymologically related formations exhibiting the consonant *t* are abundant in Indo-

[418] For the discussion of the semantics, see section IV.9 below.
[419] Hirt, loc. cit.

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European (see IEW.1122). However, whereas in several instances we probably have original *ti-* or *tu*-stems, cf. Gk. ἰτέα (Hesych. γιτέα /Fῑτέα/) 'willow' [< FῑτεFᾱ < *μih$_1$-teu-), La. *vītis* 'id.' (< *μih$_1$-ti-), in several other instances it is difficult to tell whether one is dealing with reformed original *t*-stems or with original *ti-* or *tu*-stems, cf. Lith. *vytìs* 'twig', OIr. *feith* 'fibra' (< *-*t*-, *-*ti-* or *-*t-i-*), OCS **вѣтвь** 'twig' (< *μoi[h$_1$]-t-μ-i- or *-tμ-i-), Gk. οἶσος 'withy', Old Prussian *witwan* 'willow' (< *-t-μ-o- or *-tμ-o-). Special is Av. *vāeiti-* 'willow', which could be reconstructed as a *ti*-stem, but the full grade of the root is unexpected (cf. reconstructed *μoi[h$_1$]-ti- or *μeih$_1$-ti-), because *ti*-stems with *CeRC* roots commonly exhibit the zero grade.

Given the comparative evidence, the interpretation of **vétas-* and *srótas-* as old *t*-stems reformed as *s*-stems appears quite attractive, and one should probably add the noun *rétas-* to this group as well. These three *s*-stems could also be compared with Indo-Iranian **pṛt-anā-* discussed in section IV.6 above. In both instances, we have secondary derivatives built to old *t*-stems reinterpreted as roots, cf. ***pṛ-t-* → **pṛt-* → **pṛt-anā-*; ***srau-t-* → **sraut-* → **sraut-as-*.

However, this comparison raises the following questions: first, why the root **pṛt-* in **pṛt-anā-* is in the zero grade, but the root of **sraut-as-* and the other two *s*-stems is in the full grade instead of †*srut-*? Is the full grade of **sraut-* inherited from the original ablauting paradigm of the *t*-stem **sro/eu-t-* (as per Hirt 1913: 276), or is it secondary (i.e. did **sraut-* receive the full grade as it became used in an *s*-stem)? Second, how

[420] AiGr II,2.615; Kazlauskas (1968: 285). Kazlauskas compares Skt. *srótas-* with Lith. *sriautas* (< **sreu-*). Skardžius (1941: 321) assumes a *to*-formation in *srautas*. Fraenkel (p. 888) compares Lith. *srautas* with Skt. *srota-* (*to*-stem).

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should the simplex *t*-stem *rít-* be interpreted? Is it inherited as well, or was it enlarged by *-t-* secondarily?

IV.9. On Hirt's Interpretation of the Ablaut of Ved. *srótas-* and *rétas-*

As was said in the preceding section, Hirt suggested that the full grade in the roots of the *s*-stems *srótas-* etc. was inherited from the stage when the corresponding *t*-stems still ablauted (Hirt, loc. cit.). According to Hirt, these *s*-stems would provide the only evidence for the full grade in *t*-stems (or "enlarged root nouns"). Such an analysis is problematic for *pṛt-*, because if it is inherited from the protolanguage, it perhaps should have preserved the full grade of the root as well. It has to be noted that *pṛt-* on the one hand and *srótas-*, *rétas-* and **vétas-* on the other hand all have the same *-eR(C)-* root structure, and such roots generally appear in the zero grade when they are used to build root nouns and *t*-stems. The zero grade in root nouns and *t*-stems built with *-eR(C)-* root usually can be traced back at least to the Proto-Indo-Iranian stage, and possibly is even older, because the same development can be observed in other Indo-European dialects (see chapter III). Therefore, between the zero grade of *pṛt-* on the one hand and the full grade of *srót-*, *rét-* and **vét-* on the other hand it is the full grade that appears not normal.

The roots *srót-*, *rét-* and *vét-* most probably acquired their full grade as they became used in the respective *s*-stems, since *s*-stems normally have roots in the full grade, cf. Ved. *tápas-* 'heat' (< **tép-es-* ~ La. *tepor* 'warmth'), *vácas-* 'speech, word' (< **u̯ék^{u}-es-* ~ Gk. ἔπος 'word') etc. Also in those instances when the underlying root has the

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CeR(C) structure, the *s*-stems commonly have the full grade, cf. Ved. *dvéṣ-as-* 'hatred' (< IE **dueis-*), *ój-as-* 'strength' (< **h₂eug-*), *péś-as-* 'ornament' (< **peiḱ-*), *śráv-as-* 'fame' (< **ḱleu-*) etc.[421] Therefore, the development of the nouns *srótas-*, *rétas-* and **vétas-* should probably be reconstructed in the following way (using *srótas-* as an example): **sróu-t-/*sréu-t-* (*t*-stem) → **sru-t-* → **srut-* → **sraut-as-* (*s*-stem) > *srótas-*. The full grade of the root would be not inherited from early Indo-European (contra Hirt 1913), but newly introduced as the neuter *s*-stems **sráut-as-*, **ráit-as-* and **wáit-as-* were formed.

IV.10. Ved. *rít-* 'flowing' or 'stream' (RV, vi.57.4a)

In the historical analysis of the *t*-stem *rít-*, two main questions need to be answered: first, is it a noun or an adjective? Secondly, is it a enlarged root noun or a derivative *t*-stem?

The *t*-stem *rít-* is attested in the fourth stanza of the Rigvedic hymn vi.57 to Indra and Pūṣan, cf. *yád índro ánayad ríto mahír apó vṛ́ṣantamaḥ | tátra pūṣā́bhavat sácā* 'when the most vigorous Indra led the [...] great waters, Pūṣan came along'. The form *rít-as* syntactically is the direct object of the verb *ánayad*, standing in accusative plural case.

The root of the *t*-stem *rít-* has been identified as *ri-* 'flow', and this substantive has been interpreted in two different ways. In his translation of the Rigveda, Geldner

[421] Exceptions occur, cf. Ved. *uṣ-ā́s-* 'dawn' (fem.; ← IE **h₂eus-os-*), *úr-as-* 'breast' (neut.; < **[h₁]ur-es-*), *júvas-* 'quickness' (< **ĝu[H]-es-*), etc.

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translated it as an agentive 'flowing', cf. Geldner's translation of this passage: 'Als Indra, der Bullenhafteste, die strömenden großen Gewässer (in ihre Bahnen) leitete, dar war Pūṣan darbei'.[422] A similar intepretation is to be found in a number of dictionaries, cf. *rí-t-* "etwa *entrinnend*" in the Sanskrit dictionary by O. Böhtlingk and R. Roth,[423] as well as 'running, flowing' in M. Monier-Williams,[424] 'rinnend' in Grassmann's Rigvedic dictionary,[425] and 'rinnend, strömend' in EWAia II.437. The word-form *rítas* has also been translated as 'coulantes' ('flowing') by L. Renou,[426] and 'проточные' ('id.') by T. J. Jelizarenkova.[427] However, Griffith translated it as a noun, viz. 'stream', cf. Griffith's translation of this passage: 'When Indra, wondrous strong, brought down the streams, the mighty waterfloods, Pusan was standing by his side'.[428]

Which of the two interpretations is preferable?

If one chooses the agentive meaning 'flowing', the *t*-stem *rí-t-* becomes semantically similar to the Greek nouns φώς 'hero' and πλώς 'swimmer; mullet' discussed in sections III.6.1 and III.6.4 – III.6.4.1.1 above. As was said there, these two nouns are *t*-stems, but, unlike most other *t*-stems, they secondarily developed an agentive meaning. One could say that the same thing happened to *rí-t-*, except that whereas the two Greek nouns eventually became fully nominalized, Ved. *rí-t-* remained adjectival in character.

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[422] Geldner, II.159.
[423] Böhtlingk/Roth (1855-75, VI.347).
[424] Monier-Williams (1899: 880c).
[425] Grassmann (1872: 1164).
[426] See Renou (1966: 151).
[427] See Елизаренкова (1995: 157).
[428] See Griffith (1896-7 vol. II, p. 627).

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Semantically this is possible, but in such a case *rí-t-* would be the only non-compound *t*-stem of this type, i.e. it would be the only non-compound Sanskrit adjectival *t*-stem.

It is perhaps more plausible to interpret *rí-t-* as a noun meaning 'stream' as per Griffith. Although, when *rí-t-* is translated in this way, one gets an apposition, cf. '...streams, the great waters...' as opposed to Geldner's '...the streaming great waters...', this is not a problem since apposition is not uncommon in Rigveda, cf. the following passages: *tásmā ắpo ghṛtám arṣanti síndhavas* 'to him the rivers, the waters stream ghee' (RV, i.125.5c); *ắpo arṣanti síndhavaḥ* 'the waters, the rivers stream' (RV, ix.2.4b and ix.66.13b), etc.

The interpretation of *rí-t-* as a noun meaning 'stream' is much more attractive from the historical point of view, as in such a case, this noun can be analyzed as a concretized old action noun, 'flowing'. The concretization of the original verbal abstract 'flowing' to 'stream' is quite banal, and needs not to be explained in any more detail here.

There are reasons to believe that the noun *rít-* was formed quite early. One of the reasons for this claim is the shortness of the vowel *i*. The shortness of *i* is not regular, because the underlying verbal root is normally reconstructed as **h₃réiH-* 'flow', and the reflexes of a root-final laryngeal can be clearly seen in the following forms: Ved. *riṇắti* 'flow' (3. sg. pres. indic.; < **h₃ri-ne-H-ti*), *ríyate* 'id.' (3. sg. pres. middle; < virtual **h₃riH-ie-*), Gk. ὀρίννω (Lesb.) and ὀρίνω (Att.) 'flow', Go. *rinnan* 'run' (< **h₃ri-n-H-*)

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etc.[429] If the Vedic root noun *rít-* had been built to the zero grade of this root, it should have become †*rīt-* (cf. also Ved. *rī-tí-* 'course' < **h*$_3$*riH-ti-*). The most simple way to explain the shortness of the root vowel is to reconstruct an original *ó/é* acrostatic paradigm for *rít-*, in which probably both laryngeals would have been lost due to "Saussure's law", cf. ***h*$_3$*roiH-* > **roi-*.[430] The weak stem would have been analogically remade into **rei-t-*. As the original *ó/é* ablaut pattern would have been remade into *ó/z*, the root of the weak stem would have been **ri-*.[431]

As to whether this acrostatic noun would have been a root noun or a derivative *t*-stem, it would not be possible to tell based solely on the internal reconstruction of *rít-*, since the acrostatic pattern is attested both among root nouns and derivative *t*-stems. However, since the *t* of *rétas-* is most probably old, and since *rít-* itself must have been built before the operation of the "Saussure's law", the reconstruction of an old derivative *t*-stem becomes quite attractive.

IV.11. The Summary of the Sections IV.4 – IV.10

In the preceding sections, I have tried to search for traces of old derivative *t*-stems in Indo-Iranian. The conclusion I would like to draw here is that such traces can indeed

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[429] LIV.305.

[430] The deletion of the root-initial laryngeal is indicated by the Vedic compound *su-rétas-* 'having much seed' (masc.; < virtual *[*h*$_1$]*su-h*$_3$*róiHt-os-*), because otherwise one should expect ˣ*sū-rétas-*, cf. also Ved. *sū-máya-* 'well-shaped' (< *[*h*$_1$]*su-h*$_2$*me̯i̯-o-*; cf. Catsanicos 1986). Also Gk. λοιμός 'plague' would provide evidence for the deletion of both laryngeals in *HRoRH* environment, if it should be reconstructed as **h*$_2$*loiH-mo-* (~ **h*$_2$*leiH-* 'beschmieren', LIV.277-8, cf. **loi-mo-* in IEW.661).

[431] A similar deletion of a laryngeal due to "Saussure's law" would have most probably taken place in **wí-t-* (← **u̯ói̯h*$_1$*-t-*/**u̯éi̯h*$_1$*-t-*).

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be seen in several Indo-Iranian formations. The reconstruction of the Sanskrit root nouns *nákt-*, *pṛ́t-* (Av. *pərət-*), *rít-* as historical acrostatic derivative *t*-stems **nókʷ-t-*, **pṛ́-t-*, **rí-t-* on the one hand, and the *s*-stems *rétas-*, *srótas-* and **vétas-* as secondary formations built to historical derivative *t*-stems **rí-t-*, **srú-t-* and **wí-t-* on the other hand is quite attractive. To these simplex derivative *t*-stems, one could probably also add the Avestan compounds *dāmi-dāt-*, *tara-dāt-* and *para-δāt-*. If all three indeed are athematic, their *t* has to be interpreted as a true suffix, and not a secondary Indo-Iranian enlargement, since it is added to a root ending in the long vowel *ā* (< **-eh*$_1$-).

It needs to be stressed that already at the late stages of common Proto-Indo-European, simplex derivative *t*-stems had been reinterpreted as root nouns and adopted the inflection and ablaut characteristic of root nouns. The old *t*-stem *nákt-* preserved the full grade normal for root nouns with *-eC-* roots, whereas the *t*-stems built to *-eR(C)-* roots generalized the zero grade of the root, just like the root nouns of the same structure. Furthermore, all the *t*-stems lost their original action noun semantics and became concretized. As a result, the suffix *-t-* became reanalyzed as a semantically and morphologically empty root enlargement.

IV.12. Further Parallels between Indo-Iranian *t*-Stems and the *t*-Stems of Other Indo-European Dialects

The number of old derivative *t*-stems mentioned in the preceding paragraphs could perhaps be extended, as some of the Indo-Iranian roots with the determinative *-t-* have corresponding *t*-stems in other Indo-European languages, cf. the following pairs: Ved.

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pari-kṣí-t- 'dwelling around, surrounding' ~ Gk. περι-κτί-τ-αι 'neighbours' (< **peri-tk̂i-t-*) ~ Av. *hu-ši-t-* 'providing good home' (< **-tk̂i-*), Ved. *-śrí-t-* (cf. *kṛchre-śrí-t-* 'providing support in danger') ~ OIr. *clí* 'post' (< **k̂li-t-*),[432] Av. *i-t-* 'course' ~ La. *-i-t-* (cf. *com-i-t-* 'companion', *ped-i-t-* 'foot soldier') ~ Ved. *-i-t-* (cf. *divít-* 'going to the sky') and possibly Ved. *-bhṛ́-t-* (cf. *vajra-bhṛ́-t-* 'carrying *vajra*') ~ Av. *-bərə-t-* (cf. *ā-bərə-t-* 'water carrier' [< **ăp-bhṛ-t-*]) ~ Hi. *-par-t-* (cf. *ka-par-t-* 'mouse' [if < **kṃ-bhṛ-t-*]).[433] To be sure, one cannot entirely exclude the possibility that the enlarged roots in Indo-Iranian on the one hand and their correspondences in the other dialects on the other hand could be independent creations. However, since the forms of the other Indo-European dialects are certainly old, it is tempting to interpret their Sanskrit and Avestan correspondences as inherited as well.

A handful of enlarged formations are attested both in Indic and Iranian but not in other Indo-European languages, e.g. Ved. simplex *stú-t-* and Av. *stū-t-* 'praise', Ved. compound *-stú-t-* (cf. *deva-stú-t-* 'praising the gods') ~ Av. *-stū-t-* 'praising' (cf. *aṣəm-stū-t-* 'praising *aṣa*'), Ved. *havana-śrú-t-* 'hearing the invocations' and Av. *zauuanō-srū-t-* 'listening to the prayers' (<**-k̂lu-t-*), Ved. *-pru-t-* 'swimming' (cf. *uda-prú-t-* 'swimming in the water') ~ Av. *-fru-t-* 'flying' (cf. *dunmō-fru-t-* 'flying in the clouds') and Ved. *-kṛ́-t-* (cf. *su-kṛ́-t-* 'performing well') ~ Av. *-kərə-t-* (cf. Av. *ātrə-kərə-t-* 'making the fire'). However, due to the absence of clear cognates in other Indo-European languages, it is impossible to tell whether these simplex *t*-stems and final

[432] The details of the history of Gk. κλί-τ-ος and κλεῖ-τ-ος are not entirely clear (see section IV.7 above).
[433] For the interpretation of Hi. *kapart-*, see section III.4.4 above.

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members of compounds originally were derivative *t*-stems or have a secondary determinative *t*.

IV.13. A Hypothesis about the Origin of the Indo-Iranian *t*-Enlargement Rule

Given the existence of several old derivative *t*-stems in Indo-Iranian, all of which function as root formations from the synchronic point of view and most are built to verbal roots ending in the short vowels *-u*, *-i*, *-ṛ*, one can postulate that the Indo-Iranian *t*-enlargement rule originated namely in such *t*-stems reformed as enlarged root nouns, and in the course of development was extended to multiple nominal roots and even to primary stems ending in these short vowels.[434] In order to explain how the *t*-enlargement spread throughout the system, one could imagine a possible scenario which I would like to describe in the following paragraphs.

Before the *t*-enlargement rule started to operate in Proto-Indo-Iranian and after the ancient acrostatic *t*-stems had merged with root nouns adopting the root noun ablaut and losing the original verbal abstract meaning, the Proto-Indo-Iranian root nouns could be classified into the following subclasses:

[434] As examples of enlarged primary stems, one could adduce Ved. *yák-ṛ-t* 'liver' and *śák-ṛ-t* 'excrement', both of which are old an *r/n*-heretoclites.

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1) ancient derivative *t*-stems (**pr̥t-*, **rít-* etc.);[435]

2) historical root nouns ending in the short vowels *-r̥*, *-u* and *-i*;[436]

3) historical root nouns ending in other sounds than *-r̥*, *-u* and *-i* (PIIr. **piś-*, **yúdh-*, **dwís-* [> Ved. *dviṣ-*], **bŕǰh-* [> Av. *bərəz-*] etc.).

The situation was the same among the Proto-Indo-Iranian compounds. There most probably existed compounds with old derivative *t*-stems as their final members (e.g. **pari-kpít-* [< IE **-tḱi-t-*]),[437] root noun compounds with the roots of their final members ending in the short vowels *-r̥*, *-u* and *-i*, and root noun compounds with the roots of their final members ending in other sounds (e.g. **diwas-ruć-* 'shining from the sky' [← **diu̯es-luk-*], **apsu-sad-* 'dwelling in waters').

This system with some roots enlarged with a *-t-* and others not was perceived as irregular, and was reformed so that the *t*-element of the original derivative *t*-stems was extended also to other roots ending in the short vowels *-r̥*, *-u* and *-i*. The enlargement rule had certain constraints which are not fully clear yet; however, in some instances the insertion of the consonant *-t-* may have been blocked by analogy. Thus, for example, the Sanskrit compound *su-gú-* 'having good oxen/cows' did not receive the enlargement because the corresponding simplex noun *gaú-* 'ox/cow' did not have it either. The *t*-enlargement was not added to polisyllabic *u-* or *i*-stems either. Thus, Ved. *purú* 'many' (< **pl̥h₁-u*; neut.) did not become ˣ*purú-t*, *mádhu-* 'sweet' did not become ˣ*mádhu-t-* etc. The enlargement was not added to those words in which the vowel *i* had

[435] The roots of most end in the short vowels *-u*, *-i*, *-r̥*.

[436] No secure example can be adduced, because any old root noun of such a shape was affected by the enlargement rule.

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developed due to internal Sanskrit phonological developments either, e.g. Skt. *ásthi* 'bone' (< *-h_2)[438] did not become [×]*ásthi-t*.

Judging from the existing evidence, one can observe that the *t*-enlargement is in most instances added to verbal roots, and most often occurs in compounds. In addition to this, the same verbal roots that acquire the *t*-enlargement in nominal compounds, also exhibit it when they occur as second members of compound gerunds, cf. °*-stú-tya* (√*stu-*), °*-śrú-tya* (√*śru-*), °*-kṛ́-tya* (√*kṛ-*), °*-í-tya* (√*i-*), °*-gá-tya* (< *-$g^{u}ṃ$-; √*gam-*), °*-mi-tya* (√*mi-*), etc.

As a result of the operation of the *t*-enlargement rule, the Indo-Iranian root nouns were re-classified into two major classes: those with roots ending in the vowels *-ṛ*, *-u* and *-i* and enlarged with a *-t-* vs. unenlarged root nouns ending in other sounds than the vowels *-ṛ*, *-u* and *-i*. The root noun *nákt-* theoretically would have been an exception, but since it was not associated with any verbal root any longer, the consonant *-t-* probably was not even treated as an enlargement. From the synchronic point of view, the root of this noun was *nakt-*.

IV.14. The Reinterpretation of the Suffix *-t-* as a Root Enlargement

The scenario for the *t*-enlargement rule described in the preceding section requires a more detailed discussion of two major points of this process: first, how and why the original acrostatic *t*-stems became roots nouns, and second, how the old bahuvrīhi

[437] Originally, bahuvrīhis (see a more detailed discussion below).
[438] Old collective?

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compounds with derivative *t*-stems as their final members became reformed as determinative compounds.

The first issue, viz., how and why the derivative *t*-stems became reinterpreted as root nouns was briefly touched in section III.4.1.2 above, where I considered the question why the ablaut of the *t*-stem **ĝroi-t-* 'flood' developed not like the ablaut of other derivative acrostatic formations (e.g. *u*- or *i*-stems), but rather like that of root nouns. As a probable reason, there I mentioned the fact that unlike the suffixes *-u-/-i-*, the suffix *-t-* could not form its own syllable, and was more closely connected with the root than the (semi-)vocalic suffixes or the suffixes of the *VC*- or *CV*(*C*)-structure. Whereas in the course of the reformation of the acrostatic ablaut in other formations, the full grade of the root moved to the suffix, i.e. the following morpheme (cf. **ĝen-u-* > **ĝn-eu-*), in nouns like **ĝroit-* (< **ĝroi-t-*), the morpheme following the root was the ending, as *t* was already treated as a part of the root. Therefore, in the historical acrostatic *t*-stems, the full grade of the root "skipped" the suffix (i.e. the "enlargement") and moved to the ending.

The reinterpretation of *t*-stems as root nouns could hardly depend solely on the suffix *-t-* being unable to form an independent syllable. The reinterpretation process was triggered and influenced by several internal and external reasons, the main of which most probably was the loss of the original action noun semantics and the development of concrete meanings for all acrostatic *t*-stems. After a semantic analysis of all the dialectal reflexes of Indo-European acrostatic *t*-stems one can make an observation that most of them denote concrete concepts, e.g. La. *nox*, Lith. *naktis* etc.

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'night', La. *dōt-* 'dowry', Hi. *šašt-* 'rest', *kutt-* 'wall', Gk. δαιτ- 'meal', etc. In a single case, viz. the Hittite noun *maišt-* 'glitter, brilliance' it is difficult to tell whether it is an pure action noun denoting the act of radiating (like the English noun *sleeping* which denotes the act of sleeping), or whether it should be interpreted as a concretized noun denoting a condition (e.g., like E. *sleep* 'condition of the body and the mind at rest, with the eyes closed'). However, when these primary *t*-stems were built in the proto-language, they must have been verbal abstracts, and not agent nouns (e.g., La. *dōt-* < **dóh₃-t-* *'giving', not *'giver') or individualizing formations (like some *n*-stems). The concrete meaning of the old *t*-stems must then be interpreted as a secondary development.

As *t*-stems became concretized, the suffix *-t-* lost its original function of forming verbal abstracts. The formation of verbal abstracts in Indo-European was not really compromised in a significant way because the protolanguage possessed other verbal abstract suffixes, such as *-men-*, *-ti-* or *-es-*, to mention a few. At the later stages of common Indo-European, the suffix *-t-* had probably become unproductive and eventually the formation of verbal abstracts was taken over by the other suffixes. In fact, it may even be that some of old *t*-stems became replaced by *ti*-stems. If this indeed was the case, one could explain the common occurrence of *ti*-stem simplex nouns beside *t*-stem compounds, e.g. Ved. compound °*-jí-t-* 'winning' vs. simplex *ji-tí-* 'victory' (no simplex *t*-stem **jí-t-*), Ved. °*-kṣí-t-* 'dwelling' vs. *kṣi-tí-* 'abode', Gk. °-νη-τ- (in χερ-νη-τ-) vs. νῆσις 'spinning', Gk. °-γνω-τ- 'knowing' vs. γνῶσις

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'knowledge', Av. °*-bərə-t-* vs. *bərə-ti-* 'carrying' etc. These *ti*-stems could be reformed old *t*-stems.

At that stage, i.e. when historical *t*-stems had already acquired the new concrete meaning and the formant *-t-* did not have ability to form verbal nouns any more, it became reinterpreted as a root enlargement, and the *t*-stems themselves became reinterpreted as root nouns.

The transfer of derivative *t*-stems to root nouns was probably crucial for the development of the old *t*-stem compounds. As was proposed by Gerschewitsch in his study of the Indo-Iranian compounds, *t*-stem compounds originally were bahuvrīhis.[439] The fact that they became associated with determinatives should in such a case be interpreted as a later phenomenon. The classification of *t*-stem compounds as determinatives in the Indic linguistic tradition is on the one hand based on the interpretation of the final members of most of these compounds as mere verbal roots enlarged by a *-t-*,[440] since corresponding simplex *t*-stems in most instances do not exist. On the other hand, *t*-stem compounds are classified together with determinatives because of their accentuation. Whereas bahuvrīhis are normally accented on the first member, *t*-stem compounds are generally accented on the final member.

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[439] See Gerschewitsch (1938: 150-61).

[440] For the definitions of determinative compounds, see AiGr II,1.174 ("Unter den Komposita mit Verbalnomen als Hinterglied wird eine erste Gruppe durch diejenigen gebildet, deren Hinterglied ausschließlich nur als solches, nicht auch als Simplex verwendet wird, oder wenigstens nach einem Typus geformt ist, der in Regel nur für Hinterglieder dient"), and Macdonell (1910: 159; "...the final member is a verbal noun which governs the first member in meaning, and often even in form, as a verb governs its case. [...] The final member is often a **verbal noun** either **not occurring** as a **simple word** or at least not belonging to a type used as simple words: either the **bare root** or a verbal derivative formed with the suffixes ***-a***, ***-ana***, ***-i***, ***-in***, ***-man***, ***-van***...").

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That at least those *t*-stem compounds which were built with derivative *t*-stems as their final members could have been bahuvrīhis at an earlier stage, can be demonstrated by their comparison with synchronic bahuvrīhis from the semantic, structural and functional point of view. Both synchronic bahuvrīhis and *t*-stem compounds with derivative *t*-stems as their final members are used as adjectives and mean "possessing X" or "characterized by X". Also, just like *t*-stem compounds, bahuvrīhis can have action nouns as their final members. In the following two columns several examples of synchronic bahuvrīhis with action nouns as their final members and *t*-stem compounds are compared:

BAHUVRĪHIS		*T*-STEM COMPOUNDS
Ved. *urvy-ū̆-ti-* 'giving wide aid' (< *'characterized by wide aid')	~	Gk. ὠμο-βρω-τ- 'eating raw flesh' (< *'characterized by the eating of raw flesh')
Ved. *áśvaṃ-iṣ-ṭi-* 'desiring horses' (< *'possessing a desire for horses')	~	La. *sacer-dō-t-* 'priest' (< *'posessing a gift of *sacra*')
Ved. *máno-jav-as-* 'swift as thought' (< *'characterized by the hurrying of thought')	~	Gk. προ-βλη-τ- 'forestretching' (< *'characterized by throwing forth')
Ved. *viśvá-kṛṣ-ṭi-* 'dwelling with all peoples' (< *'characterized by dwelling with all peoples')	~	Ved. *pari-kṣí-t-* 'dwelling around, surrounding', Gk. περι-κτί-τ-αι 'neighbours' (< *'characterized by dwelling around')

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From this comparison, it appears that *t*-stem compounds are semantically and structurally hardly distinguishable from bahuvrīhis.[441] Therefore, also the Indo-Iranian *t*-stem compounds, especially those beside which there exist simplex *t*-stem noun (e.g. *stú-t-* 'praise' beside *deva-stú-t-* 'praising the gods' etc.), could be interpreted as original bahuvrīhis, cf. *deva-stú-t-* ← *'characterized by the praising of the gods', or *abhi-hrú-t-* 'causing a fall' ← *'characterized by directing toward stumbling-block, trap'.

Now if some of the Indo-Iranian *t*-stem compounds were built at an early stage as bahuvrīhis with derivative *t*-stems as their final members, one needs to raise the question why and how the form of these original bahuvrīhis became changed. The main formal change involved here is the reformation of the accentuation. Since in bahuvrīhis the stress normally falls on the initial member, the accent falling on the final members of the attested *t*-stem compounds must be an innovation.

The reason for the reformation of the accentuation was probably the following: as simplex *t*-stems lost their original verbal abstract meaning and *t* became reanalyzed as a root ennlargement, a similar reanalysis must have taken place in *t*-stem compounds. In those instances when no simplex *t*-stems would have existed beside the *t*-stem compounds in the common Indo-Iranian protolanguage, e.g. no ×**kŕ̥-t-* beside *°-*kŕ̥-t-*, no ×**śrú-t-* beside *°-*śrú-t-* etc., the enlarged final members of compounds would have become reanalyzed as mere enlarged verbal roots. Since compounds with verbal roots

[441] As was said earlier, Sanskrit *t*-stem compounds and bahuvrīhis differ in accentuation, but for this, see below in this section.

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as their final members are generally determinatives, the *t*-stem compounds were step by step reanalyzed as determinatives as well, and subsequently also adopted the accentuation of determinative compounds.

IV.15. The Ablaut of the Indo-Iranian *t*-Stems

The last question that needs to be addressed is the ablaut of the Indo-Iranian *t*-stems. As one can see from the examples shown in section IV.1 above, the simplex *t*-stems and most of the *t*-stems occurring as final members of compounds show the zero grade of the root. The only *t*-stems with roots in the full grade are Ved. *nákt-* and most probably the final members of the Avestan compounds *dami-dāt-*, *tara-dāt-* and *para-δāt-*. As was concluded in section IV.7 above, the full grade of the Vedic secondary *s*-stems *srótas-*, *rétas-* and **vétas-* is not probative, and most likely is secondary.

When the ablaut of the Indo-Iranian *t*-stems is compared with the ablaut of the unenlarged root nouns like Ved. *píś-* 'ornament', *yúdh-* 'fight', *páś-* 'sight', *nábh-* 'injury, injurer', Av. *bərəz-* 'high; hill' etc. and the root noun compounds like Ved. *dvi-pád-* 'two-footed', etc. on the one hand and the ablaut of non-compound and compound derivative *t*-stems, such as La. *dō-t-*, *cō-t-*, *noc-t-*, *grā-t-*, Gk. λῑ-τ-, Hi. *šaš-t-*, *kar(a)i-tt-*, La. *sacer-dō-t-*, perhaps Gk. χερ-νη-τ- and ἀ-γνω-τ- etc. on the other hand, all these formations show the same behaviour. Whereas both root nouns and derivative *t*-stems built to roots of the *-eR(C)-* structure appear in the zero grade, the root nouns and *t*-stems with *-eC-* roots show the full grade whether they are compound or not. Thus, Ved. *nákt-*, which should be interpreted as an old derivative acrostatic *t*-stem due to

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good comparanda in other languages, as well as the Avestan compounds *dāmi-dāt-*, *tara-dāt-* and *para-δāt-*, the final members of which are perhaps best interpreted as historical *t*-stems as well, have preserved the full grade of the root because all have the root structure *-eC-*. Originally, the ancestors of both *nakt-* and *-d/δāt-* had acrostatic inflection, but whereas *nákt-* most probably reflects the *o*-grade of the strong forms, *-d/δāt-* probably reflects the *e*-grade of the weak forms.

The reconstruction of the ablaut for the rest of the Indo-Iranian *t*-stems depends on whether they can be securely reconstructed as original derivative *t*-stems. Such an interpretation is quite attractive for the Vedic simplex *t*-stems *rít-*, *pṛ́t-* and Av. *pərət-*, as well as the Vedic stems °*-sru-t-*, possibly also °*-kṣi-t-* (Avestan °*-ši-t-*), °*-śri-t-* and °*-i-t-*.[442] In addition to this, also Skt. *vetasvant-* has preserved traces of an old *t*-stem **wi-t-*.[443] Since all these stems have roots of the *-eR(C)-* structure, they followed the tendency of such stems (roots) to generalize the zero grade throughout the paradigm. This generalization must have taken place at a fairly early stage, because, first, no securely old full grades of *-eR(C)-* are anywhere attested, and second, when the noun **pṛ́t-anā-* was built in Proto-Indo-Iranian, its root *pṛt-* (< **pṛ-t-*) was already in the zero grade.

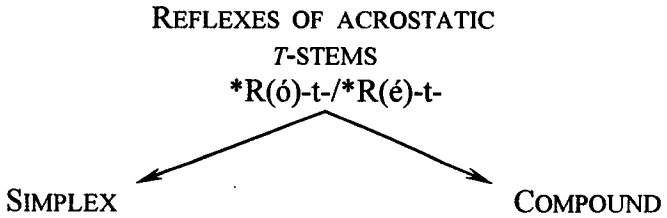
The development of the ablaut in the rest of the Indo-Iranian *t*-stems is ambiguous, as it is impossible to tell which of them are true derivative *t*-stems and which are enlarged historical root nouns.

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[442] See section IV.10 above.
[443] See section IV.7 above.

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The preceding discussion of the distribution of ablaut grades in different types of the Indo-Iranian *t*-stems and the comparison of the ablaut of the Indo-Iranian *t*-stems and the *t*-stems of other Indo-European dialects can be illustrated by the following scheme:



R(o)-t-	R(z)-t-	°-R(e)-t-	°-R(z)-t-
Ved. *nak-t-* (< **nok*$^{\mu}$*-t-*)	Ved. *rí-t-* (< **ri-t-* ~ **h*$_3$*reiH-* 'flow')	Av. *tara-dā-t-* (< **-dheh*$_1$*-t-*)	Ved. *pari-kṣí-t-* (< **-kp̂i-t-* ~ **tk̂ei-*)
La. *cō-t-* (< **k̂oH-t-*)	La. *grā-t-* (< **g*$^{\mu}$*ṛH-t-*)	La. *sacer-dō-t-* (< **-deh*$_3$*-t-*)	La. *com-i-t-* (< **-h*$_1$*i-t-*)
TB *kes-t-* (< **Kos-t-*)	Gk. λῑ-τ- (< **liH-t-*)	Gk. χερ-νη-τ- (< **-*[*s*]*neh*$_1$*-t-*)	Gk. ὠμο-βρω-τ- (< **-g*$^{\mu}$*ṛh*$_3$*-t-*)
Hi. *šaš-t-* (< **sos-t-*)	Hi. *k(a)ri-tt-* (< **ĝri-t-*)	Gk. ἀ-γνω-τ- (< **-ĝneh*$_3$*-t-*)	Gk. προ-βλη-τ- (< **-g*$^{\mu}$*l̥h*$_1$*-*)

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V. Primary Amphikinetic *t*-Stems

In the Indo-European languages, there exist a small number of *t*-stem nouns which reflect earlier formations with the following structure: *e/z* alternation in the root and *ot/ōt/t* alternation in the suffix,[444] cf. La. *nepōt-*, Ved. *nápāt* 'grandson' (< **nép-ōt-*; nom. sg.) vs. YAv. *naptō* 'id.' (< **nep-t-ós* or **nep-t-és*; gen. sg.), Go. *mitaþ-* 'measure' (< **méd-ŏt-*), Go. *menoþ-* 'month' (< **méh$_1$n-ōt-*), etc. Such vocalic alternations are reconstructed by different scholars for different Proto-Indo-European nominal ablaut classes, a brief overview of which will be presented in the following sections.

V.1. Schindler's Reconstruction of Amphikinetic Formations

Among the ablaut types, as described by Schindler (1975b), such alternations are found in the amphikinetic class which is reconstructed in the following way: strong stem R(e)-S(o)-E(z) vs. weak stem R(z)-S(z)-E(e) vs. loc. sg. R(z)-S(e).[445] Such ablaut is very well attested in the Indo-European nominal system, cf. IE **dhéĝh-ōm* 'earth' (nom. sg.; *m*-stem) vs. **dhĝh-m-és* (gen. sg.), **u̯éd-ōr* 'water' (nom./acc. coll.; *r/n*-stem) vs. *ud-n-és* (gen.), **h$_2$éus-ōs* 'dawn' (nom. sg.; *s*-stem) vs. ***h$_2$us-s-és* (> **h$_2$us-*

[444] In a single case, the root is in the *a*-grade, cf. pre-Celtic and pre-Germanic **magh-ot-* 'maiden, girl'. Since, as will be shown below, these nouns used to be inflected according to the amphikinetic pattern, beside the *o*- and zero grade of the suffix provided here, one should also assume *e*-grade (*-et-*) which theoretically would have occurred in the locative singular case. However, due to early analogies the amphikinetic *t*-stem locatives were reshaped after other, more influential cases.

[445] The locative case was originally endingless. See reconstructions in Schindler (1975b: 262) and Friedman (2004).

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és; gen. sg.),[446] etc. Amphikinetic formations often possessed collective meaning, and in a number of instances co-existed with singulative neuter nouns, cf. the above-mentioned collective **u̯éd-ōr* beside singulative **u̯ód-r̥* 'water', or **séh₁-mō(n)* beside **séh₁-mn̥* 'sowing' (→ 'seed', like in La. *sēmen*), etc. However, a number of amphikinetic nouns are non-collectives, cf. IE **h₃ér-ō(n)* 'eagle', **ĝhđém-ō(n)* 'earthling, man', etc., and some appear to be non-primary, cf. **su̯é-sōr* 'sister' (< **su̯e-* 'own' + **sor-* 'woman'?),[447] **di̯é-u̯-ōt* 'day' (see section V.10.2), etc.[448]

V.2. Hoffmann's Reconstruction of the "Amphikinetic II" Type and Tremblay's "Rhizokinetic/Acrokinetic" Type;

In more recent years, beside Schindler's amphikinetic class presented in the previous section, the existence of a variant of this amphikinetic pattern exhibiting similar vocalic alternations has been proposed. In *Aufsätze zur Iranistik*, K. Hoffmann

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[446] Rasmussen interpreted the early proto-Indo-European genitive form as **h₂us-és-s*, ascribing proterokinetic inflection to it (2001: 22). The main reason for this was the reflection of the full grade in the suffix of the weak forms, cf. the Vedic dat. sg. *uṣ-ás-e*. According to Rasmussen, the full grade of the suffix indicates an original proterokinetic inflection (loc. cit.). Such an interpretation cannot explain the lengthened grade of the suffix in the nominative form, cf. *uṣ-ā́s*. The lengthened grade in the suffix occurs only in hystero- and amphikinetic formations, and whereas the Vedic *-ā-* vocalism is ambiguous, Greek ἠώς (Aeol. ἄυως) 'dawn' indicates an original *-ō-* and, accordingly, an original amphikinetic inflection. As for the full grade of the suffix in the weak forms, it can be explained as analogical (e.g. from the locative **h₂us-és* ± *-i* or **h₂us-ér*).

[447] See IEW.1051 (with further literature), more recently also P. W. Brosman (1982) and Hamp (1988). This interpretation was dismissed by H. Bjorvand & F. O. Lindeman as a "plain speculation" ("...men all dette er tom spekulasjon..." 2000: 910-11), although no serious arguments for this dismissal were adduced. An alternative interpretation of **su̯esōr* as **su-* + **esōr* 'own blood' (Linke 1985: 353ff.) creates difficulties for the semantical interpretation of the Anatolian feminizing suffix *-šar-*, like in Hi. *ḫaššušara-* 'queen' (cf. *ḫaššu-* 'king'), *išḫaššara-* 'lady' (*išḫa-* 'lord'), Luv. *nāna-šr-i(i̯a)-* 'of a sister' (cf. *nāni[i̯a]-* 'of a brother'; see Melchert 1993: 154-5).

[448] See a brief overview also in Widmer (2003: 54).

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reconstructed a new ablaut type characterized by the shift of the accented *é*-grade between the root in the strong forms and the endings in the weak forms. According to this scheme, the suffix always stays in the zero grade.[449] Since such shifts of the accented *é*-grade are to be found also in Schindler's amphikinetic inflection, this new type has been dubbed "amphikinetic II".[450]

Essentially the same type was also reconstructed by Tremblay (1998), but in Tremblay's nomenclature it was called "rhizokinétique" or "acrokinétique".[451]

Although Hoffmann did not reconstruct any *t*-stems which would have belonged to this class, Tremblay assumed namely such an ablaut for the protoforms of the Old Irish noun *clí* 'house-post', reconstructing the nominative singular as **k̂lei-t-s* and the genitive singular as **k̂li-t-es*.[452] As was discussed in section III.8.3 above, the OIr. *clí*, which semantically is an ancient verbal noun, is more plausibly interpreted as an original acrostatic (ó/é) *t*-stem, the ablaut of which was transformed into *ó/z* and further eliminated in favour of the zero grade. This pattern must be assumed to have been regular for nouns with roots of the *-eR(C)-* structure.[453]

The reconstruction of this "amphikinetic II" or "rhizokinetic" type has not received much support, as all the examples which were claimed by Hoffmann and Tremblay to have had the "amphikinetic II" inflection, cf. IE **péi-tu-/pi-tṷ-* 'meal', **pér-tu-/pṛ-tṷ-*

[449] Hoffmann (1992: 845f.).
[450] Cf. Schaffner (2001: 84).
[451] See Tremblay (1998: 203, fn. 36).
[452] Tremblay (1996: 34, fn. 9).
[453] Also Nussbaum (2004a).

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'passage', **réh$_1$-i-/ṛh$_1$-i-́*, 'wealth', etc. can be explained using the traditional ablaut system consisting of four classes (acrostatic, protero-, hystero- and amphikinetic).[454]

V.3. Pedersen's System of Mobile Formations

H. Pedersen assumed that the Proto-Indo-European nominal system possessed two patterns of mobile inflection: protero- and hysterokinetic,[455] the proterokinetic formations being nouns like Ved. *sū-nú-ḥ* 'son' (gen. *sū-nó-ḥ*), and the hysterokinetic formations being those like Gk. πα-τήρ 'father' (gen. πα-τρ-ός). According to Pedersen's system, the nouns which nowadays are reconstructed with the R(*e/z*) and S(*o/ō/z*) vocalic alternations, cf. **pent-oh$_2$-/*pṇt-h$_2$-́* 'road' or **h$_2$eus-os-/*h$_2$us-s-́* 'dawn', were classified as hysterokinetic, i.e., were considered to be comparable to nouns like πατήρ.[456]

Although Pedersen's system of two mobile inflectional patterns never sank into oblivion, being later accepted by F. B. J. Kuiper (1942), and having influenced Rasmussen (see below in section V.4), it contains a number of problems. This theory immediately ran into accentual difficulties, being unable to provide an account for the "irregular" radical stress of Sanskrit nouns like *svásā* 'sister' (instead of the expected **svasā́*), *sákhā* 'friend' (instead of **sakhā́*), etc.[457] In addition to this, also the vowel alternation in different forms of some of Pedersen's hysterokinetic (= Schindler's

[454] For a critical discussion see S. Schaffner (2001: 84-6), Neri (2003: 35-6), in detail Widmer (2003: 56-60).
[455] In Pedersen's terminology, "amphidynamic" and "hysterodynamic" (1926: 24).
[456] Pedersen (1926: 24).
[457] Pedersen, ibid.

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amphikinetic) nouns proceeds according to principles different from those characteristic of *true* hysterokinetic formations. Whereas true hysterokinetic nouns exhibit non-ablauting roots in the zero or the *o*-grade on the one hand and *e/z* alternation between the suffix and the ending on the other hand, Pedersen's "irregular" hysterokinetic formations show *e/z* alternation between the root and the ending, whereas the suffix shows *o/z* alternation between the strong and the weak forms.

V.4. Rasmussen's Simplification of Pedersen's System

Although all the problems described above could be solved by assuming the existence of two independent inflectional patterns (like Schindler did), in recent years Pedersen's system of two mobile types was once more brought into discussion by Rasmussen. In his article "The Indo-European Amphikinetic Paradigm Type" (1999b), Rasmussen rejected the amphikinetic inflection pattern, interpreting it as a version of the proterokinetic type.[458] According to Rasmussen, some di- and trisyllabic proterokinetic nouns were at some stage reformed after monosyllabic nouns with mobile accentuation in such a way, that the marginal accentuation of monosyllabic nouns was adopted by the above-mentioned di- and trisyllabic proterokinetic formations (1999b: 576).

Although Rasmussen's derivation of the amphikinetic pattern from the original proterokinetic pattern partially solves the accentual problems which Pedersen faced deriving amphikinetic formations from the hysterokinetic type namely in that it

[458] NB that in the original Pedersen's account, amphikinetic formations were classified together with the hysterokinetic nouns (see the preceding section).

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explains the accented roots of Sanskrit forms like *svásā*, *sákhā*, etc. as typically proterokinetic, this new interpretation cannot provide a reasonable account for the rest of the problems raised in section V.3 above, as well as creating some of its own.[459] The "old" problems were briefly touched upon in the preceding section. Among the new difficulties, it is worthwile to ask the following question: why did the marginalization of the accent affect only a part of the original proterokinetic class whereas a large number remained as regular proterokinetic (i.e., showing non-marginal accent)? The parallel of the Balto-Slavic mobile accentuation nouns with a marginal accent pattern adduced by Rasmussen,[460] to my mind, can only work against Rasmussen's interpretation, because whereas *all* Balto-Slavic mobile nouns have adopted the marginalized accent pattern (as per Rasmussen), clear evidence for the existence of *stricto sensu* protero-kinetic formations beside the "reformed proterokinetic formations with a marginalized accent pattern"[461] in Indo-European indicates that in fact either there is no parallel between the Balto-Slavic and the Indo-European phenomena in question, or that the marginalization of accent only in a part of the proterokinetic formations is somehow not normal.

V.5. Beekes' System (1995)

Contrary to Rasmussen's attempt to diminish the number of Proto-Indo-European ablaut types, R. S. P. Beekes presented a rather complex system of ancient ablaut

[459] Vaguely admitted by Rasmussen himself (loc. cit.).
[460] Viz., the accentual pattern as seen in Lith. *duktẽ* 'daughter' vs. *dùkterį* (acc. sg.; Rasmussen 1999: 576-7).
[461] Corresponding to Schindler's amphikinetic type.

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patterns in his *Comparative Indo-European Linguistics* (1995). Beekes assumed two types of mobile ablaut patterns, viz. the protero- and the hysterokinetic.[462] According to Beekes, the hysterokinetic type possessed four different subtypes, two of which resemble the traditional amphikinetic pattern, cf. the following abridged paradigms adopted from Beekes (loc. cit.):

Beekes' hysterokinetic ablaut pattern:			Schindler's amphikinetic pattern:
	Type II	Type III	
Nom.	*CéC-ōR*	*C(e)C-ōR*	*CéC-ōR*
Acc.	*CC-éR-ṃ*	*C(e)C-oR-ṃ*	*CéC-oR-ṃ*
Gen.	*CC-R-ós*	*C(e)C-R-ós*	*CC-R-és*
	IE **dhégʰ-ōm*	Gk. δώτωρ	IE **dhégʰ-ōm*

When these individual cases from the three paradigms are compared, it appears that Schindler's nominative looks exactly as Beekes' whereas Schindler's accusative is somewhat similar to Beekes' type III accusative. The genitives differ in the colour of the vowel.[463] However, it is noteworthy that whereas it is Beekes' type III that is structurally closer to Schindler's amphikinetic pattern, Beekes reconstructs the hysterokinetic type II (and not type III) for Schindler's amphikinetic **dhégʰ-ōm*.

Beekes' system contains various problems. The Greek noun δώτωρ 'giver' may have acquired its final shape as a result of certain reformations which are not entirely

[462] Beekes (1995: 174-6).
[463] The problem of the reconstruction of the vowel in the genitive singular is briefly discussed in fn. 147, in section III.3 above.

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well understood yet. In the paradigm of its Sanskrit cognate *dā́-tar-* 'id.' one can see acrostatic features, cf. Skt. *dātúr* (< **-tr̥-s*; Rix 1976: 123; E. Tichy 1995: 17-20, 47 et passim; Tichy 2000: 78; Nussbaum 2004d). However, the reconstruction of an acrostatic **dóh₃-tō(r)* with an unexpected lengthened grade in the suffix (cf. Tichy, loc. cit.) is only a partial solution to the problem. Nouns of this type (cf. also Gk. φράτωρ 'member of a brotherhood') need further investigation.

As for the reconstruction of the *e*-grade in the accusative singular of hysterokinetic type II (Beekes' **CC-éR-m̥*), there is no evidence for it. Beekes' claim that hysterokinetic type II with its peculiar accusative derives from "hysterokinetic type IV" in which such an accusative was normal[464] is hard to prove because there is no good evidence for this type either. The Gothic noun *bandi* 'bond, fetter' which Beekes considers to have had "hysterokinetic type IV" inflection can be plausibly analysed as a *devī*-noun. Moreover, the root of Go. *bandi* does not show the *e*-grade required for Beekes' "hysterokinetic type IV".

Also the short Latin ending *-ă* in the nominative of *ā*-stems (Latin class I) cannot be used as evidence for the existence of the "type IV" in Indo-European, as the shortness of this *-a* can also be explained in a different way.[465]

V.6. Tremblay's "Anakinetic" Type

The alternation between *e*- and zero grade in the root, as well as the *o/ō/z* alternation in the suffix are also assumed by Tremblay for the so-called "anakinetic"

[464] Beekes' "type IV": nom. **CéC-R*, acc. **CC-éR-m̥*, gen. **CC-R-ós* (1995: 175, 193).
[465] See Meiser (1998: 132), Mayrhofer (1986: 149).

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ablaut class, discussed in detail in Tremblay (1996b). The "anakinetic" ablaut pattern, as reconstructed by Tremblay, would have had the following shape:

Nom. sg.	R(e)-S(ó)-E(z)
Acc. sg.	R(e)-S(ó)-E(z)
Weak forms	R(é)-S(z)-E(z)

Tremblay assumed such an inflection for the Indo-European noun **nepōt-* 'nephew; offspring' based on the indirect reflection of the Proto-Indo-European genitive singular form **nép-t-s* in the Etruscan word-form *nefts* 'grandson' which appears to be an Umbrian borrowing.[466] Besides the indirect Etruscan evidence, according to Tremblay the reconstruction of a genitive **nép-t-s* is also required in order to account for the Old Irish genitive *nieth* 'nephew, sister's son' on the one hand (< **nepetes*), and the Avestan genitive *naptō* 'id.' on the other hand.[467]

The "anakinetic" ablaut did not receive support of other scholars, and, as has been demonstrated by Mottausch (2000: 41-4) and more recently by Widmer (2003: 60), all "anakinetic" examples can be explained using Schindler's system of four ablaut types. As far as the alleged genitive **nép-t-s* is concerned, see section IV.11.1 – IV.11.1.3 below.

[466] Tremblay (1996: 50-1). Beside the form *nefts*, in Etruscan there also are attested *nefiś* and *nefš* 'id.'.

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V.7. Oettinger's System of Mobile Formations in Rieken's *Untersuchungen* and Irslinger's *Abstrakta*

In a still different system, Oettinger attributed the vocalic alternations *e/z* and *o/ō/z* to three Indo-European ablaut patterns: the acrostatic, the protero- and the amphikinetic.[468] Although Oettinger's reconstructions emerged in print after the appearance of the more commonly accepted Schindler's system of four ablaut types, Oettinger's system was to some extent adopted by Rieken (1999). Following Oettinger's system, Rieken reconstructed the above-mentioned vocalic alternations *e/z* and *o/ō/z* for the protero- and the amphikinetic Indo-European *t*-stems.[469]

The ablaut of Rieken's amphikinetic *t*-stems is exactly the same as described by Schindler (see above), and needs no further comments. However, a fairly similar ablaut pattern is reconstructed by Rieken for the proterokinetic *t*-stems, the only difference between the amphikinetic and the proterokinetic formations being the ablaut of the weak forms. Whereas the amphikinetic *t*-stems, as is expected, in the weak forms show a zero grade suffix *-t-* and full grade endings, the proterokinetic *t*-stems exhibit a full grade suffix *-et-* and zero grade endings in Rieken's system. The strong forms for both types are reconstructed by Rieken as R(é)-S(ŏ)-E(z),[470] and are different from the more widely accepted Schindler's system according to which proterokinetic formations do not have any *o*-grade.[471] Rieken based the reconstruction of the *o*-grade in the suffix on

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[467] Tremblay, loc. cit. NB Tremblay (1996b: 91) and Tremblay (2004: 579), where he assumes pre-Celtic ***nepetos*** (and not ***nepetes***, as in 1996a).
[468] Oettinger (1980: 46ff.). Criticism in Widmer (2003: 56).
[469] Rieken (1999: 91-3).
[470] Rieken, loc. cit.
[471] See Schindler (1972b).

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the Germanic and Celtic evidence, cf. Go. *mitade* '(Trocken)maß' (gen. pl.; < PGmc. **metaƀ-*), OE *mægeþ* 'Jungfrau' (< **magaþ-*) and OIr. *ingen maccdacht* 'junges, erwachsenes Mädchen' (< **magot-aktā*).[472]

According to Rieken, a fair number of such "proterokinetic" *t*-stems can be reconstructed. Among them, she mentions Skt. *sravát-* 'stream', *saścát-* 'pursuer', *vehát-* 'barren cow', La. *seges* 'sown field' and *teges* 'mat', Go. *mitaþs* 'measure', OE *mægeþ* 'maiden', as well as OIr. *maccdacht* 'childhood; youth', *luch* 'mouse', and a number of Hittite abstracta with the suffix *-att-*, such as *tarn-att-* 'distribution', *kartimmiy-att-* 'anger', *duškar-att-* 'joy', etc. Accepting Rieken's reconstructions, later Irslinger added the Old Church Slavic noun **печать** 'sigillum' to this group, reconstructing its protoform as a "proterokinetic" **pekʷ-ōt-*.[473]

Such a reconstruction of proterokinetic *t*-stems, besides Irslinger (2002), was also adopted by Olsen (2004: 425-6), but has not been followed by anyone else and has been criticized by Widmer (loc. cit.). A large number of well attested Indo-European proterokinetic formations point to a different stem shape, viz. the one as reconstructed by Schindler, cf. IE *séh$_1$-mn̥* 'seed' (nom./acc. sg.; R[é]-S[z]-E[z]), **kréu̯h$_2$-s* 'raw flesh' (nom./acc. sg.; R[é]-S[z]-E[z]), etc. In spite of Rieken's observations of stress shift in the protoforms of the Germanic forms listed above,[474] nothing in these words points to an original "proterokinetic" inflection of the type as reconstructed in her *Untersuchungen*. Although some of these words show certain similarities to Schindler's

[472] See Rieken (1999: 91).

[473] See Irslinger (2002: 44). Although Rieken herself discussed this noun in a section about *hysterokinetic t*-stems, she considered the possibility that the suffix of OCS **печать** could be reconstructed as **-ōt-* (1999: 86, fn. 412).

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proterokinetic pattern whereas others exhibit amphikinetic features, these features are normally not combined in a single ablaut pattern.

As for the OCS noun **печать** 'sigillum' and any other Slavic (or Russian) words ending in *-atь*, such as OCS **тать** 'thief', Ru. крова́ть 'bed', знать 'nobility', по́пать 'mosque; non-Orthodox church', по́кать 'steep slope', etc., none of them reflects historical amphikinetic formations. Whereas the nouns кровать and попать are loanwords, respectively borrowed from Greek κραββάτιος 'couch, mattress' resp. Arabic *ribāt* 'monastery' (via late Greek ῥαπάτιον),[475] the noun знать probably reflects an ancient *ti*-stem **ĝneh₃-ti-*,[476] the noun по́кать is an *i*-stem built to the Slavic verbal root *kat-* 'roll' (cf. Ru. ката́ться 'drive; ride', кати́ться 'roll'),[477] and the noun **тать** 'thief' is an ancient *ti*-stem **teh₂ti-* rebuilt in Slavic as a *jo*-stem. The OCS noun **печать** does not reflect an amphikinetic formation either, as the protoform like **pékʷ-ōt-* would have lead to **pekāt-* in Proto-Slavic, with no palatalization of the root final stop **k*. The palatal *č* could only develop before a front vowel during the first Slavic palatalization,[478] and, because of the Slavic outcome *a* in the suffix of **печать**, that front vowel has to be reconstructed as **ē*.[479] Thus, the protoform of **печать** can only be

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[474] Rieken (1999: 91).
[475] See Vasmer I.665 and II.536.
[476] As has been shown by Vine, *ti*-stems with the structure *CeH-ti-* and *CeC-ti-* commonly maintain the full grade of the root (2004: 366ff.), unlike the *ti*-stems with the structure *CeR-ti-*, which usually generalize the zero grade of the root.
[477] See Vasmer I.540.
[478] Schenker (1993: 68-70).
[479] Cf. also **часъ** 'hour; period of time' (< **kēs-*; cf. OPr. *kīsman* 'once; at a time'), **чаръ** 'charm' (< **kēr-*; probably related to Av. *cārā* 'Hilfsmittel'), etc. (see Vasmer III.303-4; V. Mažiulis II.200-4). The short Indo-European **e* remained as **e* in Proto-Slavic in this position, and was preserved as such in the Slavic daughter languages, cf. IE **kʷe-* > early PS **keso* 'what' (gen.

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reconstructed as **pek^{u}-ēt-* (or **pek^{u}-e-et-*), which makes it structurally more similar to Greek nouns of the ἕρπ-ητ- type, discussed in section I.3.[480]

V.8. The Survey of the Indo-European Primary Amphikinetic *t*-Stems

In the following sections V.11 – V.11.7, I am going to study only those *t*-stems which reflect the amphikinetic inflection pattern as described by Schindler (1975b). The development of the Old Irish noun *clí* for which an "amphikinetic II" inflection had been reconstructed by Tremblay, is discussed in section III.8.3 above. The Vedic nouns *sravát-*, *saścát-* and *vehát-*, as well as the Latin nouns *teges* and *seges*, which had been reconstructed by Rieken as R(e)-S(o)- "proterokinetic" *t*-stems and possibly are morphologically related to each other, are discussed in section I.3.

For now setting aside certain details, an original amphikinetic inflection can be securely reconstructed for the following primary *t*-stem nouns: Go. *mitaþs* 'measure' (< **med-ot-*),[481] *liuhaþ* 'light' (< **leuk-ot-*), Proto-Germanic **witōþ-* 'law', PGmc. **magaþ-* 'maiden, girl', and its Proto-Celtic cognate **magot-* 'id.' (both reflecting **magh-ot-*), as well as the dialectal reflexes of the Indo-European noun **népōt-*

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sg.) > OCS чесо, IE **kes-* 'comb' > OCS чесати, IE **(s)kerp-* > Ru. чéреп 'skull', etc. (Vasmer III.323, 329).

[480] Rieken discusses such nouns in the section about the "hysterokinetic" *t*-stems, and does mention the noun печать there (1999: 86, fn. 412). However, she tentatively reconstructs the suffix of this noun as **-ōt-*, which was to be later adopted by Irslinger (2002).

[481] Irslinger implausibly reconstructs the root as **meh$_1$-*, without explaining how it would have developed into Gothic *mit-* (2002: 44). Hajnal reconstructs the root as **met-* (2004: 110) which would have yielded Go. **miþ-*.

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'offspring' (> La. *nepōt-*, Ved. *nápāt-*, Av. *napāt-*, etc.),[482] and possibly Hi. *karāt-* '?'[483] and *ekt-* 'hunting net'.

Beside the above-mentioned primary *t*-stems, the protolanguage possessed a certain number of amphikinetic *t*-stems which have the shape of primary *t*-stems, but are secondary, cf. **meh₁nōt-* 'month', **u̯eidu̯ōt-* 'witness', and **di̯eu̯ōt-* 'day; light'. Their old age is indicated either by their attestation in several Indo-European dialects, cf. Go. *weitwoþs* 'witness', OIr. *fiadu* and Gk. εἰδώς 'id.', all in one way or another reflecting IE **u̯eidu̯ōt-*, or is betrayed by their archaic appearance, like in Proto-Germanic **mēˈnōþ-* 'month'. Several words about the development of these non-primary *t*-stems will be said in sections V.10.1 – V.10.2 below.

According to some scholars, the number of the Indo-European amphikinetic *t*-stems was somewhat larger. In different works, one can also find reconstructed the following primary and secondary amphikinetic Indo-European *t*-stems: **tepnōt-* 'heat' (> OIr. *tene* 'fire'),[484] **dhéuh₂-i̯-ōt-* (> OIr. *dé* 'smoke'),[485] **k̂li-u̯ōt-* or **k̂léi-ōt-* (> OIr. *clíu* 'body' or 'some kind of territory'),[486] **bhe-bhidh-u̯ōt-* (> OIr. *bibdu* 'culprit'),[487] **k̂om-med-u̯ōt-* (> OIr. *coimdiu* 'lord'), **bhr̥gh-u̯ōt-* (> OIr. *briugu* 'host'), **pluk-ōt-*

[482] The meaning 'grandson' is probably secondary (see Beekes 1976).
[483] Possibly 'entrails'; for the problems connected to the interpretation of the semantic of this noun, see Rieken (1999: 139-43).
[484] Rasmussen (1994: 326).
[485] Watkins (1966: 104, 118), Tremblay (1996b: 34, fn. 9), Irslinger (2002: 48, 61, with further lit.). Pedersen reconstructs **dijet-s* or **dijat-s* (VGkS II.102).
[486] Irslinger assumed the meaning 'body' and reconstructed **k̂li-u̯ōt-* (2002: 48, 52, with further lit., see also DIL.236). Tremblay assumed the meaning 'eine Art Landes', reconstructing **k̂léi-ōt-* (Tremblay, loc. cit.; see also DIL.245). See a further discussion below.
[487] Irslinger (2002: 48). Pedersen reconstructs **bhi-bhid-u̯ōt-* (VGkS II.102).

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(> OIr. *luch* 'mouse'),[488] **k$^{\mu}$riH-ōt-* (> OIr. *cré* 'clay'), **pleu̯-ot-* 'swimming' (> Lith. *plaũčiai* 'lungs'),[489] **leng$^{\mu}$h-ot-* 'light' (adj.; → PGmc. **linχ-t-a-* 'id.').[490] As can be immediately seen, with the exception of the last two items, all of these allegedly Indo-European amphikinetic *t*-stems have left traces exclusively in Celtic (and, in most instances, in Old Irish alone). This poses several problems. First of all, although all these nouns can be reconstructed as ancient *t*-stems mechanically applying morphological and phonological rules, their Proto-Indo-European status cannot be assured, since it cannot be supported by secure evidence from other Indo-European languages, especially those in which the element *-t-* would be visible, e.g. Sanskrit, Avestan, Latin or Hittite. As for the development of Lith. *plaučiai* and the Proto-Germanic adjective **linχta-*, they almost certainly were not ancient amphikinetic *t*-stems, and their development is briefly discussed in the following sections V.8.1 and V.8.2.

V.8.1. On the Reconstruction of **pleu-ot-* 'lungs'

Ancient amphikinetic formations routinely develop in the following way in Lithuanian: the nominative singular preserves the ancient suffix with the lengthened *ō*-grade, cf. *akmuõ* 'stone' (< **h$_2$eḱ-mō[n]*), *juosmuõ* 'waist' (< **i̯eh$_3$s-mō[n]*), *vanduõ* 'water' (ultimately < **u̯éd-ōr*), *mė́nuo* 'month' (< **méh$_1$n-ōs*),[491] etc., whereas in the rest of the paradigm, the suffix is in the *e*-grade, cf. *ak-meñ-s*, *juos-meñ-s*, *vand-eñ-s*, *mė́n-es-io*, etc. (all gen. sg.). The same structure can be observed in the noun *sesuõ*

[488] VGkS II.101, Rieken (1999: 91), Irslinger (2002: 61-2).
[489] Heidermanns (1993: 374), based on Seebold (see next fn.).
[490] E. Seebold (1981: 295-6)

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'sister' (gen. *seseřs*), which is believed to have been a compound **sụe-sōr* originally, but because of its shape adopted the amphikinetic inflection. There exist some rare exceptions, cf. OLith. *žmuo* 'man' (← **dhĝhém-ō*[*n*]), acc. *žmunį* instead of the expected nom. **žemuo*, acc. **žemenį*. Also Lith. *nepuotis* 'grandson' has developed in a different way, adopting *i*-stem and *į̯o*-stem inflection.[492] Thus, a putative *t*-stem **pleu̯ot-* could have either yielded the following forms in (Old) Lithuanian: nom. sg. **plevuõ*, acc. **plẽvetį*, gen. **plevetès*, nom. pl. **plẽvetes*, or, more likely, would have developed like Lith. *nepuotis*, generalizing the suffix **-ōt-* throughout the paradigm, and adopting the *i*/*į̯o*-stem inflection. The reconstruction of an amphikinetic **pleu̯-ot-* as the ancestor of *plaučiai* (or sg. *plaũtis*) is problematic also because the *au* vocalism in the root of the noun *plaučiai* points to an earlier **plou-*, not **pleu-* contra Seebold (1981: 294), and the *o*-grade did not occur in the roots of true amphikinetic formations.

Although the reconstruction of Lith. *plaučiai/plautis* as an amphikinetic **pleu̯-ot-* is not very plausible, this noun may still reflect a *t*-stem, but an acrostatic one, and not an amphikinetic. From the synchronic point of view, the consonant *t* is a part of the root; however, as has been said in chapters III and IV of this dissertation, the non-alternating suffix *-t-* of the acrostatic *t*-stems usually became unified with the roots, becoming a "root enlargement". The same thing could be said of the consonant *t* in the root of *plautis*.

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[491] See section V.9.1 below.
[492] See section V.10.1.6 below.

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It should also be noted that the noun *plautis* structurally is identical to the nouns of the *samtis* type, for which I suggested an original acrostatic inflection as well.[493] The noun *plautis*, just like *samtis* and other nouns of this type, is a synchronic *i̯o*-stem built to a verbal root in the *o*-grade and exhibiting a formant *-t-*. Therefore, one could reconstruct the noun *plautis* as an original *t*-stem abstract **plou-t-* 'swimming', which later would have acquired the concrete meaning 'lungs', and eventually become a *i̯o*-stem.

V.8.2. On the Reconstruction of **leng^{u}h-ot-* 'light'

Objections of a different type can be raised against Seebold's reconstruction of the amphikinetic adjective **leng^{u}h-ot-* 'light'[494] which he claimed to have been the ancestor of the Proto-Germanic adjective **linχta-* 'id.'. Although Seebold did not describe the details of this development precisely, according to his interpretation, **linχta-* must have developed from the weak stem of the Proto-Indo-European adjective **h$_1$leng^{u}h-ot-* in the following way: the original weak stem **h$_1$lṇg^{u}h-t-* would have adopted the root **h$_1$leng^{u}h-* (*e*-grade) of the strong stem, and then the stem **h$_1$leng^{u}h-t-* would have been thematized eventually to yield PGmc. **linχta-*. Although the Proto-Germanic stem **linχta-* can be mechanically derived in such a way, the reconstruction of the adjective **h$_1$leng^{u}h-ot-* itself is quite implausible as the protolanguage did not have any amphikinetic *t*-stem adjectives. As the investigations of other scholars and my own study (presented in greater detail below) show, all amphikinetic *t*-stems were nouns.

[493] See section III.8.5 above.

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Alternatively, one could say that the original amphikinetic *$h_1leng^{u}h$-*ot*- was a noun meaning 'light thing'. In the course of development, a thematic adjective *$h_1l\underset{\cdot}{n}g^{u}h$-t-o- could have been built from the weak stem of this noun, and later the adjective *$h_1l\underset{\cdot}{n}g^{u}h$-*to*- would have been v$\underset{\cdot}{r}$ddhied to form another noun *$h_1leng^{u}h$-*to*- 'light thing', which could have become adjectivized again. As one can see, such a scenario involves a number of nominalizations and adjectivizations, and is extremely complex. In fact, it may be simplified, as will be shown in the following section. The simplified version of the derivation of the Proto-Germanic adjective *$linχta$- involves the elimination of the implausible Indo-European adjective *$h_1leng^{u}h$-*ot*- which most probably did not exist at all.

V.8.3. Heidermanns' Historical Interpretation of Several Germanic Adjectives in -*t*- (*$ƀerχ$-*t*- 'bright', *$leuχ$-*t*- 'light', *$melχ$-*t*- 'milch' etc.)

In F. Heidermanns' *Etymologisches Wörterbuch der germanischen Primäradjektive* (1993), a small number of Germanic adjectives are interpreted as reflexes of ancient Indo-European *t*-stems. In a systematic overview of verbal adjectives, Heidermanns reconstructs the following Germanic adjectives as reflexes of Indo-European formations built with the suffix -*t*-: *$ƀerχ$-*t*- 'shining', *$ferχ$-*t*- 'decent, proper', *$lenχ$-*t*- 'light' (of weight),[495] *$leuχ$-*t*- 'light, bright', *$melχ$-*t*- 'milch, giving milk', *$senχ$-*t*-

[494] The root is more correctly reconstructed as *$h_1leng^{u}h$-, cf. Gk. ἐλαχ-ύς 'light' (< IE *$h_1l\underset{\cdot}{n}g^{u}h$-; see also Hamp 1999: 117-9).

[495] Heidermanns reconstructed this and the preceding adjectives with a "labialized" h^{w}, cf. *$ferh^{w}t$- resp. *lenh*^{w}*t- (1993: 70). However, the labialization was most probably lost between two consonants at a rather early stage (cf. fn. 149 above). These adjectival stems are

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'seeping away', **penχ-t-* 'dense'.[496] Although these adjectives generally appear as *a*-stems in the Germanic dialects, Heidermanns proposed that they were athematic in Indo-European. His main reason for the reconstruction of an original athematic inflection for these adjectives is that other Germanic *to*-stems built to verbal roots of the same *CeR(C)* shape always show the zero grade of the root, cf. PGmc. **wissa-* 'knowing, wise' (< **u̯id-tó-*; √**u̯eid-* 'know'), **sliχta-* 'even' (< **sliĝ-tó-*; √**sleiĝ-* 'smear; make even'), etc.[497]

Although Heidermanns does not clearly present his scenario for the development of these *a*-stem adjectives from the older *t*-stems, it should probably be as follows: the remodelled ancient weak stem **lenχ-t-* + *-a-* → **lenχ-ta-*; **melχ-t-* + *-a-* → **melχ-ta-*, etc.

What do the stems **lenχ-t-*, **melχ-t-* etc. represent?

In his discussion of the prehistory of these ancient athematic adjectives, Heidermanns in most cases compares them with the adjective **lenχt-*, for the etymology of which he refers to an older interpretation suggested by Seebold (see Heidermanns 1993: 380). As has been shown in the previous section, Seebold had reconstructed the Proto-Indo-European ancestor of PGmc. **lenχt-* as an amphikinetic *t*-stem adjective **leng^{u}h-ot-*.

In the previous section, I claimed that in Indo-European there existed no adjectives of this type. How, then, should the structure of these Germanic adjectives be explained?

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reconstructed without labialization also by Seebold (see the preceding section) and Bammesberger (1990: 256-7).

[496] Heidermanns, ibid.

[497] Heidermanns (1993: 70, 74-5).

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At least two of the adjectives for which Heidermanns reconstructed ancient athematic inflection have been earlier analysed by Bammesberger as vr̥ddhi formations for which the *e*-grade of the root is normal. In his *Morphologie* (1990), Bammesberger interpreted the adjectives **þerχta-* 'bright' and **lenχta-* 'light' as vr̥ddhied ancient participles **bhr̥h₁ĝ-tó-* resp. **h₁lṇgʷh-tó-*.[498] Although Bammesberger did not expand on the semantic development of these vr̥ddhi derivatives, it may have been as follows: by means of vr̥ddhiization, the original past passive participles **bhr̥h₁ĝ-tó-* resp. **h₁lṇgʷh-tó-* via the intermediate stage of possessive adjectives 'possessing brightness' resp. 'possessing lightness' could have become neuter nouns **bherh₁ĝ-to-* 'bright thing' resp. **h₁lengʷh-to-* 'light thing'. These thematic neuter nouns subsequently were reinterpreted as neuter adjectives **bherh₁ĝ-to-* 'bright' resp. **h₁lengʷh-to-* 'light', and they have yielded the Proto-Germanic **þerχ-ta-* resp. **lenχ-ta-*.

Although the process whereby nouns become reinterpreted as adjectives is much rarer than the reverse, i.e. when adjectives (or participles) become nouns, several examples of such a development can be indicated, cf. the Middle English adjective *chepe* (> Modern English *cheap*) which derives from the Old English noun *cēap* 'bargain' (← 'thing bought', cf. OIc. *kaup* 'purchase'),[499] or the Old Icelandic adjective *hljóðr* 'silent' (< virtual **k̂leu-to-s*) which derives from the neuter noun *hljóð* 'silence'

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[498] Bammesberger (1990: 256 [notation mine]).

[499] In Old English, *cēap* was used as a noun, cf. the following excerpt from *Wulfstan's Address to the English*: *...his āgene cēap, þe hē dēore gebohte* '...his own purchase that he'd bought dearly' (H. Sweet 1967: 89) etc. However, during the Middle English period, *chepe* (< *cēap*) starts being used as an adjective, cf. the excerpt from Stephen Hawes' *The Passetyme of Pleasure*: *...of mete and drynke / there was grete pleynte / nothynge I wanted / were it chepe or dere* (iv.125-6).

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(< *$\hat{k}$leu-to-m). Both the Modern English *cheap* and the Old Icelandic *hljóðr* (Modern Icelandic *hljóður*) have been fully adjectivized, and the adjective *hljóðr* is fully inflected as an adjective, cf. fem. nom. sg. *hljóð*, neut. *hljótt* (< *hljóð-t*), comparative degree *hljóð-ari* (masc. and fem.), *hljóð-ara* (neut.), superlative *hljóð-ast-*. The adjective *hljóðr* also has the weak adjectival inflection, cf. masc. *hljóð-i*, fem. and neut. *hljóð-a*. The ancestor of the Old Icelandic adjective *hljóðr* was most likely built at the prehistoric stage, and in the course of development, it developed a full adjectival paradigm, and became isolated from the old noun *hljóð*.

The situation was different with the Old Icelandic noun *gaman* 'joy, delight' which came to be used also adjectivally perhaps only in Modern Icelandic, cf. Modern Icelandic *þetta er mjög gaman* 'this is a lot of fun' in which *gaman* can only be interpreted as an adjective. Icelandic nouns may not be modified by the adverb *mjög* 'very', whereas adjectives may, cf. *í Reykjavík er mjög kalt* 'it is very cold in Reykjavik'. However, in the phrase *nú er að byrja gaman-ið* 'now the fun begins', *gaman* is a noun modified by the neuter definite article *-ið*, cf. *nú er að byrja rok* 'the wind is starting to blow'.

Icel. *gaman* is different from the adjective *hljóðr* in that *gaman* does not have adjectival inflection, and may not be used attributively, i.e. a phrase $^{\times}$*Sigga er gaman stelpa* 'Sigga is a fun girl' is grammatically incorrect, whereas *Sigga er hljóð stelpa* 'Sigga is a silent girl' is grammatically correct.

The same adjectivization has taken place in the history of the French noun *marché* 'market', which becomes an adjective when combined with *bon* 'good' into *bon marché* 'cheap' (← 'good market', cf. OE *cēap* and ME *chepe* above).

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The reinterpretation of some nouns as adjectives was probably triggered by their ability to be used predicatively as adjectives, cf. the Icelandic sentences shown above. In some cases, the full adjectivization has not taken place yet, cf. Icel. *gaman* which may still be used as a noun meaning 'fun'. However, in other instances, cf. the Modern English *cheap* or the Old Icelandic *hljóðr*, full adjectivization has already taken place.

The old nouns **bherh$_1$ĝ-to-* resp. **h$_1$leng^{u}h-to-* probably were reinterpreted as adjectives at an early stage, because, just like OIc. *hljóðr*, they have been fully adjectivized in all the Germanic languages. Their old age is also indicated by the factitive verbs **þerχt-ijan-* 'make bright' (> Go. *bairhtjan* 'announce', OHG *berahten*, OIc. *birta* 'light up, brighten') and **lenχt-ijan-* 'make light' (> OHG *līhten* 'make light', OIc. *létta* 'make light, stop [a trip]', OE *līehtan* 'dismount a horse'). Both of these verbs have acquired their *t* from the *to*-adjectives **þerχta-* 'bright' and **lenχta-* 'light', which implies that they were perceived as adjectives already in Proto-Germanic.

V.8.3.1.

The origin of the Proto-Germanic adjectives**þerχ-ta-* and **lenχ-ta-* explained in this way is more plausible than Heidermanns' analysis, since, whereas the amphikinetic *t*-stem adjectives reconstructed by Heidermanns as the ancestors of the adjectives of the **þerχta-* type most probably never existed, Bammesberger's analysis of these adjectives as vr̥ddhi derivatives from old *to*-participles operates with well-known Indo-European rules of nominal derivation. In addition to this, as will be shown

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below, Indo-European amphikinetic *t*-stems generally developed in a different way in Germanic.

As for the rest of the adjectives belonging to this morphological class, theoretically one could apply the same interpretation, i.e. explain all of them as vr̥ddhi derivatives. However, such an interpretation is not needed for all of these adjectives, since a couple of them have been formed in different ways.

The adjective **leuχta-* 'light, bright' ultimately indeed happens to derive from an amphikinetic *t*-stem **leuk-ot-*. However, as will be shown in sections V.10.5.2 and V.10.5.2.1 below, the original *t*-stem **leuk-ot-* was not an adjective, and the adjectival semantics of the Western Germanic **leuhta-* 'light, bright' is secondary.

Also the adjective **melχta-* should be explained in a different way from the one suggested by Heidermanns. As one can see, the hypothetical Proto-Germanic adjective **melχt-* only has a single reflex, viz. the Old Icelandic adjective *mjaltr* 'milch' (fem. *mjǫlt*). This adjective can be entirely plausibly analysed as a late dialectal deverbal formation, built from the verb *mjalta* 'milk' in the same way as the adjective *mjolkr* 'milch' (fem. *mjolk*) is built from the verb *mjolka* 'milk',[500] cf. the following scheme:

v. *mjolka* → adj. *mjolkr*

v. *mjalta* → adj. X, X = *mjaltr*

[500] Respectively, the verbs *mjolka* and *mjalta* are both denominative, built from the nouns *mjolk* 'milk' resp. *mjǫlt* 'milking; milking time' (gen. *mjaltar*; nom. *mjǫlt* ← **meluχ-ti-*).

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V.8.3.2.

Different problems arise with the reconstruction of the adjective **senχt-* 'seeping away' which is based on an incorrect etymology. Heidermanns proposed that this adjective is built to the root **sek-* which he glossed as 'seep away, dwindle' (G. 'versickern, versiegen').[501] According to Heidermanns, the adjective **senχt-* is reflected in OE *sīht* 'wet',[502] Middle High German *sīhte* 'shallow' and Swiss German *sīht* 'very wet'. However, a number of problematic issues arise. Whereas there is no clear evidence for the meaning 'seep away' for the verbal root **sek-*,[503] the semantics of OE *sīht* 'wet' and the Swiss German *sīht* 'very wet' is not compatible with the meaning 'dwindle' (see above). The Middle High German adjective *sīhte* (better *sīht*[*e*], because both forms occur in the texts)[504] is probably related to the common Germanic verb **sinkwan-* 'sink' (< **seng^{u}-* 'fall down, sink', LIV.531) and is best explained as a reflex of PGmc. **sīχta-* (< **sinχta-* < **seng^{u}-to-*).[505] The *e*-grade can be explained in the same way as in PGmc. **þerχta-* etc. (see above). Originally, there would have existed a past participle **sn̥g^{u}-tó-* 'with the level of water lowered', which would have been vr̥ddhied to **seng^{u}-to-* 'an area with low water', and further remade into an adjective 'shallow', whence MHG *sīht*(*e*) 'shallow' (> Modern German *seicht* 'id.').

[501] See Heidermanns (1993: 479).
[502] Heidermanns spells this form as *sīht*. However, this form seems to be a hapax, and is translated as 'drained' in BTS.703. The only time it occurs in a texts, this adjective is spelled *siht* (see section V.8.4.3 below).
[503] Cf. the examples adduced by Heidermanns on p. 479, and LIV.523-4.
[504] For selected examples, see M. Lexer (1876: 920).

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V.8.3.3.

The history of the Swiss German (or Alemanian) adjective *sīht* 'very wet' is not entirely clear, but the etymology proposed by Heidermanns requires one to assume a lot of things for which there is no evidence. Heidermann's reconstruction of an "infixed *senk-*" from the original **sek-* is an oversimplification of very complex matters. Germanic possesses a number of reflexes of Indo-European verbs that took a nasal infix in their present tense forms. From the Indo-European point of view, those roots had to appear in the zero grade, and therefore one should reconstruct pre-Germanic **s-ne/n̥-k-* and so forth. However, in a number of instances the full grade was restored after the preterite (← perfect), and there appeared forms like Go. *rinnan*, OIc. *rinna/renna* 'run, flow' (< **renn-* ← **h₃ri-n-H-*), OE *dwīnan*, OIc. *dvína* 'diminish' (< **đwein-* ← **dhgʷhi-n-*)[506] etc. One could say that the same thing could have happened with the root **sek-*; however, it is unlikely that the Proto-Germanic strong verb **senχ-* would have built a *to*-participle, since the Proto-Germanic participles of the strong verbs took the suffixes *-ina-* (< **-eno-*) or *-ana-* (< **-ono-*). Other *to*-formations built to "strong" roots, such as PGmc. **đau-þa-* 'dead' (~ **đaujan-* 'die'), **kal-đa-* 'cold' (~ **kalan-* 'freeze') etc. were built and functioned as participles only at the Indo-European stage. In Proto-Germanic they became dissociated from the verbs, and new participles **đauj-ana-* (or **đauj-ina-*) resp. **kal-ana-* (or **kal-ina-*) were formed. The *to*-formation

[505] Contra Heidermanns, who claimed that "zu den starken Verben **seiga-/*seihʷa-* [< IE **seikʷ-*; A.V.] 'sinken, tropfen, seihen' [...] oder **senkʷa-* 'sinken' besteht angesichts von Stammbildung, Bedeutung und außergerm. Etymologie keine direkte Verbindung".

[506] The verb *dvína* mostly occurs as weak, but a more archaic strong form has been preserved, cf. the saying *það dvín* (instead of *dvínar*) *ekki eftir þann dauða* '(the population) diminishes not after one has died' (i.e. after one person has died, a new one is born instead, cf. ÍO.153).

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**senχ-ta-* now has to face chronological problems. In order to be able to have been formed, it would have had to be Indo-European. However, in all likelihood, the root **senχ-* with the new *e*-grade is Germanic, and the participle should have been **senχ-ana-* or **senχ-ina-*.

In order to form a participle **senχta-* from this root, one would have to assume that there would have existed a weak verb **senχ-*, which would have formed an irregular *i*-less past participle **senχ-ta-* (like PGmc. **sōχ-ta-* from **sōk-ija-*, or **wurχ-ta-* from **wurk-ija-*). However, no evidence for such a weak verb exists.

It may be that the adjective *sīht* 'very wet' is etymologically related to the adjective *sīht(e)* 'shallow' discussed earlier, since both are semantically associated with water and moisture. One could say that both ultimately go back to the noun **seng^{u}-to-* 'area with low water', and that it underwent a semantic split into the adjective 'shallow' (> MHG *sīht*) and a noun 'wetland', which eventually turned into an adjective '(very) wet' as well.

V.8.3.4.

As far as the Old English adjective *sīht* is concerned, the story is perhaps entirely different. First of all, one cannot be entirely sure that it means 'wet' as per Heidermanns.[507] In BTS.703, this adjective is glossed as 'drained'. Second, the root vowel is not necessarily long. This adjective is attested only once, and although the length of vowels is not always marked accurately in Old English texts, the attested dative singular form *sihtre* does not bear the length mark.

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If the root vowel is long, and if the meaning of this adjective really is 'wet', one could reconstruct it in the same way as the Middle High German *wīht(e)* discussed earlier. However, if the root vowel is short and if the meaning is 'drained', an entirely different explanation may be necessary.

The adjective *sihtre* occurs in a description of landmarks and boundaries in the following passage:

...of þã treowe on þone elebeam styb þonon on ceolbaldes wylle of þã wylle on cytasihtes ford of þã forda to wulfrices gemære þonon to hordhlince ufeweardum of þã hlince on sihtre mæde norþeweardre swa forþ on cenelmes stān... '...from that tree till the olive-tree stump, from there till Ceolbald's well, from that well till Kite-stream's (?) ford,[508] from that ford till Wulfric's boundary, from there till the upper part of Hoard-linch, from that linch till the *s.* meadow in the north, and then forward till Cenelm's (?) stone...' (CD III.430)[509]

From this passage alone, it is not possible to determine the exact meaning of the adjective *siht*. However, the meaning 'drained' is possible, since both ground water and bodies of water were present, as implied by the words "well", "stream", and "linch" (i.e. a ridge or hills nearby a body of water, a link).

If the translation 'drained' (cf. BTS.703) is correct, one could interpret the adjective *siht* as an old *to*-participle built from the verbal root **seik^{u}-* 'spill out; seep through',

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[507] Loc. cit. A similar translation is to be found in IEW.906, viz. 'sumpfig'.
[508] The place-name *Cytasiht* is probably a compound of *cȳta* 'kite' (or 'buzzard') and *siht* 'stream, flowing', cf. *blōd-siht* 'flowing of blood', *ūt-siht(e)* 'diarrhea'.
[509] Translation mine.

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despite Heidermanns.[510] The verbal root **seikʷ-* is well attested in Germanic, cf. OHG *sīhan*, OE *sēon*, OFr. *sīa* 'strain' (< PGmc. **sīχwan-*), OIc. *sía* 'id.' (< **sīχwōn-*). The ancestor of the Old English adjective *siht* could then be reconstructed as **siχ-ta-* (< **sikʷ-tó-*, with an expected zero grade of the root).

The noun *siht* 'stream' which occurs in this passage as the final member of the compound *Cȳta-siht-* 'Kite-stream' (vel sim.) probably derives from the same root, and may be an old *ti*-stem **sikʷ-ti-*.[511] The original meaning of this *ti*-stem would have been 'trickling, seeping through', and at least in the protolanguage, this noun may have been used of small streams or springs. However, since the occurrence of the noun 'ford' beside *Cȳtasihtes* in this passage implies a larger stream, one should probably assume that in the course of development, the noun **siχ-ti-* (< IE **sikʷ-ti-*) came to be used to refer to any stream.

V.8.4. Widmer's Remarks about the Proliferation of *t*-Stems in Old Irish

Going back to the Old Irish *t*-stems listed earlier in section V.8, as has been shown by Widmer, it happens in Celtic that nouns which originally belonged to other stems adopt *t*-stem inflection, cf. the Proto-Celtic **tangw-āt-* 'tongue' and **te(p)n-et-* 'fire', both deriving from PIE **dn̥ĝu̯-eh₂-* resp. **tep-n-es-*.[512] According to Widmer, both nouns must have assumed the shape of *t*-stems within Celtic, since well-attested cognates in other languages, such as Old La. *dingua* 'id.', Toch. B *kantwo*, Toch. A

[510] See footnote nr. 505 above.
[511] The *n*-stem *sihte* would be a Germanic (or English) innovation.
[512] Widmer (1997: 122, 125).

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käntu 'tongue', etc., resp. Av. *tafnah-* 'fever', Ved. *tápas-* 'heat', etc. show no traces of any ancient *t*-stems.[513] Although Widmer himself claimed that *t*-stems were still not a productive morphological class in Proto-Celtic,[514] the results of his study imply that in fact it was at least becoming one. Then, given the largest amount of *t*-stems in Irish, one could draw a preliminary conclusion that this Proto-Celtic trend to attract nouns from various classes to *t*-stems became especially productive in Irish.[515]

V.8.5. Several Considerations about the Reflexes of Indo-European Amphikinetic *t*-Stems in Proto-Celtic

In the light of what has been said above, one could also say that perhaps not all of the synchronic Old Irish *t*-stems shown above were *t*-stems also in Proto-Indo-European. A certain part of them may be analogical. Against the interpretation of all of them as original, i.e. Proto-Indo-European amphikinetic *t*-stems speaks the very sparse attestation of such formations in other archaic Indo-European dialects. However, the interpretation of *some* of the Old Irish "amphikinetic" *t*-stems as analogical implies that a certain number of *t*-stems declined in this pattern must be old.

The best candidate for archaic amphikinetic *t*-stems is the Old Irish *niæ/nie/nia* 'sister's son, nephew', which has clear cognates in other dialects, cf. La. *nepōt-*, Old Lith. *nepuotìs* (or *nepúotis*; also *neptis*),[516] Ved. *nápāt-*, Av. *napāt-*, etc. Other two Old Irish *t*-stems which have corresponding cognates in other languages are *fiadu* 'witness'

[513] Widmer (1997: 122-6). This view was accepted by Irslinger (2002: 47-8).
[514] Widmer (1997: 122).
[515] Also Rieken speaks of "certain productivity of *t*-stems in Irish" (1999: 183).
[516] See a more detailed discussion below in section V.10.1.6.

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and the stem **magd-* as in OIr. *maccdacht* 'childhood, youth'. In Germanic, the correspondence to OIr. *fiadu* is to be seen in Go. *weitwoþs* 'witness', whereas Go. *magaþs*, OE *mæg(e)þ*, OHG *magad* and OFr. *megith* 'maiden, girl' are cognate to the Celtic **magd-*.[517]

There are also a small number of amphi-looking *t*-stems which are not attested as *t*-stems outside the Celtic language group, but occur in several languages within the Celtic group, cf. OIr. *luch*, Welsh *llygoden* 'mouse', *llyg* 'field mouse', Old Cornish *logoden*, Brittonic *logodenn* 'mouse', and Gaulish Λουκοτιος (PN), all reconstructed as pre-Proto-Celtic **luk-ōt-* or **pluk-ōt-*.[518] In addition to these words, one could also mention the Celtic term for 'clay', which is probably to be reconstructed as PC **k^{u}rīi̯-o/at-* and is reflected by OIr. *cré*, MW *prid*, MCorn. *pry* 'clay', etc.[519]

Although the good attestation of the words for 'mouse' and 'clay' in the Celtic language group implies that these words are likely to have been *t*-stems already in the Proto-Celtic era, due to the absence of clear cognates in other Indo-European languages, it is impossible to find out whether these words were amphikinetic *t*-stems also in Proto-Indo-European.

V.8.6. On the Reconstruction of OIr. *bibdu* 'culprit', *coimdiu* 'lord', *briugu* 'host', *dé* 'smoke' and *clíu* '?'.

Returning to the list of the allegedly "amphikinetic" *t*-stems shown in the previous section, it needs to be said that whereas the reconstruction of an original amphikinetic

[517] See a more detailed discussion in sections V.10.4 – V.10.4.1 below.
[518] VGkS II.101, Rieken (1999: 91), Irslinger (2002: 61-2).

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inflection is unlikely for some nouns, such a reconstruction cannot be totally disproved although it remains quite problematic for several other nouns.

The Old Irish noun *bibdu* 'culprit' (gen. sg. *bibdad*), has been reconstructed by Irslinger as **bhe-bhidh-u̯ōt-* and interpreted as an ancient perfect participle. The OIr. *bibdu*, if really an ancient participle, can be much more plausibly reconstructed as **bhe-bhidh-u̯ōs-*, exhibiting typical features of Indo-European active perfect participles built to reduplicating perfects, viz. C(e)-C(z)- reduplication and the **-u̯os-/*-us-* suffix.[520] One should also bear in mind that strictly amphikinetic Indo-European formations do not show reduplication. Further, one should also notice that the reconstructed **bhe-bhidh-u̯ōt-*, as well as the compound **k̂om-med-u̯ōt-* (> OIr. *coimdiu*, gen. *coimded*) and **bhr̥gh-u̯ōt-* (> OIr. *briugu*, gen. *brughadh*) are reconstructed with a suffix **-u̯ōt-*, whereas for a regular primary amphikinetic *t*-stem one would rather expect **-ōt-*.

Further uncertainties arise with the reconstruction of the Indo-European amphikinetic **dhéuh₂-i̯-ōt-* and **k̂li-u̯ōt-/*k̂léi̯-ōt-* which allegedly are the direct ancestors of OIr. *dé* 'smoke' resp. *clíu* meaning 'body, Körper' or 'some kind of territory'.

According to Tremblay, the noun **k̂léi̯-ōt-* (the ancestor of OIr. *clíu* 'some kind of territory') is an ancient collective of the Proto-Indo-European noun **k̂lói̯-t-/*k̂léi̯-t-*, the

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[519] Irslinger (2002: 63).

[520] Despite the common belief that the Indo-European perfect active participle had *s/t*-heteroclitic inflection, as has been shown by Rau (1998: 157-9) such participles were *s*-stems. The forms exhibiting *-t-* or its reflexes, such as Gk. εἰδ-οτ-, Go. *weit-wod-* etc. have a different origin.

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ancestor of another Old Irish noun, viz. *clí* 'house-post'.[521] Tremblay recosntructed the original meaning of the collective **k̂lé̯i-ōt-* as 'wickerwork, Flechtwerk'.[522] Although Tremblay did not expand on this interpretation, one would most probably have to assume that this original meaning 'wickerwork' would have further developed into 'sth. with a wickerwork', e.g. 'an area encircled with wickerwork', i.e., 'a fenced area'. If this interpretation were correct, this would also help define the rather obscure meaning of *clíu* as presented in DIL, viz. 'some kind of territory'. However, the textual Old Irish evidence provides no clues as to whether the territories called "*clíu*" were or were not fenced. In the more obvious instances, it is used as a proper noun (place name), and, more importantly, shows oblique forms with a guttural, cf. *bellum Cairn Feradhaigh i Cliu* 'battle of C. F. at Cliu'; *Clāre i Clīu-Ara i mullach Clīach* and *nisrogab as Chlíu chlemnach / oenfer bad fiu Tairdelbach* (all passages from DIL.245). The evidence for the original *t*-inflection appears to be non-existant.

Although one could say that the noun *clíu* switched from the dental class to the guttural class later, Tremblay's interpretation of the OIr. *clíu* as a *t*-stem **k̂lé̯i-ōt-* still contains other problems. First, wickerwork is a structure formed by intertwining some flexible things, such as twigs or wire. However, the meaning of the Indo-European verbal root **k̂lé̯i-* was 'lean'.[523] An ancient noun for 'wickerwork' would have rather

[521] Tremblay (1996: 34, fn. 9). Tremblay spells *cliu*, but in DIL, the spelling is *clíu* (DIL.245). See a more detailed discussion of *clí* in section III.8.3 above.
[522] Tremblay, loc. cit.
[523] See LIV.332. The meaning 'sleep' in Albanian, cf. *fjetunë* 'asleep', as well as 'stand' in Tocharian, cf. Toch. B *kaltär*, Toch. A *kälytär* 'stands' can be easily explained as secondary.

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been built to the verbal root **pleK-* 'plait',[524] and one would expect to have a verbal noun **plóK-t-/*pléK-t-* 'plaiting' and its derivative **pléK-ot-* 'wickerwork, fence'. Alternatively, one could try to say that the noun **k̂lé̯i̯-ōt-* meant not 'wickerwork' but 'fence made from upright poles or logs', i.e. things one can lean at, but for this there is no evidence either.

Although Irslinger discussed an Old Irish noun *clíu* in her study of dental stems as well, it was not the noun *clíu* 'some kind of territory' presented in the preceding paragraphs, but rather a different *clíu* meaning 'body'.[525] The noun *clíu* 'some kind of territory' is not considered in Irslinger's *Abstrakta*.

Irslinger reconstructed the ancestor of *clíu* 'body' as an Indo-European perfect participle **k̂li-u̯ōt-*.[526] Such an interpretation contains problems as well. First, the regular form of this noun is *clí*, and it is not clear whether *clíu* really is the more archaic variant.[527] Second, the meaning of such a participle should have been 'having leaned' ($\xrightarrow{?}$ 'leaning'), but it is unclear how the meaning 'body' should have developed from it. Finally, perfect participles were *u̯es*-stems, and the consonant *t* in the reconstructed **k̂li-u̯ōt-* would have to be understood as secondary.

Similar things can be said about the derivation of the noun *dé* 'smoke' from an ancient collective **dhéuh₂-i-ōt* or allegedly more directly from a somewhat reformed **dhuh₂-i-ot-* or **dhu̯h₂-i̯-ot-*,[528] a derivative of the verbal root **dheuh₂-*.[529] As a

[524] It is not clear, whether the velar should be reconstructed as **-k-* (as implied by Slavic), or as **-k̂-* (as implied by Indic, see LIV.486).
[525] Cf. DIL.236.
[526] Irslinger (2002: 48, 52).
[527] See DIL, loc. cit.
[528] Tremblay, loc. cit., Irslinger (2002: 61).

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corresponding singulative *t*-stem, Tremblay sets up a (secondary?) acrostatic *t*-stem **dhóuh$_2$-i-t* which, according to him, is reflected by La. *foetus* 'stench'. However, this noun appears to be unattested.[530] Instead, the Latin noun for 'stench' is *foetor*, and beside it there is attested an adjective *foetidus* 'stinking, fetid'. Both in all probability derive from the verb *foetēre* 'stink'.[531] The consonant *-t-* of the verb *foetēre* possibly has its origin in a *to*-formation, cf. *fatērī* 'admit, confess' (← **bhh$_2$-to-* ~ **bheh$_2$-* 'speak'), *nitēre* 'glitter, shine' (**ni-to-*).[532]

Alternatively, one could perhaps say that the alleged Proto-Indo-European **k̂lé̯i-ōt-* and **dhéuh$_2$-i-ōt-* were non-collective amphikinetic *t*-stems comparable to such non-collective amphikinetic formations as **nepōt-*, **meh$_1$nōt-*, **di̯eu̯ōt-*, **su̯esōr-*, etc. The merit of this interpretation would be that in such a case, one could derive the Old Irish *clíu* resp. *dé* from ancient amphikinetic *t*-stems without having to deal with the problems of internal derivation described above. However, reconstructing the Indo-European ancestors of *clíu* and *dé* as amphikinetic *t*-stems, one has to admit that such reconstructions are solely based on their synchronic inflection which follow the dental stem pattern. Since *t*-stems were becoming a productive category in Old Irish (and, to some extent, already in Common Celtic, cf. section V.9.5 above), it can equally be the

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[529] The two similar-looking verbal roots reconstructed in LIV as **dheuh$_2$-* 'smoke' resp. **dheuH-* 'move' should probably be connected into one root **dheuh$_2$-* 'move', since smoking represents the movement of smoke.

[530] Tremblay did not provide any semantic and derivational details about this form in this article, but see Tremblay (2004: 581). However, La. *foetus* 'stench' does not seem to exist at all, whereas the familiar Latin noun *foetus* meaning 'fetus' (a variant form of *fētus* 'id.') is definitely not related.

[531] Cf. La. *acēre* 'be sour' → *acidus* 'sour' and *acor* 'acid', *pūtēre* 'stink, rot' → *pūtidus* 'rotten' and *pūtor* '(the process of) rotting', *tepēre* 'be warm' → *tepidus* 'warm' and *tepor* 'warmth' etc.

[532] Cf. IEW.760.

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case that the nouns *clíu* and *dé* acquired the dental stem inflection in post-Indo-European times, and do not reflect any Proto-Indo-European amphikinetic *t*-stems.

V.9. The Secondary Amphikinetic *t*-Stems **meh₁nōt-* 'month, moon', **u̯eid̯u̯ōt-* 'witness' and **di̯eu̯ōt-* 'day'

As far as the secondary amphikinetic *t*-stems **meh₁nōt-*, **u̯eid̯u̯ōt-* and **di̯eu̯ōt-* mentioned in section V.8 above are concerned, their Indo-European status is much more likely. However, the reasons for such an assumption are not the same for all three. Whereas the Indo-European status of **u̯eid̯u̯ōt-* and **di̯eu̯ōt-* is assured by their good attestation in several Indo-European dialects (see a more detailed discussion in the following sections), the *t*-stem **meh₁nōt-* is only attested in Germanic.[533] Although the non-attestation of comparable *t*-stems elsewhere potentially becomes a problem, the case of pre-Germanic **meh₁nōt-* is different. Whereas *t*-stems were a productive morphological class in Old Irish, and multiple nouns from other classes were (or could have been) attracted to this class, *t*-stems were not productive in Germanic, with only less than a handful being attested. Moreover, *t*-stems eagerly moved to other, more productive stem classes in Germanic. Therefore, the Proto-Germanic **mēnōþ-* must be considered an archaism, inherited from the protolanguage.

A vast amount of literature exists about these three problematic nouns. In the following section I would like to provide an overview of recent scholarship concerning the history of these secondary amphikinetic *t*-stems, as well as analyze several details of their development, absent from other treatments.

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V.9.1.

As shown by Rieken, the noun **meh₁nōt-* is an *ot*-derivative of an older, also amphikinetic *n*-stem **meh₁-on-* 'time period' reflected by the Hittite *meiani* 'season' (frozen locative in the phrase *uitti meiani* 'at the annually recurring point in the year').[534] It is also necessary to mention that in a number of languages, the reflexes of the Indo-European noun for 'month' instead of the suffix *-ot-* exhibit *-es-* or *-s-*, cf. Lith. *mė́n-es-io* (gen. sg.; < **meh₁-n-es-*), OCS мѣсѧць (< **meh₁-n-s-n̥-k-*),[535] Skt. *mā́s-* 'month, moon' (< **meh₁-n̥-s-*), La. *mēnsis* 'month' (< **meh₁-n-s-*), etc. The appearance of *-s-* in some words and of *-t-* in others has caused some scholars to assume an original *s/t*-heteroclite, **meh₁-not-/*meh₁-nes-*.[536] According to a still different interpretation proposed by Rasmussen, the Indo-European word for 'month' originally was a special "*c*-stem", in some forms of which the stem-final early PIE **c* developed into classical PIE **t*, whereas in others it became **s*.[537]

The problems of Rasmussen's theory according to which early Proto-Indo-European possessed certain "*c*-stems" have been discussed in sections II.9 – II.9.2 above, and need not to be repeated here.

As for the "heteroclite" theory, one needs to notice that clear indication for the stop *t* can only be seen in the Germanic branch, viz. Go. *mēnōþs*, OHG *mānōt*, OE *mōnað*,

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[533] As correctly stated by Rau (1998: 158).
[534] Rieken (2001: 73).
[535] On the development from **meh₁n-s-n̥-k-* to **mēs-n̥-k-* see Beekes (1982: 55).
[536] No attention paid to ablaut and accentuation. See Rieken (1999: 95), Beekes (1982: 55), etc.
[537] Rasmussen (1994: 335).

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OFr. *mōnath*, OS *mānuth* and OIc. *mánaðr/mánuðr/mǫ́nuðr* 'month'.[538] The claims that also the Gothic noun *mēna* 'moon' as well as its North- and West-Germanic relatives, such as OIc. *máni*, OHG *māno* etc. directly reflect a *t*-stem, are false.[539] Both *mēna* and the North/West Germanic nouns are secondary but obvious *n*-stems built in the Germanic dialects according to standard rules, cf. other Gothic *n*-stems like *guma* 'man', *hana* 'rooster', OHG *gomo*, *hano* (< PGmc. **-ō̃*), vs. OIc. *gumi*, *hani* 'id.' (< **-ēn*).[540] Neither does the Lithuanian *mėnuo* continue **mēnōt* as was supposed by Beekes (1982: 54-6),[541] or is a later reformation of an original *s*-stem **mēn-os* as assumed by Kazlauskas (1968: 285).[542] The Lithuanian form could easily continue an older **mēnōs* (< **méh₁-n-ōs*). The final spirant **-s* would have been dropped at the Proto-Baltic stage by analogy from other nominal formations which at an older stage had ended in **-V̄C*, such as PB **sesō* 'sister' (< **s[u̯]é-sōr*), **akmō* 'stone' (< **h₂éḱ-mō[n]*), **paimō* 'herdsman' (← **poh₂i-mḗn*), etc. The putative Proto-Baltic **mēnōs* could not have survived as such, and it would have developed into an awkward ˣ*mėnuos* in

[538] The Old Icelandic forms have been reformed in the course of development, partially adopting features of *a*-stems, and partially of *u*-stems. The correct interpretation of the vowel *u* in the Old Saxon *mānuth* is impossible due to confusion of vowel quality in unstressed syllables (see J. H. Gallée 1910: 95-108).

[539] Cf. a similar view of Beekes (1982: 53).

[540] For the derivation of the Germanic *n*-stems see Bammesberger (1990: 158).

[541] Beekes also provided an overview of previous research.

[542] Kazlauskas assumed PB **mēnos* basing his theory on an alternative nominative form *mė́nas* and comparing it with Lith. *véidas* 'face', which he derived from an *s*-stem **u̯eid-os* (cf. Gk. [Ϝ]εῖδος 'shape'), or with Lith. *sriaũtas* 'stream' (< **sréut-os*, cf. Ved. *srótas-* 'id.', Kazlauskas, loc. cit., but see section IV.8 above for a different interpretation). The existence of a "proterokinetic" **méh₁n-os/*-és-* in Indo-European is not supported by any other evidence, and it seems more likely to me that the nominative *mėnas* mentioned previously is a secondary analogical formation.

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Lithuanian.[543] One could also add here, that OLith. *nepuotìs* 'nephew', which clearly reflects an ancient *t*-stem, has, unlike *mėnuo*, preserved its *-t-*.

As implied by Rau (loc. cit.), most of the dialectal Indo-European words for 'month' can derive from an ancient amphikinetic *s*-stem **méh$_1$-n-os-/*mh$_1$-n-s-*. The Proto-Germanic **mē^{1}nōþ-* would then have to be explained as a reflex of a *t*-stem, **méh$_1$-n-ot-/*mh$_1$-n-t-*, which would have co-existed with the *s*-stem mentioned above.

V.10.2.

Respectively, the nouns **di̯eu̯ot-* resp. **u̯eidu̯ot-* have now been explained as secondary *ot*-derivatives of two old Indo-European *u*-stems.

According to an interpretation proposed by Rau (1998), the noun **u̯eidu̯ot-* 'witness' is ultimately a substantivized possessive *ot*-derivative of an old acrostatic *u*-stem neuter noun **u̯óid-u-/u̯eid-u-* 'knowledge', the meaning of the derivative having been 'the one possessing knowledge', which works well both semantically and from the derivational point of view.[544] Although in some languages, the reflexes of the word **u̯eidu̯ot-* instead of the formant *-t-* show the formant *-s-*, cf. Ved. *vidvás-* 'knowing; sage', Av. *vīduuāh-* (OAv.), *vīδuuāh-* (YAv.) 'id.', as opposed to Go. *weitwoþs* 'witness', OIr. *fiadu* 'id.',[545] Rau explained this discrepancy by proposing the existence

[543] It is also unnecessary to assume an ancient Proto-Baltic *n*-stem *mēn-en-* vel sim. (as per Stang 1966: 224). For criticism see Beekes (1982: 53-4).
[544] See Rau (1998: 159-60).
[545] Cf. also the similar problem with **meh$_1$-n-ot-/*meh$_1$-n-os-* discussed above.

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of two morphologically and semantically close formations, viz. an original *s*-stem perfect participle in **-u̯os-* and the *t*-stem noun in **-u̯-ot-*.[546]

As for the history of the noun **di̯eu̯ot-* 'daylight, sun', according to a recent interpretation proposed by Nussbaum (2004a), it is an *-ot-* derivative formed from the word-form **di̯-éu̯*, the locative singular form of the hysterokinetic *u*-stem nominalized adjective **di̯-éu̯-/*di-u̯-* 'day, sky', built from the verbal root **dei-* 'shine'.[547] However, although Nussbaum's proposal is totally plausible, theoretically the formation of **di̯eu̯ot-* can also be explained in a different way. Since there exist some amphi-looking non-primary *t*-stems built from adjectival stems, cf. Ved. *nivát-* 'depth' (< **ni-u̯e-t-*), Hi. *karuiliiatt-* 'previous stage' (← *karuili-* 'old' + **-ot-*), one could say that **di̯eu̯ot-* was built from the adjectival stem **di̯-eu̯-* 'bright' with the nominalizing suffix *-ot-*. The original meaning of this noun would have been 'brightness', but the semantics would have been altered in different ways in the different Indo-European dialects.

Reflexes of the Indo-European noun **di̯eu̯ot-* can be found in several Indo-European branches. As was pointed out by Ivanov, different forms of this noun are to be recognized in the Vedic nominal stem *dyú-t-* 'shining; ray of light' on the one hand and the Hittite *šiu̯-att-* 'day' on the other hand.[548] As was later shown by Yoshida, the Hittite stem *šiu̯-att-* reflects **di̯eu̯ot-*, viz. the generalized old strong stem of the original

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[546] Rau, loc. cit.

[547] On the interpretation of the root **di̯eu̯-* as *di̯-eu̯-* (← **dei̯-* 'shine' + *-eu̯-*) already P. Kretschmer (1924: 109-10), also Schindler (1973: 149), Nussbaum (2004a). An original amphikinetic inflection for the noun **di̯eu̯-ot-* was also assumed by K. Yoshida, although he did not discuss the derivation (2000: 182).

[548] Ivanov (1961: 324-8). Later also Yoshida, loc. cit.

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amphikinetic paradigm, whereas the Vedic *dyú-t-* reflects the generalized weak stem of this paradigm, viz. **di̯-u-t-*.[549]

The etymological connection between these Anatolian and Indic lexical items has special significance for Indic. First, it is important for the correct etymological interpretation of the noun *dyú-t-*, which earlier was considered to be a historical root noun, derived from the implausible Indo-European verbal root **di̯eu̯t-* 'shine'.[550] This interpretation was based on the existence of the verbal root *dyut-* 'shine' in Indic, cf. *dyót-a-te* 'shines', *di-dyót-a* (perf.), *dyut-áya-ti* (caus.), etc. However, Ivanov's and subsequently Yoshida's studies have indirectly shown that in fact the verbal root *dyut-* itself is denominal.

The interpretation of the noun *dyú-t-* as a historical *t*-stem rather than an ancient root noun also shows that monosyllabic Indic *stút-* type *t*-stems, all of which synchronically function as root nouns, could have a different historical background. Whereas some of them are probably best interpreted as original acrostatic *t*-stem action nouns and others may be original root nouns enlarged by a *-t-* secondarily, the noun *dyú-t-* was of a still different origin. However, in the course of the development of the Indo-Iranian nominal system, all these nouns merged together and eventually were reinterpreted as root nouns.

From the discussion in the previous paragraphs one can see that the two nouns, **di̯eu̯ot-* and **u̯eidu̯ot-*, despite their identical shape could have been formed in two different ways. Whereas **u̯eid-u̯-ot-* is a nominalized possessive built with the help of

[549] Yoshida (2000: 182).

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the nominalizing suffix *-*ot*-, the noun *$d\underline{i}$-e$\underline{u}$-*ot*- either is a delocatival adnominal (using Nussbaum's terminology, see Nussbaum 2004b), or started out as a de-adjectival abstract 'brigthness'. At any rate, these examples show that at some stage of the Indo-European protolanguage, it was possible to use the suffix *-ot-* not only in order to derive primary amphikinetic *t*-stems (e.g., like **leuk-ot-*), but also to build secondary amphikinetic *t*-stem nouns from different stems and possessing different semantic nuances.

V.10. The Historical Interpretation of the Primary Amphikinetic Indo-European *t*-Stems

The number of primary amphikinetic *t*-stems which are likely to have been formed in Proto-Indo-European is rather small and they are quite poorly attested in the Indo-European dialects. Only the PIE noun **nép-ŏt-* 'nephew' is attested in many Indo-European dialects, whereas PIE **magh-ŏt-* 'girl' is attested only in Germanic and Celtic (see section V.10.4 and below). Others are attested only in a single dialect, cf. Go. *mitaþs* 'measure' (< **med-ŏt-*), as well as Hi. *ekt-* 'net' and *karāt-* 'entrails; appearance' if these really reflect amphikinetic formations *$\underline{i}$ek-ōt- resp. *$\hat{g}$her-ōt-.[551] Certain problems arise with Hi. *luk(k)at(t)-* 'at dawn' and Go. *liuhaþ* 'light' (< **leuk-ŏt-*), as well as Go. *witoþ* 'law' (< *$\underline{u}$id-ōt-) and its West Germanic cognates, which have been interpreted in several different ways.

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[550] Schindler (1972a: 26), LIV.125 (however, see fn. 1 on the same page).
[551] See Rieken (1999: 139-46).

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All of the nouns from this list have been to a smaller or larger degree studied by many scholars, and their interpretations have been presented in multiple works.[552] There is no need to repeat all the existing theories about the interpretation of these *t*-stems, and therefore each individual interpretation will not be investigated here. Several older interpretations of some of these nouns are briefly discussed in Rieken (1999: 91-3). In the following sections, I would like to focus my discussion on more recent theories, as well as investigate some less frequently discussed issues.

From the following discussion of primary amphikinetic *t*-stems, I will exclude the Greek noun Εἵλως 'Helot, Spartan serf' (Herodotus +; stem Εἵλω-τ-), since the etymology of this noun is uncertain, and no cognates in any other languages are attested.[553]

V.10.1. PIE **nép-ŏt-* 'offspring' (?)

The Indo-European noun **nép-ŏt-* 'offspring?'[554] is both the most widely attested and the most widely discussed primary-looking amphikinetic *t*-stem throughout the Indo-European language family.[555] It is attested in Vedic, cf. *nápāt-* 'grandson', younger Avestan, cf. *napå* 'id.' (nom. sg.), *napātəm* (acc. sg.) and *naptō* (gen. sg.), *nafsu* (loc. pl.; < **naft-su*),[556] Old Persian *napat-* 'id.', Latin, cf. *nepōt-* 'id.', Old

[552] For the most recent studies, see Rieken (1999) and Irslinger (2003).
[553] See section II.11.1.
[554] On the semantic interpretation, see Szemerényi (1977: 48-53), also Beekes (1976).
[555] See Rieken (1999: 92, fn. 431) for some references. Not included in Rieken's list are C. S. Littleton (1973: 423-40), Beekes (1976: 43-63), F. Bader (1986: 39-77).
[556] K. Hoffmann & B. Forssman (1996: 139-40).

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Lithuanian *nepuotìs*, *nepúotis* and *neptis* 'id.',[557] OIr. *nia* 'sister's son' (Ogham **niotta** [gen. sg.]), Middle Welsh *nei*, Middle Cornish *noi*, Middle Brittonic *ni* 'id.',[558] OHG *nefo* 'grandson; kinsman',[559] OE *nefa* 'nephew; grandson, step-son',[560] OIc. *nefi* 'kinsman', Gk. νέποδες 'offspring' (pl.), Albanian *nip* 'grandson; nephew'.[561] This Indo-European term has also been borrowed by some neighbouring non-Indo-European languages, cf. Etruscan *nefts/nefś/nefiś* 'grandson' and Lemnian *naφoϑ* 'id.'.[562] In a number of Indo-European dialects, various derivatives of **nep-ŏt-* are attested, some meaning 'niece' or other female relative, cf. Latin *neptis* 'niece', Ved. *naptī́-* 'granddaughter', Av. *naptī-* (< **nept-ih₂-*), OLith. *neptẽ/nepuotė* 'id.',[563] OIc. *nipt* 'sister; sister's daughter; woman, valkyrie (poet.)' (← PGmc. **neftijō-* ← IE **nept-ih₂-* or an original *i*-stem **nepti-*?),[564] OIr. *necht* 'niece', whereas others refer to some male

[557] LKŽ VIII.670 only provides *nepuotìs* and *neptis*, but *nepúotis* must be assumed as well (see a more detailed discussion below in section V.10.1.6).
[558] Irslinger (2002: 63).
[559] According to EIEC, 'sister's son, paternal/maternal cousin' (EIEC.392).
[560] According to Bosworth (1898: 713). According to de Vries, 'nephew, brother's son' (1977: 406). EIEC: 'grandson, sister's son' (p. 392). L. Lancaster: 'brother's or sister's daughter' (1958: 237).
[561] Hieroglyphic Luvian *ham(a)s(a/i)-* 'grandson' and Cuneiform Luvian *ḫamša/i-* 'id.' cannot be related to the word-forms listed above, despite Szemerényi (1977: 49-50). See also below.
[562] See Kretschmer (1940: 266), IEW.764, Beekes (1976: 50), Steinbauer (1993: 288-92). See a more detailed discussion below.
[563] The accentuation of *nepuotė* (Bretkūnas, I Moses:36.2) is unclear.
[564] Synchronically, OIc. *nipt* behaves like an abnormal *i*-stem, exhibiting an irregular acc. sg. form *nipti* (*Helgakviða hundingsbana* II.30; the expected form is *nipt*). The interpretation of OIc. *nipt* as a reflex of **neptih₂-* is complicated by the OHG noun *nift* 'id.', which structurally looks like an *i*-stem (Indo-European *ī*-feminines yielded Old High German forms in *-e[a]* and *-in*). However, from the Indo-European point of view, the interpretation of the Proto-Germanic **nefti-* 'niece', a feminine derivative of **nef-an-* 'nephew', as an *i*-stem rather than a more expected *ijō*-stem is problematic, and requires a special study.

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relatives, cf. OCS **нєтии** 'nephew' (< **nept-iio-*), Gk. ἀνεψιός 'sister's son' (< **sm̥-nept-iio-*).[565]

There exist several opinions concerning the original structure of IE **népōt*. According to one interpretation, IE **nép-ŏt-* originally was an amphikinetic *t*-stem, and in early idealized Proto-Indo-European had to be declined in the following way:[566]

Nom.	**nép-ōt*
Acc.	**nép-ot-m̥* or **nep-ót-m̥*
Voc.	**nép-ot*
Dat.	**n̥p-téi*
Instr.	**n̥p-t-éh*$_1$
Loc.	**n̥p-ét*
Gen.	**n̥p-t-és*

According to a hypothesis proposed by Tremblay, the noun **nep-ŏt-* originally was anakinetic or acrostatic, and accordingly had an accented *é*-grade in the root of the weak forms, cf. gen. **nép-t-s*.[567]

Tremblay's arguments for the reconstruction of such an original paradigm rest on the Etruscan word-forms *nefts/nefś/nefiś* 'grandson' (a loanword from Italic) on the one hand and, if I understand him correctly, on OIr. *nieth* 'nephew' (gen. sg.), arguably the

[565] Schwyzer (1939: 270), Rieken (1999: 93). Critically Szemerényi (1977: 49) and Schrijver (1991: 21). On the basis of this form, Schrijver reconstructs **h$_2$nep-*, but this creates a problem for Gk. νέποδες.

[566] See Rieken (1999: 93).

[567] Tremblay assumes an original anakinetic inflection in Tremblay (1996a: 50-1 and 1996b: 101-2), but acrostatic in Tremblay (2004: 579).

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oldest attested genitive singular form of the Old Irish noun *nia* 'sister's son; nephew' (*Annals of Ulster*, 692), and YAv. *naptō* 'id.' (gen. sg.) on the other hand.[568]

The Younger Avestan form *naptō* can be very easily explained in a different way, namely, by assuming that the ending *-ō* reflects an original amphikinetic ending **-és*. The full grade of the root vowel can be explained as a trivial analogy from the strong forms, cf. acc. sg. *napātəm* (< **nep-ot-m̥*). As one can see from the reflexes of IE **népōt-* in other Indo-European dialects, the root vocalism of the strong forms was generalized throughout the entire family (see examples above in this section).

The problems concerning the correct interpretation of the Old Irish form *nieth* will be discussed in the following subsection.

V.10.1.1. Problems Related to the Historical Interpretation of OIr. *nieth* (gen. sg.)

The historical interpretation of the Old Irish genitive form *nieth* involves many difficulties. First of all, not everybody agrees that this form really is a genitive form of OIr. *nia*. In two different editions of the annals of Ulster, it is taken to be a personal name, viz. *Nieth* (*Nieth-Neill*), cf. W. M. Hennessy (1887: 142-3), or *Nia* (*Nia-Néill*), cf. S. Mac Airt & G. Mac Niocaill (1983: 155). As noted by Hennessy (1887: 142, fn. 5), nothing is known about this person. On the contrary, both in OIG and DIL it is assumed that this Old Irish form, *nieth*, indeed is a genitive form of *nia* 'nephew' and, moreover, an archaic one.[569]

[568] See Tremblay, loc. cit.
[569] Sic Thurneysen (1946: 207), see also DIL.43.

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Against the interpretation of *nieth* as a common noun could speak the fact that the annals of Ulster were written in Latin. The passage in which the form *nieth* occurs reads as follows:

"Kl. Ian*air*. Anno Domini.dc°.xc°.ii°. Bruide m. Bili, rex For*t*rend, mori*tur*, ⁊ Alphin m. Nectin. Mors Dirath epi*scopi* Fernann, ⁊ Bran nepos Faelan, rex Lagen-entium, mortui sunt. Ceallach m. Ronain Moir mor*itur*. Bel*lum inter* Osraighi ⁊ Laigniu in quo cecidit Faelcar nepos Maele Odrę. Iugul*atio* Ainftigh ⁊ Nieth[570] Neill ⁊ filiorum Boendo.[571] Mors Doergairt m. Finguine. Bel*lum* contra Pante." '...the killing of Ainfthech and Nia-Néill and the sons of Boendo...' (Mac Airt & Mac Niocaill 1983: 154-5).

As can be seen, the passage is written in Latin, which would seem to support the interpretation of the different editors of the annals that *nieth* is not a genitive form of the Old Irish common noun *nia*. However, it has to be noted that Old Irish common nouns and sometimes even entire phrases in Old Irish do occur in the annals of Ulster,[572] cf. *Gabail Eliuin mic Cuirp* ⁊ *Conamail filii Canonn* 'the capture (OIr. *gabail*) of Elvin son (OIr. *mic*, gen. of *mac*[*c*]) of Corp, and Conamail son (La. *filii*, gen. of *filius*) of Cano' (AU, 672), *loch Ęchach do shoudh hi fhuil hoc anno* 'Loch Echach was turned into blood this year (*hoc anno* written in Latin; AU, 683)', *bellum Liacch Moelain* ⁊ *doirad Eilind* 'the battle (La. *bellum*) of Lia Maeláin and the

[570] In the earlier edition, this form is printed as nıeċ /nieθ/, cf. Hennessy (1887: 142).
[571] Underlined by me. Only this sentence is translated below.
[572] Not mentioned by Tremblay.

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enslavement (OIr. *doirad*) of Elend' (AU, 677), *mes mor* 'abundant (harvest of) acorns' (AU, 671), *fames ⁊ mes mar* 'famine (*fames* written in Latin) and abundance of acorns' (AU, 759), etc.

It seems that Irish words, especially when they occur individually, were used in those cases when the scribes might not have known the appropriate Latin word, e.g. 'acorn'. This may also be true of words which probably described fairly rare events, such as 'enslavement' or 'capture' (as opposed to 'death' and 'killing', which occur very often in the annals, and are written in Latin). If one assumed that also 'nephew' was one of the words that the scribe did not know how to translate into Latin, Tremblay's interpretation of *nieth* as a common noun might be right. However, there are further problems – throughout the annals, people are usually described in terms of direct (vertical) lineage, using either the Old Irish term *mac(c)* or La. *filius* 'son', or La. *nepos*, which is often used in the sense 'grandson' (AU 688, 692, 696, etc.). To have someone referred to as a 'nephew' of somebody would be awkward, unless one also assumed that the person described thus was, for example, an orphan raised by his relatives, or that nothing is known of the true parents of this person. Neither of these choices is really appealing. Moreover, as stated in the earlier edition of the annals of Ulster (Hennessy 1887), nothing is known of this person registered as *nieth Neill* (or *Nia-Neill*), which makes all such speculations entirely unprovable.

An alternative way to interpret the form *nieth* would be to assume that it means 'nephew', but is the nickname of a person whose real name was Neill. This is possibly the assumption made in both editions of the annals of Ulster, although in neither of them the rendering of the original *nieth Neill* as 'Nieth-Neill' resp. 'Nia-Néill' is

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commented upon. Such an interpretation would again imply that this person would have been an orphan or a foster-son, but these would be mere speculations.

There exists one additional theoretical way to interpret this problematic passage. As often happens with ancient copied texts, scribes make various mistakes, such as using modernized spelling (or, on the contrary, deliberately making it more archaic), misspelling words which are no longer clear to them, changing the syntax of unclear passages, etc. Thus, one could imagine that this problematic passage from the annals of Ulster originally looked differently, and was changed by the scribe (copyist), who did not understand it properly. As it appears, most names in the annals of Ulster are used along with some modifier, in most cases the noun *mac(c)* 'son' (usually abbreviated to *m.*), cf. *Cerball m. Maele Odrę* 'Cerball son of Mael Odar' or *Conghal m. Maeile Duin m. Aedho Bennain* 'Congal son of Mael Dúin son of Aed Bennán', etc. Sometimes, the name is modified by the Latin noun *nepos* 'grandson', cf. *Faelcar nepos Maele Odrę* 'Faelchar, grandson of M. O.', or the person's occupation is mentioned, cf. *Becc Fhola episcopus* 'bishop Becc Fhola', etc. However, in the passage AU 692, the name *Ainfthech* appears without any such modifier, wherefore one could postulate that originally this passage was written in the following way: **Iugulatio Ainftigh nieth Neill ⁊ filiorum Boendo* 'the killing of A. [who was a] nephew of N., and of the sons of B.'. In the process of copying, the phrase "Ainfthich nephew of Néill" would have been mistakenly interpreted as "Ainfthech [and] N.-Néill", and the connector ⁊ would have been inserted. If this had been the case, Tremblay's interpretation of the word-form *nieth* in this passage as a genitive singular form of the noun 'nephew' could be saved.

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However, although in studies of ancient texts which have been frequently re-copied it is always worthwhile considering scribal errors, in this case, such an interpretation is not very attractive. In order to accept it, one has to make poorly founded assumptions, such as believing that some copyist (by all probability, an Irishman)[573] did not understand the Irish word-form *nieth* 'nephew', and that the text in this manuscript is corrupt. Neither of these assumptions is appealing. Furthermore, it has to be added that the absence of a modifier for the personal name Ainfthech is not really a problem at all, as quite a few other names appear without a modifier in this text, cf. *Nechtan m. Canonn ┐ Aedh obierunt* 'Nechtan, son of Canu, and Aedh died' (AU 621), *Itarnan ┐ Corindu apud Pictores defuncti sunt* 'Itarnan and Corindu died among the Picts' (AU 668), *iugulatio Muirmin* 'the killing of Muirmen' (AU 681), *iugulatio Rotachtaigh* 'the killing of Rothachtach' (AU 685), *Ainfceallach iugulatus est* 'Ainfcellach was slain' (AU 718), *bellum Fernmuighi in quo cecidit Cetomun* 'the battle of Fernmag in which Cétomun fell' (AU 729), etc. All these factors make the hypothesis proposed above of little value.

V.10.1.2. An Alternative Interpretation of OIr. *nieth*

In fact, an entirely different interpretation could be proposed for the word-form *nieth*. It has to be noted that in Old Irish, there also existed a noun *nia* (*nía*) meaning 'warrior'. It was homonymous with the noun *nia* 'nephew', and was sometimes used in

[573] The main scribe of the main surviving manuscript (*H*) was an Irishman named Ruaidhrí Ó Luinín, but he was apparently using some older sources (for the history of the manuscripts of AU, see Mac Airt & Mac Niocaill 1983: viii-ix).

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personal names as well, cf. the following excerpts from various ancient Irish documents: *...Baidanus, gente nepos Niath Taloirc, benedici á Sancto petiuit...*; *...et babtitzauit filios Nioth Fruich...*;[574] *...mac Nioth...*(all underlined forms are genitive singular),[575] further cf. OIr. *Nía-Corb, Nía-Cuilind, Nía-Segamain.*[576] Thus, one could argue that also in the named passage from the annals of Ulster, the word-form *nieth* in fact is the genitive form of the proper name *Nia* (← *nia* 'warrior'), and has nothing to do with the common noun *nia* 'nephew'.

One should also be aware of certain phonological problems with the interpretation of the word-form *nieth* as a putative reflex of **nepetos*. First of all, in the Oghamic inscriptions, it is written **niotta** 'nephew' (gen. sg.), and together with an alternative Old Irish genitive of *nia*, written *niath*, they point towards a different protoform, namely **nep-ot-os*.[577] Towards a suffix *-ot-* also point other Indo-European dialects, cf. La. *nepōt-*, Lith. *nepuot-*, and Gk. νέποδες.[578] The reconstruction of a Proto-Celtic **-ot-* is also supported by the shape the suffix in OIr. *lochad* 'mouse' (gen. sg. of *luch*), which reflects **(p)luk-ot-os*.[579]

To maintain that the reconstruction of **nepetos* is required by the vowel *-e-* in the only once attested word-form *nieth* is not secure enough also on a different ground, namely because the vowel *e* in this word can easily be an instance of imprecise

[574] W. Stokes & J. Strachan (1975: 269, 274).
[575] DIL.43.
[576] The last three examples adduced from P. Sims-Williams (2003: 139).
[577] Cf. Szemerényi (1977: 49), Schrijver (1995: 387, 449, et passim), Irslinger (2002: 63).
[578] The letter δ is secondary.
[579] VGkS II.101.

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spelling. As the pronunciation of *nieth/niath* was [niʲəθ],[580] i.e., contained a schwa-sound, in a position after *j* it is hardly possible to tell whether the sound realized as [ə] represents an underlying /a/ or /e/. It could be mentioned here for comparison that also the genitives of the noun *nia* 'warrior' are spelled in more than one way, cf. *nioth, niath, neth* (and, perhaps, *nieth*, if my interpretation presented above is correct).[581]

To summarize, it can be said that the word-form *nieth* occurring in the annals of Ulster is too unclear to be used as proof for any morphological changes that would have taken place in Proto-Celtic (or Proto-Indo-European itself). Although the possibility that it really is a genitive form of the noun *nia* 'nephew' cannot be entirely ruled out, in the previous paragraphs I have discussed a number of serious problems concerning this interpretation, as well as an alternative way to interpret it.

V.10.1.3. Etruscan *nefts, nefś* and *nefiś* 'grandson' (?)

As far as the Etruscan word-forms *nefts*, *nefś* and *nefiś* are concerned, they can hardly be used as evidence for an original Indo-European genitive **nép-t-s*. As shown by D. Steinbauer, the East Italic (Umbrian?) noun **nefts* 'grandson' that the Etruscan word-forms were based on possessed a reformed paradigm, entirely different from the Proto-Indo-European one reconstructed on the basis of various Indo-European dialects. According to Steinbauer, in the course of the development of the noun **nepōt-* in East Italic, the *o*-grade of the suffix was slowly eliminated from the strong forms (contrary

[580] Cf. Schrijver (1995: 449).
[581] See DIL.43.

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to Latin, in which the lengthened *ō*-grade was generalized), and the resulting consonant cluster *-pt-* underwent a change to *-ft-* before the noun was borrowed into Etruscan.[582]

V.10.1.4. The Reformation of the Paradigm of IE **nep-ŏt-*

The amphikinetic paradigm of the noun **nep-ŏt-*, as presented in section V.11.1 above, did not survive as such due to its allomorphy, and was reformed already in the protolanguage, generalizing the *e*-grade of the root throughout the paradigm, levelling out the original accented *e*-grade in the locative case, and providing the locative case with the *i*-marker. As a result of these changes, the new paradigm must have looked the following way:

Nom.	**nép-ōt* (or **nép-ōs*?)
Acc.	**nép-ot-m̥* or **nep-ót-m̥*
Voc.	**nép-ot*
Dat.	**nep-téi*
Instr.	**nep-t-éh*$_1$
Loc.	**nep-t-í*
Gen.	**nep-t-és* or **nep-t-ós*

The reconstruction of the nominative Auslaut is not entirely clear, as it may have also been **-ōs*. An older **-ās* (< **-ōs*) is indicated by the Av. *-å̊*. The cluster **-ōs* could reflect an early assimilation of the suffix *-ōt-* and the nominative marker *-s*.[583] Similarly, also the Latin ending *-os* indicates an underlying assimilation of *t* and *s*. The consonant *-t-* shown by both Sanskrit and Baltic could have been reintroduced into the

[582] See Steinbauer (1993: 288-91).

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nominative case at a dialectal level by simple analogy from the other cases. The Germanic situation is not entirely clear (see a more detailed discussion in the following section).

The paradigm as shown above was preserved well in Proto-Indo-Iranian. The ablaut of the suffix can be readily seen in both in Vedic and Avestan, cf. the word-forms as shown earlier in this section. The Vedic strong forms also show the preserved original accentuation, cf. *nápāt* < **népōt*.

The suffix ablaut *-ōt-/-t-*, preserved in Indo-Iranian, was apparently also preserved in Proto-Baltic and Proto-Italic, as indicated by Lith. *nepuotis* vs. *neptis* on the one hand, and La. *nepōt-* vs. East-Italic **nept-* on the other hand.[584]

V.10.1.5. The Development of IE **nepōt-* in Proto-Germanic

If one assumed that the pre-Germanic nominative form of this noun had been **nepōt*, the transfer to *n*-stems would have been triggered by the loss of the word-final *t* in the nominative singular,[585] whereby it would have become somewhat comparable to *n*-stems, cf. IE **népōt* > PGmc. **nefōþ* > **nefō* (> OE *nefa*, OHG *nefo*) ~ **gumõ* 'man' (> OE *guma*, OS *gumo*, OHG *gomo*), **wigõ* 'warrior' (> OE *wiga*), etc. As one can see, the historical *n*-stems have a "schleiftonig" **-õ*, whereas **nefō* has a "stoßtonig" **-ō*. However, since the attested OE *nefa*, OHG *nefo* have the same ending as the historical *n*-stems, one has to assume that the "stoßtonig" **-ō* had become

[583] This is rejected by Beekes, who calls "nominatives of this [i.e. PIE **nepōt-*] type [...] asigmatic" (cf. Beekes 1986: 58), and assumes late analogical origin for the Avestan *-å*.
[584] For the discussion of the Lithuanian forms, see section V.10.1.6 below.

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replaced by a "schleiftonig" *-$\tilde{o}$ by analogy from the *n*-stems. The reason for this replacement of *-$\bar{o}$ by *-$\tilde{o}$ would have been the gender similarity between the masculine **nefō̃* and the masculine *n*-stems, because otherwise **nefō̃* should have become inflected like feminine $\bar{o}$-stems (< IE < *-eh_2-).

Following the example of *n*-stems, the Proto-Germanic **nefō̃* acquired a new stem characterized by the suffix *-an-*, viz. **nef-an-* (cf. **gum-an-*).

A different interpretation is preferred by Bammesberger who assumed that the early nominative should have been **nepōs*.[586] According to his opinion, pre-Germanic **nepōs* would have become Proto-Germanic **nefōz*, which would have lost the word-final **z* and subsequently developed an *n*-stem inflection (Bammesberger, loc. cit.).

Neither of the two interpretations can be entirely ruled out, and neither is completely free of problems, although I find Bammesberger's interpretation less plausible than the one proposed earlier. If one assumed that my interpretation were correct, it would require the reintroduction of *t* into the nominative singular form, which would have thence also been asigmatic. This in itself is not a big problem, as many Proto-Indo-European animate nouns were asigmatic in the nominative singular case. One could compare IE **népōt* to **$h_2$ék̂-mō(n)* 'stone', **ph_2tḗr* 'father', **$h_2$uksḗn* 'bull', etc.[587]

According to Bammesberger's interpretation, one does not need to assume a re-introduction of *t*, but then the loss of the word-final Proto-Germanic **z* has to be

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[585] See Krahe/Meid (1969, I.126).
[586] Bammesberger (1990: 186).
[587] Also noted by Bammesberger, loc. cit.

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explained as analogical, since in other locations it was preserved until the dialectal era, cf. the *ō*-stem genitive singular **-ōz* (> Go. *-os*, OIc. *-ar*) or *u*-stem genitive singular **-auz* (> Go. *-aus*, early Scandinavian runic -ᚮᚱ).[588]

V.10.1.6. The Development of IE **nepōt-* in Lithuanian

A few words need to be said about the development of the Indo-European noun **nép-ŏt-* in Baltic. IE **nép-ŏt-* is reflected in Old Lithuanian by *nepuotis* and *neptis* 'grandson' (both masc.). These forms have developed from the late Indo-European paradigm of the noun **nép-ŏt-*, shown in section V.10.1.4 above. The noun *nepuotis* has generalized the suffix **-ōt-* throughout the paradigm, whereas *neptis* shows the generalization of the zero grade suffix **-t-* which originally occurred in the weak forms. The preservation of both suffixes indicates that the late Indo-European ablauting paradigm of the noun **népōt-* was still preserved in Proto-Baltic.

However, the nouns *nepuotis* and *neptis* are not consonantal stems. Probably at a fairly early date they merged with historical *i*-stems, following the general trend of Proto-Baltic consonantal stems.[589] As an *i*-stem, the noun *nepuotis* is preserved in Bretkūnas's texts, cf. *nepuotims* (dat. pl.).[590] Side by side with the *i*-inflection, the nouns *nepuotis* and *neptis* also developed forms with features of *jo*-stems, as is

[588] Bammesberger's claim that also the loss of a word-final **t* would have had to be analogical is not based on any evidence. Both the Proto-Germanic spirant **þ* (< IE **t*) and the stop **t* (< IE **d*) were lost in absolute final position is non-monosyllables (cf. Krahe/Meid, loc. cit.).
[589] Stang (1966: 219).
[590] LKŽ VIII.670b.

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betrayed by forms like *nepuočius* (acc. pl.; instead of **nepuotis*, more appropriate to *i*-inflection) and *nepčio* (gen. sg.; instead of **nepties*).[591]

Although, as can be seen from the earlier discussion of the Lithuanian nouns of the *samtis* type,[592] the nouns *nepuotis* and *neptis* were not the only *i*-stems to have transferred over to *į̯o*-stems,[593] the transfer of the latter two nouns may have been triggered by their semantics. Lithuanian nouns ending in *-tis* are predominantly feminine, cf. *mirtìs* 'death', *sagtìs* 'buckle', *naktìs* 'night' (all feminine), etc. Since the nouns *nepuotis* and *neptis* denote masculine beings, they were isolated among other (*t*)*i*-stems, and gradually developed *į̯o*-stem forms, following the pattern of masculine *į̯o*-stems like *mẽdis* 'tree' (← **medhį̯os*), *kir̃tis* 'strike' (← PB **kirt-į̯as*),[594] etc.

V.10.1.7. On the Etymology of IE **nepŏ̆t-*

As one can see from what has been said in the preceding chapters, the ablaut of the noun **nep-ŏ̆t-* in Proto-Indo-European and its development in the various dialects is fairly clear. It originally was an amphikinetic *t*-stem, but in all dialects, except for Indo-Iranian, slowly lost its amphikinetic inflection.[595]

However, the etymology and the formation of **nepŏ̆t-* remain unclear. As pointed out by Szemerényi, the interpretation of **nepŏ̆t-* as **ne-pŏ̆t-* 'Unselbständiger; not a

[591] LKŽ VIII.670a-b. The asterisked *i*-stem acc. pl. and gen. sg. forms are not attested.
[592] See sections III.8.4 and II.8.4.1 above.
[593] In fact, this list can be extended further, cf. Lith. *lūšis* 'lynx', *vinis* (beside *-ys*) 'nail', *dantis* (*-ys*) 'tooth', *vagis* (*-ys*) 'thief', etc., all of which can be declined either as *i*-stems or *į̯o*-stems.
[594] The ending *-is* is most probably analogical after *i*-stems, cf. Kazlauskas (1968: 178-85).
[595] Also in Sanskrit it eventually became infected by *ter*-stems (kinship terms).

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master' cannot be correct.[596] However, Szemerényi's own etymology **Hn-e-pot-* '(little) master grandfather, Herr Großväterchen'[597] faces various problems as well. As noted in Rieken (loc. cit.), the proto-form **Hnepot-* should show reflexes of the laryngeal in Greek, but the Greek word-form νέποδες shows no laryngeal vocalization. Another problem is the original meaning as well as the structure of the compound **Hn-e-pot-*. According to Szemerényi, the original meaning of **Hn-e-pot-* was 'the master (**-pot-*) belonging to grandfather (**Han-*; **Han-* > **Hn-e-* in the compound), or "simply '(little) master grandfather, Herr Großväterchen'". This interpretation is hard to accept on the semantic grounds. Also Szemerényi's claim that the vowel *-e-* following the root **Hn-* should be interpreted as an adjectival suffix (vel sim.)[598] is based on nothing.

According to the communis opinio, **népŏt-* is a primary formation **nép-ŏt-*.[599] However, the history and the meaning of the root **nep-* is unknown, as other words that have it are either equally unclear, e.g. OIr. *Nechtan* (divine name; < **nept°* or same as La. *Neptūnus*? [see below]),[600] or are etymologically unrelated, e.g. La. *nepeta* 'mint (?)' and *nepa* 'scorpion'. The Latin divine name *Neptūnus*, which synchronically

[596] Szemerényi (1977: 50), similar criticism in Rieken (1999: 92, fn. 431).
[597] Szemerényi (1977: 51).
[598] Szemerényi, ibid.
[599] Schindler (1975a: 61), Nussbaum (2004a), Rieken, loc. cit., Irslinger (2002: 53).
[600] These two deities have often been compared to the Iranian deity *Apąm Napāt-*, see references in fn. 557 above. However, at least *Neptūnus* is clearly unrelated to *Apąm Napāt-* (see also H. Petersmann 1995: 257, fn. 20). See also EIEC.203-4.

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appears to have a root *nep-*, has been interpreted as a reflex of an original **nebh-tu-* (or virtual **nebh-tu-H-no-s*), and is unrelated.[601]

V.10.2. PGmc. **mét-ōþ-/mét-aþ-* 'measure'

Reflexes of this noun are preserved in several Germanic languages: on the one hand, as Gothic *mitaþs* 'measure, μέτρον, χόρος',[602] directly reflecting a Proto-Germanic consonantal stem **met-aþ-* (~ IE **med-* 'measure'), and on the other hand, by a *u*-stem derivative 'measurer; fate (> secondarily 'God, Creator')', as reflected in OIc. *mjǫtuðr* 'fate; death', OE *me(o)tod/metud* and OS *metod* 'fate; Creator' (< 'measurer'?), all reflecting Proto-Germanic **met-ōþ/đ-u-*.[603]

Although no reflexes of a putative Indo-European noun 'measure' which could be reconstructed on the basis of Germanic evidence as an amphikinetic **méd-ōt-* (acc. **méd-ot-* or **med-ót-*) are attested anywhere else, the word is probably fairly old, as it both shows archaic ablaut of the suffix typical of Indo-European amphikinetic formations and traces of Verner's law, cf. the consonant *d* in both the Old English and the Old Saxon forms, which reflects a Proto-Germanic **đ* (< IE **t* via Verner's voicing). The alternation between the consonants *þ* and *d* in the paradigm of Go. *mitaþs* (*vs.* gen. pl. *mitade*, St. Luke 16:7) is another indication of the operation of Verner's law.

[601] See Petersmann (1995: 257 and further). Petersmann also provides references to some earlier studies.
[602] Written *mitads* in Wulfila's text (St. Luke 6:38).
[603] Cf. de Vries (1977: 390), Lehmann (1986: 258).

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On the basis of the Germanic word-forms discussed in the preceding paragraphs, one can reconstruct a theoretical Indo-European stem **méd-ōt-/*méd-ot-* 'measure', built from the Indo-European verbal root **med-* 'measure' (cf. Gk. μέδομαι, OE/OS *metan* 'measure') with an ablauting suffix *-ōt-/-ot-*. These features are characteristic of the strong forms of amphikinetic formations. No traces of the putative weak stem **m̥d-t-* have been preserved. However, the phonotactically awkward sequence **m̥d-t-* would have been levelled out already in Indo-European, and the full grade of the root would have been generalized throughout the paradigm.

V.10.2.1. On the Semantics of **médōt-*. The Comparison with Amphikinetic *t*-Stems of the **nepōt-* Type and the Anatolian Action Nouns in *-att-*

As far as the reconstruction of the original semantics of the amphikinetic noun **méd-ōt-* is concerned, there exist two possibilities: either to interpret it as an original action noun *'measuring' which would have become concretized to 'measure' (cf. acrostatic *t*-stems like **sós-t-*, **ĥóH-t-* etc. in chapter III above), or assume that **méd-ōt-* from the very beginning denoted a concrete noun 'measure, measuring unit', cf. **méh₁n-ōt-* 'month; moon' or **di̯éu̯-ōt-* 'day-light', neither of which is likely to have been an original action noun.

Given the semantics of the last-mentioned *t*-stems it might be tempting to interpret also the noun **med-ŏt-* in such a way, assuming that it never had the semantics of an action noun. It should also be borne in mind that the protolanguage already possessed a fairly large number of other suffixes which had the ability to form action nouns, such as

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-men-, *-es-*, *-ti-*, and, notably, *-t-*. To add one more suffix to this list of action noun suffixes might appear quite uneconomical.

However, despite these considerations, the interpretation of **med-ŏt-* as an original action noun is really worth considering. First of all, it has to be noted that **med-ŏt-* is different from **méh$_1$n-ōt-*, **di̯éu̯-ōt-*, etc. in that it is built to a pure verbal root whereas the latter formations are built to secondary nominal resp. adjectival stems. Therefore it is quite possible that the original semantics of the differently built **med-ŏt-* could have been different, too. Since the root of the *t*-stem **med-ŏt-* is verbal, the semantics of an action noun certainly should come into consideration.

As mentioned earlier in this section, the semantic development from *'measuring' to 'measure, measuring unit' is quite easily imaginable. The resulting noun 'measure, measuring unit' could be interpreted as a result noun, or simply as a concretized action noun, vel sim. As is often the case, the semantics of derivatives can be interpreted in more than one way, all being equally plausible, and the ultimate choice being mostly a matter of preference.[604]

To my mind, there is no particular reason not to interpret the noun **med-ŏt-* as an original action noun. As has been shown above, the noun **med-ŏt-* is derivationally different from other amphikinetic *t*-stems discussed hitherto, and therefore it may have had different semantics. One should not be troubled by the fact that all these nouns share a single suffix, *-*ŏt-*, since polysemy of derivational morphemes is not so

[604] See also Melchert (1983: 15-6) and Nussbaum (2004a).

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uncommon in the Indo-European languages, cf. the derivatives with the Indo-European suffix *-*eto*-,[605] the Lithuanian verbal suffix -*au*-,[606] E. *wit-ness* vs. *white-ness*, etc. Thus, it should not strike anybody as abnormal that the Indo-European amphikinetic *t*-stems could have different semantics.

Additional support for the reconstruction of action noun semantics for **med-ŏt-* can be drawn from the Anatolian *att*-formations, many of which have the semantics of action nouns, cf. Hi. *tarnatt-* 'distribution', *karšatt-* 'cutting', *duškaratt-* 'rejoicing, joy', or are best interpreted as concretized original action nouns and not as concrete nouns from the very beginning, cf. Hi. *aniiatt-* 'performance' (← *'performing', cf. verbal stem *aniia-* 'perform'), CLuv. *ariiatt(i)-* 'mountain' (← *'raising' ← *ariia-* 'raise'),[607] etc. Although all of these Anatolian examples shown here are non-primary formations,[608] and probably were formed at the dialectal stage, they apparently must have been built according to some already existing pattern. While forms like *tarnatt-*, *aniiatt-*, *duškaratt-* and *ariiatt(i)-* are ambiguous (**R-S-ot-* or **R-S-o-t-*), the structure of the noun *karšatt-*, built to the athematic *karš-* 'cut (off)', would point towards a protoform **R-ot-* (or **R-S-ot-*), comparable to the structure of amphikinetic formations. Also Hi. *karuiliiatt-* 'previous state, old age', built to the athematic adjectival stem

[605] See Vine (1998: 10-12, 17-21, et passim).
[606] The suffix -*au*- can be used in order to derive *a*) verbs meaning 'be *X*', cf. *valkatáuti* 'be a vagabond' (← *válkata* 'vagabond'), *elgetáuti* 'be a beggar' (← *elgeta* 'beggar'); *b*) verbs meaning 'collect/hunt *X*', cf. *uogáuti* 'collect berries' (← *úoga* 'berry'), *žuváuti* 'fish' (← *žuvìs* 'fish'); *c*) verbs meaning 'engage in *X*', cf. *atostogáuti* 'be on vacation' (← *atóstogos* 'vacation'), *pamoksláuti* 'preach' (← *pamókslas* 'sermon'); *d*) verbs with iterative meaning, cf. *rékauti* 'shout repeatedly' (← *rékti* 'shout'), etc.
[607] Rieken (1999: 101-2), Melchert (1993: 26-7). The Luvian form has been morphologically reformed.
[608] As was stressed by Hajnal (2004: 110).

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karuili- 'old' points towards the suffix **-ot-* rather than a plain **-t-*. If the suffix in all these derivatives had consisted of a plain *-t-*, *karuiliiatt-* would have probably been formed as **karuilit-*.

Ironically, primary formations built according to this pattern are extremely sparse in Anatolian. Not a single one is known from Luvian, and the two Hittite examples adduced by Rieken, viz. *garāt-* 'entrails; appearance' and *ekt-* 'hunting net', do not have the same shape as the *att*-formations shown above. Some problematic issues concerning the development of these nouns will be briefly discussed in the following section.[609]

V.10.3. Hi. *garāt-* 'entrails; appearance' (?) and *ekt-* 'hunting net'.

The Hittite noun *garāt-* is attested since the Old Hittite period, cf. acc. sg. *ka-ra-a-ṭa-an* (KBo XXII 2 Vs. 16; OH/OH).[610] The reconstruction of the Hittite noun *garāt-* as an original Indo-European amphikinetic **ĝher-ōt-* and the genetic comparison of the root **ĝher-* with Gk. χορδή, Lith. *žárna* (or *žarnà*), Latvian *zarnа̃*, OIc. *gǫrn* 'intestine', OIc./OHG *garn* 'yarn', etc.,[611] based on the Hittite textual evidence is quite attractive. However, several problematic issues remain. First, it has to be determined whether the root **ĝher-* is verbal or nominal. In IEW, this root is considered to be

[609] Also see Rieken (1999: 139-46).
[610] Four Old Hittite nominative singular forms are attested in New Hittite copies of Old Hittite texts, cf. *ka-ra-az-* (KUB XLI 23 II 19′), *ka-ra-az-za-* (KUB XLI 23 II 21′), *ga-ra-az-* (KUB LVII 86, 2′, 5′, 7′), *ka-ra-a-az-* (KUB XVII 10 II 14′; see Rieken [1999: 141-2] for several additional late Hittite and plural forms).
[611] See Rieken, loc. cit.

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nominal, meaning exactly 'intestine'.[612] If this were true, the *t*-stem **ĝher-ōt-* would be structurally more similar to the noun **méh₁nōt-*, which was built to a nominal stem **meh₁-(o)n-*. However, unlike **meh₁nōt-*, the noun **ĝher-ōt-* would be primary, and isolated as such.

I think that it might be more attractive to interpret the root **ĝher-* in the Hittite noun *garāt-* and its genetic relatives in other dialects as verbal. The Germanic and the Lithuanian examples, both of which are built with the suffix *-no-/-nā-* would support this claim, as quite a few *no*-formations are ultimately verbal formations. Their original meaning is not quite clear, though. Nussbaum considers *no*-formations with a full grade of the root verbal or deverbative nouns, cf. **pet(h₂)-no-* 'wing' ("deverbative noun" ← *pet[h₂]-* 'fly'), and **por-no-* 'id.' ("verbal noun" ← **per-* 'traverse').[613] However, theoretically, both **pet(h₂)-no-* and **por-no-* could also be interpreted as participial/ agentive formations, viz. 'flying' resp. 'traversing' which eventually were nominalized. Whether one chooses the participial, the nomen actionis or any other meaning, often is a matter of preference, cf. PGmc. **ƀar-na-* 'child', PB **ber-na-* 'child, boy' (~ IE **bher-* 'carry'), which can be interpreted as a nominalized action noun **'carrying', a result noun, or a nominalized past participle, **'the carried one, the born one', or cf. Lith. *sie-na* 'wall' (~ IE **seh₂[i]-* 'bind'), which could be interpreted as **'binding' or **'the bound, plaited thing, wickerwork', further OIc. *mein* 'harm', Lith. *maĩ-na-s* 'exchange' (< **moi-no-* ~ IE **mei-* '[ex]change'). Some *no*-formations are adjectives,

[612] IEW.443. Schrijver reconstructs this root as **ĝherH-* on the basis of La. *haruspex* 'sooth-sayer' (<**ĝhr̥H-u-*), Ved. *hirā́-* 'vein' (<**ĝhr̥H-éh₂*), etc., cf. Schrijver (1995: 208). Watkins reconstructs **ĝherh₂-* (Watkins 2000: 30), on which see below in this section.
[613] See Nussbaum (1986: 14).

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cf. OIc. *gjarn* 'eager' (< PGmc. **ger-na-* ~ IE **ĝher-* 'take pleasure in sth.'), Latvian *pẽrns* 'last year's' (< **per-na-*; ~ IE **per-* 'traverse'), etc.[614] These examples show that the Indo-European suffix **-no-* could be used in various formations, such as /R(e)-no-/, /R(o)-no-/, and /R(z)-no-/. Also the semantics of these different *no*-stems were not always the same. The formations of the *R(z)-no-* type were generally (originally: only?) past participles, whereas the semantics of *no*-stems with a full grade in the root varies, and is not easy to classify into strict classes.

Contra IEW and Schrijver (1995), OHG/OIc. *garn*, Lith. *žarna*, Latv. *zarna* and OIc. *gǫrn* could be interpreted not as *no*-derivatives of the nominal root **ĝherH-* 'intestine' (which semantically is quite implausible), but rather as ancient participial/ nominal formations built to some verbal root **ĝherH-*, the meaning of which needs to be clarified.

In LIV, one can find a verbal *seṭ*-root **ĝherH-*,[615] but its meaning 'be wrathful' sounds rather distant. The root **ĝherH-* underlying the nominal formations **ĝhorH-neh$_2$-*, **ĝhorH-no-*, etc. may have been a separate root. The meaning of this root could have been 'twist' or 'curl'. There are several possible and independent reasons for the reconstruction of such semantics. First, intestines are curled in the bodies of both animals and humans. Second, as hinted at in EIEC, the semantics of dialectal derivatives like Gmc. **garna-* 'yarn' or Gk. χορδή 'string' implies twisting, or something twisted.[616] Thus, one could postulate that the original meaning of at least

[614] See Skardžius (1941: 215-9), Krahe/Meid (1967: 103-6).
[615] LIV.178.
[616] EIEC.180.

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*ĝhorH-no-/-neh$_2$- would have been 'twisted' or 'the twisted thing', and the meaning of the root *ĝherH-, respectively, 'twist, curl'.

As for the quality of the root-final laryngeal, it may have been h_1 or h_3. In those word-forms where the root *ĝherH- would been in the *o*-grade, viz., *ĝhorH-, the laryngeal would have been lost via "Saussure's law". The Hittite situation is less clear. In a protoform like *ĝherH-ōt- one would expect the development into *garratt-.[617] However, in the weak cases, where the consonant cluster *-rH- would have (ideally) *not* been intervocalic, the laryngeal would have been lost, and then the root allomorph without the laryngeal could have spread to the strong cases as well. However, all this is extremely speculative, and the truth can be beyond reach now. The reconstruction of h_2 (as per Watkins, loc. cit.) cannot be entirely ruled out, as the development of h_2 in sequences of the type *CR̥HC- in Anatolian is not known. However, h_2 appears to be preserved in sequences of the type *-RHC-,[618] cf. Hi. *tarḫzi* 'overcomes' (3. sg.), *u̯alḫzi* 'strikes' (3. sg.), etc.

If this interpretation is correct and the root *ĝherH- is verbal, the Hittite noun *garāt-* would structurally be comparable to *med-ōt-. However, as has been noted by Rieken, whereas *att*-formations constantly show an unlenited suffix *-att-*, in the noun *garāt-*, the suffix never shows a geminate. The reason for this must have been the synchronic analysis of this noun as /grā́t-/ (< *gŕ̥t-).[619] Since the consonant *t* followed a stressed long vowel, it became lenited.

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[617] See Melchert (1994: 79-80).
[618] See Melchert (1994: 70).
[619] Following Rieken (1999: 143).

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The story is quite different with the noun *ekt-* 'hunting net' (attested since Middle Hittite).[620] The etymology of this noun has been studied by many scholars, and various hypotheses have been put forward.[621] From the multiple etymologies, the one proposed by Hamp (and subsequently accepted by Rieken), whereby *ekt-* is interpreted as a re-make of IE **i̯ek-ōt-* 'hunting' looks at least from the point of view of the root etymology quite attractive.[622] However, if one accepts this reconstruction, one also has to admit that in the paradigm of this noun, not the *o*-grade of the suffix must have been generalized, but rather the zero grade (cf. the nom. sg. *e-ik-za* /ekts/ which implies *ek-t-* + *-s*). This would apparently be different both from what is assumed to have happened to the *att*-abstracta on the one hand and the noun *g(a)rāt-* which shows the zero grade of the root and the lengthened grade of the suffix on the other hand, i.e., is the opposite of *ekt-*.

The proposed reconstruction of the amphikinetic is attractive for both nouns, although the reasons are different. The lengthened grade *-āt-* in *g(a)rāt-* may reflect the amphikinetic suffix *-ōt-*, whereas the *e*-grade in the root of the noun *ekt-*, as well as the attractive connection with the verbal root **i̯ek-* 'to hunt' forces one to interpret *-t-* as a suffix. However, *ekt-* probably cannot reflect an ancient acrostatic formation (like *nekuz*), because whereas *nekuz* is a frozen form without a living paradigm if its own, *ekt-* is a fully inflected noun, and, as far as we can see, ancient acrostatic *t*-stems

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[620] Nom. sg. *ik-za-* (sic Rieken 1999; KBo II 16), *e-ik-za* (KBo XVII 61 Vs. 17); acc. sg. *e-ek-ta-an* (KUB XXXI 68 Vs. 27′; Oettinger has *e-ik-ta-an*, see Oettinger 1976: 22), *e-ek-za-an* (Oettinger, loc. cit. has *e-ik-za-an*; KBo XIII 101 Rs. 6′ [NH]; KUB XLV 26 II 2 [NH]); gen. sg. *ik-ta-aš-* (KBo II 17 2′, 3′, 33′).

[621] For overviews, see M. Poetto (1992: 161-4), Rieken, loc. cit.

[622] See Hamp (1978: 119-20), Rieken (1999: 145).

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usually generalized the *o*-grade of the root. The reconstruction of the amphikinetic pattern for *ekt-* solves the problem of the *e*-grade of the root; however, in such a case one has to deal with the generalization of the zero grade of the suffix. Thus, in conclusion one could say that these two nouns contain various morphological, phonological and etymological problems, and they cannot be used to build any elaborate theories.

V.10.4. IE **magh-ŏt-* 'maidenhood' or 'maiden'

This noun with preserved reflexes only in Celtic and Germanic (see below) has been reconstructed in two different ways: in IEW, it is reconstructed as a *ti*-stem **magho-ti-* (IEW.696) whereas Rieken and Irslinger reconstruct it as a *t*-stem, viz. **magh-ōt-*.[623] In the Celtic languages, the original stem **maghot(i)-* has been secondarily extended, cf. Middle Welsh *machteith*, Old Cornish *mahtheid*, Middle Cornish *maghteth/maghtyth* 'maiden', Brittonic *matez* 'maid', Old Irish *ro-macdact* 'superadulta', Middle Irish *mac(c)dacht* 'youth' (< virtual PIE **maghot-aktā-* or virtual Proto-Celtic **mak^{u}k^{u}ot-aktā/ī-*).[624] The Germanic languages show both forms with secondary suffixes and without them, cf. Gothic *magaþs* (*-þ-*), Old English *mæg(e)þ* and *mægd-en*, Old Saxon *magath*, Old High German *magad* and *magat-īn*, Old Frisian *megith* 'girl, maiden'.

[623] See Irslinger (2002: 62-3); Rieken does not provide the actual reconstruction, but discusses this noun in the chapter about amphikinetic (called "proterokinetic" by Rieken) *t*-stems (1999: 91).
[624] IEW.696, Rieken (1999: 91), Irslinger (2002: 62-3).

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Of the two reconstructions shown above, the reconstruction **magho-ti-* (= **magh-o-ti-*?) is less preferable for several reasons. Although there are attested a certain number of *ti*-stems in Indo-European that are built to verbal thematic stems (cf. Lith. *aũgatis* 'growth', Skt. *dṛś-a-tí-* 'appearance', Greek Λάχεσις [divine name], etc.), and they theoretically could be reconstructed as, resp., IE **h₂eug-o-ti-*, **dṛ̂k-o/e-ti-* and Proto-Greek **lakh-e-ti-*, the reconstruction **magho-ti-* leaves the *i*-less Proto-Celtic **makᵘkᵘot-* without an explanation. Secondly, the fact that in most Germanic languages the reflexes of this noun (i.e. **maghot-*) are declined as *i*-stems is not enough to reconstruct the original noun as such, too. As noted in Irslinger (2002: 62), the Old English noun *mæg(e)þ* is declined as a consonantal noun, whereas in those languages in which it is declined as an *i*-stem, this can be a dialectal innovation. In addition to this, it has to be noted that beside the *o*-stem **magh-o-* (assumed in IEW), there existed a *u*-stem built to the same root, cf. **magh-u-* (for examples, see below in this section). The co-existence of an *o*-stem and *u*-stem in Proto-Indo-European was hardly possible. Whenever a thematic stem exists beside a *u*-stem, it generally results from the thematization of the latter, cf. IE **dór-u* 'tree' → **dr-eu̯-o-m* > PGmc. **trewa-* 'id.', etc. In random instances, the formant *-u/u̯-* is totally gone, cf. Lith. *sūnaĩ* 'sons' (nom. pl. [poet.]; the regular form is *sū́nūs*), but such reformations are certainly very late.

The reconstruction **magh-ōt-* is more plausible, although, as one can see from the dialectal forms adduced above, none of them actually reflects it. Instead, all the forms point towards **magh-ŏt-* with a normal *o*-grade. Still, although the stem **magh-ōt-* has not been preserved anywhere, it is correctly assumed by Rieken and Irslinger that it

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must have existed. The attested stem **magh-ŏt-* can easily be interpreted as an original accusative singular stem of an original amphikinetic *t*-stem.

The reconstruction of an ancient amphikinetic *t*-stem **magh-ŏ̆t-* raises a further question, viz., why the root is not in the more expected *e*-grade, as is common practice with Indo-European amphikinetic formations. The simplest thing to assume here is that this noun was built to a root which had *a*-vocalism and perhaps did not ablaut.

This root **magh-* – with *a*-vocalism – is also attested in some other nominal Indo-European formations, e.g., in the *u*-stem noun **magh-u-* 'boy'[625] (> Go. *magus*, OE *magu* 'boy', dialectally also 'son' or 'servant', cf. OIc. *mǫgr* 'son', OIr. *mug* 'servant', *macc* 'son'),[626] **magh-u̯-ih₂* 'girl' (> Go. *mawi*, OIc. *mær* 'girl'),[627] etc.[628] Theoretically, the Germanic forms could also reflect the *o*-grade of this root, but since Celtic requires an **a*, for reasons of economy also the Germanic forms most probably should be reconstructed with an **a*.

Several etymological interpretations of this root have been proposed. According to Brugmann, the root **magh-* in these nominal formations is the same root as that of the Germanic verb **magan-*, OCS мощтн 'may, be able' (root мог-), Lith. *magėti* 'want', and possibly Gk. μάχομαι 'fight' (1. p. sg.), all of which reflect **magh-* as well.[629] Brugmann did not discuss the semantics of the nouns **magh-u-* and **magh-ot-* in any detail, but his adduced parallels of La. *vir* 'man' ~ *vīs* 'power, strength', Lith. *vaĩkas*

[625] Sic also Szemerényi (1977: 19).
[626] On *-cc-* in OIr. *macc*, see OIG.92f.
[627] OIc. *mær* has been reformed in the same way as all other *ijō*-stems, adopting the ending *-r*.
[628] See more derivatives in IEW.696.
[629] See LIV.422.

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'child' ~ *viẽkas* 'strength' imply a similar relationship between the verbal root **magh-* and its derivatives, whereby **magh-u-* could be interpreted as *'the (physically/ sexually) able one' → *'the young one' → 'boy' (→ 'son', 'servant', etc.).

In IEW, no etymological connection is made between the verbal **magh-* and the nominal formations listed above. Instead, an Indo-European stem **magho-* (i.e., **magh-o-*?) is reconstructed, and assigned the meaning 'young'.[630] The stem **magho-* is not explained in any detail, and no clear evidence for its existence is adduced. The Avestan adjectival stem *mayauua-* 'not married, single' more probably reflects **magh-eu̯-o-* (← **magh-u-*),[631] whereas the Latvian form *mač* 'small' (< *maǵš*) and the Albanian *makth* 'young hare' are not entirely clear, and may or may not be related.[632] Other problems related to the reconstruction of IE **magh-o-* were discussed earlier in this section.

Of the two opinions, the former one (i.e. the one presented by Brugmann) is more attractive, as, first of all, admitting that the nominal **magh-* formations have a verbal root, one would not have to reconstruct two independent roots **magh-* with somewhat similar semantics and rather rare structure (viz., *a*-vocalism instead of the much more common *o/e*-vocalism). Secondly, the semantic development from 'be able' to 'young' (← 'being able; possessing ability, physical/sexual potency') can be easily accounted for, as has just been shown.

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[630] IEW.696. The *ti*-stem **magho-ti-* discussed above would have been built to this thematic stem.
[631] A putative **magh-o-u̯o-* both looks morphologically awkward, and would have probably yielded **mayāuua-* in Avestan.
[632] See LIV.696.

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However, there also exists a third etymology of the root **magh-*. According to Lehmann, the root **magh-* in the formations **magh-u-*, **magh-ŏt-* and the verbal **magh-* are probably not related to each other.[633] Lehmann considered the root of **magh-ŏt-* to be unclear, whereas the term **magh-u-*, according to him, was an Indo-European term for a 'son of the maternal lineage' (as opposed to **suH-nu-* 'son of the paternal lineage'). This theory is quite improbable, both because of its *ad hoc* flavour, and because the meaning 'son' can very easily develop secondarily from an older meaning 'boy' which was shown above, and which is in favour of Brugmann's etymology.

V.10.4.1. Semantic Analysis

The semantics of the Indo-European noun **magh-ŏt-* are not discussed in detail anywhere. In IEW, the reconstructed noun **magho-ti-* is assigned the meaning 'young femininity' ('junge Weiblichkeit'), whereas Rieken and Irslinger do not talk about the original semantics at all. The attested dialectal forms point towards the meaning 'maiden, girl' or 'maid' (see the examples adduced in the previous section). How should all these meanings be reconciled? Which of them is the original meaning of the *t*-stem **magh-ŏt-*?

The meaning 'maid' (i.e. 'servant') is very unlikely to have been the original meaning of this noun. As one can see, this meaning is only attested in Brittonic,

[633] Lehmann (1986: 240). Here he also dismisses Brugmann's interpretation, although he does not provide any arguments for this dismissal.

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whereas in other languages it usually means 'girl, maiden' (except Irish, see the examples shown above). Also, as has been said earlier, the meaning 'maid, servant' can easily develop out of the meaning 'girl', cf. also E. *boy* 'child of masc. gender' and 'servant', or Lith. *mergà* 'female servant, female hired labourer in a farm' (← 'girl', cf. Lith. *merg-áit-ė*, *merg-in-à*, *meřg-ė* 'girl'), etc.

It is more difficult to choose between the other two meanings, viz. 'maiden, girl' and 'junge Weiblichkeit' (probably better interpreted as 'maidenhood'). The meaning 'maidenhood' is only attested in Irish, and, as was shown above, the Irish noun has been extended with a secondary suffix *-*aktā*-. In all other languages, the meaning of this noun is 'maiden, girl'. I am inclined to believe that at the stage immediately preceding Proto-Germanic and Proto-Celtic, the meaning of the noun **magh-ŏt*- would have been 'maiden, girl'. Such a reconstruction would not raise any structural or semantic problems, and would even have a parallel, cf. IE **nép-ŏt*- 'offspring' (→ 'nephew; grandson'). However, it is not impossible that an older meaning of this noun could have been 'maidenhood', as assumed in IEW. The meaning 'maiden' could have developed secondarily from an original 'maidenhood' (IEW.696). Such semantic changes are indeed attested, cf. E. *youth* 'young age' and 'young people, young person', G. *Jugend* 'young age' and 'young people', La. *juventūs* and *juventa* 'young age' and 'young people, young men', Lith. *jaunìmas* 'young people' and 'young age', etc. In a similar fashion, one could assume that a more archaic meaning of the noun **magh-ŏt*- would have been 'maidenhood', and in the course of development it would have been transformed into 'maiden, girl'. Although the reconstruction of the original

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meaning as 'maidenhood' makes the semantic development in Celtic very complex, cf. IE **magh-ŏt-* 'maidenhood' → 'maiden' → PC **magot-aktā-* 'maidenhood' → (dialectally) 'maiden', cf. MW *machteith*, OCorn. *mahtheid* 'virgo', etc., there are certain reasons why this theory still is worth considering.

In Indo-European there existed another name for a 'girl', built to the same root, viz. **magh-u̯-ih₂*. Since it is an obvious feminization of masc. **magh-u-* 'boy', the meaning of **magh-u̯-ih₂* must have certainly been 'girl'. If one wanted to give the Indo-European noun **magh-ŏt-* the meaning 'girl', it may be asked why Indo-European would have possessed two nouns with an identical meaning, but built in rather different ways? This question can be answered in two ways: either Indo-European indeed possessed two synonymous words **magh-ŏt-* 'girl' and **magh-u̯-ih₂* 'girl' from the very beginning, or that one of these nouns had a different meaning. In this case, the uncertain noun would apparently be **magh-ŏt-*.

The meaning 'maidenhood' for the *t*-stem **magh-ŏt-* can be easily accounted for; however, a series of semantic changes need to be assumed. Since the noun **magh-ŏt-* is built to a verbal root, i.e. in the same way as the nouns **med-ŏt-* and **leuk-ŏt-* discussed in sections V.10.2 and V.10.5, also its original semantics may have been similar. This being the case, **magh-ŏt-* should be interpreted as an action noun, 'being able', cf. IE **magh-* 'be able'. In the course of development, this action noun underwent several semantic changes, essentially quite similar to the ones assumed for the *u*-stem **magh-u-* (see previous section). Thus, **magh-ŏt-* 'being able' would have

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first been narrowed down to 'being physically/sexually able'. From this step, the meaning of this *t*-stem could have been further transformed to first 'young' (cf. **magh-u-*), and then to 'sexual/reproductive ability', from where it would be already a short way to 'womanhood' (and thence, to 'girl'). However, it is quite possible that the step 'young' could have been skipped. The exact details of this chain of semantic changes are now not easy to retrieve.

To summarize, it may be said that *t*-stem status for the noun **magh-ŏt-* is secure, despite the different reconstruction in IEW. According to such an interpretation, the late pre-dialectal Indo-European meaning of this noun would have probably been 'girl, maiden'. However, the original meaning of this noun may have been 'maidenhood' (as assumed in IEW).

In a way, the reconstruction of the original meaning of **magh-ŏt-* as 'maidenhood' rather than 'maiden' could prompt one to re-evaluate the reconstruction of the original meaning of the noun **nép-ŏt-* 'offspring' as well, but since the meaning of the root **nep-* is not known, any hypotheses about it would be extremely speculative.

V.10.5. IE **léuk-ŏt-* 'light; dawn, morning'

The reconstruction of a Proto-Indo-European amphikinetic *t*-stem **léuk-ŏt-* is controversial. It is probably reflected in the Germanic branch, cf. the Gothic noun *liuhaþ* 'light', as well as OE *lēoht*, OHG/OS *lioht* and OFr. *liāht* 'id.'.[634] One could say that also the Hittite word-forms *luk(k)at(t)i* and *luk(k)at* 'in the morning / at dawn' are

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reflexes of this Proto-Indo-European noun, and they have been interpreted by some scholars as such.[635] The Hittite forms would be reflecting **luk-ot-*, a version of the stem **leuk-ot-* with the zero grade of the root generalized.

The stem **leuk-ot-* reconstructed on the basis of the Germanic and the Hittite evidence could be interpreted as an Indo-European amphikinetic *t*-stem built to the verbal root **leuk-* 'shine' with the help of the nominalizing suffix *-ŏt-*. Since the root of this noun is verbal again, the original meaning of the noun **leuk-ŏt-* could be interpreted as having been 'shining' (action noun). The original meaning 'shining' would have eventually developed into a concrete/result noun 'light' in the Germanic language branch and 'morning, dawn; morrow' in Hittite.

However, according to a more wide-spread opinion, the Hittite word-forms *luk(k)at* and *luk(k)at(t)i* which occur as such only in late Hittite (New Hittite), do not represent anything archaic, although, concerning the question what it is that they represent, it is not always clearly explained. In the following paragraph, I will provide a brief survey of various interpretations.

H. Kronasser considered the forms *luk(k)atta* and *luk(k)at(ti)* to be ancient dative/ locative forms of a noun which eventually became confused with forms of a similar-looking verb *lukkatta*.[636] Without providing a more detailed account, H. M. Kümmel treated *lukkatta* and *lukkatti* as a single item and translated it as 'bei Tagesanbruch' (E.

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[634] All these Germanic nouns are neuter *a*-stems from the synchronic point of view.
[635] Indecisively B. Hrozný (1917: 182), considering it an adverb from the synchronic point of view (NB that he also translated it as 'afterwards, [t]hereupon'). Krahe/Meid (1967: 141), Puhvel (see HED I-II.435), Leukart (1994: 169) and Tischler (2001: 95) considered it to be a noun. Irslinger calls it a "nominalized adjective" (Irslinger 2002: 44, fn. 54).
[636] Kronasser (1966: 347).

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'at daybreak'); yet he assumed verbal origin.[637] Similarly, without expanding on his interpretation, E. Neu made a distinction between the verbal form *lukkatta* and the denominal adverb *luk(kat)ta/lukkatte/lukkatti*.[638] Melchert interpreted the form *lukat* as a late innovation, without providing a more detailed explanation.[639] In a number of works, the *i*-form *lukkatti* is not considered at all, cf. Kammenhuber (1961), A. Götze (1967), whereas Rieken (1999) and Irslinger (2002) do not discuss any of the Hittite forms shown above.

Only in CHD, the description of this problem is more or less clear: in CHD, it is assumed that the adverb *lukkatta* 'at dawn, in the morning' (somehow) derives from the verb form *lukkatta*, the middle form of the verb *lukzi* 'shines' (3. p. sg.), whereas the adverb *lukkat(ti)* 'id.' is a late Hittite innovation, a reinterpretation of a nominal dat./loc. form *lukkat(ti)*, belonging to a noun **lukat-*.[640] It remains, however, unexplained in CHD what exactly the noun **lukat-* is, and how it developed. Neither is it explained how the verb *lukkatta* became an adverb in Old Hittite.

The question of *luk(k)atta* and *luk(k)at(ti)* will be briefly discussed in the following section.

V.10.5.1. Hi. *luk(k)atta* and *luk(k)at(ti)*

In order to illustrate the usage of the word-forms *luk(k)atta luk(k)at(ti)* in Hittite, a number of sentences in which these word-forms occur are presented below. The

[637] Kümmel (1967: 10-11, 32-3).
[638] Neu (1968: 109-10).
[639] Melchert (1994: 149).
[640] CHD.76-7.

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sentences are classified into five groups: in the first group, the word-form *lukkatta* is a verb, in the second group, *lukkatta* is used as an adverb, in the fourth group, the word-form *lukkatti* is used as an adverb, in the third group, a couple of examples are shown where in different copies of the same text, either *lukkatta* and *lukkatti* occurs (both used adverbially):

1. *lukkatta* as a verb: *maḫḫan-ma lukkatta na-aš-kán*...'when [it] dawns, X...';[641] *maḫḫan-ma-áš lukkatta nu INA* VII *MUŠI*... 'when [it] dawns, for seven nights X...';[642] *maḫḫan-ma lukatta nu ú̯ātar e[kuu̯anzi]*... 'when [it] dawns, they d[rink] water...';[643] *maḫḫan-ma lukkatta nu INA 5 MUŠI penniškizzi* 'when it dawns, he drives [the horses] for five nights';[644]

2. *lukkatta* as a temporal adverb (← verb? Cf. 1.): *lukkatta-ma anda* ^{URU}*Kuliu̯išna unnāi* 'but in the morning / at dawn [he] drives into K.';[645] *lukkatta-ma-kán* ^{d}UTU-*uš upzi*... 'but in the morning / at dawn, the Sun-god rises...';[646] *maḫḫan-ma-aš-kán lukkatta IŠTU* É ^{LÚ}IŠ[647] [*parā*] *u̯u̯adanzi*... 'when on the following day they bring them out of the stable...';[648] *lukkatta-ma-aš namma tūriu̯anzi* 'in the morning / on the following day they hitch them up again';[649] *maḫḫan-ma-aš lukkatta tūriu̯anzi na-aš* 2 DANNA *pennai* 'when in the morning they hitch them up, he drives them two miles';[650] *maḫḫan-ma lukkatta u̯*[*ātar*] *akuu̯a*[*nzi*] *na-aš tūriu̯anzi* 'when in the morning they [the horses] drink water, they hitch them

[641] KUB I 13 Vs. I 37.
[642] KBo III 5 Vs. II 49-50.
[643] KUB I 11 Rs. III 49.
[644] KBo III 5 Rs. III 29.
[645] KBo 15.34 ii 17-21.
[646] KUB 7.1 + KBo 3.8 ii 25.
[647] Read "É ^{LÚ}KUŠ$_7$" as per Tischler (2001: 220)?
[648] KUB I 13 iv 71.
[649] KBo III 5 Vs. I 21.
[650] KBo III 5 Rs. III 47-8.

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up';[651] *nu* GIM-*an* *lukkatta* ^{d}UTU-*uš-kan kalmaraz uit* ^{l}*Kiššiš* [*šuppi*]*ia-az*[652] *šaštaš*[653] *arāiš nu-za* GE_6-*andaš tešḫuš* [EGIR-*pa ANA*] AMA-*ŠU memiškiuan dāiš* 'when[654] in the morning / at dawn the Sun god came from [...],[655] Kešši arose from the clean[656] bed, and started to tell his mother [his] dreams of the night';[657]

3. *lukkatta* or *lukkatti* as temporal adverbs: *lukkatta-ma-kán* UD-*az ištarna paiz*[*zi*]... 'on the next day, [when] the day reaches the middle (i.e. noon)...',[658] *na-aš-kán* *lukkatti* DINGIR.MEŠ-*aš* [*adann*]*aš meḫuni ḫūdāk aru* 'and in the morning, during the gods' eating time let him quickly arrive',[659] etc.

4. *lukkatti* (← noun?) as a temporal adverb: *nu-ua* *lukkatti* [*kiššan na-aš-ma-ua ki*]*ššan*... 'in the morrow / tomorrow, will it be this way or this way?';[660] UN-*aš-ua* *lukkatti* EGIR-*anda uiddu* 'let a man come after me in the morrow / tomorrow';[661] *kinun-aš* *lukkatti* KASKAL-*aḫta* 'now he took the road at dawn / in the morning'; *lukkatti-ma-za* LUGAL-*uš* SISKUR [*ḫalalē*]*nzi* DÙ-*zi* 'but in the morning, the king performs the pure rituals';[662] ...*nu* *lukkatti* UD-*ti ANA*

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[651] KUB I 11 Vs. II 1-2.
[652] Sic Friedrich (1960a: 157). In Friedrich (1960b: 55), the text is to be read [*ša-ne-iz-zi-*]*ia-az* 'sweet, aromatic (?)'.
[653] Genitive *šaštaš* instead of the expected ablative **šaštaz*, cf. Friedrich (1960a: 157, 1960b II. 68).
[654] In CHD.75, instead of GIM-*an* *lukkatta* the text is reconstructed as G[E_6-*anza l*]*ukkatta*, which semantically and syntactically seems awkward.
[655] Friedrich (1960a: 157) translates *kalmaraz* as 'Berg', i.e., 'Als nun am nächsten Morgen der Sonnengott aus dem Berge (?) kam...'; Tischler (2001: 70) translates it as 'sun-ray', which makes the meaning of this passage a little awkward, cf. '...when the Sun god came from the sun-ray...' (i.e., 'started to shine'?).
[656] See fn. 654 above.
[657] KUB 17.1 ii 14-6.
[658] KBo 23.1 + KUB 30.38 i 58. Var. *lukkatti* (CHD.77).
[659] KUB 13.4 iii 71-3. Var. *lukkatta* (CHD.77).
[660] KUB 6.43 5-7.
[661] This and the following examples are from KBo 9.82 obv. 4-9.
[662] KUB XXIV 5 + IX 13 Vs. 29.

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EZEN-*KA* *eḫu* '[in/on] the dawning day (i.e. tomorrow's day, tomorrow) come to your festival' *or* 'in the morning [on the next] day come to your festival'.[663]

As can be seen from 1., when the conjunctions *maḫḫan* and *mān* are used, the word-form *lukkatta* functions as a verb in a subordinate temporal clause. The forms *lukkatti* and *lukkat* do not occur in such sentences. However, in the sentences listed under 2., the word-form *lukkatta* cannot be interpreted in the same way, as it does not appear to have a subject. The word-form *lukkatta* is clearly used as an adverb here. In some of these examples, *lukkatta* occurs in a subordinate temporal clause as well, but this time, it functions as an adverb, whereas the verb of the clause is a different word, viz., *akuuanzi, tūriianzi, uuadanzi.*

In the sentences listed under 4. (all New Hittite),[664] the word-form *lukkatti* is used, and *lukkatta* does not occur. The function of the occurring *lukkatti* is clearly adverbial. The word-form *lukkatti* looks certainly non-verbal, but rather nominal, cf. *šiuatti* 'day' (dat./loc.), *aniiatti* 'work, performance' (dat./loc.).

In the sentences listed under 3., both the "deverbative" *lukkatta* and the "denominal" *lukkatti* occur interchangeably, which implies that their meaning was equivalent.

We face the following distribution: the verb *lukkatta* and the adverb *lukkatta* existed both in Old and New Hittite, but the adverb *lukkatti* only in New Hittite. Now, the question is what the adverb *lukkatta* represents historically. If one wanted to do

[663] KUB 36.90 obv. 18-9.

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away with the theory according to which the adverb *lukkatta* would have developed directly from the verb *lukkatta*, one could argue that the adverb *lukkatta* is an old directive of an otherwise unattested *t*-stem noun **lukkatt-*. The reconstruction of an old *t*-stem noun could be supported by the Germanic evidence, viz. Go. *liuhaþ* 'light', as well as the West Germanic forms, listed in section V.10.5 above.

However, this is hardly correct, since directives are generally not used in the temporal sense in Hittite.[665] Instead, one generally makes use of the dative/locative, cf. *nekuz meḫuni* 'in the evening', DINGIR.MEŠ-*aš adannaš meḫūni* 'during the meal of the gods' (KUB XIII 4 III 72-3), or accusative, cf. *nekuz meḫur-ma-aš namma tūriyanzi* 'but in the evening, they hitch them [the horses] up again' (KUB I 13 Vs. II 5), *nu* GE_6-*an ḫumandan uzuḫrin* ḪÁD.DU.A *pitalun azzikkanzi* 'then the entire night they keep eating pure dry [grass] (i.e. hay)' (KBo III 5 Vs. I 19-20). It is quite unlikely that a noun **lukkatt-* existed in Old Hittite at all, as it would have most probably been used, either in the dative/locative, or the accusative case. Therefore this implies, that first, the Old Hittite adverb *lukkatta* must derive from the verb *lukkatta*, and second, that the adverb *lukkatti* is not a dat./loc. form of a really existing noun (see below).

Let us first consider the question how the verb *lukkatta* could start being used adverbially. The key here probably are the subordinate temporal clauses as listed under 1. above, in which the the phrase *maḫḫan lukkatta* 'when it dawns' probably came to be used in an adverbial sense 'in the morning'. In the course of time, the word-form *lukkatta* became lexicalized as an adverb 'in the morning', whereby the necessity to add

[664] CHD.76-7.

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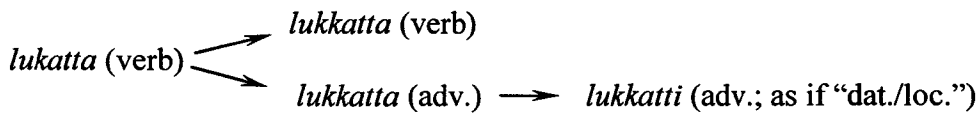
maḫḫan to it became smaller and smaller, until eventually *maḫḫan* was entirely dropped. It may well be, that *lukkatta* started being used adverbially without *maḫḫan* in those subordinate temporal clauses where besides *lukkatta*, there was a verb that required *maḫḫan* as well. Theoretically, the word *maḫḫan* would have been used twice in such clauses, cf. **maḫḫan$_i$-ma maḫḫan$_j$ lukkatta$_j$ u̯ātar$_i$ akuu̯anzi$_i$ na-aš tūriia̯nzi* (lit.) 'but when$_i$ when$_j$ it$_j$ dawns$_j$ they$_i$ [= the horses] drink$_i$ water$_i$, they [= the men] hitch them up'. In order to avoid using *maḫḫan* twice consecutively, one would have been dropped, thereby strengthening the adverbial sense of *lukkatta* even more. In the course of time, *lukkatta* started being used adverbially alone also in sentences in which *maḫḫan* did not occur at all, cf. *lukkatta-ma-aš namma tūriia̯nzi* 'in the morning / next morning they hitch them up again'.[666] The new adverb *lukkatta* was not a black sheep among other Hittite adverbs, as there were more adverbs with a similar shape in the language, cf. *zilatiia̯* 'in the future', *apiia̯* 'then', *menaḫḫanda* 'in front of', *araḫz(and)a* 'outside, around'.

At later stages of Hittite, the adverb *lukkatta* 'in the morning' became re-interpreted as a nominal form, based on its similarity to fairly numerous *att*-abstract nouns, many of which were built to verbal roots (or stems), cf. *tarnatt-* 'distribution' (cf. verbal *tarna-*), *karšatt-* 'division' (verbal *karš-*), etc. Since temporal sense is not seldom conveyed by the dative/locative case in Hittite (cf. *nekuz meḫuni* 'in the night-time, at night', *adannaš meḫuni* 'during the time of eating', *išpandi* 'at night' etc.), probably

[665] The only exception that I have found is *appa-šiu̯atta* 'in the future', which consists of two old directives, both frozen.
[666] See group 2. above.

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also *lukkatta* 'in the morning' was reformed as a "dative/locative" *lukkatti* with a more regular dative/locative ending *-i*. However, from this new dative/locative *lukkatti*, a full new paradigm of a *t*-stem noun **lukkatt-* 'morning' was never back-formed, and one should not assume that such a noun ever existed in Hittite. The development from the verb *lukkatta* to the adverb *lukkatti* can be presented in the following scheme:



The development of the Hittite adverbs *lukkatta* and *lukkatti* presented in this way excludes the possibility that *lukkatti* corresponds to Go. *liuhaþ*, as *lukkatti* appears not to be inherited from Proto-Indo-European. Having eliminated *lukkatti*, the Germanic forms remain the only evidence for a putative Indo-European *t*-stem **leukōt-*.[667]

It must be noted that if one assumes that *lukkatti* is verbal by origin, and that it is not a dative/locative form of an unattested noun **lukkaz* (stem **lukkatt-*), this leaves us with no Hittite noun for the concept of 'morning'. However, this is not a serious problem, as Hittite did not really have the noun 'evening' either. The phrases *nekuz meḫur* as well as *nekuz meḫuni*, which are generally translated 'in the evening', literally mean 'at the time of darkening, at the time when it becomes dark'. This is curiously similar to *maḫḫan lukkatta* 'when it dawns', which, as is commonly believed, could also be used in Hittite in the meaning 'in the morning'.[668] Thus, if *nekuz meḫuni* and

[667] The history of the Germanic forms will be investigated in the following section.
[668] See multiple Hittite sentences above.

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nekuz meḫur are not counted, we are only left with the words for 'day', viz. *šiu̯att-* (= UD[KAM]), and 'night', viz. *išpant-* (= GE_6[KAM]). It may well be that, unlike many modern European languages which have four nouns meaning 'morning', 'afternoon/ day', 'evening' and 'night', Hittites distinguished only two periods, namely 'day/light' and 'night/dark'. What in many other languages is called 'morning' and 'evening', was considered by Hittites as transitional periods between the light period, i.e. 'day', and the dark period of 'night'. Accordingly, these transitional periods were called *maḫḫan lukkatta* 'when it dawns, becomes light' resp. *nekuz meḫur* 'the time of darkening'.

V.10.5.2. IE **léuk-ot-* in Germanic

In this section, I am going to study the development of the East and West Germanic terms for 'light', introduced in section V.10.5 above, viz. Go. *liuhaþ* (*-d-*), OHG/OS *lioht*, OE *lēoht* and OFr. *liāht*. Instead of the formant *-t-*, the North Germanic word for 'light' exhibits a formant *-s-*, cf. OIc. *ljós*, OSw. *liūs* 'light' (< PGmc. **leuχsa-*). A few words about the possible origins of this distribution will be said below in this section.

From the synchronic point of view, all the West and the East Germanic nouns listed above are thematic neuter nouns, and it might be tempting to reconstruct them mechanically as pre-West Germanic **leuk-to-* (> WGmc. **leuχta-* > OHG *lioht* etc.) resp. pre-East Germanic **leuk-o-to-* (> EGmc. **leuχaþa-* > Go. *liuhaþ*).[669]

One could interpret the West Germanic stem **leuχta-* in a similar way as OHG *kind* 'child' (< **kinþa-* < virtual **ĝénh₁-to-*) or OIc. *hljóð* 'sound' (< **hleuð/þa-* < virt. **k̂leu-to-*), whereas Go. *liuhaþ* could be structurally compared to such formations as

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Go. *naqaþ-* 'naked' (< **nog^{u}-o-tó-*),[670] Lith. *gyvãtas*, OCS **животъ** 'life' (< **g^{u}ih$_3$-u̯o-tó-*), Lith. *plẽpatas* 'blabber, who speaks much' (~ *plepéti* 'blab'), etc.

However, although phonologically plausible, such reconstructions raise a number of unattractive implications. Such an account would imply that the West and the East Germanic nouns share only the root, but otherwise are independent formations, the West Germanic noun being formed directly from the verbal root **leuk-*, and the East Germanic noun being formed from the thematic stem **leuk-o-* (otherwise unattested in Germanic). Although, in itself, it is not impossible that these two words have independent origins, below in this section an alternative way to interpret them, according to which **leuχta-* and **leuχaþa-* could be derived from a single source, will be discussed.

The comparison of PGmc. **leuχta-* and **kinþa-* shown earlier, is at second sight neither clear nor really attractive. Whereas the nouns **kinþa-* 'child' and **hleuđa-* 'sound' have some participial sense, and originally probably meant 'the born one' resp. 'the heard one', for the noun **leuχta-*, any traces of participial sense are hard to demonstrate. If one assumed that the original meaning of **leuχta-* was 'illuminated' (past ptc.) vel sim., it would be impossible to derive the meaning 'light' from it. Often, the noun 'light' is originally formed as an action noun 'shining', built to the verbal root **leuk-* 'shine' (or any synonymous root, e.g. **bheh$_2$-* or **ĥu̯ei[t]-*)[671] with the help of

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[669] A protoform **leuk-o-to-* is assumed by Brugmann in Grundriß 401.
[670] Krahe/Meid (1967: 142).
[671] The root reconstructed above as **ĥu̯ei(t)-* often is reconstructed as **ĥu̯eit-*, cf. EIEC.641, LIV.340, etc. However, Proto-Germanic **hwīta-* 'white' clearly requires a root with a voiced unaspirated *d*, cf. **ĥu̯eid-*. Somewhat unclear is Lith. *šviesà* 'light', which might reflect **ĥu̯o/ei-s-*, or perhaps **ĥu̯o/eit-s-* or **ĥu̯o/eid-s-*. In any case, Germanic implies that the more primitive

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various suffixes, or as a root noun, cf. Ved. *róc-as-*, Av. *raoc-ah-* (< **léuk-es-*), La. *lūx* (< **lo/euk-s*), *lūmen* (< **leuk-s-men-*), Gk. φάος (< **bhéh$_2$-os*), OCS свѣтъ (< **k̑u̯oit-o-*), etc.

Otherwise, if the stem **leuχta-* (< **leuk-to-*) is not an ancient participle, or does not somehow derive from an ancient participle, it becomes quite obscure.

The reconstruction of **leuk*(*-o*)*-to-* as the ancestor of Go. *liuhaþ* raises certain problems, too. First of all, the comparison with Go. *naqaþs* is not unanimously accepted. Although nouns often secondarily derive from adjectives, *naqaþs* is an adjective with no parallel noun **naqaþ* 'nakedness' vel sim., and *liuhaþ* is a noun without a parallel adjective **liuhaþs* 'bright', creating an awkward asymmetry. Furthermore, based on the comparison with the Latin adjective *nūdus* 'id.' (< **nog^{u}-e-dho-* or **nog^{u}-o-dho-*),[672] also the Gothic adjective *naqaþs* (obl. stem *naqad-*!) can be alternatively reconstructed as **nog^{u}-o-dho-*. The Old Irish word-form *nocht* 'naked', which provides evidence for **-to-*, is most probably of little value for the analysis of Go. *naqaþs*, since *nocht* continues **nog^{u}-to-*, not **nog^{u}-o-to-*, and thus is morphologically different.

An alternative reconstruction of Go. *naqaþs* can be found in Heidermanns, who reconstructs it as IE **nóg^{u}-ot-o-*.[673] This reconstruction of a thematized ancient amphikinetic *t*-stem (with the *o*-grade in the root) is Heidermann's further development

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shape of the commonly reconstructed root **k̑u̯eit-* was ***k̑u̯ei-*, whereas the elements *-t-* and *-d-* should be interpreted as secondary additions. Whether these elements *-t-* and *-d-* were tense markers or derivational suffixes, is now difficult to determine, but the latter can be true at least for the formant *-t-*.

[672] Lehmann (1986: 263, with further refs.), Sihler (1995: 56).

[673] See Heidermanns (1993: 420).

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of Seebold's idea that a number of Germanic *þ/đ*-formations reflect ancient Indo-European amphikinetic *t*-stems.[674] However, because of the implausible structure of **nóg^{u}-ot-o-*, as well as the lack of any comparative evidence for the existence of an ancient amphikinetic **nóg^{u}-ot-*, this reconstruction is *ad hoc*.

Returning to Go. *liuhaþ*, although the existence of **-o-to-* and similar formations in Indo-European does not need to be doubted,[675] the reconstruction of **leuk-o-to-* for *liuhaþ* is still not the best choice. If one looks at the semantics and the structure of Lith. *gyvãtas* and OCS **животъ** 'life', they are most likely deadjectival abstracta, and originally meant 'the state of being alive' (thus, from the derivational point of view, the suffix *-to-* would have been added to an adjectival stem **g^{u}ih$_3$-u̯o-*). In Lithuanian, where this derivational pattern acquired some vitality, many *-o-to-* derivatives are agent or result nouns, and in most cases are built to contemporary thematic stems, cf. Lith. *mãskatas* 'person who follows others, dangles after others' (~ *maskéti* 'run'), *šìkatas* 'who defecates much' (~ *šìkti* 'defecate'), *plẽpatas* 'blabber, who speaks much' (~ *plepéti* 'blab'), *lùpatas* 'peel; rag' (~ *lùpti* 'peel'), etc.[676] Go. *liuhaþ*, which is neither an agent noun, nor is built to a known Proto-Germanic thematic stem (**leuχ-a-* is not attested), appears to be different from every other *-o-to-* formation, and this makes the reconstruction **leuk-o-to-* less attractive than before.

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[674] See sections V.8.1 and V.8.2.
[675] Cf., for example, **-u̯o-to-* formations, such as Proto-Balto-Slavic **gī-u̯o-to-* 'life' (< **g^{u}ih$_3$-u̯o-to-*).
[676] See Skardžius (1941: 335).

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Thus, to summarize the preceding discussion, the derivation of East Germanic **leuχaþa-* and the West Germanic **leuχta-* from two independent Indo-European formations, **leuk-o-to-* resp. **leuk-to-*, is not very appealing.

Alternatively, one could try to derive the two forms from a single ancestor. In the following section I would like to discuss a different interpretation of these Germanic nouns.

V.10.5.2.1. The Differences between the Development of IE **léuk-ot-* in West and East Germanic

A different, and to my mind more plausible possibility would be to assume that the thematic vowel must have been added to a more archaic *t*-formation, viz. **leuk-ot-/*leuk-t-* + *-o-*. According to such an interpretation, one could reconstruct an ancient amphikinetic *t*-stem **léuk-ot-/*luk-t-*, from which it would be possible to derive all the East and West Germanic forms listed at the beginning of the previous section.

The idea that the Germanic nouns **leuχta-* resp. **leuhaþa-* are reflexes of an ancient amphikinetic noun **leuk-ot-* is not new anymore. However, the discussion is often extremely brief, and the details of the derivation and the development are usually omitted or presented in a rather confusing way.[677]

In the case of Go. *liuhaþ*, the development was along these lines: either, still at the pre-Germanic stage, the accusative stem **leuk-ot-* was extended with the characteristic

[677] For the latter, see esp. Leukart (1994: 169).

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neuter *-o-m*, or, at the Proto-Germanic stage, **liuhaþ-* was extended with the neuter morpheme **-an*. In either case, *liuhaþ* would be a regular outcome in Gothic.

The development in West Germanic must have been more complex, as in the case of West Germanic, it is necessary to take into consideration the adjective **leuχta-* 'bright', apparently built from the same root, and exhibiting the same dental suffix. The questions we need to answer now are the following: *a*) do both the adjective **leuχta-* and the noun **leuχta-* derive directly from the ancient amphikinetic **leuk-ot-*?; *b*) if this is so, how is it to be explained?; *c*) do the adjective **leuχta-* and the noun **leuχta-* derive from each other?; *d*) if this is true, which derives from which?

The option *a* is quite implausible, as such homonyms – one noun and the other adjective – could not be built from one source simultaneously. Usually, one develops from the other, and the general trend is *adjective* → *noun*, cf. La. *senex* 'old' → *senex* 'old man', or IE **pet-n-o-* 'winged' → 'bird', etc. The latter example, adopted from Nussbaum (1986: 13), was interpreted by Nussbaum in the following way: first, a thematic adjective **pet-n-o-* 'winged' was formed from the stem of the heteroclite **pot-r̥/*pet-(e)n-* 'wing', and then the adjective **pet-n-o-* 'winged' was substantivized to 'bird'. In the case of the West Germanic **leuχta-* 'bright' (adj.) and **leuχta-* 'light' (noun), the development must have been essentially the same as that of **pet-n-o-* assumed by Nussbaum. From the stem **leuk-t-*, the adjective **leuk-t-o-* 'bright' was formed, and it developed into WGmc. **leuχta-* 'bright'. Subsequently, the adjective **leuχta-* acquired an additional substantival meaning 'light'.

As can be seen from this discussion, although East and West Germanic preserved different variants of the original suffix (*-ot-* vs. *-t-*) and in general developed quite

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differently, both branches chose to generalize the full grade of the root, cf. **leuk-*. For the Gothic form, the choice of the full grade version **leuk-* is not unexpected, since, as mentioned before, Gothic *liuhaþ* reflects the ancient accusative stem, for which the *e*-grade of the root is regular.

In West Germanic, the root **leuχ-* has the *e*-grade of the root of the adjective **leuχ-ta-*. The *e*-grade of the adjective **leuχ-ta-* has to be explained in the following way: the parallel example of **pet-n-o-* would imply that such formations are built from the weak stem of the underlying nouns, cf. **pot-r̥/*pet-(e)n-* → **pet-n-o-* shown above. However, if this is true, instead of **leuk-t-o-* we should expect **luk-t-o-* and, accordingly, WGmc. **luχta-*. The neccesity to reconstruct **leuk-t-* (or **leuχt-*) for pre-West Germanic implies that the original ablauting paradigm **leuk-ot-/*luk-t-* was reformed as **leuk-ot-/*leuk-t-* already in pre-Germanic. It is unlikely that the original **leuk-ot-/*luk-t-* would have survived into Proto-Germanic, since in Proto-Germanic they would have developed into rather different-looking **leuχaþ-/*luχt-*. Therefore, I am inclined to believe that already at the Proto-Germanic stage, the paradigm was **leuχaþ-/*leuχt-*. The East Germanic dialects (or the East Germanic dialect from which Gothic later developed) generalized the old strong stem **leuχaþ-*. Eventually, **leuχaþ-* was thematized, and absorbed by the neuter. The exact reasons for the thematization of pre-Go. **leuχaþ-* and the transfer to the neuter class, as was said before, are not entirely

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clear.[678] In West Germanic, the noun **leuχta-* developed via an intermediate adjectival stage.

As far as the the original semantics of the *t*-stem **leuk-ot-*, obviously built with the verbal root **leuk-* 'shine' and the nominalizing suffix **-ot-* is concerned, it most probably was an action noun and meant 'shining'. Thus, it would be comparable to the amphikinetic *t*-stem **med-ot-* which was built in the same way. In the course of development, the abstract meaning 'shining' would have been concretized to 'light' and possibly reinterpreted a result noun, i.e., 'light' ← 'the result of shining'.

It has to be noted that Proto-Germanic probably also possessed another action noun built to the same root, **leuχ-*, but this action noun was built using a different suffix, cf. PGmc. **leuχ-sa-*. The noun **leuχ-sa-* is reflected in Scandinavian (> OIc. *ljós*, OSw. *liūs*). The form **leuχ-sa-* most probably is a thematized *s*-stem **leuk-es-* 'shining; light' which is attested in several other Indo-European languages, cf. Ved. *rócas-* 'light', etc. (see some additional examples in section V.11.5.2 above).

The reconstruction of two synonymous action nouns should not be considered problematic, since synonyms are quite common in languages, cf. also PGmc. **geƀō* (< */ghebh-ā/) and **gifti-* 'gift' (< */ghebh-ti-/), **đau-þu-* and **đau-þ-an-* 'death',[679] etc.

To summarize, it is possible to draw the following conclusions: despite the superficial formal and semantic similarity between (especially) the Gothic noun *liuhaþ* and late Hittite *lukkatti*, these word-forms do not derive from a single Proto-Indo-

[678] In fact, also Go. *witoþ* 'law', which goes back to an old amphikinetic *t*-stem, synhcronically is a neuter *a*-stem, but the development of these two nouns was not exactly the same (see section V.11.6 below).

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European source. Hi. *lukkatti* is an innovation, and it acquired the shape of the "dative/ locative" of a non-attested (and most probably non-existing) *t*-stem **lukkatt-* by re-interpretation. The Gothic noun, on the contrary, in the beginning had consonantal declension, but in the course of development became thematized. In West Germanic, the noun **leuχta-* is deadjectival.

V.10.6. PGmc. **witōþ-* 'law'

In Gothic, there existed another noun, in its structure somewhat similar to the noun *liuhaþ* discussed in the preceding section. The noun in question is *witoþ* 'law' (*-d-*), a neuter *a*-stem, various inflected forms of which are well attested in Wulfila's translation of the Bible.[680]

The two nouns, *liuhaþ* and *witoþ*, share the following grammatical features: both are built to verbal roots, **leuχ-* resp. **wit-*, both are formed with a primary-looking dental suffix which shows the alternation *þ* ~ *d* (*-þ* in nom./acc. sg., *-d-* elsewhere), both are neuter *a*-stems. In the case of *liuhaþ*, the thematization is likely to be secondary, and the question can be raised if the same cannot be the case with the noun *witoþ*.

The nouns *liuhaþ* and *witoþ* also have several differences: whereas the suffix of *liuhaþ* reflects IE **-ŏt-*, the suffix of *witoþ* points to **-ōt-*. In addition to this, the root of *witoþ* is in zero grade (< IE **u̯id-*), but the root of *liuhaþ* is in the full grade (< **leuk-*).

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[679] The latter word-form is reflected in Scandinavian only, but the former is reflected in all the Germanic languages, cf. Go. *dauþus*, OIc. *dauðr*, OE *dēað*, OS *dōð*, OFr. *dāth*, and OHG *tōd*.

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Just like the noun *liuhaþ*, the noun *witoþ* formally resembles an old amphikinetic *t*-stem which could be reconstructed as **u̯éid-ōt-/*u̯id-ōt-* with the generalized suffix **-ōt-* (< nom. sg.). Whereas the zero grade suffix **-t-* of the weak cases probably was eliminated from the paradigm at an early stage,[681] there is no evidence for the preservation of the suffix **-ŏt-* which apparently was generalized in the proto-paradigms of both Go. *mitaþs* and *liuhaþ*. Some thoughts about the reason for the generalization of the suffix **-ōt-* in the paradigm of **u̯eid-ōt-* vs. the generalization of **-ŏt-* in *mitaþs* and *liuhaþ* will be discussed in the following section.

The generalization of the zero grade of the root is easy to account for. In this case, one can assume that the paradigm of **u̯eid-ōt-* was influenced by the corresponding verb, the root of which generally occurred in the zero grade, viz. **u̯id-* 'see' (→ PGmc. **wit-* 'know').

As implied in Lehmann (1986: 407, further referring to F. Kluge, on which see below), Go. *witoþ* should be interpreted as a formation built with the Proto-Germanic suffix **-ōþ/đu-*. This suffix was generally used to build verbal nouns, e.g. Go. *wrat-odu-s* 'travel, trip' (← *wraton* 'travel'), as well as deadjectival nouns denoting a certain quality, e.g. *mannisk-odu-s* 'human nature' (← *mannisks* 'human'), etc. However, this theory obviously could not be built on the basis of the Gothic word-form *witoþ* itself namely because whereas the derivatives of the *wratodus* type are masculine *u*-stems, *witoþ* is a neuter *a*-stem, i.e., belongs to a different morphological class. In addition to

[680] Nom. sg. *witoþ* in St. Matthew (7:12; 11:13, etc.), acc. sg. *witoþ* in St. Matthew (5:17), etc., dat. sg. *witoda* in St. Matthew (5:18), St. John (8:17), etc., gen. sg. *witodis* in St. Luke (2:27, 16:17), etc. See a full display in M. Snædal (1998).

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this, Kluge himself did not derive Go. *witoþ* from an old *ōþ/đu*-formation. In Kluge's discussion referred to by Lehmann, Kluge wrote in the following way: "Doch ist zu bedenken, dass eventuell auch andere Formen der Wortbildung in Wörtern dieses Ausgangs[682] stecken können, wie ahd. *wizzōd* = got. *witōþ* (gen. *witōdis*) neut. 'Gesetz' zeigt" (Kluge 1899: 68), which implies that Kluge was thinking that OHG *wizzōd* and Go. *witoþ* were possibly formations of some other type, i.e., perhaps *not* ancient *ōþ/đu*-formations. Kluge does not say what they could have been, but I shall return to this passage in his book below.

The main arguments for the interpretation of Go. *witoþ* as an *ōþ/đu*-formation (as per Lehmann) come from West Germanic, where apparent cognates of the Gothic word-form are attested, namely Old High German *wizzōd* 'law; eucharist, host' and Old Frisian *witat/witad* 'host, consecrated wafer'.[683] Lehmann also adduces the Old English adverb *witod-līce* 'verily' (Lehmann 1986: 407), but its formation is in fact different. The Old English *witod-līce* is built by adding a synhronic adverb suffix *-līce* to the past participle *witod* 'certain', which ultimately belongs to the verb *be-witian* 'observe'.[684] A similar past participle is attested in Old Saxon, cf. *witod* 'certain' (← *witōn* 'observe; determine'). In both instances we are dealing with independent dialectal innovations.

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[681] See pp. 293-4 below, where a possible explanation for the elimination of the allomorph *-t-* is discussed.

[682] I.e., Old High German words with the same word-end as the genuine *ōþ/đu*-formations.

[683] There is a lot of fluctuation in the spelling of dentals in Old Frisian, which sometimes complicates the historical interpretation of Old Frisian data, cf. also *dāt* ~ *dāth* 'death' (< **đauþu-*), *gabbat* ~ *gabbath* 'commotion' (< **-ōþ/đu-*), *fōgith* ~ *fōgid* 'steward' (cf. OHG *fogat/foget*, G. *Vogt*), *thingad* ~ *thingat* ~ *thingath* 'process' (< **-ōþ/đu-*), etc. (see A. H. Cummins 1881: 22; W. L. van Helten 1890: 96ff., 123ff.). Of the two variants shown above, *witat* is the one less likely to be historically correct, as it would point to an implausible Germanic proto-form **witōt-*. On the contrary, *witad* could reflect a more plausible **witōđ-* (see below).

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The Old English verb *be-witian* and the Old Saxon *witōn*, both belonging to the second weak class, are most likely transfers from the third class.[685] First of all, their cognates in other Germanic languages, cf. Go. *witan* 'observe; take care of sth.' (3. pret. *witaida* [St. Mark 6:20]), OIc. *vita* 'observe; check' (3. pret. *vitti*) and OHG *gi-wizzēn* 'understand' belong to the third class, and second, both in Old Saxon and in Old English, third class weak verbs commonly transfer to the second class.[686] After this transfer, the old past participle built according to the third class pattern (not attested) was reformed in order to conform to the formal requirements of the second weak class, and regularly became *witod.*

Let us return to the Old High German and the Old Frisian forms *wizzōd* resp. *witad.* Although OHG *wizzōd* and OFr. *witad* synchronically are *a*-stems just like Go. *witoþ*, they have been interpreted as *ōþ/đu*-formations based on their structural similarity to such nouns as OHG *klagōd* 'complaining' (← *klagōn* 'complain'), *wegōd* 'salvation; help' (← *wegōn* 'save, help'), *weinōt* 'crying' (← *weinōn* 'cry, weep'), OFr. *somnath* 'collection, gathering' (← *samnia* 'gather, collect' [< PGmc. **sam(a)nōn*]), *tilath* 'cultivation' (← *tilia* 'cultivate land' [< **tilōn*]), etc. These nouns are generally considered to be *ōþ/đu*-formations built to weak verbs belonging to the second class (*ōn*-class), exactly like Gothic nouns of the *wratodus* type mentioned previously, which are built using this suffix, and in which the vowel *-u-* can still be seen. However, at this

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[684] E. Sievers & K. Brunner (1942: 347, fn. 12).

[685] The hard-to-find Old Saxon verb *witōn* is registered in F. Holthausen's Old Saxon dictionary (Holthausen 1954), as well as de Vries' Old Icelandic etymological dictionary (see OIc. *vitaðr*). In Gallée's Old Saxon grammar, the verb *witōn* (spelled *witon*) is mistakenly classified together with the preterito-present verb *witan* as an orthographic variant (see Gallée 1910: 268).

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point the argumentation becomes circular – the Gothic noun *witoþ* is interpreted as an *ōþ/đu*-formation based on its comparison with certain West Germanic forms, which can be interpreted as *ōþ/đu*-formations only on the basis of their comparison with Gothic. But, as was mentioned above, Go. *witoþ* and nouns of the *wratodus* type are morphologically entirely different, and there is no obvious way in which *witoþ* could be explained as a reformed (neuter?) *u*-stem. From the morphological point of view, *witoþ* is isolated, and is more likely to be an archaism. On the contrary, the Old High German noun *wizzōd* and the Old Frisian *witad* can be interpreted in more than one way, namely, as *u*-stems, as *a*-stems, and equally easily as consonantal stems, since the stem marker cannot be seen any more. After the syncope of the unstressed short vowels, nouns began to fluctuate between stem classes, and members of smaller classes were usually absorbed by larger and more regular classes. The transfer of Old High German *ōþ/đu*-formations over to *a*-stems is only one out of many examples.[687] In Old Frisian, nouns ending in the consonants *-t(h)* and *-d* derive from different sources, e.g. *dāt(h)* 'death' < **đauþu-* (*tu*-stem), *megith* 'maiden' < (ultimately) **magh-ot-* (cons. stem), *fogith* 'steward' ('neo-root noun'; ultimately ← La. *vocatus* < **-to-* [past ptc.]),[688] etc.

Therefore, the three nouns *witoþ*, *wizzōd* and *witad* can in fact be interpreted in more than one way. Since they have more or less the same meaning and share a number of morphological features, it would be tempting to assume that they also derive from a single ancestor. Three options then become available: the first option would be to take

[686] Gallée (1910: 266), Krahe/Meid (1969: 124-5), Campbell (1959: 337-41), recently Jasanoff (2002-3: 134).
[687] See Braune (1963: 202-4).

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them at face value and interpret them as original *a*-stems; according to the second option, these nouns reflect ancient *ōþ/đu*-formations; and the last option is to interpret them as ancient *t*-stems.

The first theory is practically worthless, as it is merely based on the synchronic shape of these nouns, and does not explain their history. Some of the shortcomings of the second option were mentioned above, but here it should be discussed in more detail. Although the Old High German *wizzōd* and the Old Frisian *witad* formally could be interpreted as *ōþ/đu*-formations, the Gothic *witoþ* cannot. As was mentioned previously, the genuine Gothic *ōþ/đu*-formations are masculine, and *witoþ* is neuter. No neuter *ōþ/đu*-formations have ever existed in Germanic, and *witoþ* cannot be one either. On the contrary, as was shown above, the West Germanic forms can be easily interpreted as ancient consonantal formations, which speaks in favour of the last, namely the third option.

V.10.6.1. IE **u̯idōt* – an Ancient Collective?

In the previous section I proposed that the Gothic noun *witoþ* along with OHG *wizzōd* and OFr. *witad* reflect a consonantal noun **witōþ/đ-*[689] 'law' or, in terms of Indo-European, **u̯id-ōt-*. It appears to be a primary formation, built from the verb root **u̯eid-* 'observe, see' and the nominalizing suffix **-ŏt-*. From the derivational point of

[688] Possibly via Old High German, cf. OHG *foget* 'steward' (for the etymology see Kluge/ Seebold 1995: 866).

[689] The voiced variant **-ōđ-* is continued in Gothic, cf. *wit-od-is* (gen. sg.) and Old Frisian, in which *d* is a regular reflex of Proto-Germanic **đ*. On the contrary, the Old High German form *wizzōd* points to a voiceless **-ōþ-*.

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view, it looks like the nouns **leuk-ŏt-* and **med-ŏt-*, both of which are primary derivatives of verbal roots. The original meaning of the noun **u̯id-ōt* would have been 'things observed', which was eventually narrowed down to 'rules, laws', as attested in both Gothic and Old High German. The meaning 'eucharist; host', attested in Old High German and Old Frisian, is clearly secondary, and it developed after the Christianization of these peoples. The semantic change from 'law' to 'eucharist; host' can be easily understood, as the sacraments certainly were to be "observed" by practicing Christians adhering to the church law.

From the formal point of view, the common ancestor of Go. *witoþ*, OHG *wizzōd* and OFr. *witad* has undergone multiple reformations, all of which took place at a rather early date, viz., still in Proto-Germanic.

Although this word is not attested outside Germanic, it cannot be a dialectal innovation created at the Proto-Germanic stage. This is indicated both by the paucity and common reinterpretation of such formations, as well as the alternation between the voiceless *þ* and the voiced *đ* in the suffix, which indicates that the formation is old enough to have undergone Verner's voicing. Consequently, the Proto-Germanic noun **witōþ/đ-* has to go back, at the very least, to late Indo-European.

Since, as was noted above, Proto-Germanic **witōþ/đ-* shows evidence for Verner's voicing, the original paradigm must have had mobile accentuation of the type found namely in amphikinetic formations. The original (idealized) shape of the Indo-European ancestor of PGmc. **witōþ/đ-* should then be reconstructed in the following way: nom. **u̯éid-ōt*, weak forms **u̯id-t́-*. The suffix allomorph *-t-* was eliminated at an

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early stage, possibly because **u̯id-t-* would have regularly become **u̯itt-* by voicing assimilation and further to **u̯iss-*, which semantically this would have become opaque. In order to distinguish the root from the suffix, the weak stem was reformed following the example of the strong forms and adopting the suffix **-ōt-* from the nominative case. However, the weak forms did not adopt the full grade root allomorph **u̯eid-*. The allomorph **u̯eid-* was eliminated from the paradigm, probably by influence from the etymologically related verb(s), from which eventually PGmc. **witan-* 'know' (pret.-pres.) and **witan-* 'observe' (the third weak class), both having multiple forms with the root in the zero grade, developed.

As a result of these changes, at a certain point the paradigm must have contained forms like the following: nom./acc. **u̯íd-ōt*, gen. **u̯id-ōt-és*. In Proto-Germanic, the corresponding forms would have become **wít-ōþ* resp. **wít-ōđ-iz*. The word-final **þ* would have fallen off by Germanic sound laws. The voiced *đ* must have been introduced into the nominative from the oblique stem. Once *đ* was introduced, it probably became devoiced in word-final position to *þ*. This new paradigm was further reformed, adopting the thematic declension still at the stage of the protolanguage, and it is this fully reformed paradigm that is reflected in Gothic. In the West Germanic branch, the development was slightly different. In the dialect from which Old High German ultimately developed, the voiceless suffix **-ōþ-* was generalized throughout the paradigm. On the contrary, Old Frisian points towards the generalization of the voiced **-ōđ-*.

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Now the question can be raised why the noun **witōþ* did not develop like **nefōþ* 'nephew', which, as was said in section V.10.1.5 above, was absorbed by *n*-stems. The reasons for the different development may have been several. However, it is most likely that **nefōþ* and **witōþ* were morphologically different, the latter being an old collective. The existence of this category in Indo-European is nowadays widely accepted.[690] Such an interpretation is quite attractive for PGmc. **witōþ-*, and perhaps this was the main reason why it developed differently.

The interpretation of **u̯id-ōt* as collective would provide an explanation why it became neuter.[691] Since other early Germanic collectives, such as the terms for 'water' and 'fire',[692] etc. were built to neuters, as well as having in mind that in Indo-European itself collectives were normally built to neuters, one is inclined to think that also the noun **u̯id-ōt* was perceived as a neuter collective. It must have retained its neuter collective status long enough not to have undergone the same development as other *ōt*-formations (like **nep-ōt-*, **magh-ōt-*, **med-ōt-* etc.).

Although Kluge did not interpret OHG *wizzōd* and Go. *witoþ* as collective formations, he may have been one step away from such an interpretation, as in the same paragraph he raised a rhetorical question about the relationship between the Old

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[690] For an early discussion, see Schmidt (1889: 1-12 and further). For more recent work, see Eichner (1985) and Melchert (2000, especially pp. 61-8).

[691] In Old High German, *wizzōd* occassionally appears as masculine, and in Old Frisian it is always masculine. The switch to the masculine gender can be easily understood as analogical after the *ōþ/đu*-formations, which at the dialectal level were formally undistinguishable.

[692] For the development of the noun 'fire' in the Germanic languages, see J. Hilmarsson (1988-9: 38-40).

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English collective noun *weorod* 'troop' and the singulative *wer* 'man' (Kluge 1899: 68).[693]

The historical interpretation of the noun **ǫ̱id-ōt* presented in this and the previous sections provides plausible additional evidence for the existence of collectives in Proto-Germanic, and adds one more item to the host of Indo-European collectives. The special significance of the collective **ǫ̱id-ōt* for *t*-stems is that it provides so far the only piece of evidence for the existence of such formations among amphikinetic *t*-stems. The collective **ǫ̱id-ōt* is not necessarily a very early formation going back to early Proto-Indo-European. However, it is apparently old enough to be considered pre-Germanic.

V.10.7. Conclusions for Chapter V

After having studied a number of amphikinetic *t*-stems, the following features have emerged: *a*) whereas some amphikinetic *t*-stems are primary, others are secondary (and even tertiary), *b*) amphikinetic *t*-stems can be built to verbal, adjectival and nominal roots/stems, *c*) the amphikinetic *t*-stems built to verbal roots/stems either have the semantics of action nouns or can be interpreted as having had it at an earlier stage, *d*) the amphikinetic *t*-stems built to adjectival and nominal roots/stems denote either

[693] However, OE *weorod* cannot be an ancient Proto-Germanic or Proto-Indo-European collective of the noun **wer-* (IE **ǫ̱iH-ró-*) 'man' in the same way as **witōþ* (IE **ǫ̱id-ōt*) was, because the noun **ǫ̱iH-ró-* was not a *t*-stem. As far as I can tell, the suffix **-ōt-* was not used to form collectives to nouns belonging to other stem-classes. Therefore some different origin must be assumed for OE *weorod*.

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abstract nouns of quality, or concrete nouns which cannot be analyzed as original action nouns.

The analysis of data implies that the suffix *-*ŏt*- was nominalizing, but the semantics of the *t*-stems built with this suffix mostly depended on the type of the root or the primary/secondary stem to which the suffix *-*ŏt*- was added. If the root/stem was verbal, the resulting amphikinetic *t*-stem was an action noun (but could become concretized in the course of development). If the root/stem was adjectival/nominal, the resulting *t*-stem was either a concrete noun or an abstract quality noun (the latter could become concretized as well, cf. **di̯eu̯-ŏt*-).

As far as the original Indo-European ablaut is concerned, it has been partially preserved in Indo-Iranian, where the noun *nápāt*- exhibits the zero grade of the suffix in the oblique forms, cf. YAv. *naptō* 'grandson' (gen. sg.; < **nep-t-é/ós*), OAv. *nafšu* (loc. pl.; < **nep-t-su*), Ved. *náptṛbhis* (< **nep-t*-).[694] Otherwise, traces of the mobile accent can be seen in Germanic, cf. Go. *witod*- 'law', OFr. *witad* 'eucharist' (< **-ōđ*- ← pre-Gmc. **u̯id-ōt-*) vs. OHG *wizzōd* 'eucharist' (< **-ōþ*- < **u̯íd-ōt*-), etc.



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VI. Concluding Remarks

VI.1.

This dissertation was dedicated to the study of the structure, the semantics and the development of primary Indo-European simplex and compound *t*-stems. In the following sections, I would like to review the main conclusions arrived at in this study.

In chapter I (Introduction), I surveyed the classifications of the Indo-European primary *t*-stems according to their ablaut that have been presented by Rieken (1999), Irslinger (2002) and Nussbaum (2004a). Although Rieken (followed by Irslinger) had originally divided the Proto-Indo-European primary *t*-stems into four ablaut classes, viz. acrostatic, protero-, hystero- and amphikinetic, in a subsequent study of *t*-stems Nussbaum reduced their number to two, viz. the acrostatic (**nók$^{\mu}$-t-/*nék$^{\mu}$-t-* type; see sections I.3 – I.4) and the amphikinetic *t*-stems (**nepōt-* type).

The reason for the reanalysis of Rieken's classification was that the protero- and the hysterokinetic *t*-stems reconstructed by Rieken exhibited unusual morphological features not shared by other Indo-European protero- and hysterokinetic formations. The strong forms of Rieken's proterokinetic *t*-stems were reconstructed with the suffix in the *o*-grade in the strong forms, cf. **tég-ōt(-s)*.[695] However, according to Schindler's system of Indo-European ablaut (1975b), the suffix of the Indo-European proterokinetic formations has to appear either in the *e*-grade or the zero grade, cf. IE **séh$_1$-mn̥/*s(e)h$_1$-mén-* 'sowing' (> La. *sēmen* 'seed'),[696] **pér-tu-/*pr̥-téu-* 'passage' (> OIc.

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[694] The syllabic *r̥* is secondary.
[695] Rieken (1999: 92).
[696] The expected zero-grade †*sh$_1$-* in the root of the weak forms is not attested.

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fjǫrðr 'fjord', Av. *pərətu-/pəšu-* 'passage'), **péh₂-u̯r/*ph₂-u̯én-* (→ Hi. *paḫḫur/ paḫḫuen-* 'fire') etc.

Rieken's hysterokinetic *t*-stems commonly exhibited the *e*-grade of the root, cf. Gk. ἑρπ-ητ- 'shingles', ἐχ-ητ- 'wealthy man', OIr. *cing* 'warrior' (gen. *cinged* < **kengh-*) etc.[697] According to Schindler's system, the roots of hysterokinetic formations originally normally had the zero grade (or, more rarely, the *o*-grade), cf. **si̯uH-mén-* 'sewing' (> Gk. ὑμήν 'membrane'), **h₂uks-én-* (*/h₂ugs-én-/) 'calf, ox' (> Skt. *ukṣán-* 'bull', OIc. *uxi* 'ox'), **poh₂i-mén-* 'shepherd' (> Gk. ποιμήν, Lith. *piemuõ* 'id.') etc.

Since the *t*-stems that exhibit features not found among the traditionally reconstructed Proto-Indo-European ablaut types normally co-exist with thematic stems, Nussbaum proposed that some of them might be secondary de-thematic formations, built by adding the nominalizing suffix *-(o)t-* to thematic stems (cf. **serp-e-* + *-ot-* > **serpēt-* > Gk. ἕρπ-ητ-, **teg-e-* + *-t-* > La. *teget-* etc., see section I.3).

Some of Rieken's proterokinetic *t*-stems with the suffix *-ot-* are likely to reflect primary Indo-European *t*-formations; however, they are more plausibly interpreted as original amphikinetic, and not proterokinetic *t*-stems, cf. Go. *mitaþs* 'measure' (< IE **med-ot-*; see sections V.10.2 – V.10.2.1), *magaþs* 'maid' (< **magh-ot-*; see sections V.10.4 – V.10.4.1).

From the present study, the following general picture of the primary Indo-European *t*-stems can be drawn: in the protolanguage, primary *t*-stems occurred as simplex formations, cf. **nók^u̯-t-* 'night' (see section III.3), **sós-t-* 'rest' (see III.4), **nép-ot-*

[697] Rieken (1999: 86-7).

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'offspring' (see V.10.1), **léuk-ot-* 'light' (see V.10.5.2) and as final members of compounds, cf. IE *°-*h*$_1$*i-t-* 'going', *°-*doh*$_3$*-t-* 'giving', *°-*kṗi-t-* 'dwelling', *°-*dheh*$_1$*-t-* 'placing' etc. (see section IV.12). Primary *t*-stems were built to verbal roots, and originally most probably were verbal abstracts. Although the dialectal reflexes of Indo-European *t*-stems generally denote concrete nouns (cf. Gk. νυκτ- 'night', La. *dōt-* 'dowry', Go. *liuhaþ* 'light' or *magaþ* 'maiden' etc.), this has to be interpreted as a semantic innovation. The concretization of the original abstract meaning is a commonly attested phenomenon; however, the necessity to reconstruct the Proto-Indo-European primary *t*-stems as original verbal abstracts arises from the method of their derivation. An original verbal abstract meaning is reconstructed for a large number of primary nominal formations built to verbal roots, e.g., for *ti*-stems (**mr̥-ti-* 'dying' [→ Lith. *mirtìs* 'death'], **mn̥-ti-* 'thinking' [→ Skt. *matì-* 'thought']), *men*-stems (**seh*$_1$*-mn̥* 'sowing' [→ La. *sēmen* 'seed'], **g*^{u}*elh*$_1$*-mn̥* 'throwing' [→ Gk. βλῆμα 'throw']), *s*-stems (**k*^{u}*eit-es-* 'perceiving' [→ Skt. *cétas-* 'intellect']) etc. In fact, primary derivatives built to verbal roots can also have an agentive meaning, e.g. Gk. φώρ 'thief' (< **bhór-* ~ IE **bher-* 'carry'), Skt. *dā́tar-*/*dātár-*, Gk. δώτωρ/δωτήρ 'giver' (< **doh*$_3$*-te*/*or-* ~ **deh*$_3$- 'give'), Lith. *káltas* 'chisel' (< virtual **kolh*$_2$*-to-* ~ **kelh*$_2$- 'beat'), OCS млатъ 'sledge' (< virtual **molh*$_2$*-to-* ~ *melh*$_2$- 'grind down') etc. However, based on the synchronic meaning of the dialectal reflexes of the Indo-European *t*-stems, they are more plausibly interpreted as concretized original verbal abstracts rather than agentives, cf. La. *dō-t-* 'dowry' (← *'[act of] giving', not *'giver, the giving one'), Hi. *šaš-t-* 'rest' (← *'[act of] resting, sleeping', not *'the resting, sleeping one'), Go. *mit-aþ-*

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'measure' (← *'[act of] measuring', not *'the measuring one') etc. Even in those instances when the reflexes of *t*-stems are better interpreted as agentive nouns, e.g. Gk. φω-τ- 'hero' (← *'the shining one' ~ *$bheh_2$- 'shine'), the agentive meaning is better explained as secondary, and on the basis of comparison with nouns like La. *dō-t-* etc., the original meaning should be reconstructed as that of a verbal abstract (for a more detailed discussion, see section III.6.1).

Unlike simplex *t*-stems, *t*-stem compounds often have an agentive meaning, e.g. Gk. ὠμο-βρω-τ- 'eating raw flesh', Skt. *pari-kṣí-t-* 'dwelling around, surrounding', Av. *tara-dā-t-* 'prevailing' (vel sim.), etc.[698] However, all such compounds can be interpreted as original bahuvrīhis, and the final members of bahuvrīhis often are historical verbal nouns (see section IV.14). According to the present study, the *t*-stems that are used as final members of agentive compounds can be interpreted as original verbal abstracts as well. Also in those instances when a *t*-stem compound has a concrete meaning, e.g. La. *locuplēt-* 'wealthy', *pedit-* 'foot soldier', Hi. *kapart-* 'mouse, rodent' etc., the agentive meaning can be reconstructed, cf. *'filling *locus*' resp. *'walking on foot' (see a more detailed discussion in section III.5.5) resp. 'bringing [food] together' (see section III.4.4) etc. The final members of such compounds can be interpreted as *t*-stem verbal abstracts as well.

When the ablaut of the primary Indo-European *t*-stems is compared with the ablaut of other stem classes, it appears that *t*-stems behave differently from the other classes. Among all the Indo-European primary athematic formations, *t*-stems alone form only

[698] For the discussion of the dialectal reflexes of *t*-stem compounds, see sections III.4.4, III.5.5, III.6.5, IV.3.1 – IV.3.4, IV.12 – IV.14.

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acrostatic *ó/é* and amphikinetic formations.[699] With regard to ablaut, *t*-stems are quite different from *ti*-stems which both are quite close to *t*-stems semantically and even have been interpreted as descendants of *t*-stems (cf. Grundriß 428). As is well known, *ti*-stems were normally proterokinetic, although a small number may have been acrostatic (see Vine 2004: 374-6). The comparison of the ablaut of *t*- and *tu*-stems yields even more awkward results, as *tu*-stems appear to have formed only proterokinetic formations in the protolanguage.

The reason for the unique behaviour of *t*-stems is quite obscure and has not been studied in detail. The ablaut differences between *t*-stems and *ti/tu*-stems on the one hand and the differences between all the Indo-European stem classes in general require a special study.

VI.2.

In chapter II, I surveyed various historical interpretations of the suffixes *-t-* and *-ot-*. In the course of time, the suffix *-t-* has been interpreted as an original participial suffix (Bréal), a weakened original suffix *-ti-* (Schleicher), *-to-* (Streitberg), *-et-* (Meillet, Lindeman), and as an original non-alternating *-t-* of verbal abstracts (Brugmann, Schindler, Rieken, et alii). Whereas the interpretation of the suffix *-t-* as an original participial suffix, or a weakened *-ti-*, *-to-* or *-et-* meets with various morphological or phonological problems discussed in more detail in chapter II, the reconstruction of an original Indo-European suffix *-t-* as a non-alternating suffix of verbal abstracts is

[699] For a comprehensive overview, see Widmer (2004: 49ff.).

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plausible both from the phonological and the morphological point of view. Since the roots of many primary *t*-stems exhibit the *o*-grade, and in the particular case of the noun **nók^{u}-t-* 'night' traces of an earlier *ó/é* alternation in the root can be observed, this implies their original acrostatic inflection. The constant appearance of the suffix *-t-* in the zero grade agrees well with the behaviour of suffixes in the Indo-European acrostatic formations as well. In a number of cases, when *t*-stems built with the non-alternating suffix *-t-* exhibit roots in the zero grade (cf. pre-Proto-Italic **g^{u}ṛH-t-* 'praising'), the zero grade of the root can be explained as secondary (see section III.4.1 and below).

As for the historical interpretation of the suffix *-ot-*, Rasmussen proposed that at least in certain instances, *-ot-* reflects an earlier **-oc-* with an early Proto-Indo-European affricate /c/.[700] However, as has been shown in sections II.9 – II.9.2 of this dissertation, the existing data do not provide sufficient evidence for the reconstruction of an early Indo-European amphikinetic suffix **-oc-*.

The study of the data allows for the reconstruction of two suffixes, a non-alternating *-t-* and an ablauting *-ot-* which can also occur in the lengthened grade (*-ōt-*, cf. **nep-ōt* 'offspring') or the zero grade (*-t-*, cf. **leuk-t-* in West Germanic **leuχta-* 'light', or Ved. *dyút-* 'ray of light' < **dḭu-t-* ~ Hi. *šiu̯att-* 'day' < **dḭeu̯-ot-*). They probably were quite similar semantically, as both acrostatic and amphikinetic primary *t*-stems could be built to verbal roots and originally denoted verbal abstracts, cf. **dóh$_3$-t-* 'giving' (~

[700] Rasmussen (1994).

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√*deh_3- 'give'), *$sós$-t- 'resting, sleeping' (~ √*ses- 'sleep'), *$méd$-ot- 'measuring' (~ √*med- 'measure'), *$léuk$-ot- 'illuminating' (~ √*$leuk$- 'shine') etc.[701]

As the suffixes -t- and -ot- were used in secondary (non-primary) derivation, both had a more general nominalizing meaning, and they could be added to nominal, adjectival or verbal stems, cf. Hi. *išpant-* 'night' (← loc. *$k^{u}sp$-$én$ 'at night' + -t-),[702] Ved. *harí-t-* 'tawny mare' (← *hári-* 'tawny' < *$ĝhelh_3$-i-), Gk. ἕρπ-ητ- 'shingles' (← ἕρπ-ε- 'creep' < *$serp$-e-), Hi. *šiu̯att-*, Ved. *dyút-* (← *$di̯$-$eu̯$- [u-st. adjective or noun][703] + -ot-).

The similarity between the functions of the suffixes -t- and -ot- is quite puzzling, especially having in mind that the protolanguage possessed a large number of suffixes that could form action nouns. Theoretically, one could speculate that amphikinetic t-stems originally were the collectives of acrostatic t-stems. Thus, one could imagine that *$népōt$ originally meant 'offspring' collectively rather than '(single) descendant', or that *$léukōt$ 'light, shining' originally was a collective, semantically comparable to such collectives as $u̯édōr$ 'water' existing beside the acrostatic *$u̯ódr̥$, and perhaps *$séh_2u̯ōl$ 'sun' existing beside the proterokinetic *$séh_2u̯l̥$. In addition to this, the original collective meaning should probably be reconstructed for the t-stem *$u̯id$-$ōt$ (> Go. *witoþ* 'law'; see section V.10.6). If it were possible to prove that the amphikinetic t-stems were the collectives of acrostatic t-stems, one could probably say that the most archaic type of t-stems were the acrostatic formations of the *$nók^{u}$-t- type. However,

[701] However, the original meaning of the noun *$nepōt$- is not clear since the meaning of the root *nep- is uncertain.
[702] See Nussbaum (2004a).
[703] See section V.10.2 above.

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this hypothesis is both difficult to prove and raises various complications. First of all, not all of the potential *t*-stem collectives have clear etymologies, cf. **nép-ōt* 'offspring'. As was said in section V.10.1.7, it is unclear whether the root **nep-* is verbal or nominal. If it were nominal, the reconstruction of an original acrostatic ***nóp-t-* would become problematic. In addition to this, if the amphikinetic *t*-stems were to be interpreted as collectives, the respective acrostatic *t*-stems should have probably had concrete rather than abstract meanings. Since some of the amphikinetic *t*-stems are clearly old, this would mean that the acrostatic *t*-stem verbal abstracts should have become concretized at a rather early stage. Another serious issue is that no doublets of acrostatic and amphikinetic *t*-stems are attested, i.e., there are no reflexes of amphikinetic $^{\times}$*ses-ōt-*, $^{\times}$*nég^{u}-ōt-*/$^{\times}$*nék^{u}-ōt-* etc. beside the acrostatic **sós-t-*, **nók^{u}-t-* etc., or, vice versa, no reflexes of acrostatic $^{\times}$*lóuk-t-* (→ $^{\times}$*luk-t-*), $^{\times}$*mód-t-* beside the amphikinetic **léuk-ōt-*, **méd-ōt-* etc.

VI.3.

The Proto-Indo-European acrostatic and amphikinetic *t*-stems have not preserved their original ablauting inflection intact in the dialects. In the course of development, all *t*-stems underwent paradigmatic reformation, the most common reform being the elimination of vocalic alternations in the root and the suffix morphemes, which led to simplified and more uniform paradigms. The reformation of ablaut probably took place at several stages, the earliest stage being the reformation of the acrostatic locatives which ideally should have had the roots in the zero grade and an accented suffix **-ét-* in

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the full grade (see section III.3). By paradigmatic pressure, such locatives would have introduced a new full grade into the root, and deleted the full grade in the suffix. This is probably why even old frozen *t*-stem locatives like Lith. *artì*, Gk. ἄρτι do not preserve the full grade of the suffix (see section III.8.6).

The ablaut of the root morphemes was reformed in several different ways, and acrostatic *t*-stems do not always agree with amphikinetic *t*-stems (see below). In addition to this, the ablaut of simplex acrostatic *t*-stems most probably developed slightly differently from the ablaut of acrostatic *t*-stems that occurred as final members of compounds.

The ablaut of simplex acrostatic *t*-stems was in most instances reformed in the following way: *t*-stems with *-eC-* roots (including both *CeC* and *CReC* roots) eliminated the *e*-grade and generalized the *o*-grade throughout the paradigm, whereas *t*-stems with *-eR(C)-* roots (including *eRH* roots) in most dialects exhibit a secondary zero grade of the root. Baltic shows different behaviour here, cf. the discussion of the Lithuanian *samtis* type nouns in section III.8.5. However, it may be that the attested *o*-grade in the nouns of the *samtis* type is not the preserved original Proto-Indo-European *o*-grade, but is analogical after nouns with *CeC* roots (like *sagtis*, *naktis*) in which the *o*-grade would be expected.

The reformation of the original acrostatic R(ó/é) ablaut as non-alternating R(z) was most probably due to an early merger of acrostatic *t*-stems and root nouns, whereby the suffix *-t-* was reinterpreted as a root enlargement and the original *t*-stems became root nouns (see section IV.14).

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The ablaut of simplex amphikinetic *t*-stems shows less regularity than the ablaut of acrostatic formations, but also among the amphikinetic nouns, one can observe the tendency to preserve the full grade in *-eC-* roots, cf. Go. *magaþs* (< **magh-ot-*), *mitaþs* (< **med-ot-*), Ved. *nápāt* 'grandson' (< **nep-ōt*). The roots of the *-eR(C)-* structure are attested both with the full and the zero grade, cf. Go. *liuhaþ* 'light' (< **leuk-*), but *witoþ* 'law' (< **u̯id-*). Also the suffix *-ot-* has been preserved in all three variants, viz. the normal *o*-grade, the lengthened *ō*-grade, and the zero grade (for the latter, cf. Av. *nəptō* 'grandson's' [gen. sg.], OHG *lioht* 'light' [< **leuk-t-*]).

VI.3.1.

There are reasons to assume that beside simplex *t*-stems, there existed *t*-stem compounds in Proto-Indo-European. Compound *t*-stems are well attested in several Indo-European branches, viz. Greek, Italic, Indo-Iranian. Reflexes of at least one ancient *t*-stem compound are attested in Anatolian (cf. Hi. *kapa/irt-*, Lyd. *kabrd-*). All these reflexes of ancient compounds exhibit a number of structural and semantic similarities. Compound *t*-stems tend to be agentive and the final member is normally built to a verbal root (cf. Gk. ὠμο-βρω-τ- 'eating raw flesh' [~ IE **g*ʷ*erh*$_3$- 'devour'], Ved. *kṛchre-śrí-t-* 'providing support in danger' [~ IE **k̂lei-* 'lean'] etc.). In a number of instances, the final members are built to roots of the *CeRC* type and exhibit the zero grade (cf. simplex acrostatic *t*-stems built to *CeRC* roots). The *CeRC* roots of the final members of *t*-stem compounds acquired the zero grade still in the protolanguage, as can

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be demonstrated by the laryngeal reflexes in Greek *t*-stem compounds, cf. °-βλη-τ- < *-g^{u}*ḷh*$_1$-*t*-, °-ϑνη-τ- < *-*dhn̥h*$_2$-*t*-, °-βρω-τ- < *-g^{u}*r̥h*$_3$-*t*-, etc.

The study of the root ablaut in the final members of *t*-stem compounds built to *CeC* and *CReC* roots is made difficult by the ambiguity of the data. The following *CeC* and *CReC* roots occur, all with a final laryngeal: **dheh*$_1$- 'place' (cf. Av. *tara-dāt-* 'prevailing'), **pleh*$_1$- 'fill' (cf. La. *locu-plēt-* 'wealthy'), **sneh*$_1$- 'spin' (cf. Gk. χερ-νητ- 'poor'), **deh*$_3$- 'give' (cf. La. *sacer-dōt-* 'priest') and **ĝneh*$_3$- 'know' (cf. Gk. ἀ-γνωτ- 'ignorant'). All of these compounds can be reconstructed in more than one way: according to Nussbaum (2004a), the compounds *locu-plēt-* and *tara-dāt-* are secondary de-thematic *-*pl*(*h*$_1$)-*e*-*ot*- resp. *-*dhh*$_1$-*e*-*ot*-, whereas Strunk (1995: 227) reconstructed the zero grade for Gk. ἀ-γνωτ- (cf. *-*ĝn̥h*$_3$-*t*-) and for La. *locu-plēt-* (cf. *-*pl̥h*$_1$-*t*-; accordingly, one could reconstruct Gk. χερ-νητ- as original *-[*s*]*n̥h*$_1$-*t*-). La. *sacer-dōt-* could be reconstructed both as **sakro-deh*$_3$-*t*- and **sakro-doh*$_3$-*t*-.

VI.3.2.

In section III.6.5, I proposed that the final members of the *t*-stem compounds χερ-νητ- and ἀ-γνωτ- should be reconstructed with the full grade of the root rather than the zero grade (cf. *-*ĝne/oh*$_3$-*t*- resp. *-[*s*]*neh*$_1$-*t*-). I based this reconstruction on the general tendency of *CReC* roots to preserve the full grade. When reconstructed in this way, the stem *-(*s*)*neh*$_1$-*t*- provides direct evidence for the surviving *e*-grade in an archaic *t*-stem. If this reconstruction is correct,[704] then also the reconstruction of La.

[704] For alternative views, see section III.6.5 above.

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sacer-dōt- and Gk. ἀ-γνωτ- may be reconsidered, and the second members of these compounds could be reconstructed as *-*deh*$_3$-*t*- resp. *-*ĝneh*$_3$-*t*-.

Roots in the full grade should probably be reconstructed also for the Avestan compounds in °-*dāt*- and La. *locu-plēt*-. The reason for the reconstruction of the full grade in the Avestan compounds was discussed in section IV.2.2 above. The reason for the reconstruction of the full grade in La. -*plēt*- is different. The reconstruction of an de-thematic *-*pl*(h_1)-*e*-*ot*- is implausible because semantically it would have been identical to its thematic base *-*pl*(h_1)-*e*/*o*- (see section III.5.5 above).

Accordingly, one could arrive at the following conclusion about the distribution of ablaut in acrostatic *t*-stems: whereas acrostatic *t*-stems built to *CeR*(*C*) roots appear in the zero grade both in compounds and in simplex formations,[705] the acrostatic *t*-stems built to *C*(*R*)*eC* roots generalize the *o*-grade when they occur as simplex, and the *e*-grade (from the original weak forms) when they are used in compounds.[706]



[705] Baltic is different (see section VI.3 above). Certain exceptions also exist in other dialects, cf. Old Hittite *karaitt*- 'flood' (< **ĝrṓi-t*-) and Greek δαιτ- 'meal' (< **deh*$_2$*i-t*-), both of which are built to -*eR*- roots (see sections III.4.1 – III.4.1.2 and III.6.2).

[706] It is perhaps significant that also primary heteroclitic stems show a similar type of behaviour, cf. Skt. *ū́dhar*/*ū́dhan*- 'udder' vs. *an-ūdhā́* 'udderless', *rapśád-ūdha-bhiḥ* 'with swollen udders' (< *-*uHdh-n̥*-), Gk. πεῖραρ/πείρατ- (< **per-u̯r̥*/*u̯én*-) 'end' vs. ἀπείρων, -ονος 'endless' (see Nussbaum 1986: 91f.).

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LANGUAGE ABBREVIATIONS:

AAY	Alutiiq Alaskan Yupik
Arg.	Argolic
Arm.	Armenian
Att.	Attic
Av.	Avestan
CLuv.	Cuneiform Luvian
E.	English
EGmc.	East Germanic
Gk.	Greek
Gmc.	Germanic
Go.	Gothic
Hebr.	Hebrew
Hi.	Hittite
Icel.	Icelandic
IE	Indo-European
IIr.	Indo-Iranian
La.	Latin
Latv.	Latvian
Lesb.	Lesbian
Lith.	Lithuanian
Luv.	Luvian
MCorn.	Middle Cornish
ME	Middle English
MH	Middle Hittite
MHG	Middle High German
MoG.	Modern German
MW	Middle Welsh
NH	New Hittite
OAv.	Old Avestan
OCS	Old Church Slavic
OE	Old English
OF	Old French
OFr.	Old Frisian
OH	Old Hittite
OHG	Old High German
OIc.	Old Icelandic
OIr.	Old Irish
OLith.	Old Lithuanian
OPort.	Old Portuguese
OPr.	Old Prussian
OS	Old Saxon
OSw.	Old Swedish
PAu.	Proto-Austronesian
PB	Proto-Baltic
PC	Proto-Celtic
PE	Proto-Eskimo
PGmc.	Proto-Germanic
PIE	Proto-Indo-European
PIIr.	Proto-Indo-Iranian
PS	Proto-Slavic
Ru.	Russian
Skt.	Sanskrit
Toch.	Tocharian
Ved.	Vedic
WCCI	West Canadian Copper Inuit
WG	West Greenlandic
WGmc.	West Germanic
YAv.	Young Avestan

MISCELLANEOUS ABBREVIATIONS

AU	Annals of Ulster
AV	Atharvaveda
Hesych.	Hesychius
Hom.	Homer, Homeric
Il.	Iliad
Kauś	Kauśika-Sūtra
Od.	Odyssey
RV	Rigveda
S	Sīrōza
V	Vendīdād
Y	Yasna
Yt.	Yašt

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AiGr J. WACKERNAGEL (– A. DEBRUNNER), *Altindische Grammatik.* Bänder II,1 – 2. Göttingen: Vandenhoeck & Ruprecht, 1954 – 1957.

BB *Beiträge zur Kunde der indogermanischen Sprachen.* Göttingen.

BSL *Bulletin de la Société de Linguistique de Paris.* Paris.

BTS *Supplement to An Anglo-Saxon Dictionary.* Based on the manuscript collections of Joseph Bosworth. By T. Northcote Toller (with revised and enlarged addenda by Alistair Campbell). Oxford University Press, 1921.

CD *Codex Diplomaticus Aevi Saxonici.* Opera Johannis M. Kemble. Tomus III. Londini: Sumptibus Societatis, 1845 (Kraus Reprint Ltd. Vaduz, 1964).

DIL *Dictionary of the Irish Language, Based Mainly on Old and Middle Irish Materials.* Dublin: Royal Irish Academy, 1913 – 1976.

EIEC *Encyclopedia of Indo-European Culture* (edited by J. P. Mallory and D. Q. Adams). London and Chicago: Fitzroy Dearborn Publishers, 1997.

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HS *Historische Sprachforschung*. Göttingen – Zürich.

IBS *Innsbrucker Beiträge zur Sprachwissenschaft*. Innsbruck.

IESB *Indo-European Studies Bulletin*. Los Angeles.

IEW J. POKORNY, *Indogermanisches etymologisches Wörterbuch*. Band I. Bern und München: Francke Verlag, 1959.

IF *Indogermanische Forschungen*. Berlin – New York.

ÍO Á. BÖÐVARSSON (ritstjóri), *Íslensk orðabók*. Önnur útgáfa, aukin og bætt. Reykjavík: Mál og Menning, 1994.

JIES *Journal of Indo-European Studies*. Washington, D.C.

LEW A. WALDE; J. B. HOFMANN, *Lateinisches etymologisches Wörterbuch*. 3., neubearbeitete Auflage von J. B. Hofmann. B. I – III. Heidelberg: Carl Winter Universitätsverlag, 1938 – 1956.

LIV *Lexikon der indogermanischen Verben*. Zweite, erweiterte und verbesserte Auflage bearbeitet von Martin Kümmel und Helmut Rix. Wiesbaden: Dr. Ludwig Reichert Verlag, 2001.

LKŽ *Lietuvių kalbos žodynas*. Lietuvos mokslų akademija – Lietuvių kalbos institutas. Vilnius: Mokslo ir enciklopedijų leidykla, 1968 – 2002.

LSJ H. G. LIDDELL; R. SCOTT, *A Greek-English Lexicon*. A New edition, revised and augmented throughout by Sir H. S. Jones with the assistance of R. McKenzie and with the cooperation of many scholars. Oxford: Clarendon Press. New (ninth) edition completed 1940.

Mažiulis V. MAŽIULIS, *Prūsų kalbos etimologijos žodynas*. Vols. I – IV. Vilnius: Mokslas, 1988 – 1997.

MSS *Münchener Studien zur Sprachwissenschaft*. München.

OIG R. THURNEYSEN, *A Grammar of Old Irish*. Revised and enlarged edition, translated from the German by D. A. Binchy and Osborn Bergin. Dublin: The Dublin Institute for Advanced Studies, 1946.

StBoT *Studien zu den Boğazköy-Texten*. Wiesbaden.

TIES *Tocharian and Indo-European Studies*. Reykjavík.

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ThGL ΘΗΣΑΥΡΟΣ ΤΗΣ ΕΛΛΗΝΙΚΗΣ ΓΛΩΣΣΗΣ. *Thesaurus Graecae Linguae.* Ab H. Stephano constructus. Editio nova auctior et emendatior. Vol. I – X. Londini: in Aedibus Valpianis, 1816 – 1826.

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VGkS H. PEDERSEN, *Vergleichende Grammatik der keltischen Sprachen.* Vol. I: *Einleitung und Lautlehre.* Vol. II: *Bedeutungslehre* (*Wortlehre*). Göttingen: Vandenhoeck & Ruprecht, 1909 – 1913.

ZfCPh *Zeitschrift für Celtische Philologie.* Halle (Saale).

ZvS *Zeitschrift für vergleichende Sprachforschung auf dem Gebiete der indogermanischen Sprachen.* Gütersloh.

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